FPT UNIVERSITY

Capstone Project Report

Report 4 – Software Design Document

– Can Tho, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description

A - Added M - Modified D - Deleted

II. Software Design Document

1. System Design

1.1 System Architecture

Frevia follows a three-tier, deployment-separated architecture (Figure 1).

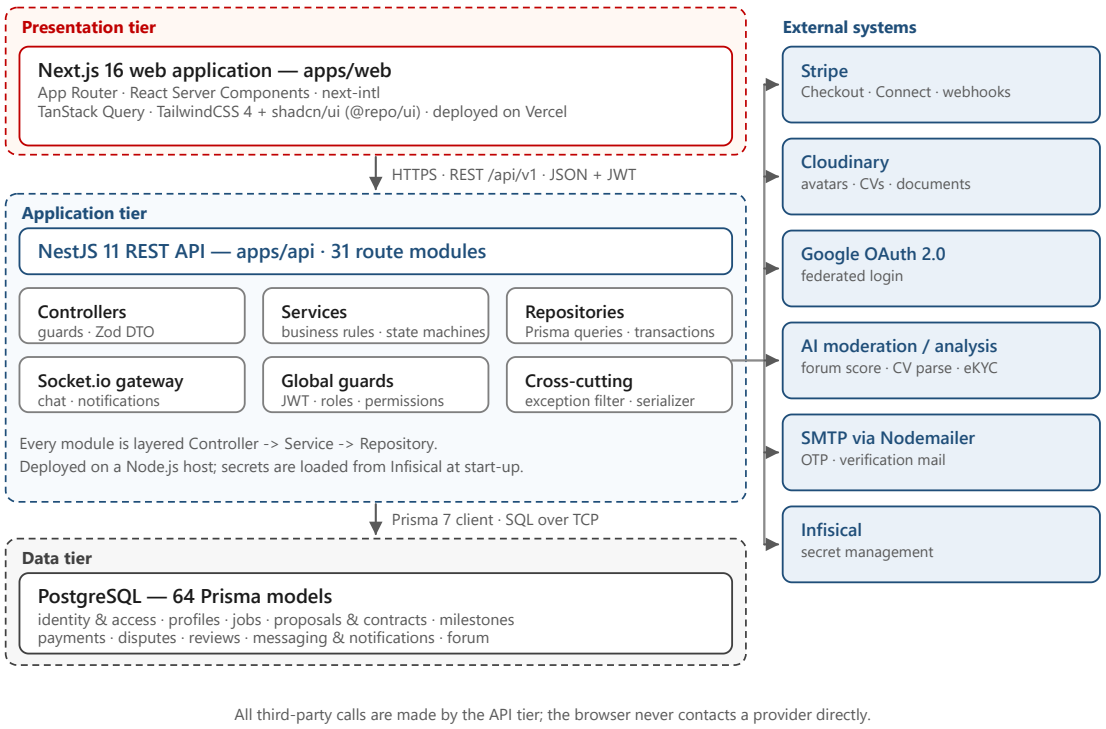

Figure 1: Overall architecture of the Frevia system

The three tiers are:

- **Presentation tier — Next.js 16 web application:** server components render public pages (job board, job detail, community forum, expert directory) and private dashboards; client components handle interactivity such as forms, likes, and chat. All server access goes through a single typed HTTP client that attaches the JWT access token and refreshes it transparently when it expires.
- **Application tier — NestJS REST API:** a modular monolith organized into 31 route modules. Every module is internally layered as Controller → Service → Repository. Controllers validate and serialize with shared Zod DTOs; services hold business rules and authorization checks; repositories encapsulate all Prisma queries.

- **Data tier — PostgreSQL via Prisma 7:** the relational store for 64 domain models. Binary files are never stored in the database but in Cloudinary; real-time transport for chat and notifications is handled by a Socket.io gateway beside the REST API.

The two applications are deployed independently (the web app on Vercel, the API on a Node.js host) and communicate only through the documented REST contract validated by the shared Zod schemas package.

Beside the three tiers, four external systems are connected to the application tier (Figure 1): **Stripe** (Checkout, Connect and webhooks) for escrow payments and payouts, **Cloudinary** for avatar, document and portfolio binaries, **Google OAuth** for federated login, and the **AI moderation/analysis service** that scores forum content, parses CVs and checks identity documents. E-mail (OTP and notifications) is delivered through SMTP with Nodemailer, and secrets are loaded from **Infisical** at start-up. None of these systems is called from the browser: the web application only talks to the API, which owns every credential and decides when a third party is contacted. The table below (Table 1) explains the API modules, and Section 1.2 explains every package of the repository.

1.1.1 Layered Architecture

The internal architecture of the API follows a strict layered pattern, enforced identically in every module (Figure 2):

- **Controller layer** — thin Routing Layer. Exposes REST endpoints, applies guards (authentication, roles, permissions), binds validated request DTOs, and delegates to a service. Controllers never query the database directly.
- **Service layer** — business logic layer. Implements use cases, ownership/authorization checks, state-machine transitions (proposal, contract, dispute), and cross-aggregation logic; it throws typed HTTP exceptions (`NotFoundException 404`, `ForbiddenException 403`, `ConflictException 409`) that a global exception filter converts to a uniform error response.
- **Repository layer** — data access layer. The only component allowed to touch Prisma; complex list queries (pagination + counting) are wrapped in `$transaction` for consistency.
- **Shared kernel** — the `@shared/types` workspace package contains Zod models, constants, permission names, and error codes shared with the web app.

Security flow: every request carries a JWT access token; the auth guard validates it and resolves the active role; role and permission guards authorize module-specific operations; refresh tokens are stored server-side in a `Session` row and can be revoked from the session management feature.

AI-assisted pipeline: user-generated content (forum posts/comments, CVs, identity documents) is passed through AI moderation or AI analysis services; forum content can be auto-approved when clean, or queued in `ModerationQueue` for administrator approval when suspicious.

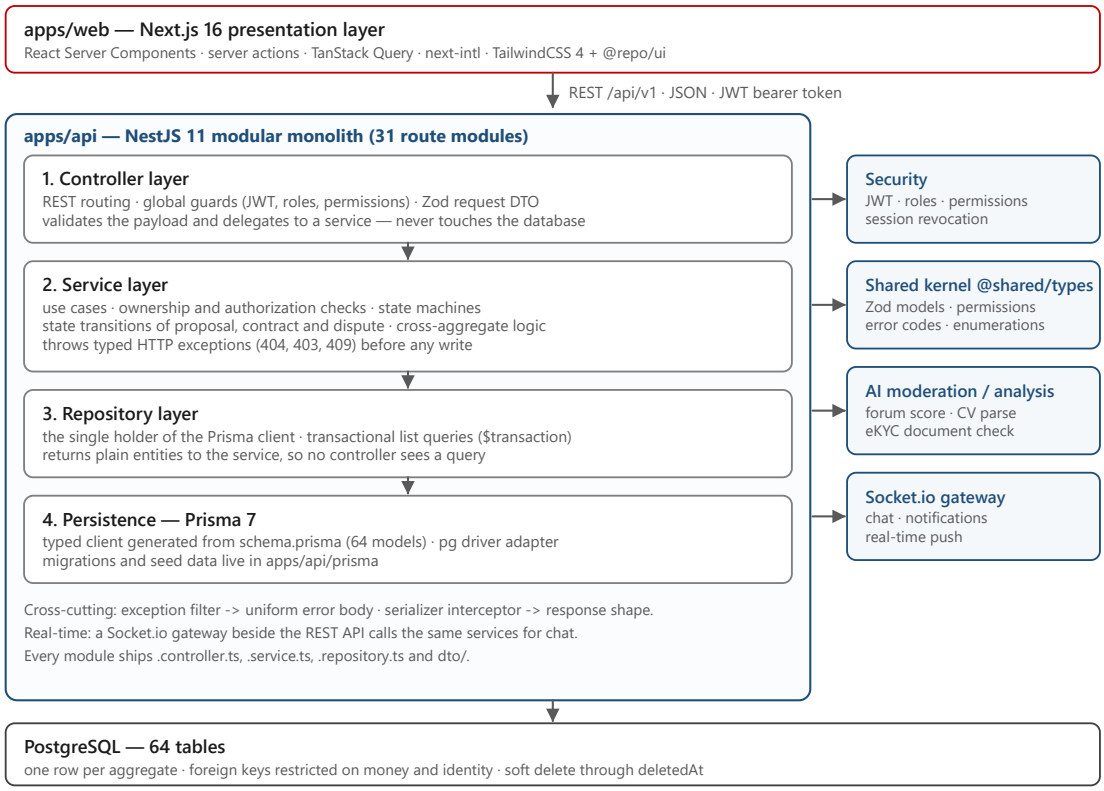

Figure 2: Layered system architecture with security and AI moderation flows

1.1.2 API Module Structure

Table 1 summarizes the 31 route modules of the API and their responsibilities.

Table 1: API route modules and responsibilities

Module	**Responsibility**
auth	registration, OTP, login, refresh token, logout, forgot password, Google OAuth, current account, role switching/joining
sessions	device session listing, detail, revocation
account-profile	general account profile, avatar, identity verification upload/AI check, client profile, social links, favorite freelancers, following
users	administrator user management and skill catalog
roles, permissions	RBAC role and permission administration
profiles	professional profile, CV upload/AI analysis, skills

Module	Responsibility
portfolios	portfolio item CRUD on profiles
expert-profile	expert profile management and public expert directory
profile-revisions	profile change request approval workflow
identity-verifications-admin	identity verification review
job-categories	public job category browsing
job-categories-admin	job category administration (soft delete + restore)
skills-admin	skill catalog administration (soft delete + restore)
browse-job	public job listing and job detail by slug
manage-job	client job management, bookmarks, skill search
proposals	proposal lifecycle: draft, submit, withdraw, interview, reject, accept
contract	contract lifecycle: create, update, sign, complete, cancel
milestone	milestone management inside a contract
milestone-file, milestone-submission	milestone attachments and work submission review
shared-file	contract shared workspace files
payments	Stripe Connect onboarding, platform fee, milestone funding, release, refund, dispute fee, transactions, webhook
disputes	dispute lifecycle (user side and admin arbitration side)
reviews	contract reviews and responses
conversations	direct messaging with real-time delivery
notifications	notification feed management
forums	forum categories and posts (public side)
forums-comment, forums-like, forums-reports	comments, likes, content reports
forums-admin	AI moderation queue, pending posts, trash restore, forum statistics
saved-search	saved job search filters
banners, banners-admin	advertisement banner delivery and administration

1.2 Package Diagram

The repository is a pnpm monorepo managed by Turborepo (Figure 3):

- `apps/web` — the Next.js application. Sub-packages group screens by actor (guest browsing, client, freelancer, admin), shared UI components (`@repo/ui`), server actions, and the i18n

message catalogs used by next-intl.

- **`apps/api`** — the NestJS application. `src/routes/*` contains the 31 domain modules described above; `src/shared` contains cross-cutting infrastructure: authentication guards and decorators, permission constants, global HTTP exception filter, Zod serializer interceptor, Cloudinary storage service, and Prisma client setup. Each domain module folder is organized as `<module>.controller.ts`, `<module>.service.ts`, `<module>.repository.ts`, DTO classes, and module-specific error factories (e.g., `forums.error.ts`).
- **`packages/shared-type`** — the single source of truth for API contracts: Zod request/response models (`model/*`), enumeration/constant definitions, and permission name lists, imported by both `apps/web` and `apps/api`.

Dependencies only point downwards: `web` and `api` both depend on `shared-type`; no route module imports another route module — cross-module data needs flow through services at the Prisma schema level.

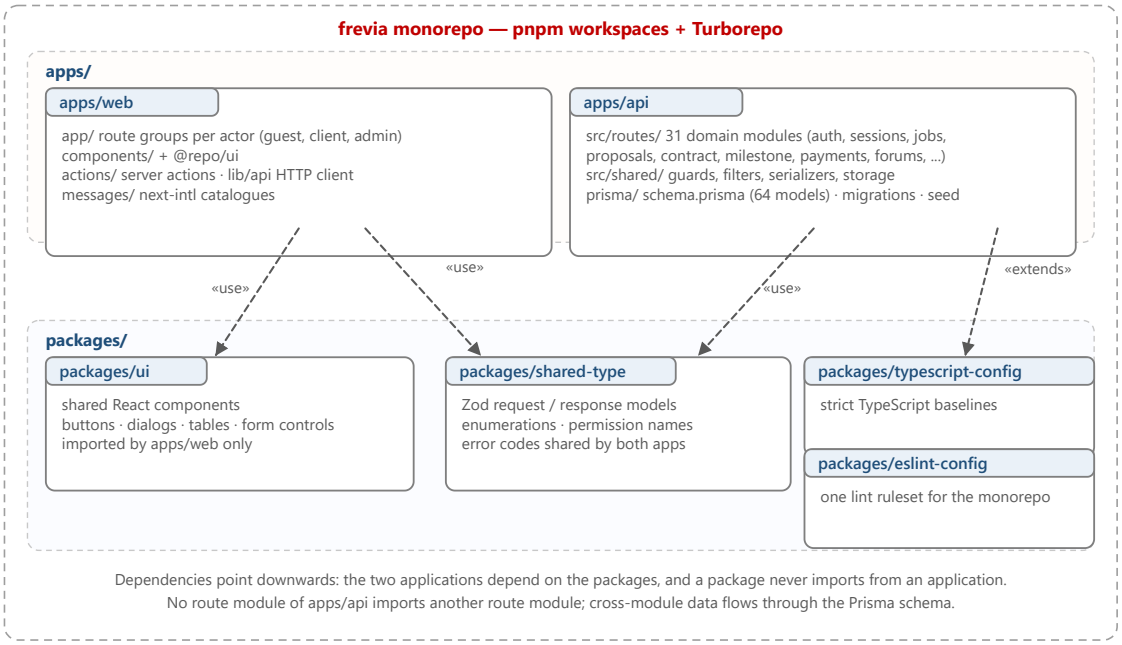

Figure 3: Package diagram of the Frevia monorepo

Package Descriptions

Table 2 lists every workspace package of the repository and its responsibility.

Table 2: Workspace packages and their responsibility

No	Package	Description
01	`apps/web`	Next.js 16 presentation layer (App Router, React Server Components, next-intl, TanStack Query, TailwindCSS 4 with shadcn/ui). Contains the screens per actor, the shared UI components, the server actions and the BFF route handlers that attach the access token.
02	`apps/api`	NestJS 11 application tier: 31 route modules laid out as Controller, Service, Repository and DTO files, together with the global guards, exception filter, Zod serializer interceptor and the Socket.io gateway used by messaging.
03	`apps/api/prisma`	Prisma 7 data layer: `schema.prisma` (64 models and their enums), migrations and seed data. The generated client is the only component allowed to query PostgreSQL.
04	`packages/shared-type`	Contract single source of truth (`@shared/types`): Zod request/response models, enumerations, permission names and error codes consumed by both applications.
05	`packages/ui`	Shared React component library (`@repo/ui`) imported by the web application so that buttons, dialogs, tables and form controls look and behave identically on every screen.
06	`packages/eslint-config`	Shared ESLint configuration (`@repo/eslint-config`) applied to every workspace so that the same rules run across the monorepo.
07	`packages/typescript-config`	Shared TypeScript configuration baselines (`@repo/typescript-config`) that the applications extend for strict compilation.

1.3 Assumptions and Risks

This system is designed based on the following assumptions:

- The Node.js runtime and the frameworks chosen for the project (NestJS 11 for the back-end and Next.js 16 for the frontend) will continue to be actively supported and maintained, ensuring long-term stability and compatibility with evolving technologies.
- The RESTful API architecture will remain a scalable and efficient choice for separating the Next.js presentation layer from the NestJS business layer.

- PostgreSQL, accessed through Prisma ORM 7, will continue to provide reliable, scalable, and secure relational data storage suitable for handling user identity, jobs, contracts, escrow payments, and forum data.
- Third-party services used by the platform — Stripe (payments and escrow), Cloudinary (file storage), Google (OAuth login), the AI moderation/analysis service, and mail delivery via Nodemailer — will stay available and API-compatible during the project lifetime.
- Modern web browsers (e.g., Chrome, Edge, Firefox, Safari) will continue to support the front-end standards required by Next.js/React 19, including HTML5, CSS3 (TailwindCSS), JavaScript ESM, and responsive rendering with React Server Components.
- The application will be deployed in environments with stable internet access to support real-time messaging (Socket.io) and online payment processing.
- Users (Guests, Clients, Freelancers, Experts, and Admins) are familiar with basic web application navigation: browsing job boards, filling forms, managing profiles, and following payment flows.

This system design is also based on the following risks:

- Any breaking change in the major versions of Next.js, NestJS, or Prisma could require refactoring of the user interface components or of the data access layer.
- Cross-browser inconsistencies or outdated device software may affect rendering or the real-time messaging experience.
- Cybersecurity threats such as credential stuffing, token hijacking, or database breaches could compromise sensitive personal and financial information if not proactively managed (the design therefore includes bcrypt-hashed passwords, JWT access/refresh tokens, session revocation, and role-based access control).
- Outages or policy changes of the integrated third-party services (Stripe, Cloudinary, Google OAuth, AI moderation) could temporarily disrupt payments, file uploads, login, or content moderation.
- Network connectivity issues could disrupt access to job proposals, contract collaboration, or checkout flows that rely on server interaction.

1.4 Design Constraints

This system should comply with the following constraints:

1.4.1 Platform Compatibility

The Frevia application must be compatible with modern web browsers (e.g., Chrome, Edge, Firefox, Safari) for all user roles including Guest, Client, Freelancer, Expert, and Admin. The frontend is implemented with Next.js 16 (App Router, React Server Components) styled with TailwindCSS 4 and shadcn/ui components; the backend is a NestJS 11 REST API. Responsive web design principles must be applied so that a consistent user experience is provided across desktop, tablet, and mobile browsers. Internationalization is achieved with next-intl so the interface can be served in multiple languages from the same code base.

1.4.2 Integration Complexity

The Frevia platform integrates a variety of components: the Next.js web application, the NestJS REST API, the PostgreSQL database (via Prisma 7 with the pg adapter), Stripe Checkout and Stripe Connect for escrow payments and payouts, Cloudinary for image and document storage, Socket.io for real-time chat and notifications, Google OAuth for federated login, e-mail for OTP delivery, and AI services for content moderation, CV parsing, and identity document verification. Communication between the client and the server is carried out through versioned REST endpoints secured with JWT access tokens and refresh tokens stored in dedicated session records; role-based access control restricts each operation to the allowed roles. Design decisions must keep these integrations loosely coupled behind services and repositories so that a provider can be replaced without rewriting business logic.

1.4.3 Performance Requirements

The application must remain responsive under concurrent usage by many freelancers and clients. Job browsing, proposal listing, forum feeds, and conversation histories must load promptly; therefore the design mandates server-side pagination for every list operation, database indexes on frequently filtered columns, optimized Prisma queries executed inside transactions for counted lists, and streaming React Server Components to reduce time-to-first-byte. Real-time features (messaging, notifications) must propagate events with minimal delay via Socket.io. Payment operations must reconcile eventual states through Stripe webhooks and explicit synchronization endpoints.

1.5 Technology Stack

Frevia adopts a RESTful API architecture to support secure, stateless, and scalable communication between the Next.js frontend and the NestJS backend. This approach provides:

- Modular, reusable API endpoints for account management, job posting, proposals, contracting, milestone escrow, dispute resolution, community forum, and administration.
- Easy extension for mobile applications or third-party integrations in the future.

- Secure access using JWT tokens with role-based authorization for every endpoint.
- Uniform use of standard HTTP methods (GET, POST, PATCH, PUT, DELETE), making the system intuitive to maintain.
- A clear separation between the presentation layer and the business logic layer.

All request/response payloads are validated and typed with shared Zod schemas packaged in `@shared/types`, which is consumed by both the frontend and the backend so that a single source of truth prevents contract drift between the two applications.

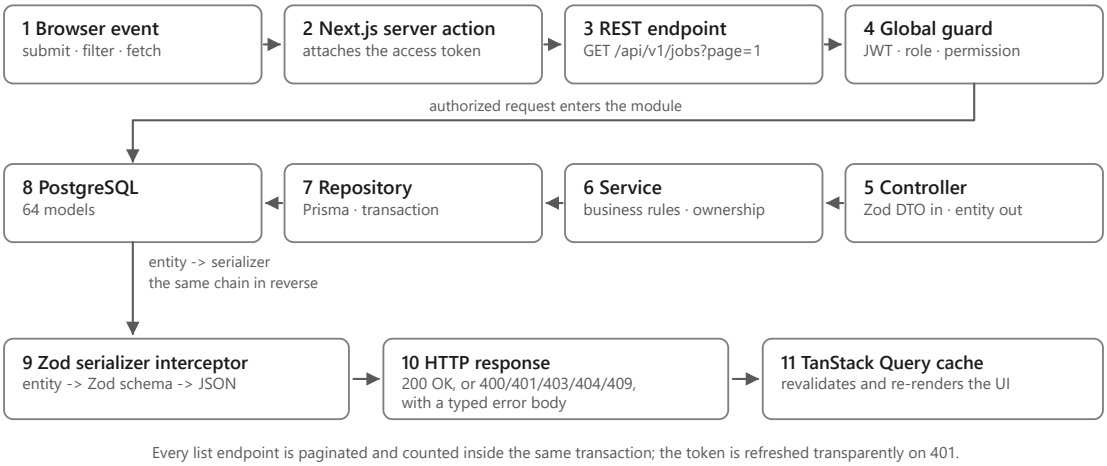

Figure 4: RESTful API communication between the Next.js frontend and the NestJS backend

1.5.1 NestJS Framework

The backend of Frevia is built with NestJS 11, a TypeScript-first, modular web framework running on Node.js. It is selected for its:

- Rich dependency-injection and module ecosystem, matching the Controller–Service–Repository layering.
- First-class decorator-based REST routing with guards and interceptors for authentication, roles, permissions, and response serialization.
- Excellent support for Zod schema validation (nestjs-zod) and typed DTOs.
- Component structure for WebSockets (Socket.io gateways) used by the messaging module.
- Straightforward integration with Prisma ORM and PostgreSQL for persistence.

1.5.2 Next.js and React 19 Frontend

The web client uses Next.js 16 with the App Router. React Server Components render public pages (job lists, job details, forum) directly from the server, while TanStack Query drives interactive islands (forms, likes, real-time chat). react-hook-form together with Zod resolvers validate user input with the same schemas the API uses.

1.5.3 Prisma ORM and PostgreSQL

Persistence relies on PostgreSQL accessed through Prisma 7. The schema defines 64 models covering identity and access, profiles, jobs, proposals, contracts and milestones, payments and disputes, reviews, messaging, the community forum, and platform configuration. Prisma's type-safe client keeps data access code verifiable at compile time.

1.5.4 Supporting Services

Payments and payouts use Stripe Checkout and Stripe Connect with webhook-driven reconciliation; binaries are stored in Cloudinary; emails (OTP, notifications) are sent with Nodemailer; Google login uses OAuth 2.0; environment configuration is loaded through Infisical so no secret is committed to the repository.

2. Database Design

Frevia persists every domain object in a single PostgreSQL database whose schema is declared with Prisma 7 in `apps/api/prisma/schema.prisma`. The schema contains 64 models together with their enumerations, every table has a surrogate integer primary key generated by the database (nine membership tables use a composite key instead), and deletion is either restricted by a foreign-key rule (identity and money-related rows are never removed) or implemented as a soft delete through a `deletedAt` column (content that an administrator must be able to restore: jobs, forum categories and posts, skills, banners, contract term templates). Binary files are not stored in the database: only their Cloudinary URL is kept in the owning row.

Figure 5 shows the entity–relationship diagram of the whole schema; the arrows read from the one-side to the many-side of each relationship. Each entity is drawn as a box that lists its key attributes, and the diamond shapes are the relationship endpoints of the foreign keys.

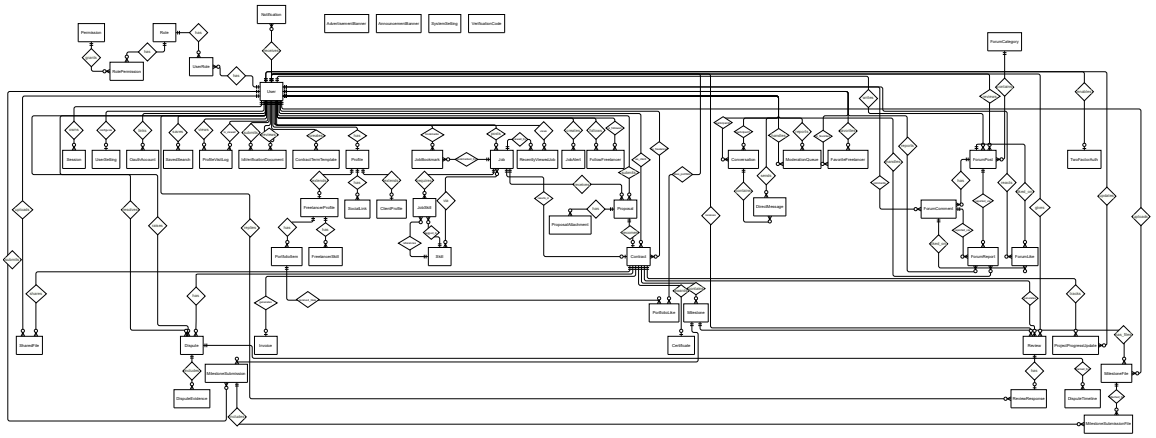

Figure 5: Entity–relationship diagram of the Frevia database

2.1 Table Relationships

The 64 tables are grouped into nine families; inside a family the relationships are one-to-many, and membership tables carry the many-to-many links:

- **Identity and access** — `User` is the root of the schema. `UserRole` links an account to a `Role`, and `RolePermission` links a role to a `Permission`, which yields the RBAC model enforced by the guards. Authentication adds `Session` (one row per device), `VerificationCode` (OTP for registration and password reset), `OauthAccount` (Google login), `TwoFactorAuth` and `UserSetting`.

- **Profiles** — `Profile` is shared by every account; `FreelancerProfile` and `ClientProfile` extend it with role-specific columns, `Expert` publishes an expert entry in the public directory, and `FreelancerSkill` joins a freelancer to the `Skill` catalogue. `PortfolioItem`, `SocialLink`, `Certificate`, `IdVerificationDocument` and `ProfileRevision` hang from the profile.

- **Jobs** — `Job` belongs to a client and is classified through `JobJobCategory` (`JobCategory`) and `JobSkill` (`Skill`). `JobBookmark`, `RecentlyViewedJob`, `SavedSearch` and `JobAlert` record the browsing behaviour of a freelancer.

- **Proposals and contracts** — `Proposal` references one `Job` and one freelancer; accepting it creates exactly one `Contract`, which carries `ContractTermTemplate` snapshots and a list of `Milestone` rows. `ProjectProgressUpdate`, `SharedFile` and `MilestoneFile` document the work in progress, while `MilestoneSubmission` (with `MilestoneSubmissionFile`) records the deliverable and its review decision.

- **Payments** — `Transaction` stores every Stripe movement (funding, release, refund, platform fee, dispute fee) with its payment-intent identifier, and `Invoice` holds the invoice issued

per contract. Money is never derived: the escrow balance of a milestone is the sum of its transactions.

- **Disputes** — `Dispute` is opened on a contract or a milestone, `DisputeEvidence` collects the attachments of both parties, `DisputeFee` records the fee charged to open it, and `DisputeDecisionReview` stores the request of a party that asks for the decision to be reconsidered.
- **Reviews** — one `Review` per completed contract, answered by at most one `ReviewResponse`.
- **Messaging and notifications** — `Conversation` groups the `DirectMessage` rows exchanged by two accounts (pin and read state are kept on the message and on the conversation), and `Notification` feeds the in-app bell.
- **Community forum and configuration** — `ForumCategory` contains `ForumPost`, a post contains `ForumComment`, and `ForumLike` / `ForumReport` attach reactions and reports to posts or comments. Administrators work through `ModerationQueue` and `BannedKeyword`; platform-wide configuration lives in `SystemSetting`, `AnnouncementBanner` and `AdvertisementBanner`.

Table Descriptions

Table 3 describes the purpose of every table of the schema, in the order in which the models are declared in `schema.prisma`, together with its primary key and its foreign keys.

Table 3: Table descriptions of the Frevia database

No	Table	Description
01	`Role`	Role catalogue of the RBAC model (admin, client, freelancer, expert); carries the role code, name and description. • **Primary keys:** `id` • **Foreign keys:** —
02	`UserRole`	Assignment of a role to an account, so one account can hold several roles and switch between them. • **Primary keys:** `userId`, `roleId` • **Foreign keys:** `userId` → `User`, `roleId` → `Role`
03	`Permission`	Catalogue of route-level permissions (`module.action` names) used by the permission guard. • **Primary keys:** `id` • **Foreign keys:** —

No	Table	Description
04	RolePermission	Permissions granted to a role; drives the effective permission list of every signed-in account. • **Primary keys:** roleId, permissionId • **Foreign keys:** roleId → Role, permissionId → Permission
05	User	Account identity: e-mail, username, bcrypt password hash, status, verification and ban flags, preferred language and timestamps. • **Primary keys:** id • **Foreign keys:** —
06	VerificationCode	One-time code (OTP) issued for e-mail verification or password reset, with its purpose, expiry and consumed flag. • **Primary keys:** id • **Foreign keys:** —
07	Session	Server-side session per device: refresh-token hash, device/user-agent information, last activity and revocation time. • **Primary keys:** id • **Foreign keys:** userId → User
08	OauthAccount	Link between an account and an external identity provider (Google) for federated login. • **Primary keys:** id • **Foreign keys:** userId → User
09	TwoFactorAuth	Two-factor authentication secret and enabled flag of an account (one row per account). • **Primary keys:** id • **Foreign keys:** userId → User
10	UserSetting	Per-account preferences: interface theme, language and the JSON notification preferences. • **Primary keys:** id • **Foreign keys:** userId → User
11	Profile	Base profile shared by all accounts: display name, avatar, cover, headline and biography. • **Primary keys:** id • **Foreign keys:** userId → User
12	FreelancerProfile	Professional data of a freelancer: professional title, overview, hourly rate, availability and total earnings. • **Primary keys:** id • **Foreign keys:** profileId → Profile

No	Table	Description
13	FreelancerSkill	Many-to-many link between a freelancer profile and a skill, with the self-declared proficiency level. • **Primary keys:** id • **Foreign keys:** freelancerProfileId → FreelancerProfile, skillId → Skill
14	ClientProfile	Client-side data: company name, company size, industry and billing information. • **Primary keys:** id • **Foreign keys:** profileId → Profile
15	Expert	Expert entry published in the public expert directory, with specialisation and verification state. • **Primary keys:** id • **Foreign keys:** profileId → Profile
16	ProfileRevision	Pending change request on a profile, reviewed and approved or rejected by an administrator. • **Primary keys:** id • **Foreign keys:** userId → User, adminId → User, profileId → Profile
17	PortfolioItem	Portfolio entry of a profile: title, description, media URL, project link and ordering position. • **Primary keys:** id • **Foreign keys:** freelancerProfileId → FreelancerProfile
18	SocialLink	External link of a profile (website, LinkedIn, GitHub, Behance) shown on the public profile. • **Primary keys:** id • **Foreign keys:** profileId → Profile
19	IdVerificationDocument	Identity document uploaded for eKYC, with the AI check result and the administrator decision. • **Primary keys:** id • **Foreign keys:** userId → User, adminId → User, duplicateOfId → IdVerificationDocument
20	ProfileVisitLog	View log of a freelancer profile (viewerId, targetFreelancerId, viewedAt); source of the profile statistics. • **Primary keys:** id • **Foreign keys:** viewerId → User, targetFreelancerId → User

No	Table	Description
21	JobCategory	Hierarchical job category (parent reference, slug, name, icon) with soft-delete support. • **Primary keys:** id • **Foreign keys:** —
22	Job	Job posting: slug, title, description, budget range, duration, deadline, visibility, status and the owning client. • **Primary keys:** id • **Foreign keys:** clientId → User
23	JobJobCategory	Many-to-many link between a job and the job categories it is published in. • **Primary keys:** jobId, categoryId • **Foreign keys:** jobId → Job, categoryId → JobCategory
24	Skill	Skill catalogue shared by freelancer profiles and jobs, with soft-delete support. • **Primary keys:** id • **Foreign keys:** —
25	JobSkill	Skills required by a job, with the required proficiency level. • **Primary keys:** jobId, skillId • **Foreign keys:** jobId → Job, skillId → Skill
26	JobBookmark	Jobs saved by a freelancer. • **Primary keys:** userId, jobId • **Foreign keys:** userId → User, jobId → Job
27	RecentlyViewedJob	Recently opened job postings of an account, with the viewing timestamp. • **Primary keys:** id • **Foreign keys:** userId → User, jobId → Job
28	SavedSearch	Stored job filter criteria (keyword, categories, budget, skills) that the freelancer can reapply. • **Primary keys:** id • **Foreign keys:** userId → User
29	JobAlert	Alert configuration attached to a saved search (frequency, channel, active flag). • **Primary keys:** id • **Foreign keys:** userId → User

No	Table	Description
30	Proposal	Proposal submitted by a freelancer for a job: cover letter, bid amount, estimated duration and lifecycle status. • **Primary keys:** id • **Foreign keys:** jobId → Job, freelancerId → User
31	ProposalAttachment	File attached to a proposal (portfolio sample, reference document) stored in Cloudinary. • **Primary keys:** id • **Foreign keys:** proposalId → Proposal
32	Contract	Contract created when a proposal is accepted: client, freelancer, total amount, terms, signature state, start/end dates and status. • **Primary keys:** id • **Foreign keys:** jobId → Job, proposalId → Proposal, clientId → User, freelancerId → User
33	ContractTermTemplate	Reusable contract term template owned by a user, with category, content and soft-delete support. • **Primary keys:** id • **Foreign keys:** creatorId → User
34	Milestone	Milestone inside a contract: title, description, amount, due date, escrow status, progress percentage and approval state. • **Primary keys:** id • **Foreign keys:** contractId → Contract
35	MilestoneFile	Reference file attached to a milestone and shared with both parties. • **Primary keys:** id • **Foreign keys:** milestoneId → Milestone, uploaderId → User
36	MilestoneSubmission	Work submitted for a milestone, with the submitted note, timestamp and the client review decision. • **Primary keys:** id • **Foreign keys:** milestoneId → Milestone, freelancerId → User
37	MilestoneSubmissionFile	Join table linking a submission to the milestone files it delivers. • **Primary keys:** submissionId, fileId • **Foreign keys:** submissionId → MilestoneSubmission, fileId → MilestoneFile

No	Table	Description
38	Invoice	Invoice of a contract (one per contract): invoice number, total amount, status, issued, due and paid dates. • **Primary keys:** id • **Foreign keys:** contractId → Contract
39	Certificate	Completion certificate generated when a contract is finished: unique certificate UID, completion date and JSON metadata. • **Primary keys:** id • **Foreign keys:** contractId → Contract, freelancerId → User, clientId → User
40	Dispute	Dispute opened on a contract or a milestone: reason, description, opened by, status, resolution and decision timestamps. • **Primary keys:** id • **Foreign keys:** milestoneId → Milestone, openedById → User, respondentId → User, decisionById → User
41	DisputeFee	Fee charged to open a dispute, with its amount, payment state and the transaction that settled it. • **Primary keys:** id • **Foreign keys:** disputeId → Dispute, userId → User
42	DisputeEvidence	Evidence attached by either party to a dispute (file, description, uploader). • **Primary keys:** id • **Foreign keys:** disputeId → Dispute, submittedById → User, fileId → SharedFile
43	DisputeDecisionReview	Request of a party to reconsider a dispute decision, with the response, the reason and the timestamp. • **Primary keys:** id • **Foreign keys:** disputeId → Dispute, userId → User
44	Transaction	Payment movement reconciled with Stripe (funding, release, refund, platform fee, dispute fee): type, amount, status, payment-intent id and timestamps. • **Primary keys:** id • **Foreign keys:** userId → User, contractId → Contract

No	Table	Description
45	Review	Review written after a contract is completed: rating, comment, author, recipient and the reviewed contract. • **Primary keys:** id • **Foreign keys:** contractId → Contract, reviewerId → User, revieweeId → User
46	ReviewResponse	Reply of the reviewed party to a review (at most one per review). • **Primary keys:** id • **Foreign keys:** reviewId → Review, userId → User
47	ForumCategory	Forum category: slug, name, description, display order, post counter and soft-delete support. • **Primary keys:** id • **Foreign keys:** —
48	ForumPost	Forum post: title, content, author, category, moderation status, AI score, reviewer and soft-delete support. • **Primary keys:** id • **Foreign keys:** categoryId → ForumCategory, userId → User, reviewedById → User
49	ForumComment	Comment on a forum post; comments are one level deep and can be soft-deleted. • **Primary keys:** id • **Foreign keys:** postId → ForumPost, userId → User
50	ForumLike	Reaction of a member on a post or a comment; a member can like a target only once. • **Primary keys:** id • **Foreign keys:** userId → User, postId → ForumPost, commentId → ForumComment
51	ForumReport	Content report filed by a member: reason, description, status and the administrator who handled it. • **Primary keys:** id • **Foreign keys:** reporterId → User, postId → ForumPost, commentId → ForumComment, adminId → User
52	Notification	In-app notification of an account: type, title, body, target link and read state. • **Primary keys:** id • **Foreign keys:** userId → User

No	Table	Description
53	AnnouncementBanner	Platform announcement shown in the application: message, link, display window, active flag and soft delete. • **Primary keys:** id • **Foreign keys:** —
54	ModerationQueue	Queue of AI-flagged or user-reported content waiting for an administrator decision, with the item type, the score and the assignee. • **Primary keys:** id • **Foreign keys:** reporterId → User, adminId → User
55	SystemSetting	Platform-wide configuration values stored as key/value rows. • **Primary keys:** id • **Foreign keys:** —
56	BannedKeyword	Keyword list used by the moderation filter before the AI service is called. • **Primary keys:** id • **Foreign keys:** createdBy → User
57	AdvertisementBanner	Advertising banner with image, target URL, position, schedule and click counter. • **Primary keys:** id • **Foreign keys:** —
58	FavoriteFreelancer	Freelancers marked as favourite by a client account. • **Primary keys:** clientId, freelancerId • **Foreign keys:** clientId → User, freelancerId → User
59	FollowFreelancer	Follow relationship between two accounts, used by the activity feed. • **Primary keys:** clientId, freelancerId • **Foreign keys:** clientId → User, freelancerId → User
60	PortfolioLike	Like given on a portfolio item. • **Primary keys:** userId, portfolioItemId • **Foreign keys:** userId → User, portfolioItemId → PortfolioItem

No	Table	Description
61	`ProjectProgressUpdate`	Progress note posted by the freelancer on a contract, with the update timestamp. • **Primary keys:** `id` • **Foreign keys:** `contractId` → `Contract`, `freelancerId` → `User`
62	`SharedFile`	File shared in the workspace of a contract (both parties can upload and read). • **Primary keys:** `id` • **Foreign keys:** `contractId` → `Contract`, `uploaderId` → `User`
63	`Conversation`	Direct conversation between two accounts, with last-message cache and the pin state of each participant. • **Primary keys:** `id` • **Foreign keys:** `participant1` → `User`, `participant2` → `User`
64	`DirectMessage`	Message of a conversation: sender, content, attachments, read state, recall (soft delete) flag and timestamp. • **Primary keys:** `id` • **Foreign keys:** `conversationId` → `Conversation`, `senderId` → `User`

3. Detailed Design

This chapter presents the detailed design of every feature of the system. Each section (3.x) covers one feature and contains the class diagram of the feature (3.x.1) followed by its sequence diagrams (3.x.2, 3.x.3, ...), one for each use case of the feature. The class specification of every class that appears in these diagrams — attributes, methods and their types — is tabulated in Chapter 4. rather than repeated under each feature. Where several functions are built from the same classes and follow the same interaction pattern, the diagrams are presented once for the representative function and the other functions refer to that section.

3.1 Forum Management

Forum Management covers the UC-14 community forum: browsing forum categories and posts, publishing and maintaining posts and comments, toggling likes on posts and comments, and reporting content that breaks the forum rules.

3.1.1 Class Diagram

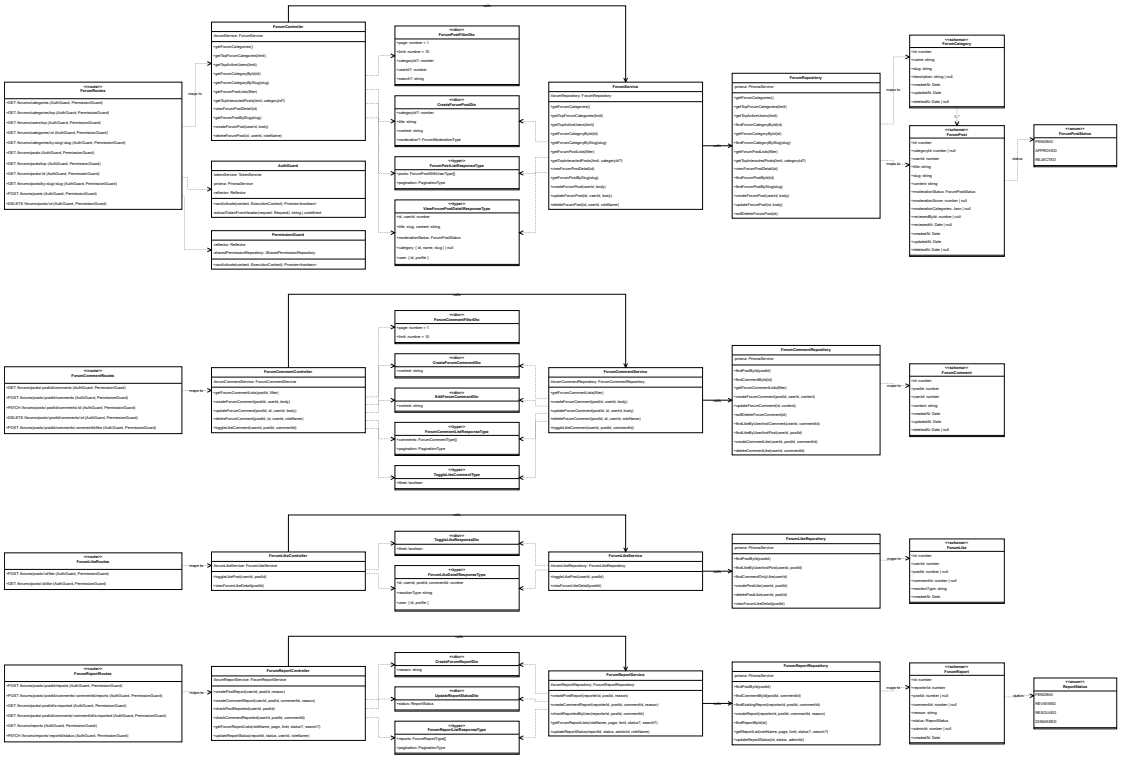

Figure 6: Class diagram of Forum Management

3.1.2 UC-14.01 – View Forum Category List

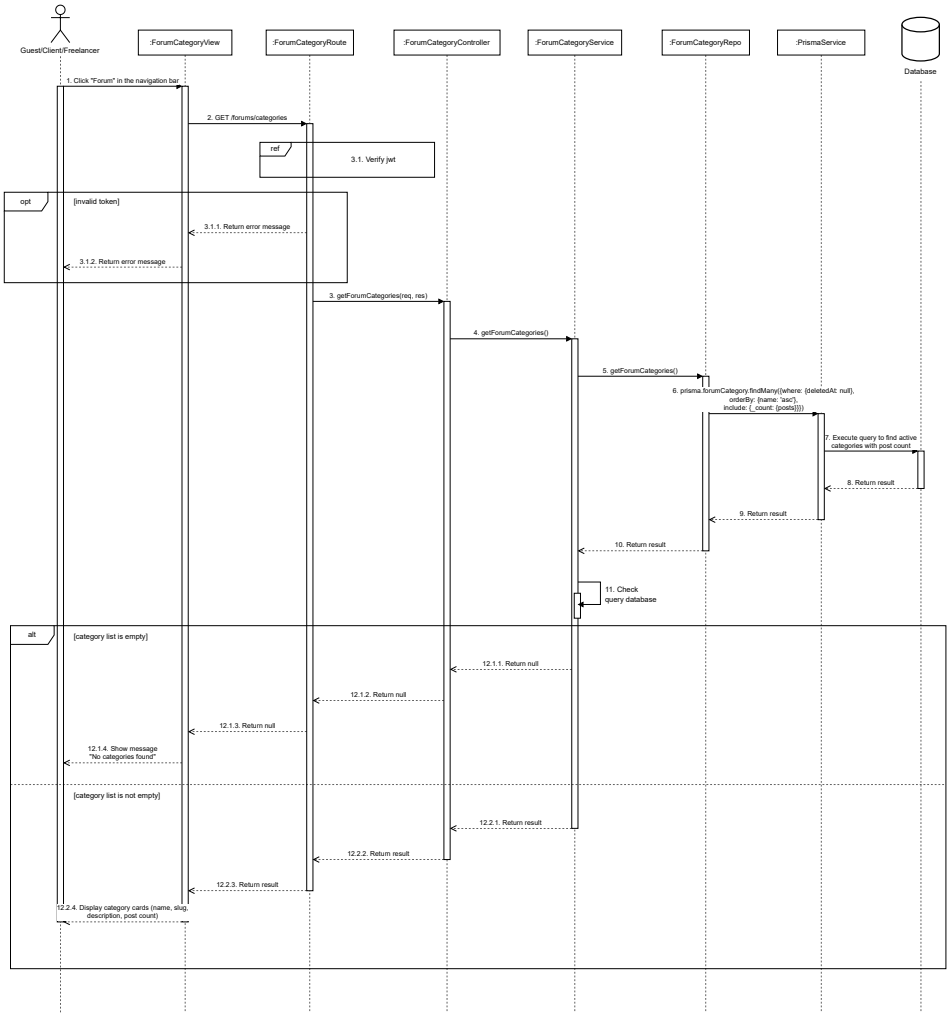

Figure 7: Sequence diagram of UC-14.01 View Forum Category List

3.1.3 UC-14.02 – View Forum Category Detail

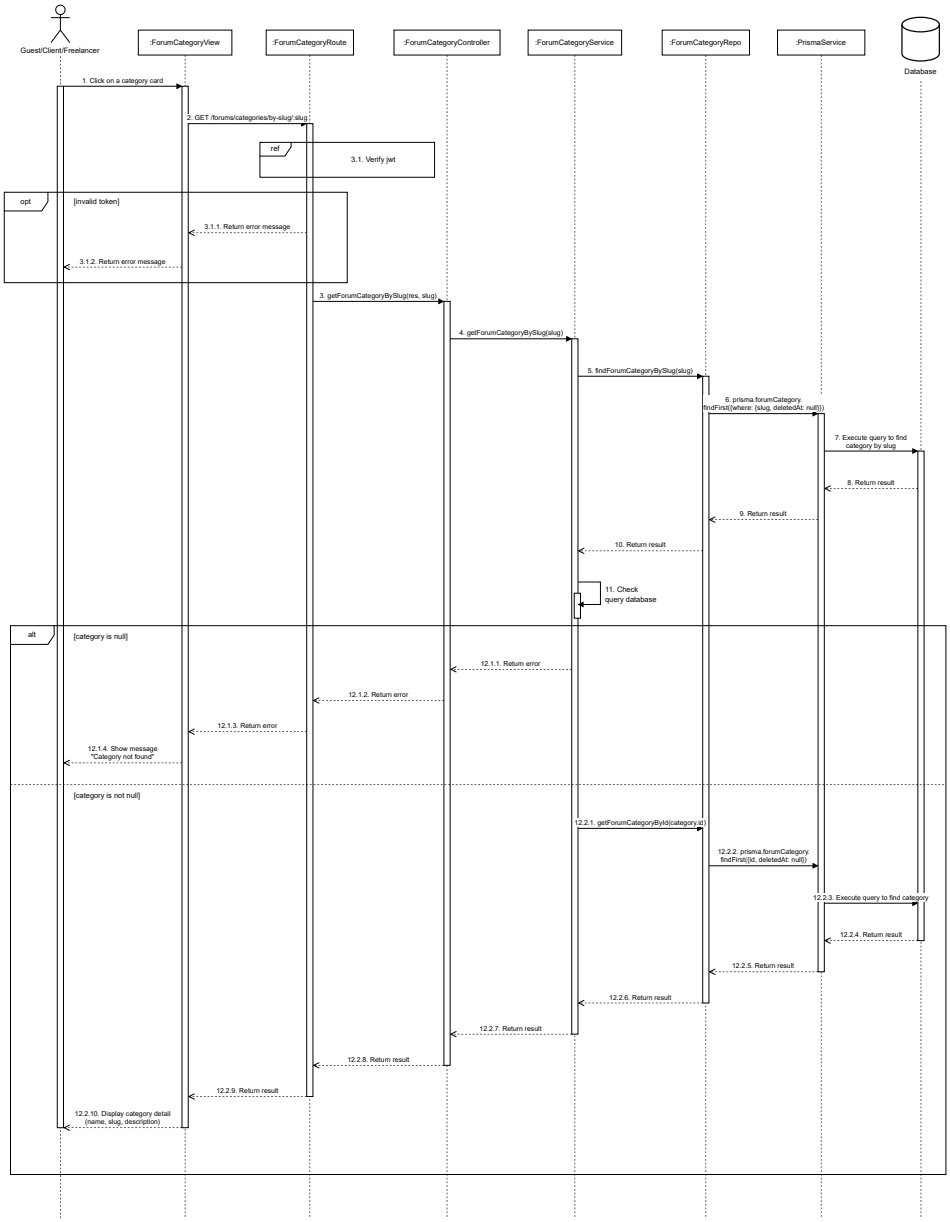

Figure 8: Sequence diagram of UC-14.02 View Forum Category Detail

3.1.4 UC-14.03 – View Post List

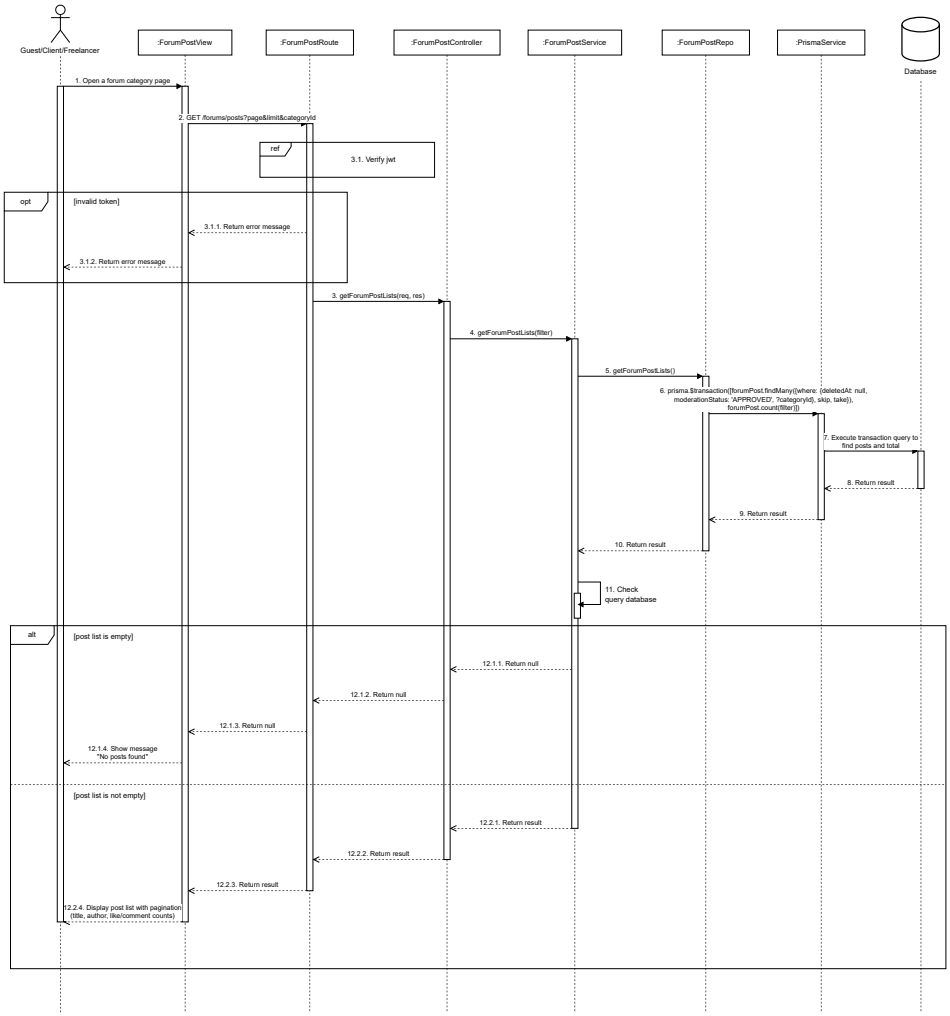

Figure 9: Sequence diagram of UC-14.03 View Post List

3.1.5 UC-14.04 – View Post Detail

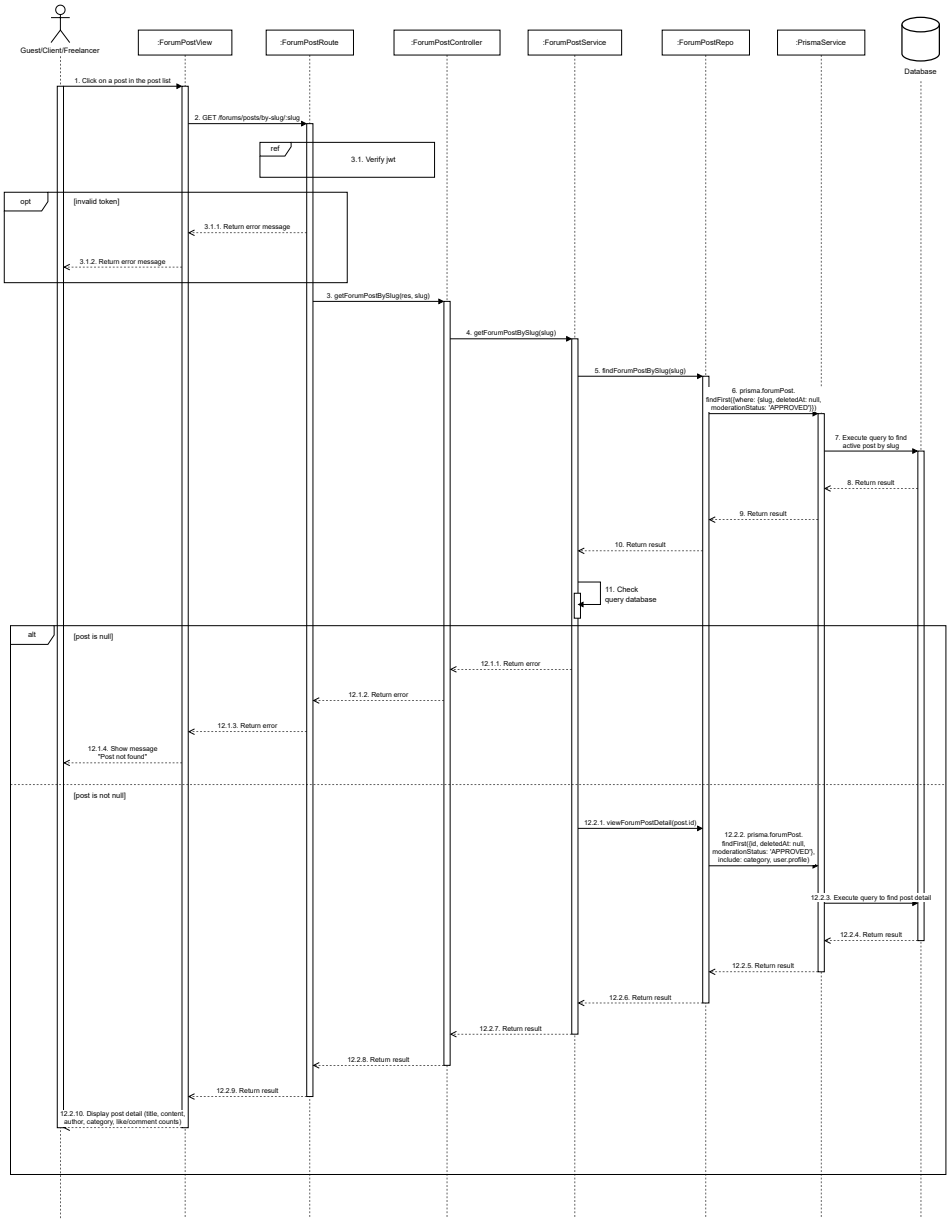

Figure 10: Sequence diagram of UC-14.04 View Post Detail

3.1.6 UC-14.05 – Create Post

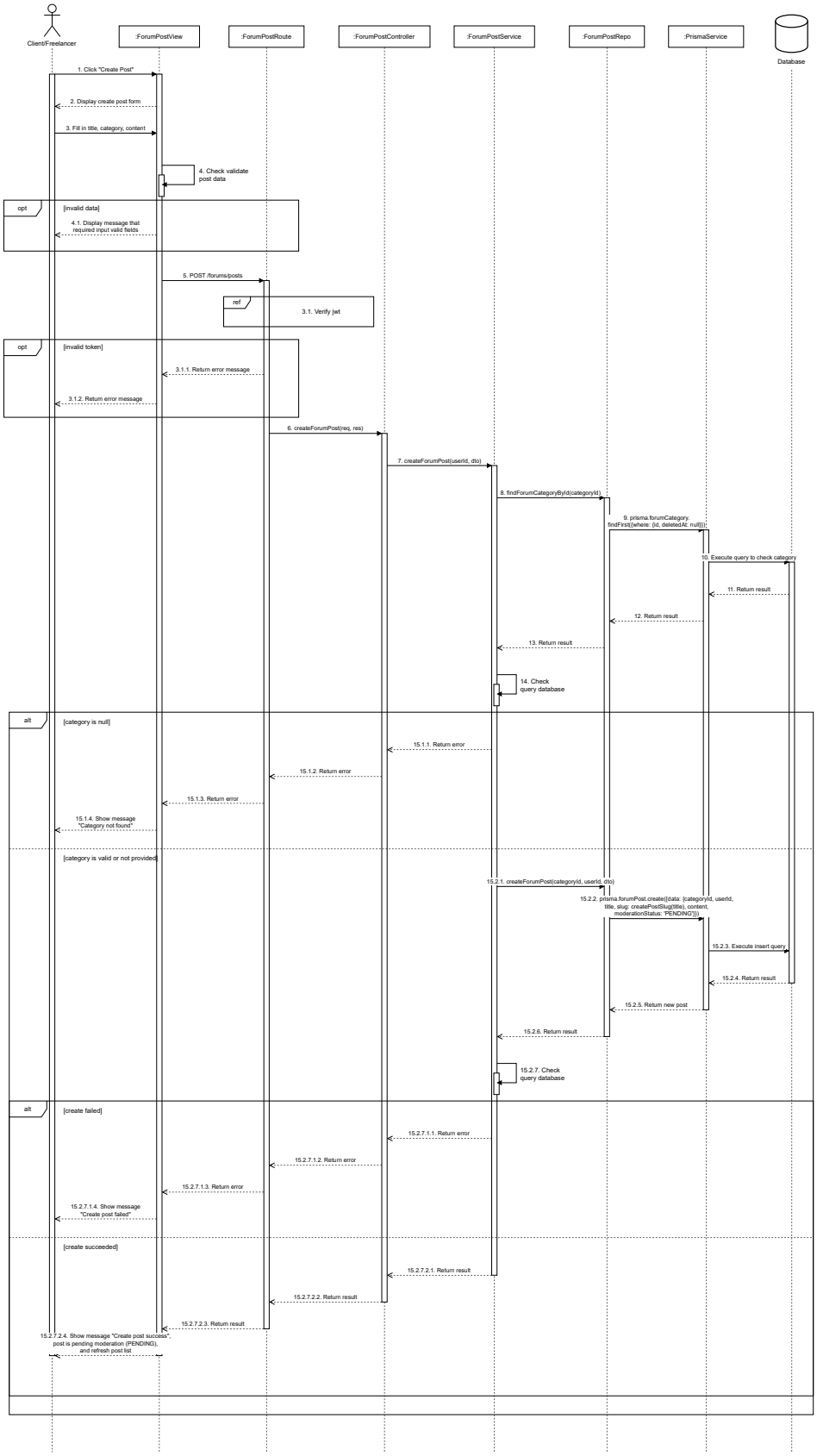

Figure 11: Sequence diagram of UC-14.05 Create Post

3.1.7 UC-14.06 – Update Post

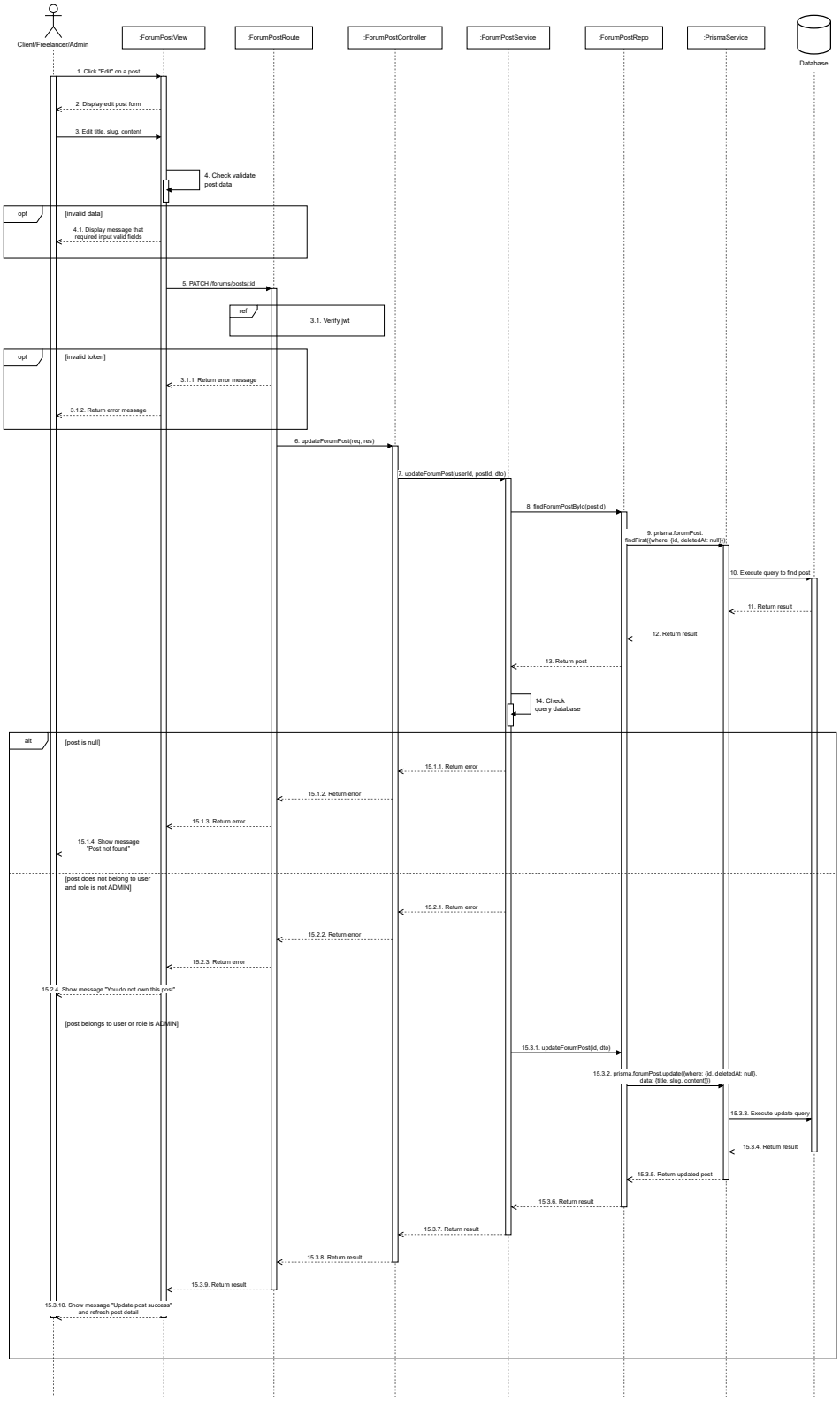

Figure 12: Sequence diagram of UC-14.06 Update Post

3.1.8 UC-14.07 – Delete Post

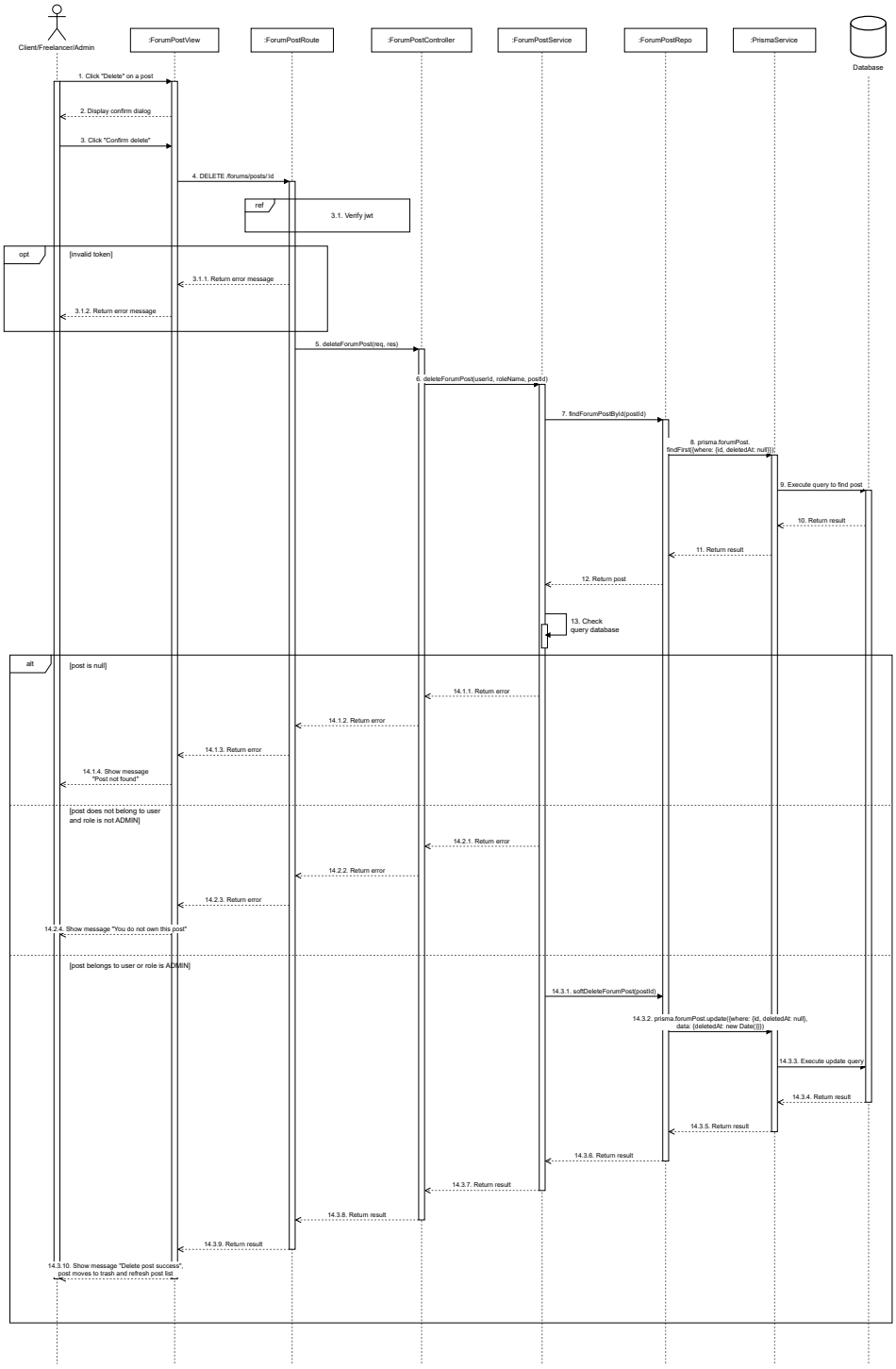

Figure 13: Sequence diagram of UC-14.07 Delete Post

3.1.9 UC-14.08 – Like Post

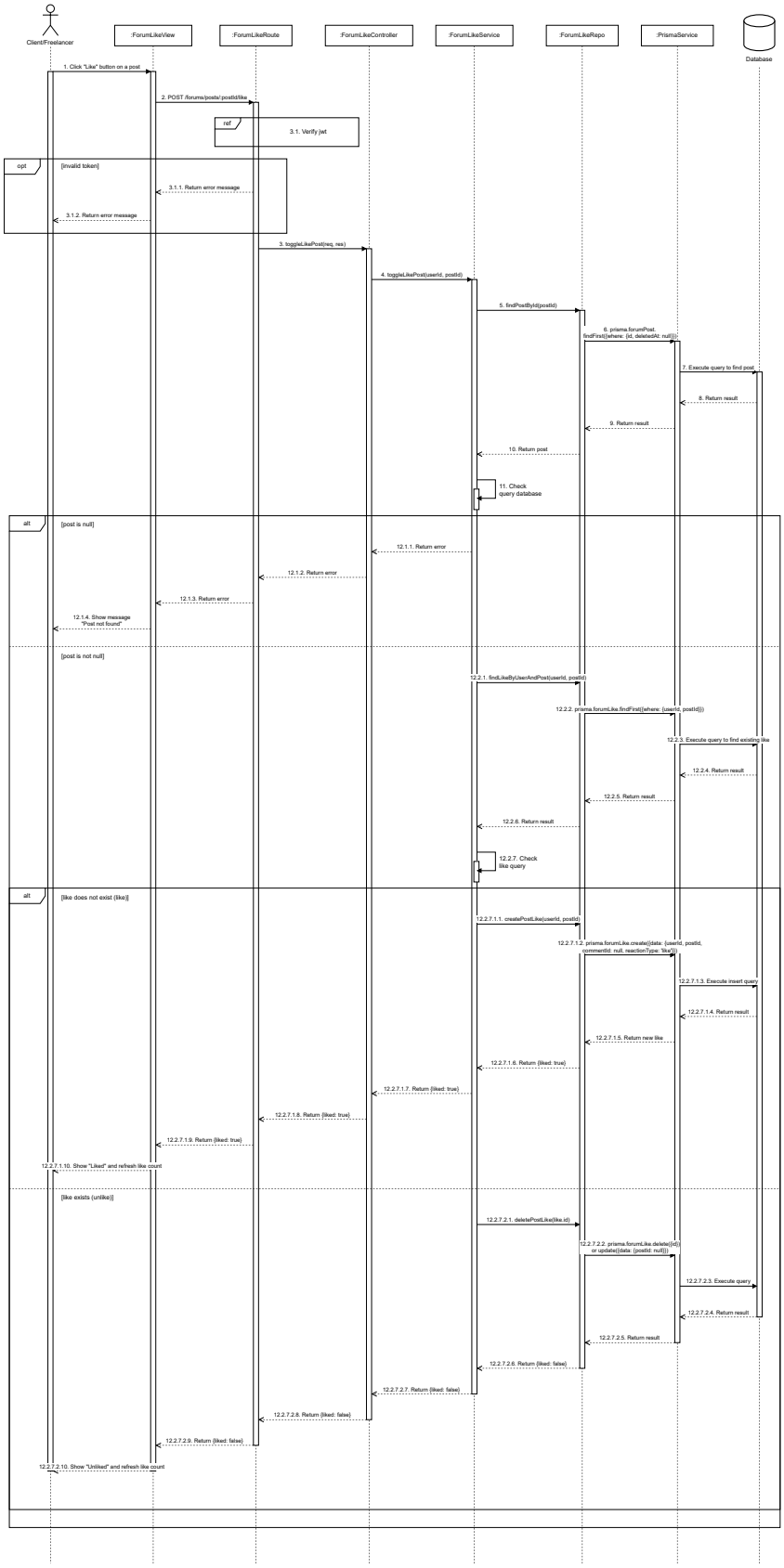

Figure 14: Sequence diagram of UC-14.08 Like Post

3.1.10 UC-14.09 – Unlike Post

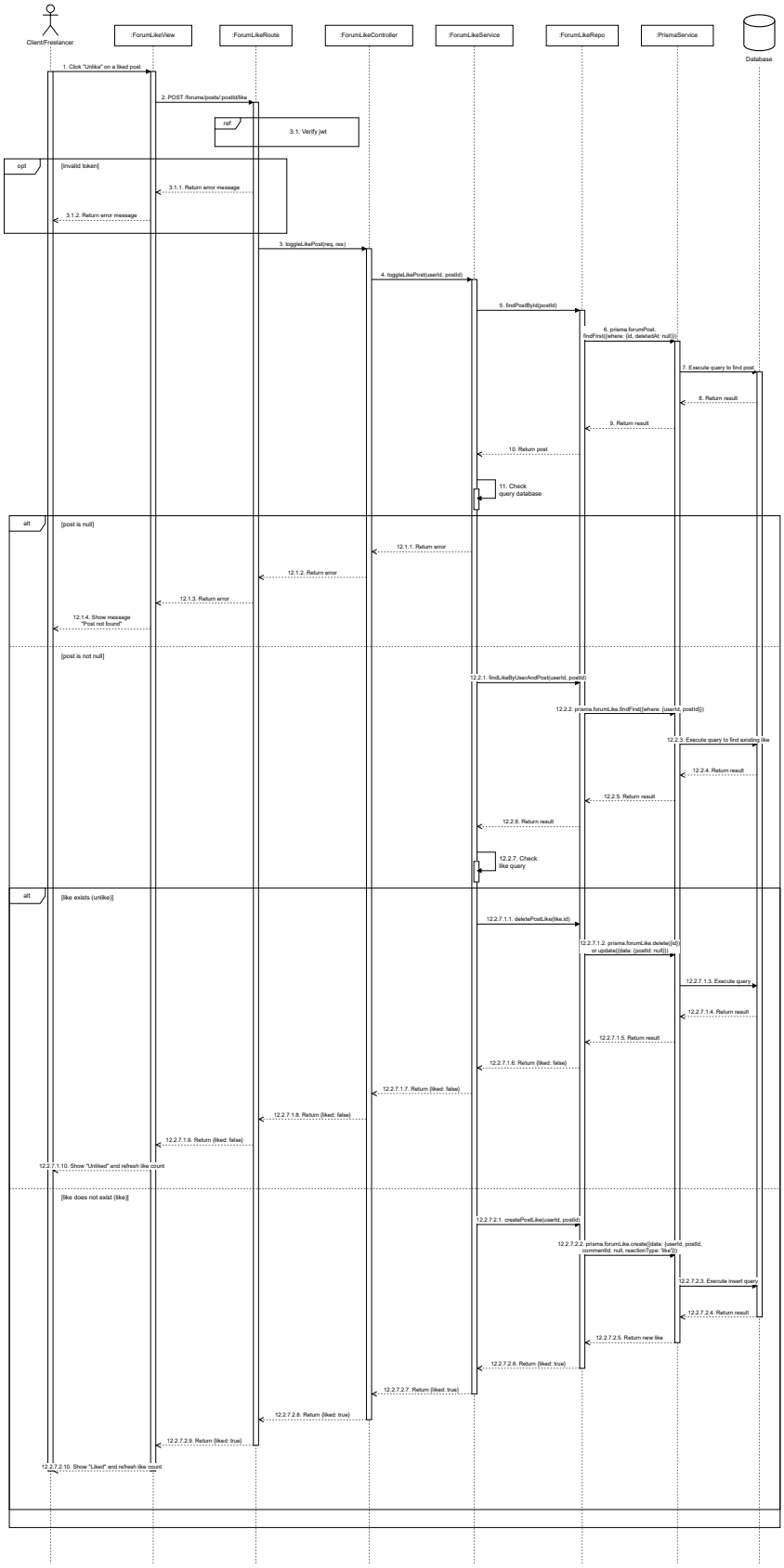

Figure 15: Sequence diagram of UC-14.09 Unlike Post

3.1.11 UC-14.10 – View Post Likes

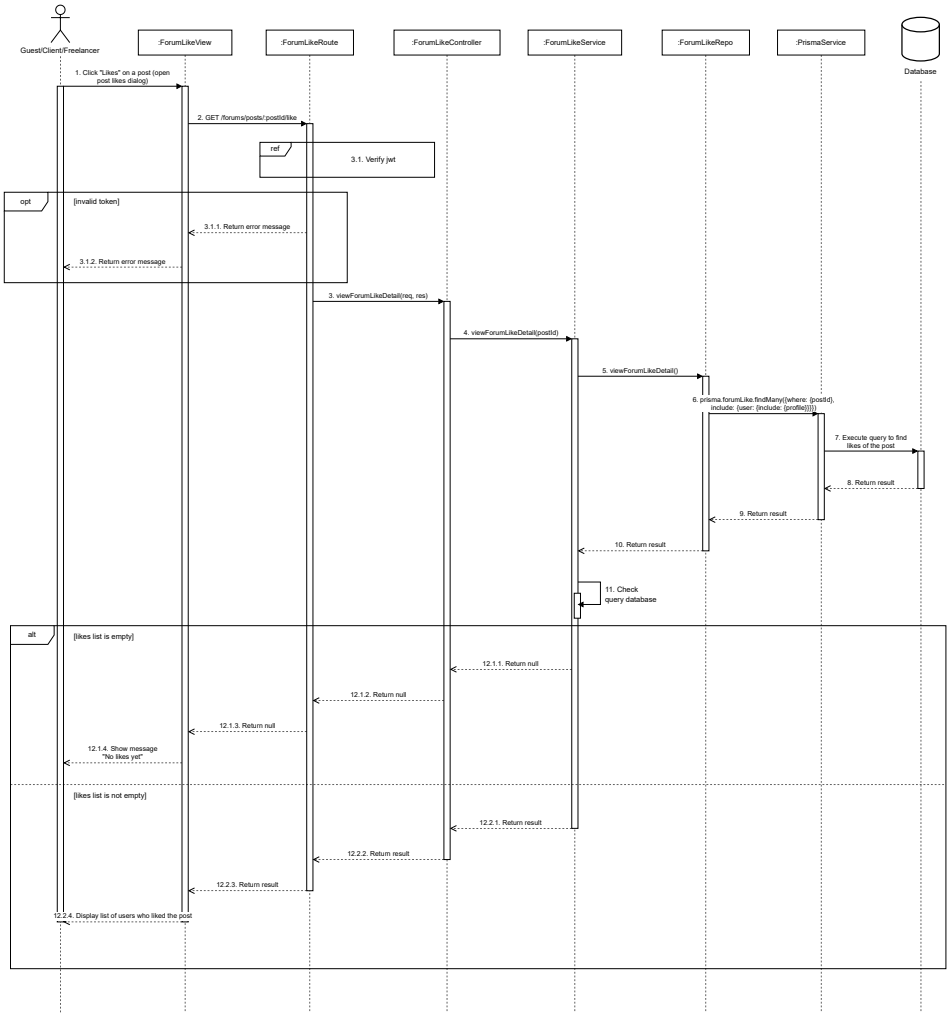

Figure 16: Sequence diagram of UC-14.10 View Post Likes

3.1.12 UC-14.11 – View Post Comments

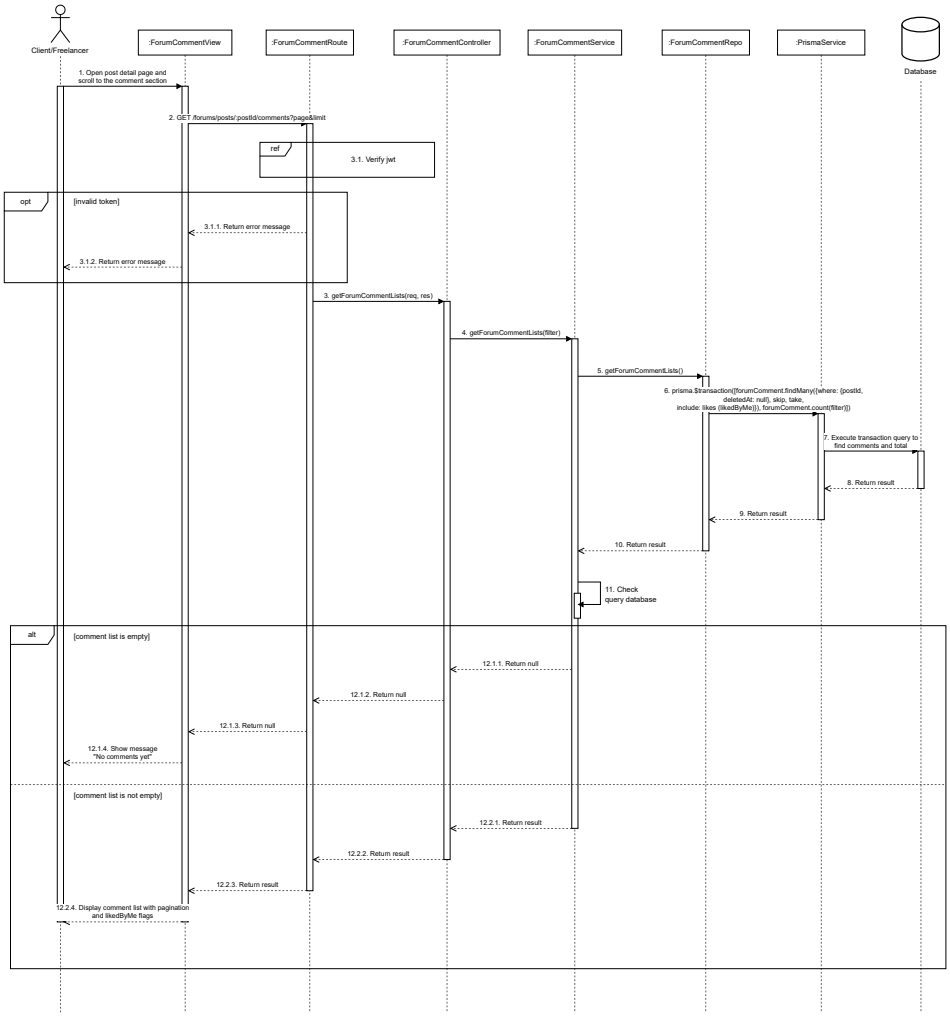

Figure 17: Sequence diagram of UC-14.11 View Post Comments

3.1.13 UC-14.12 – Create Comment

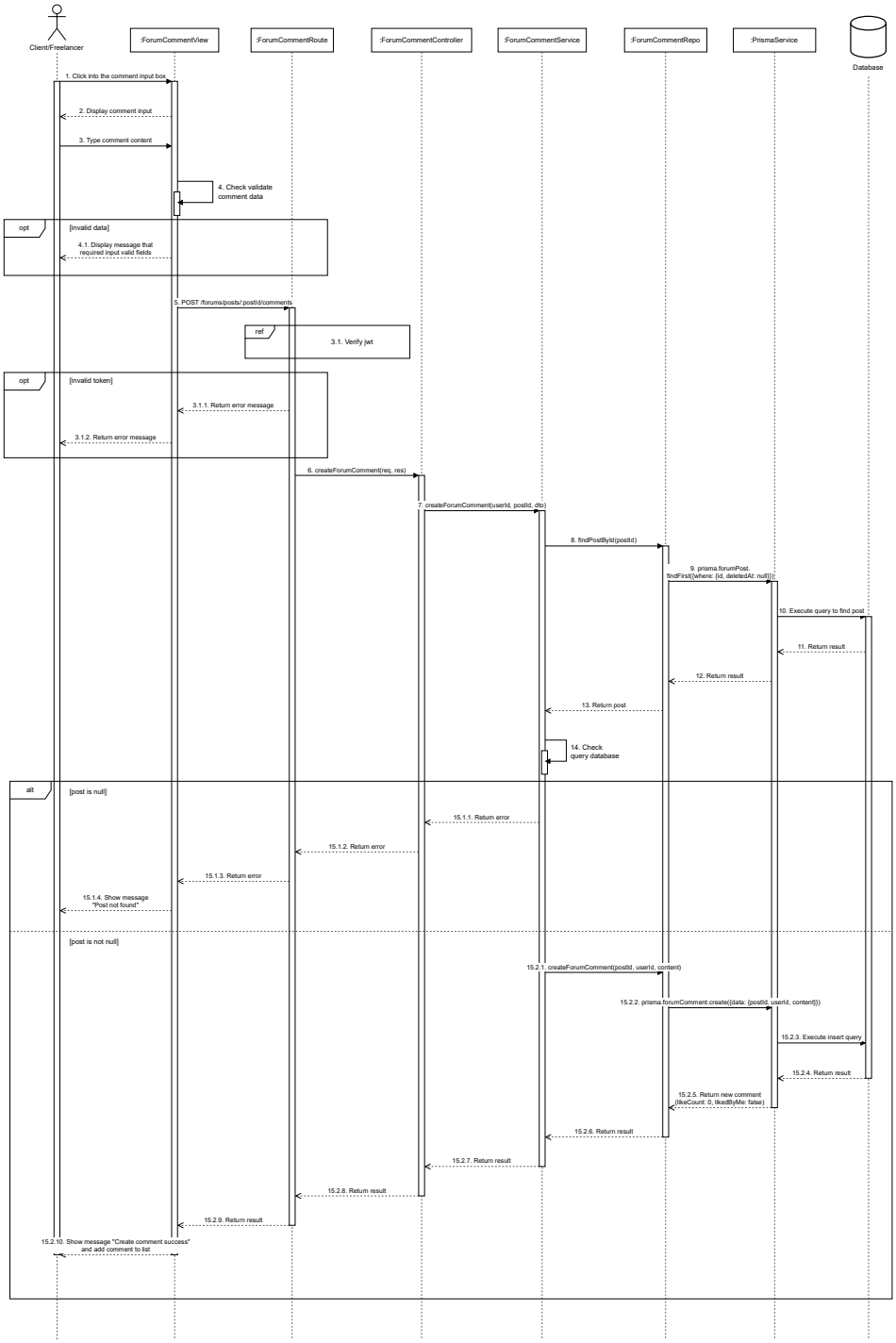

Figure 18: Sequence diagram of UC-14.12 Create Comment

3.1.14 UC-14.13 – Update Comment

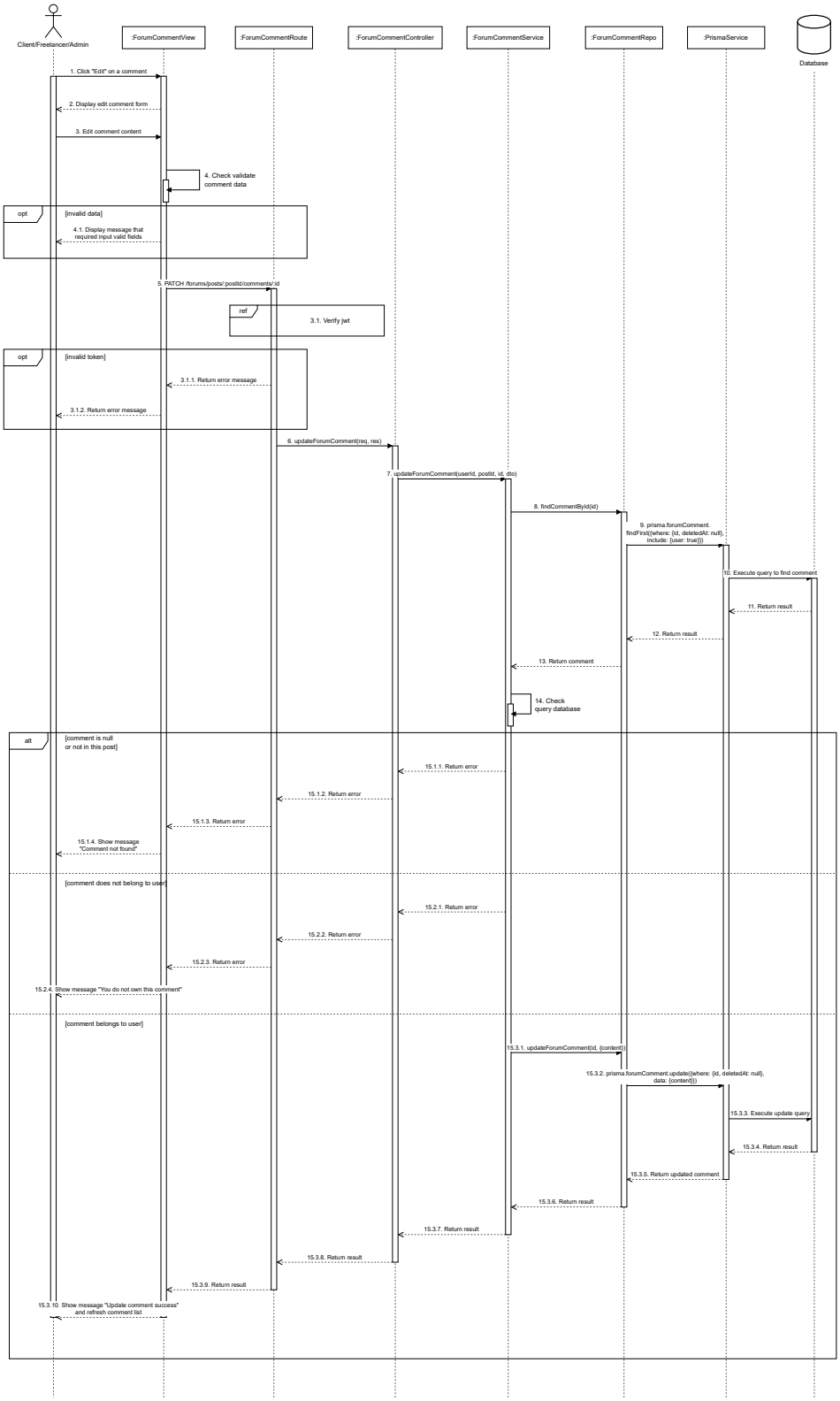

Figure 19: Sequence diagram of UC-14.13 Update Comment

3.1.15 UC-14.14 – Delete Comment

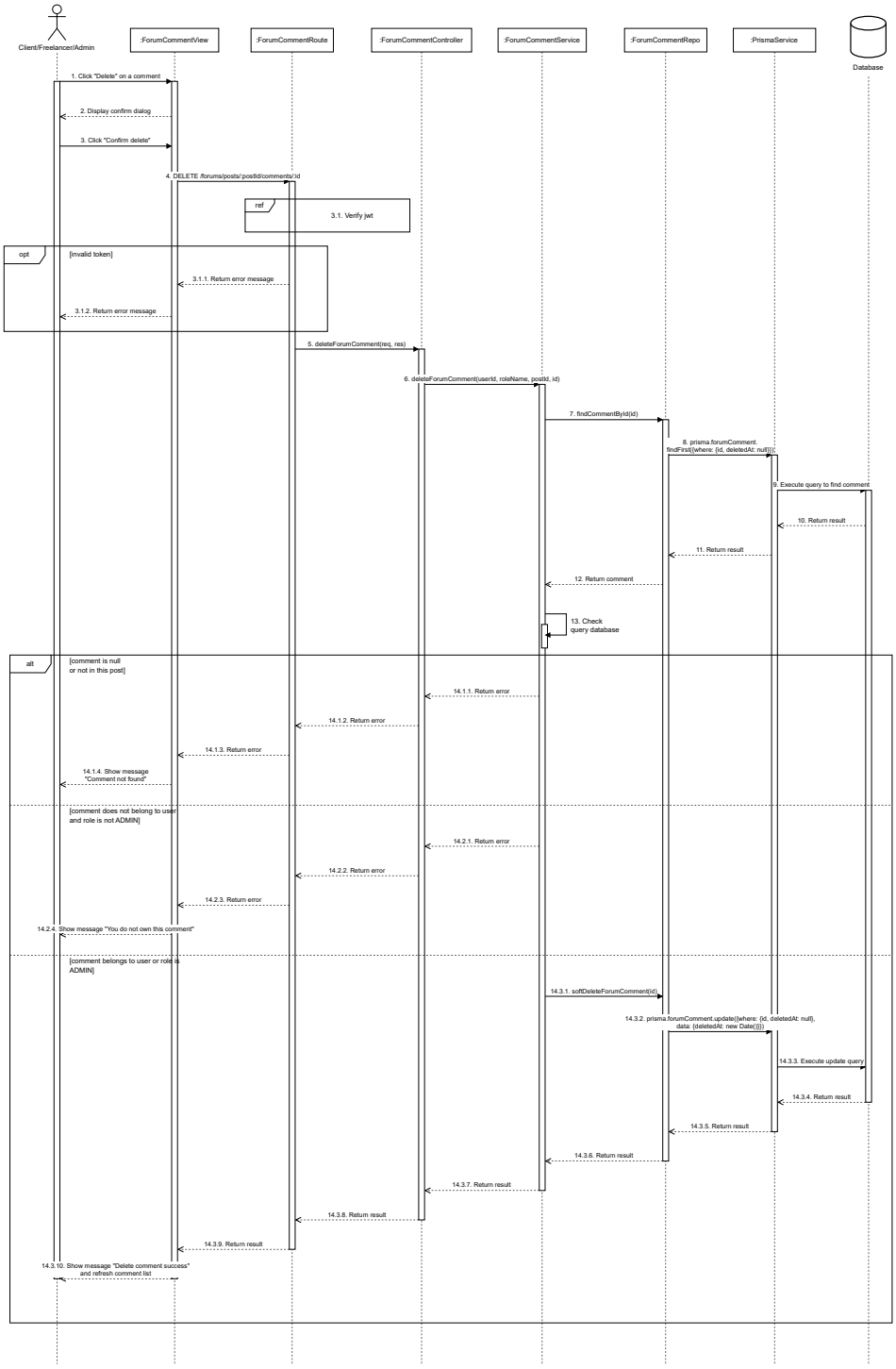

Figure 20: Sequence diagram of UC-14.14 Delete Comment

3.1.16 UC-14.15 – Like Comment

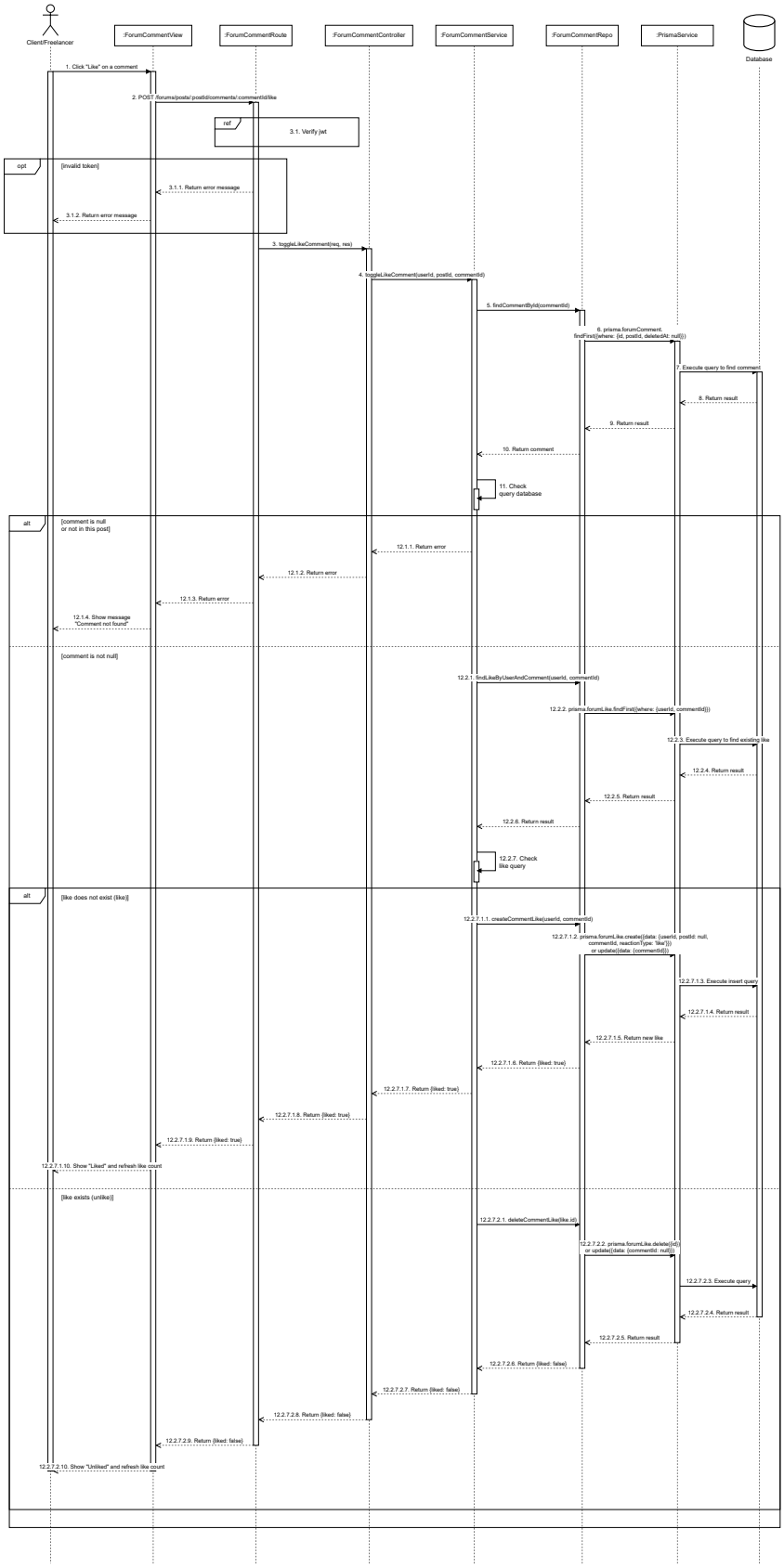

Figure 21: Sequence diagram of UC-14.15 Like Comment

3.1.17 UC-14.16 – Unlike Comment

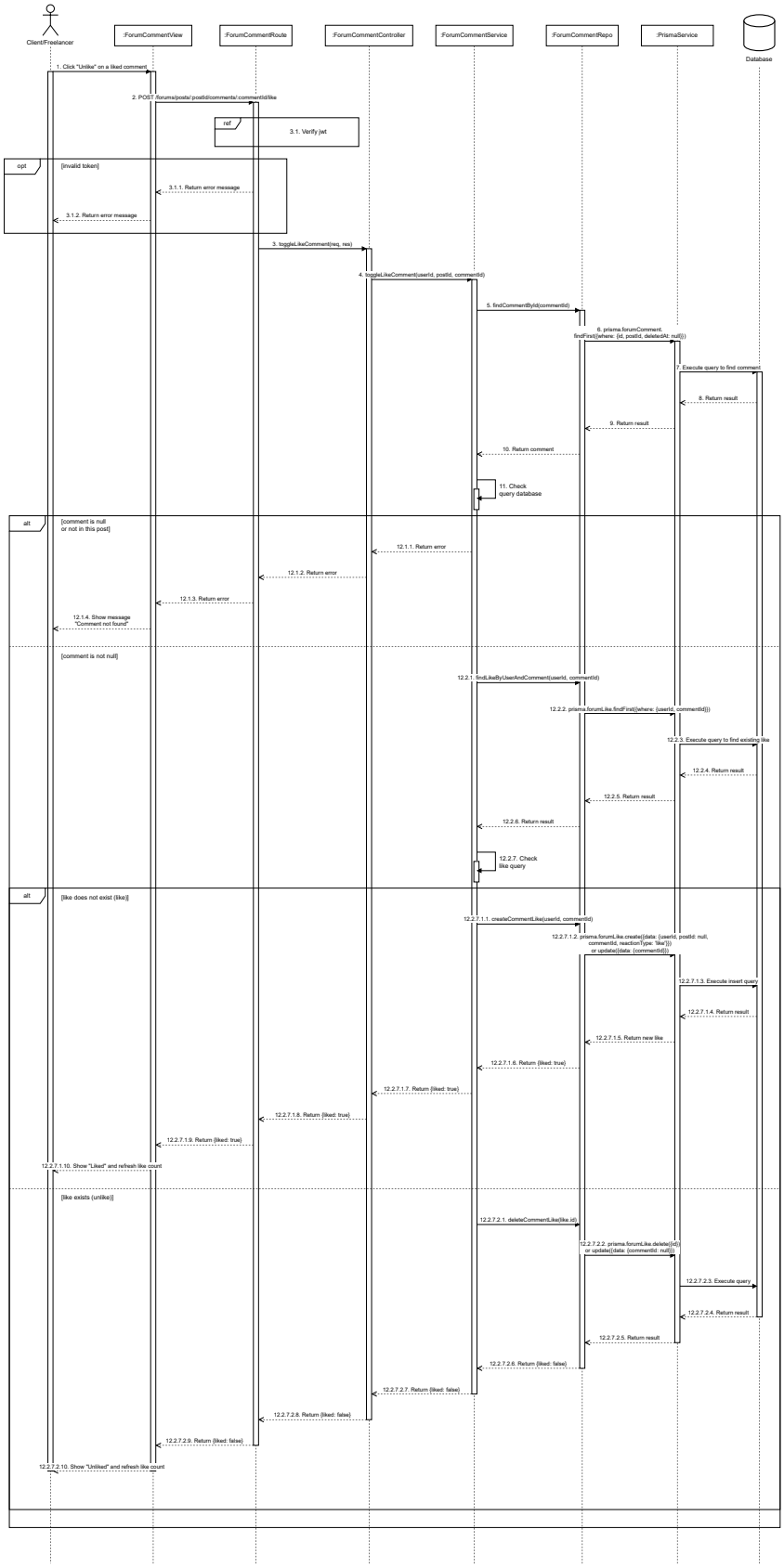

Figure 22: Sequence diagram of UC-14.16 Unlike Comment

3.1.18 UC-14.17 – Report Content

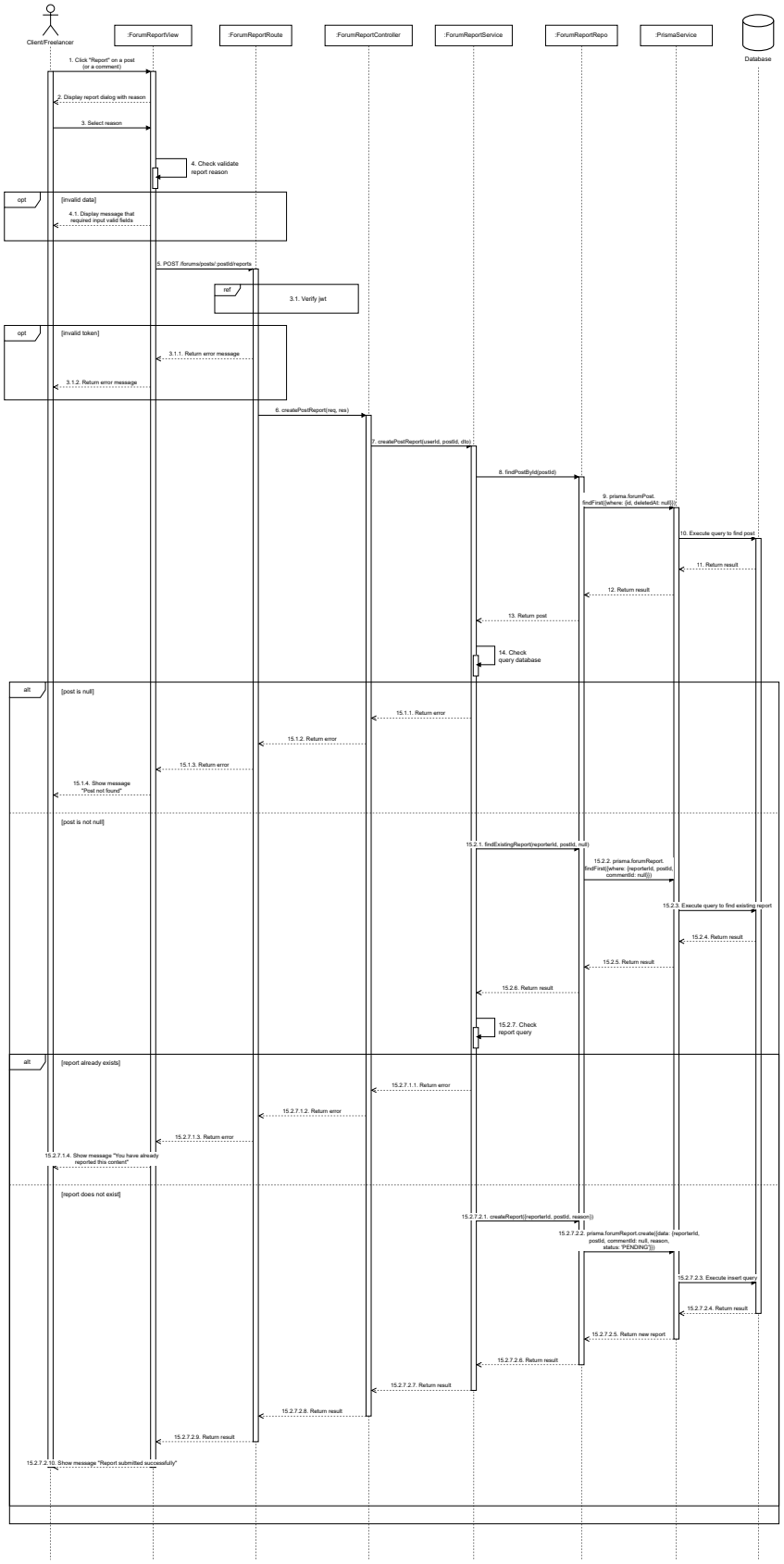

Figure 23: Sequence diagram of UC-14.17 Report Content

3.2 Freelancer Profile Management

Visitors can view a freelancer profile. Owners may submit profile edits; low-strength profiles require manual review through ProfileRevisionService.

3.2.1 Class Diagram

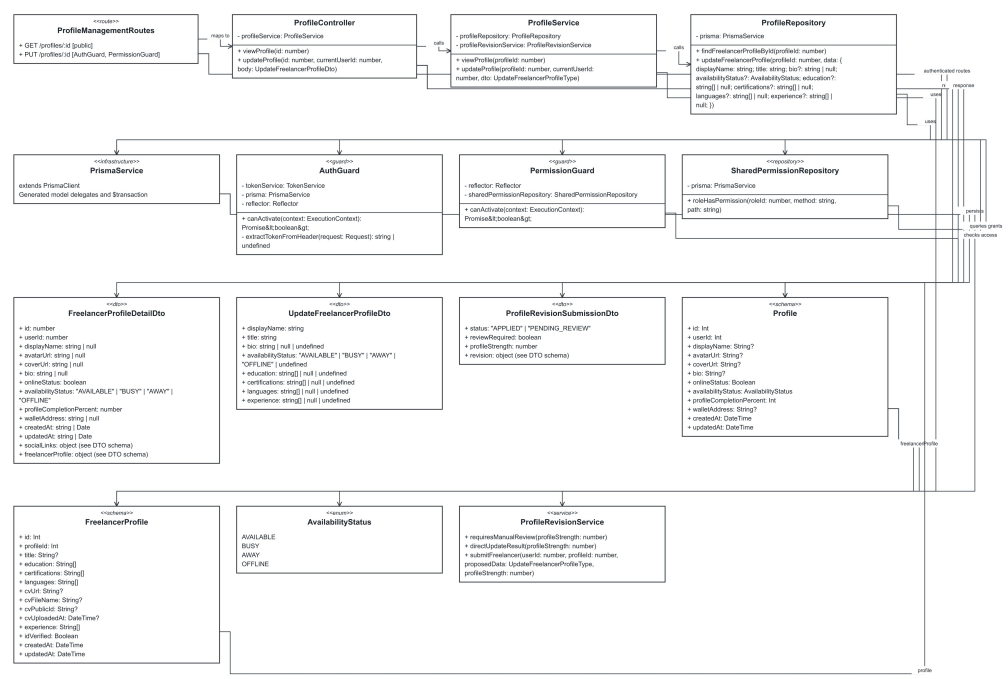

Figure 24: Class diagram of Freelancer Profile Management

3.2.2 UC-07.01 – View Freelancer Profile

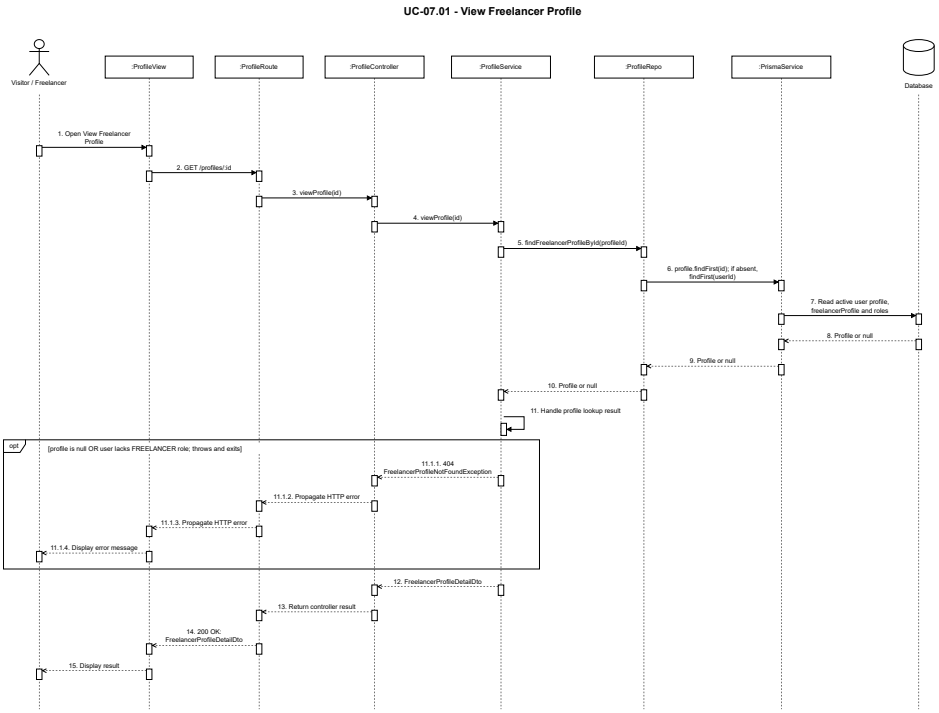

Figure 25: Sequence diagram of UC-07.01 View Freelancer Profile

3.2.3 UC-07.02 – Update Freelancer Profile

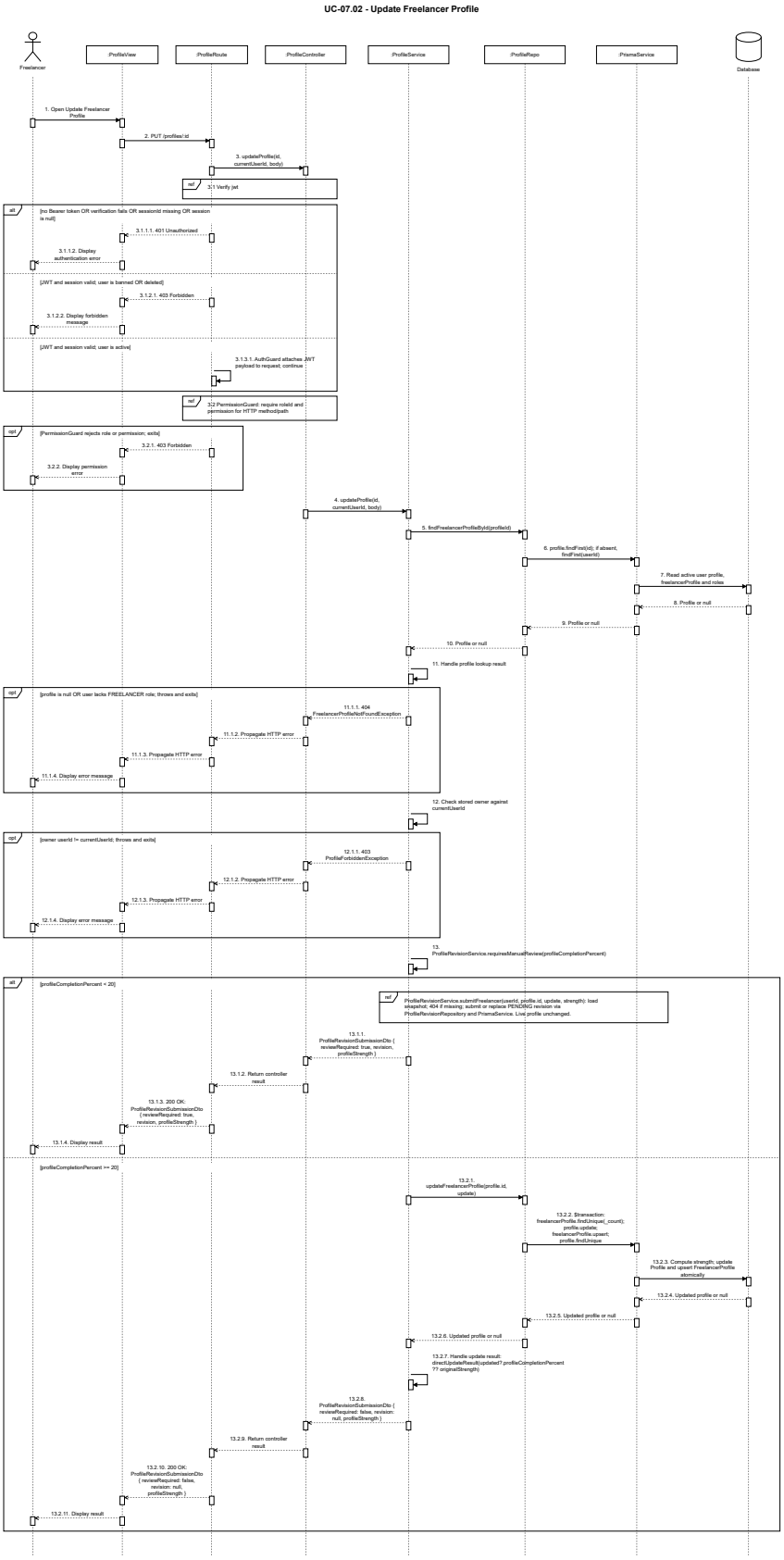

Figure 26: Sequence diagram of UC-07.02 Update Freelancer Profile

3.3 Freelancer Skill Management

Freelancers can view skills, search the catalog, add a skill with a proficiency level, and remove an owned skill association.

3.3.1 Class Diagram

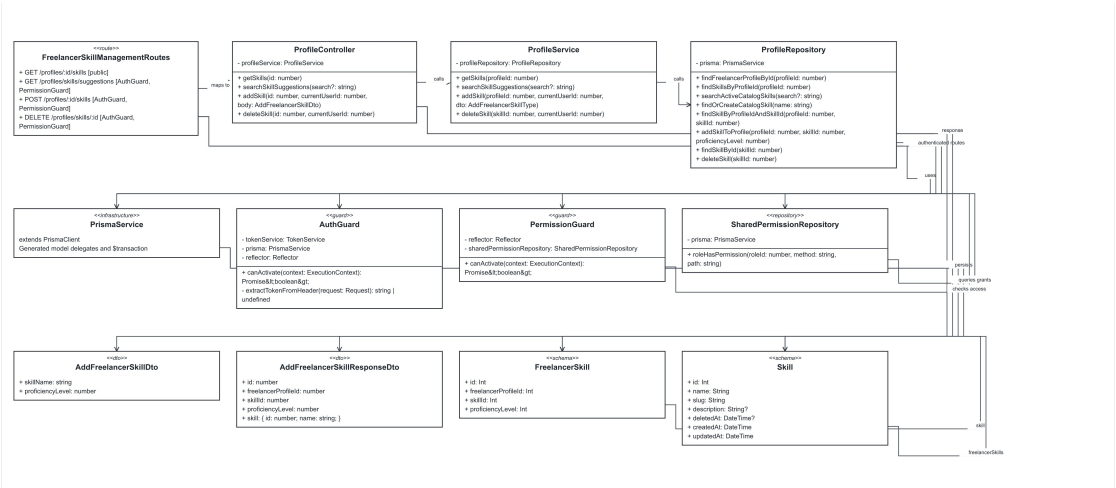

Figure 27: Class diagram of Freelancer Skill Management

3.3.2 UC-08.01 – View Freelancer Skill List

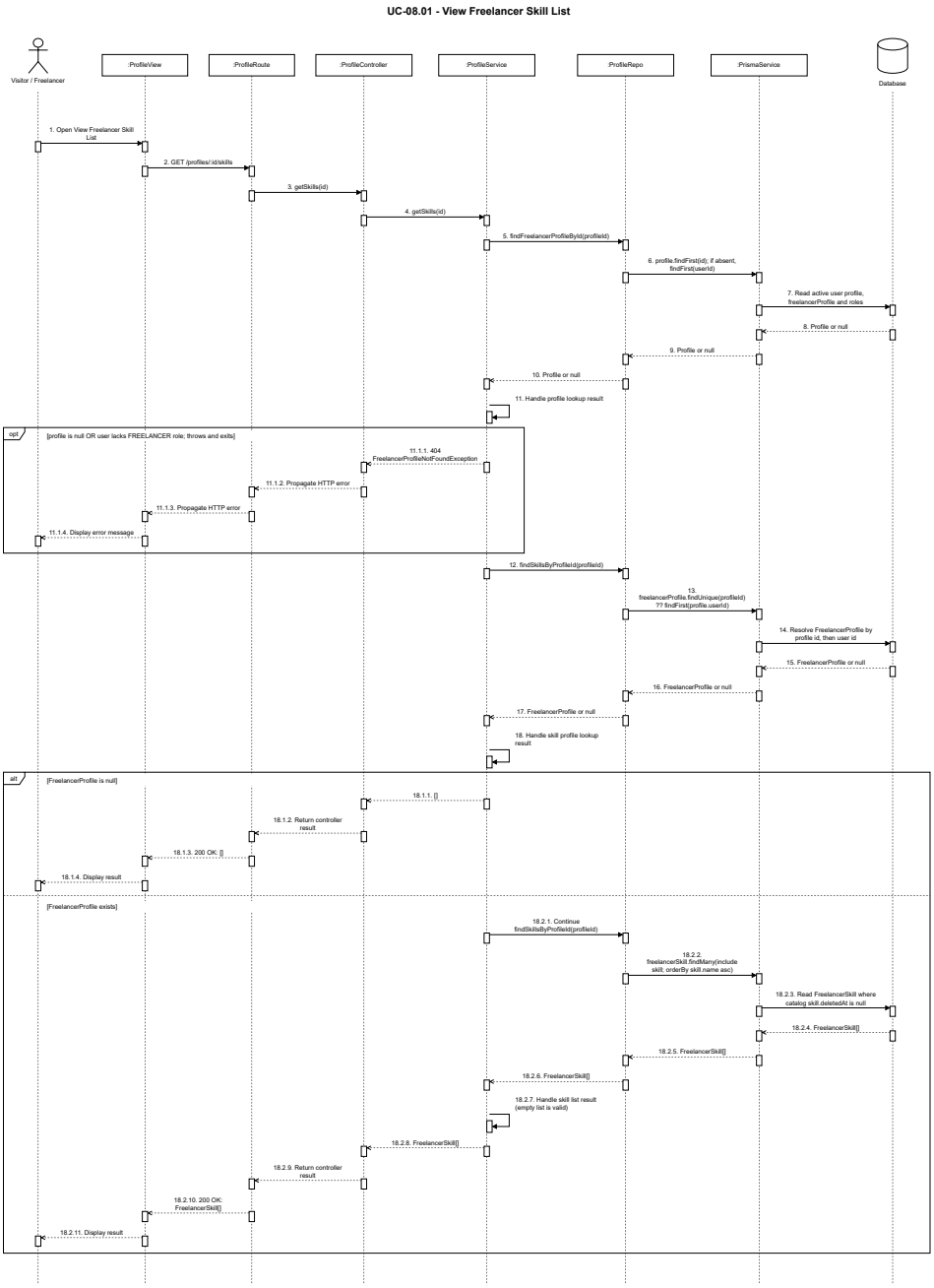

Figure 28: Sequence diagram of UC-08.01 View Freelancer Skill List

3.3.3 UC-08.02 – Add Freelancer Skill

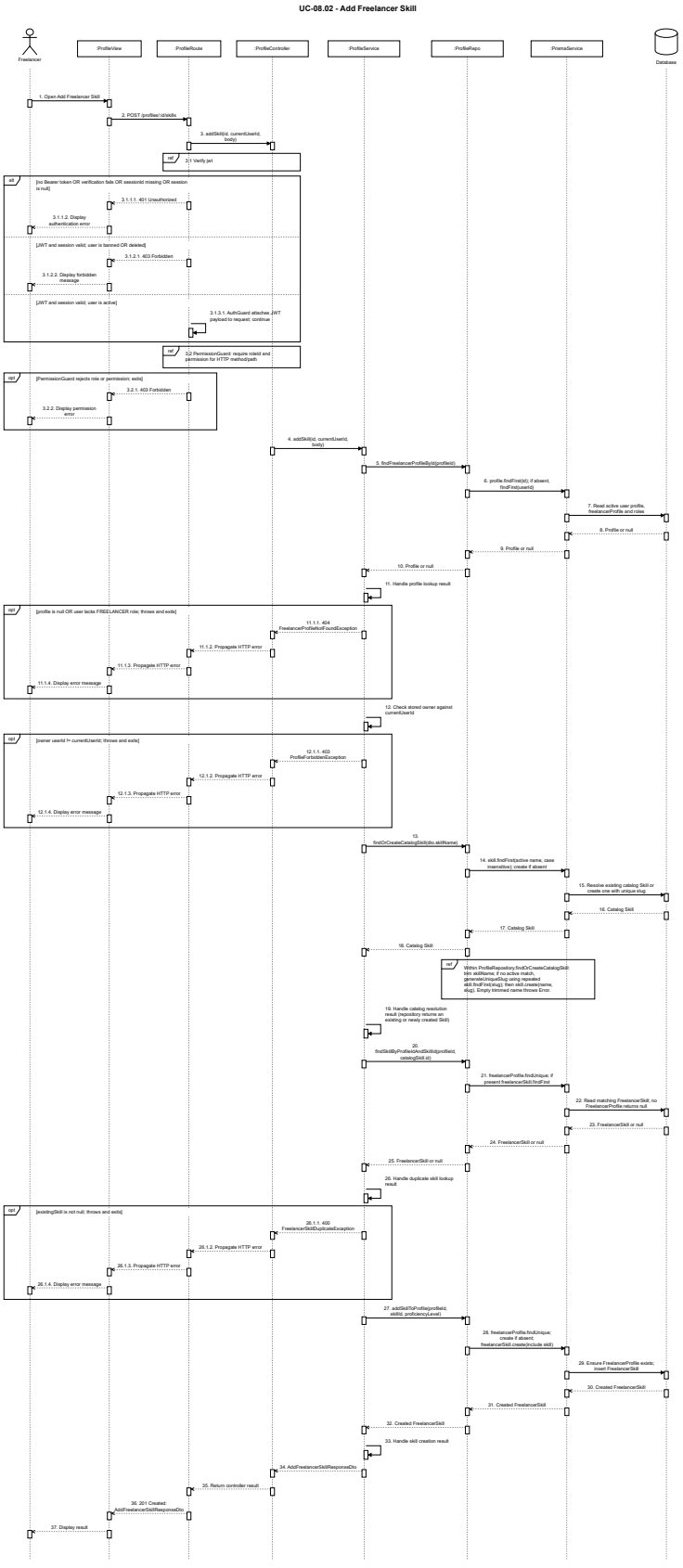

Figure 29: Sequence diagram of UC-08.02 Add Freelancer Skill

3.3.4 UC-08.03 – Delete Freelancer Skill

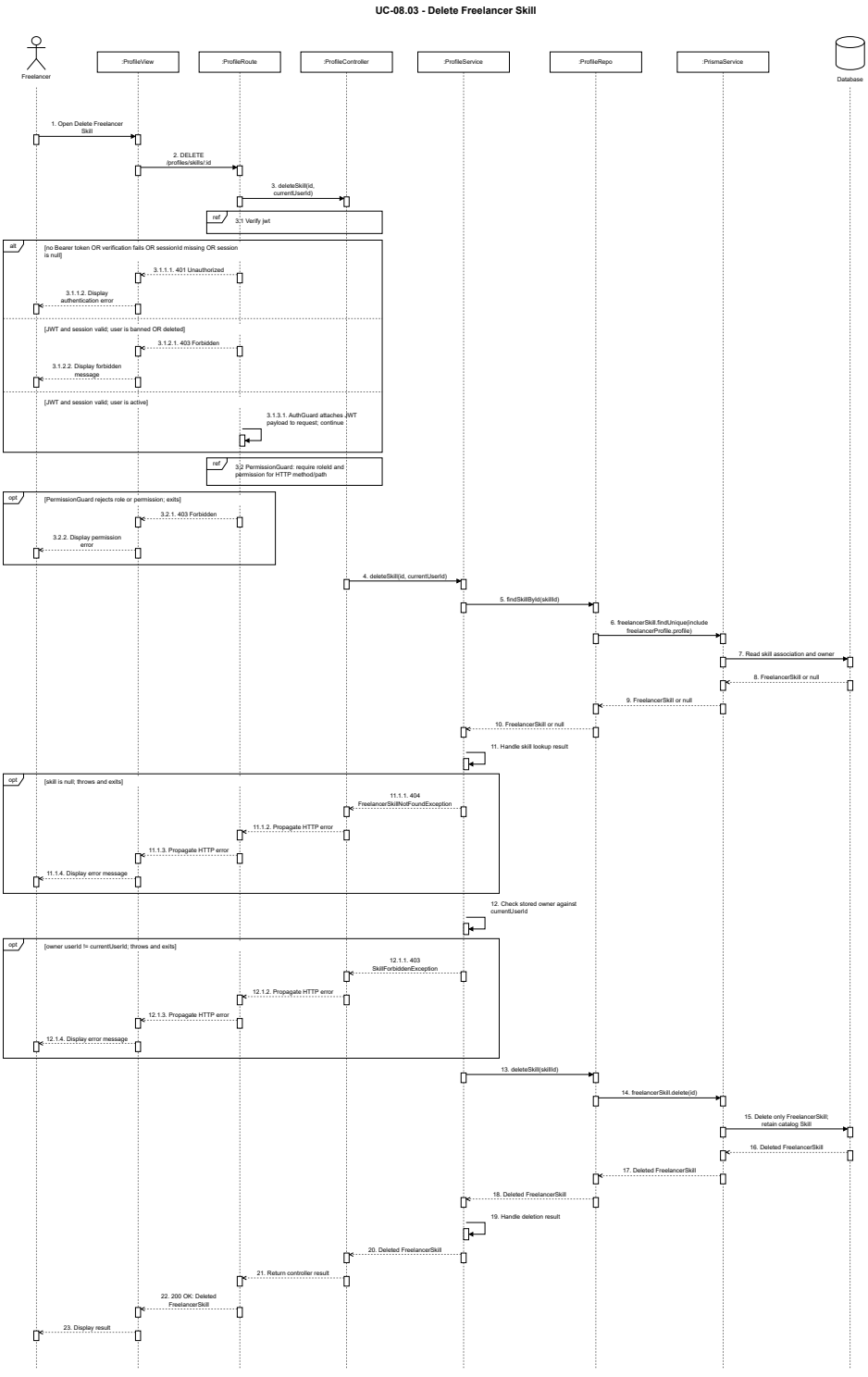

Figure 30: Sequence diagram of UC-08.03 Delete Freelancer Skill

3.4 Portfolio Management

Public portfolio list and detail queries are separated from owner-authorised creation, update, and soft deletion.

3.4.1 Class Diagram

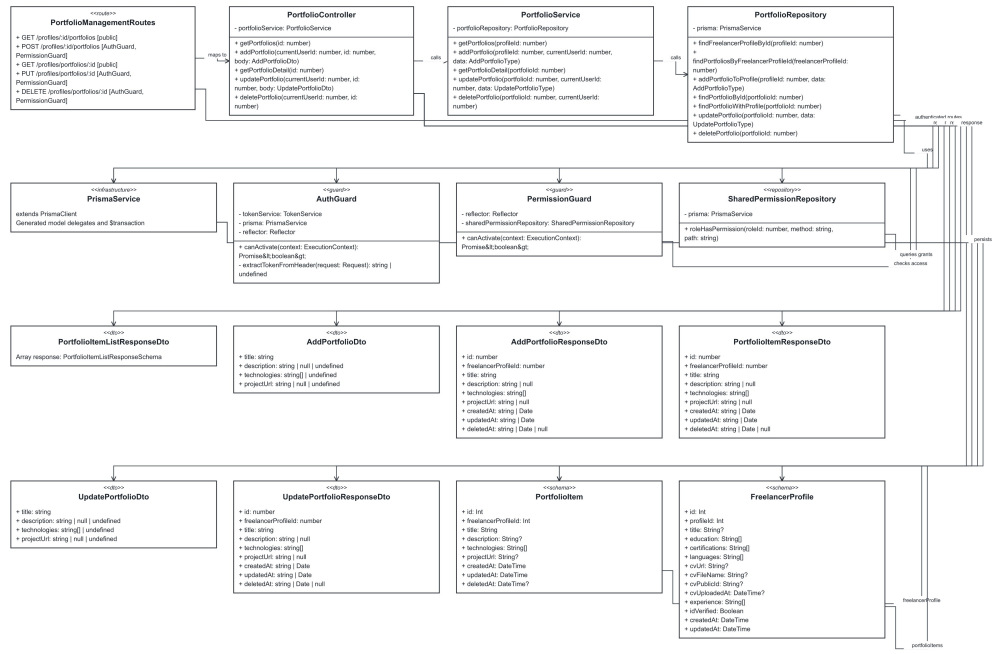

Figure 31: Class diagram of Portfolio Management

3.4.2 UC-09.01 – View Portfolio List

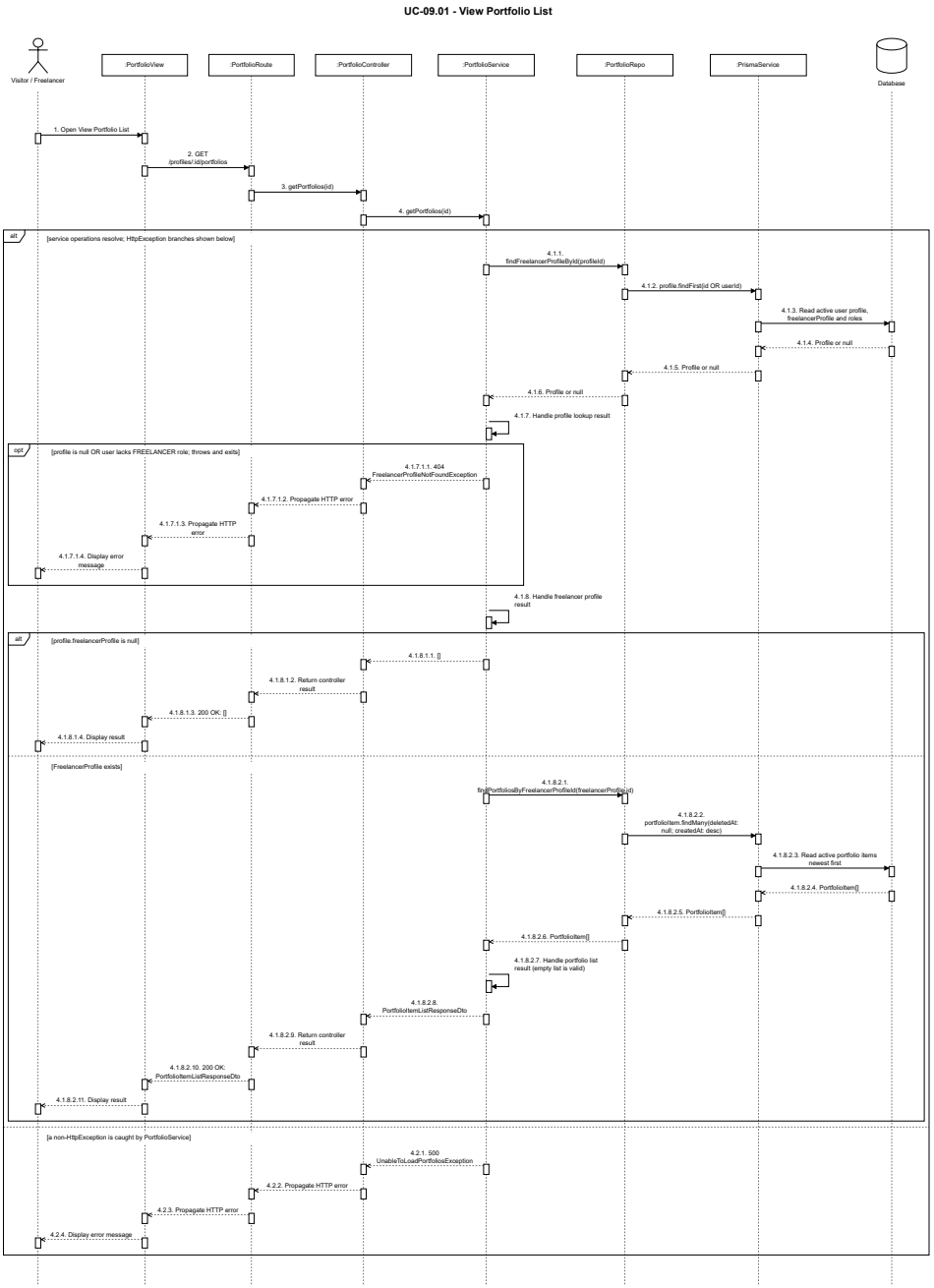

Figure 32: Sequence diagram of UC-09.01 View Portfolio List

3.4.3 UC-09.02 – Add Portfolio

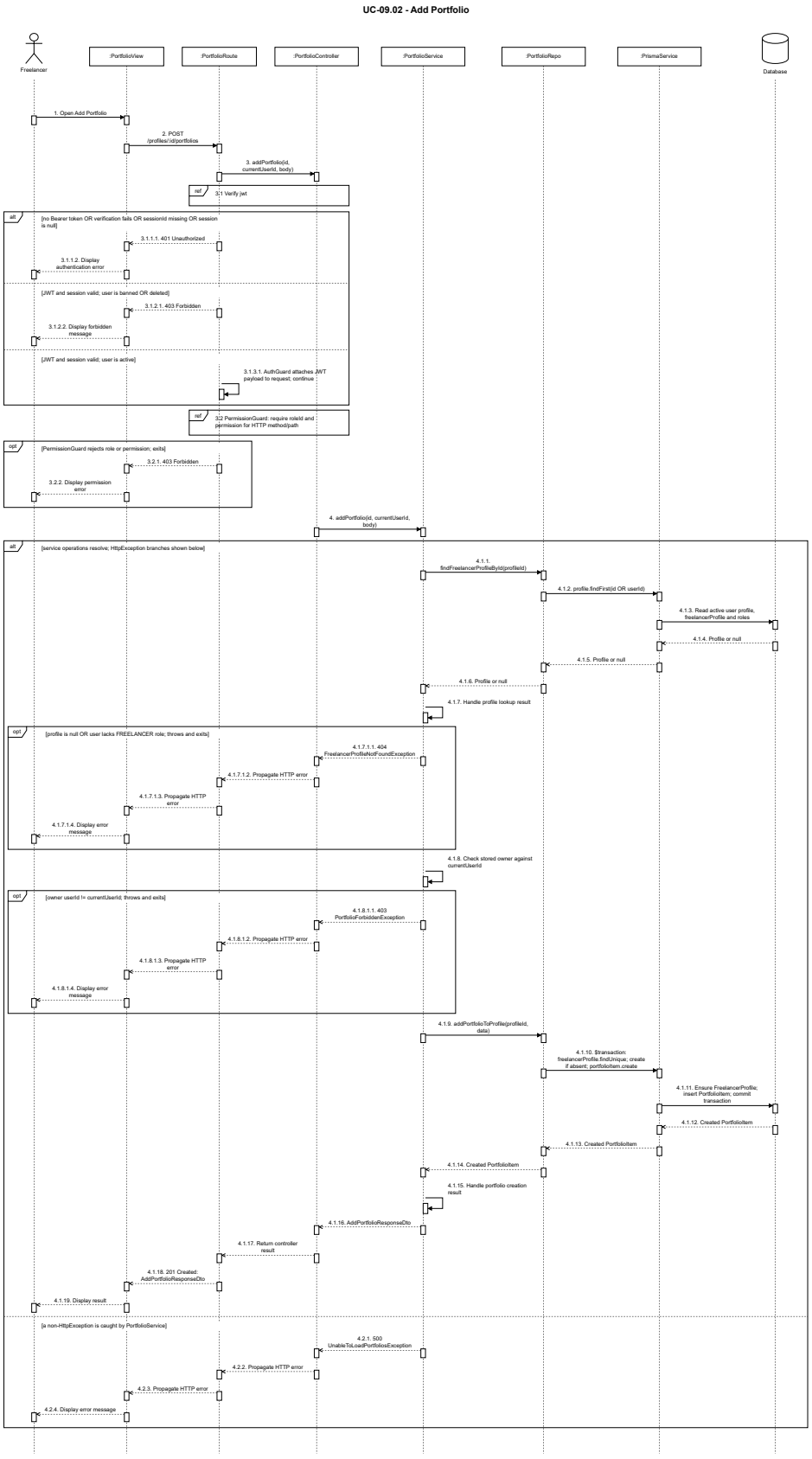

Figure 33: Sequence diagram of UC-09.02 Add Portfolio

3.4.4 UC-09.03 – View Portfolio Detail

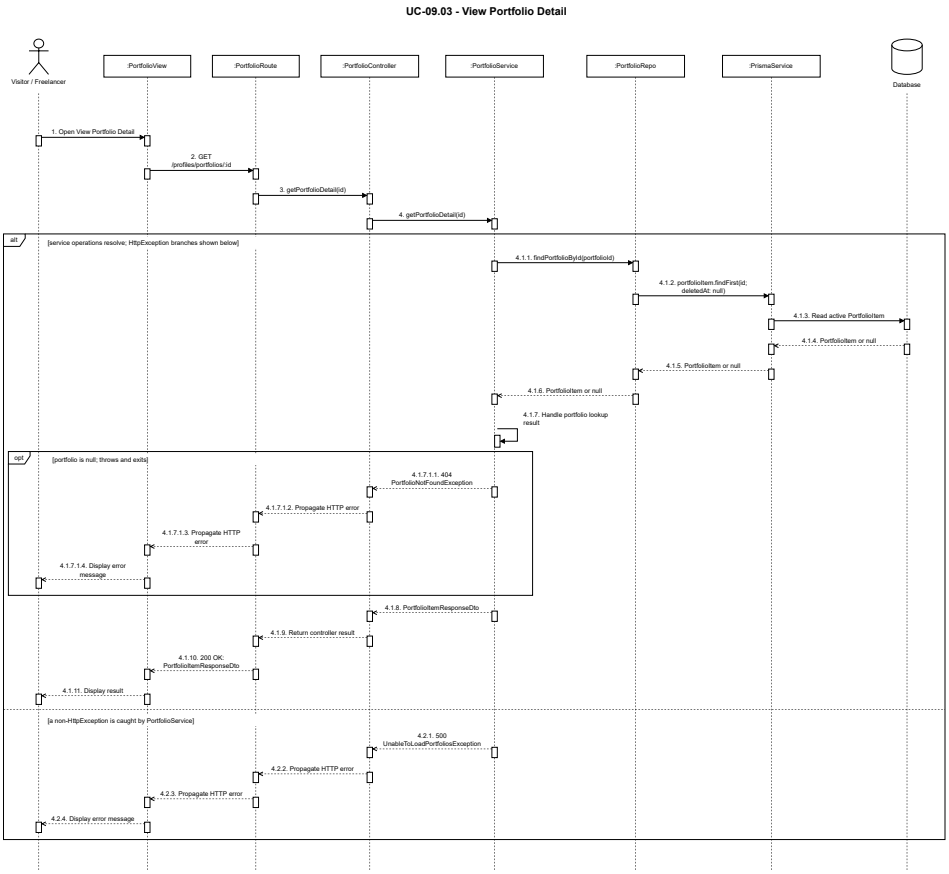

Figure 34: Sequence diagram of UC-09.03 View Portfolio Detail

3.4.5 UC-09.04 – Update Portfolio

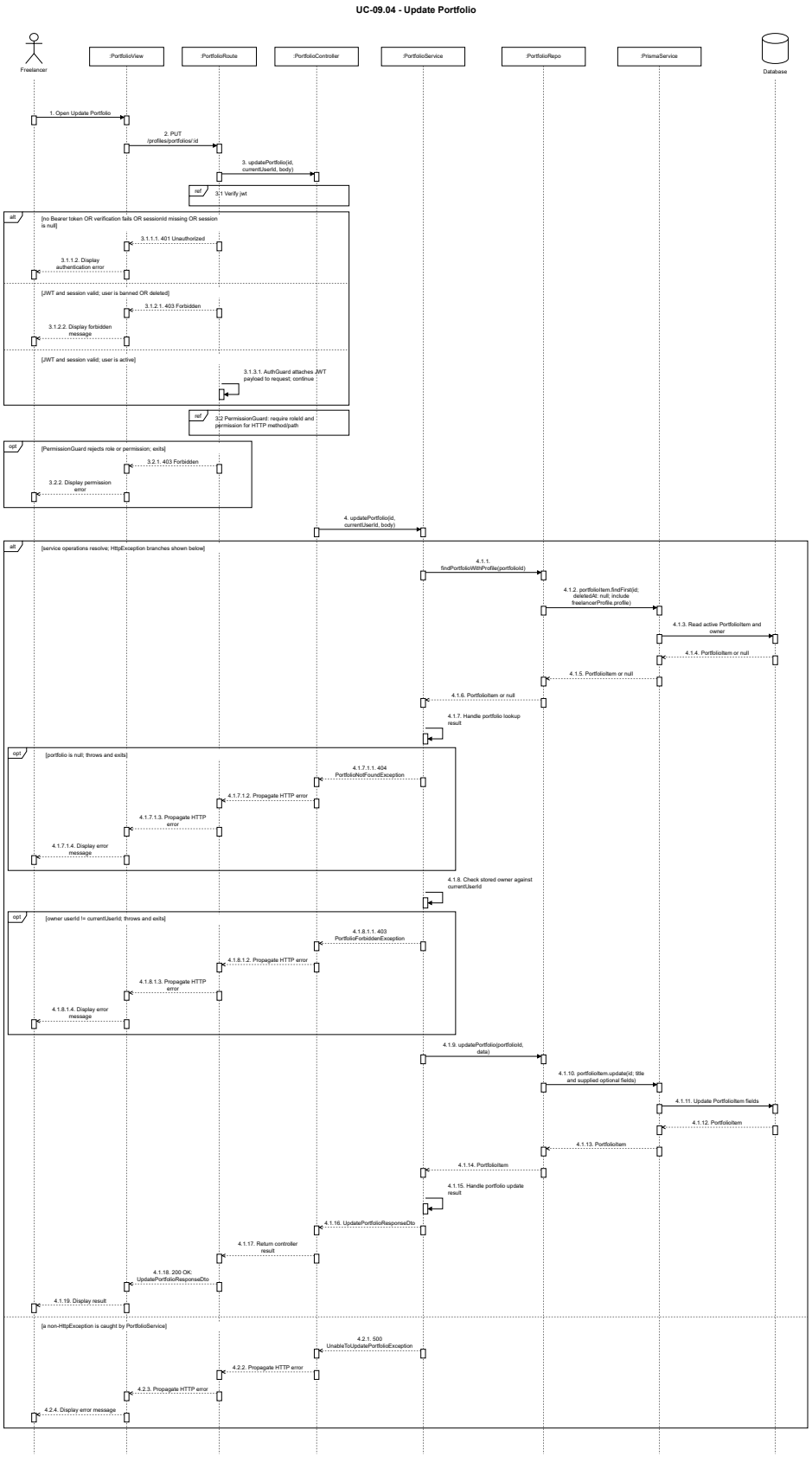

Figure 35: Sequence diagram of UC-09.04 Update Portfolio

3.4.6 UC-09.05 – Delete Portfolio

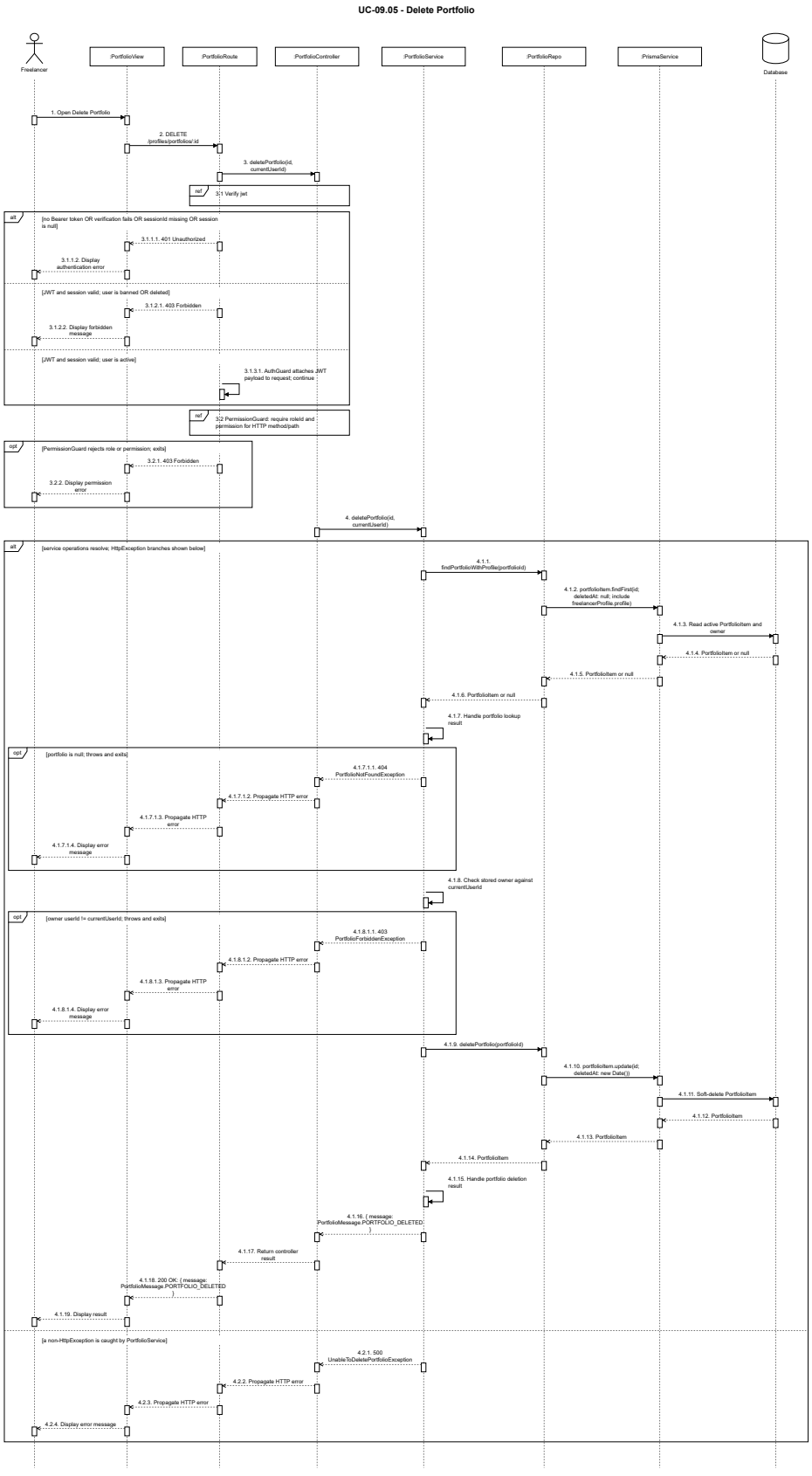

Figure 36: Sequence diagram of UC-09.05 Delete Portfolio

3.5 Job Browsing

Job Browsing allows guests and registered users to search, filter, and view the list of available jobs on the platform, as well as view comprehensive job details.

3.5.1 Class Diagram

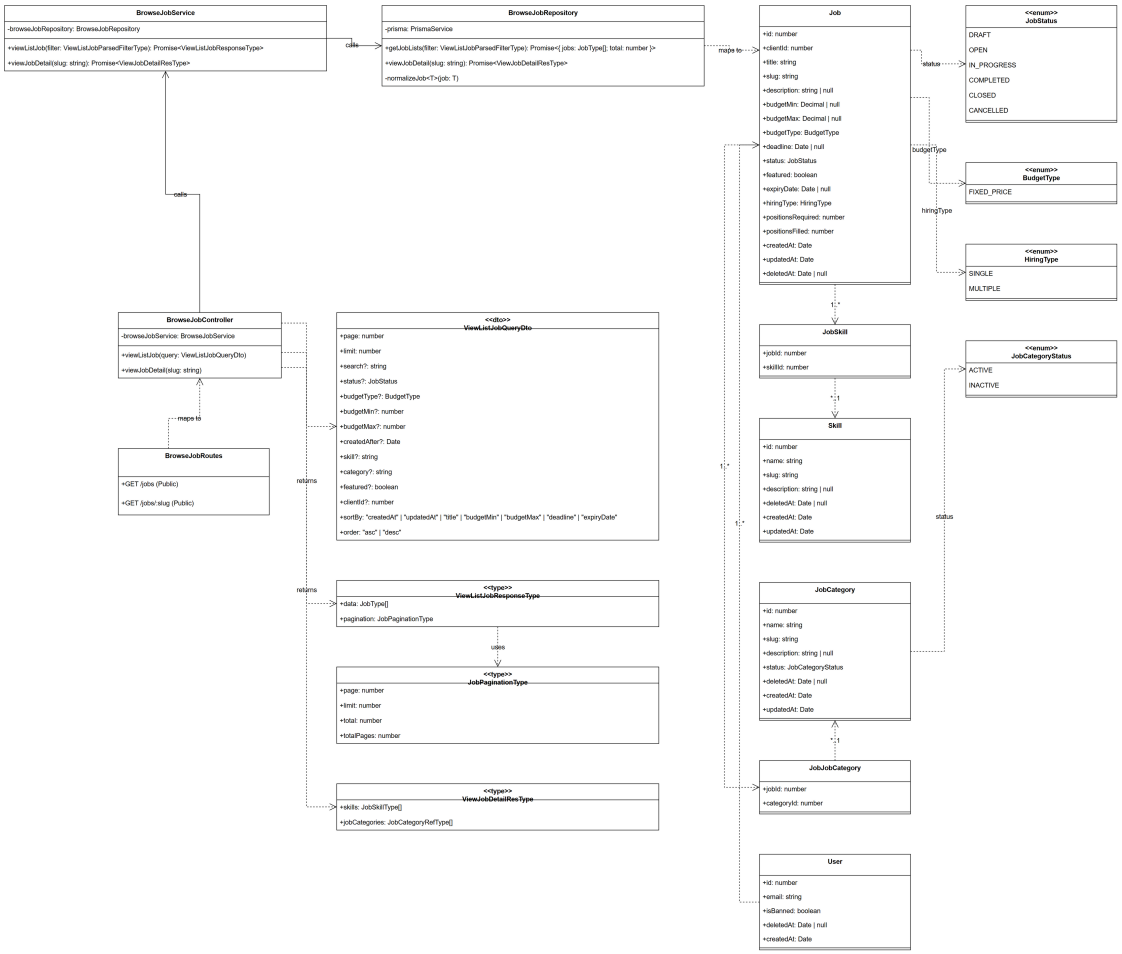

Figure 37: Class diagram of Job Browsing

3.5.2 UC-10.01 – View Job List

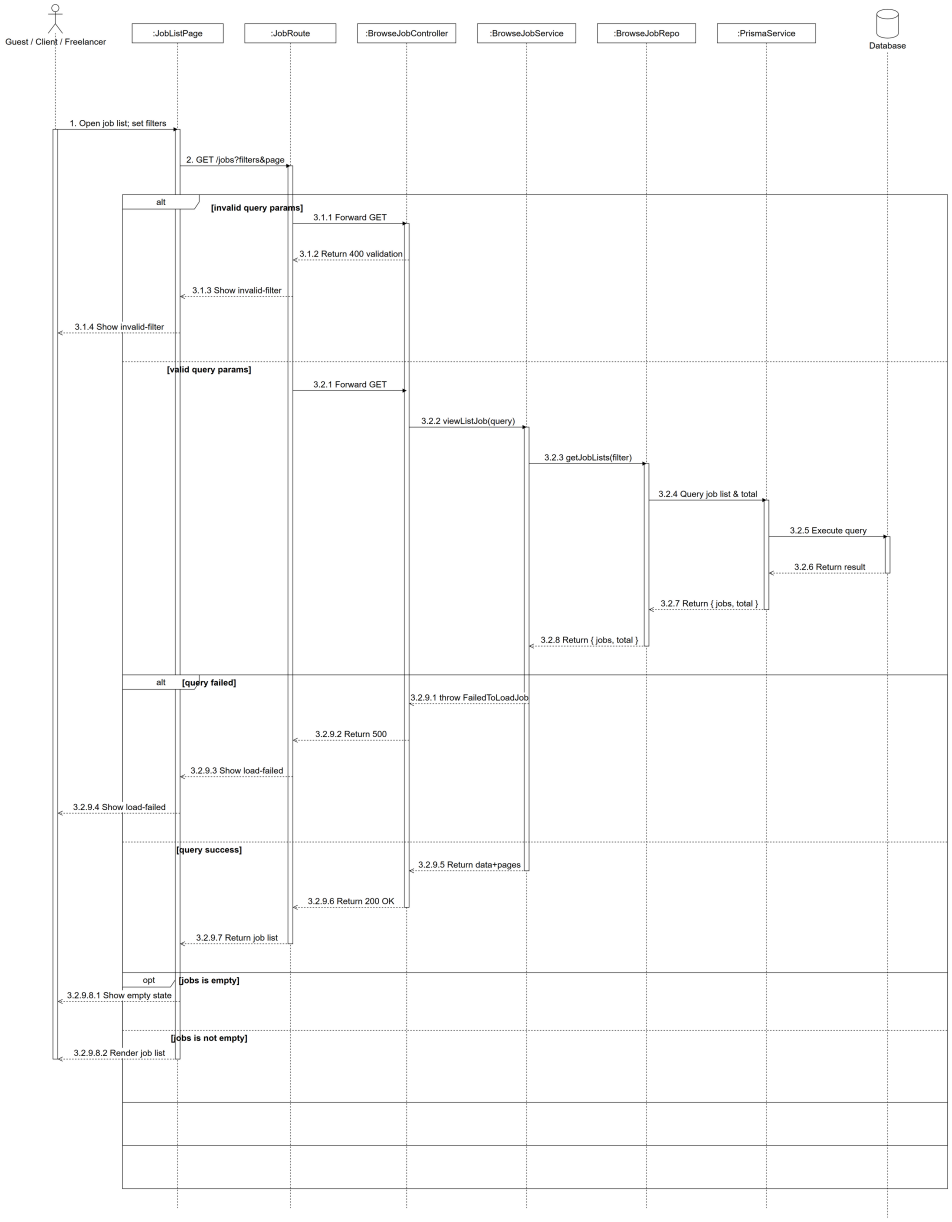

Figure 38: Sequence diagram of UC-10.01 View Job List

3.5.3 UC-10.02 – View Job Detail

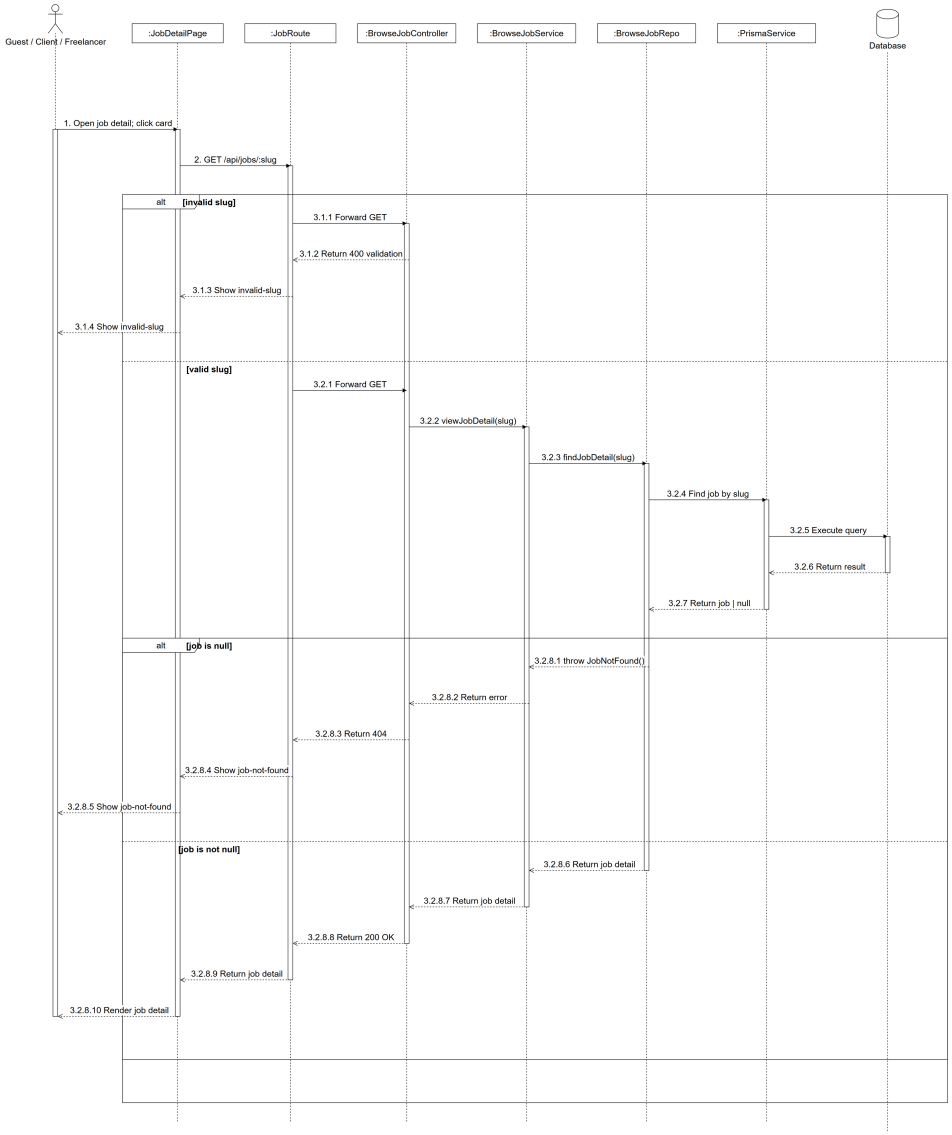

Figure 39: Sequence diagram of UC-10.02 View Job Detail

3.6 Job Management

Job Management enables clients to manage their posted jobs throughout their lifecycle, including creating new jobs, editing job parameters, publishing drafts, closing completed or cancelled jobs, and soft-deleting job postings.

3.6.1 Class Diagram

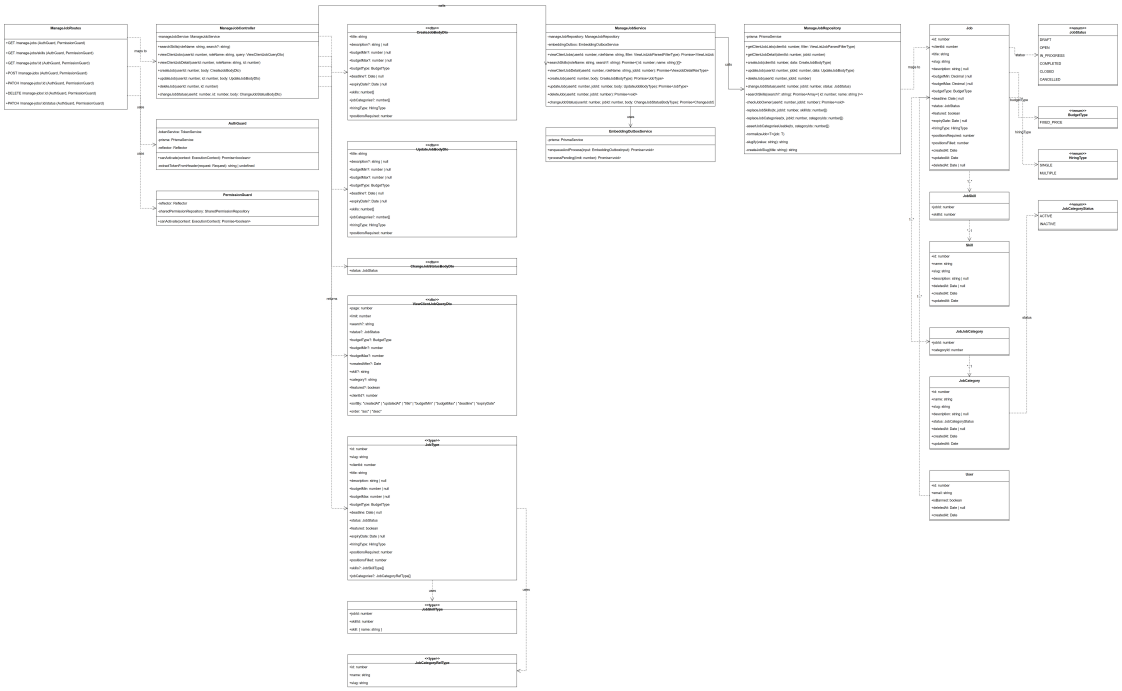

Figure 40: Class diagram of Job Management

3.6.2 UC-11.01 – View My Jobs

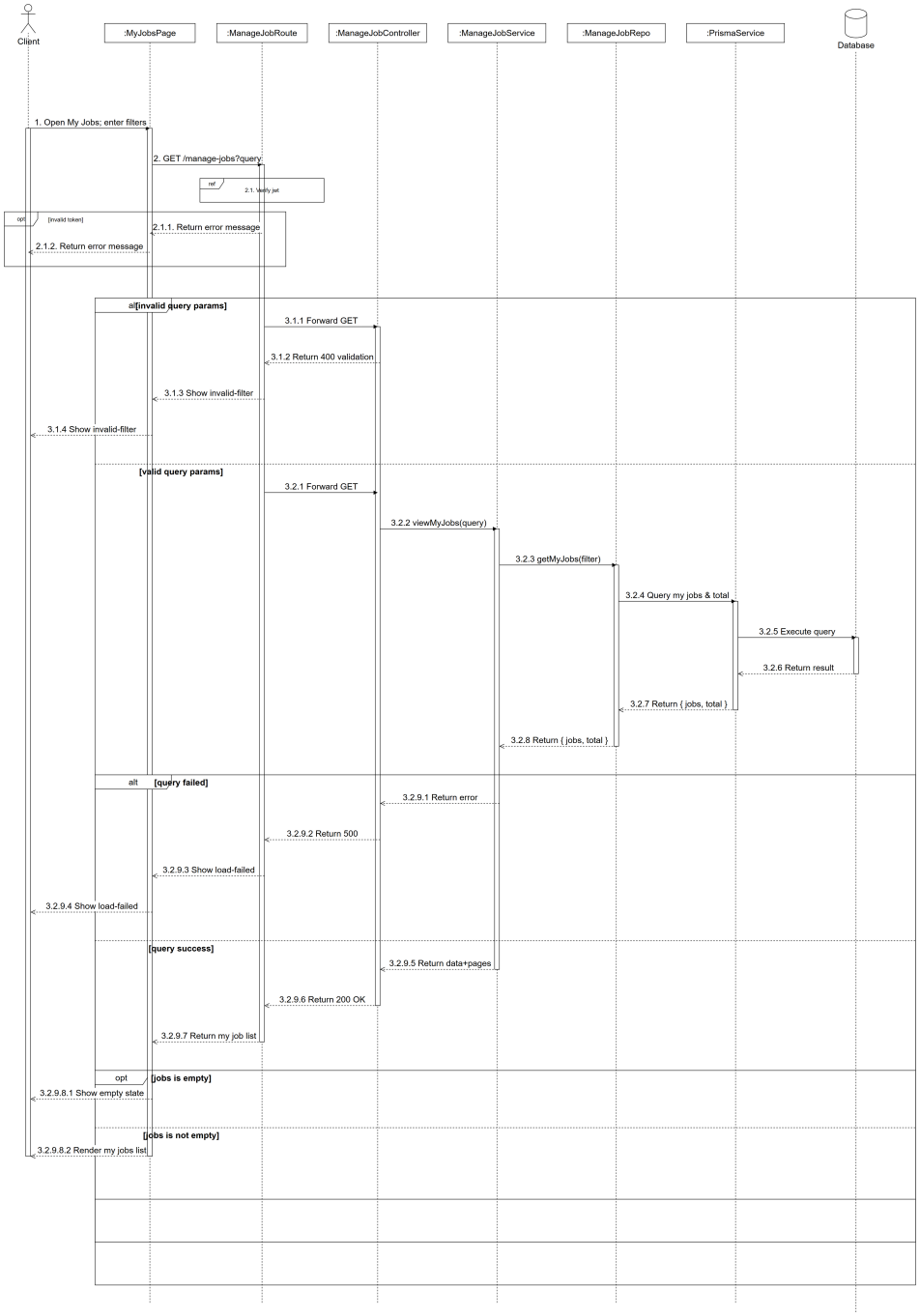

Figure 41: Sequence diagram of UC-11.01 View My Jobs

3.6.3 UC-11.02 – Create Job

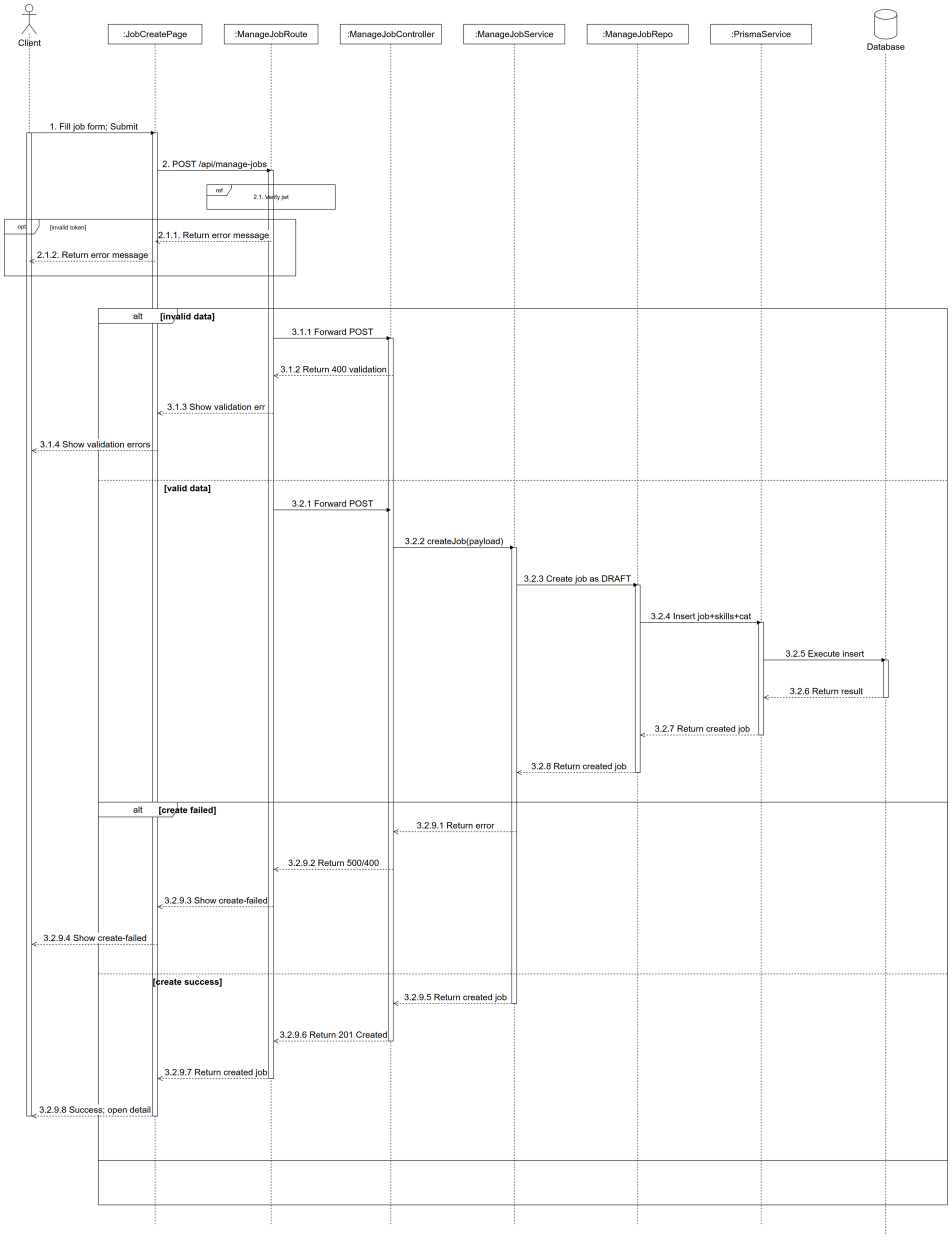

Figure 42: Sequence diagram of UC-11.02 Create Job

3.6.4 UC-11.03 – Publish Job

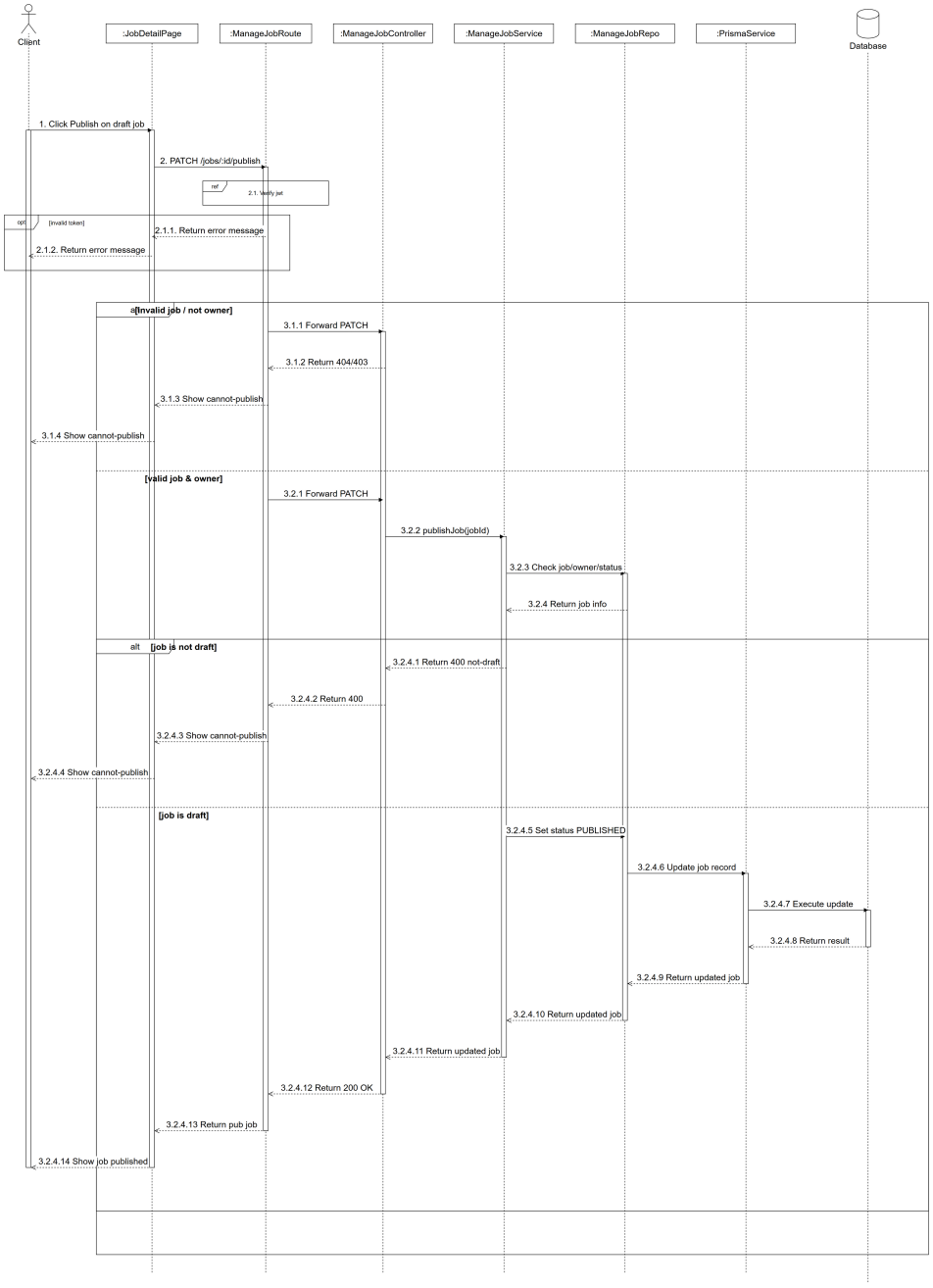

Figure 43: Sequence diagram of UC-11.03 Publish Job

3.6.5 UC-11.04 – Update Job

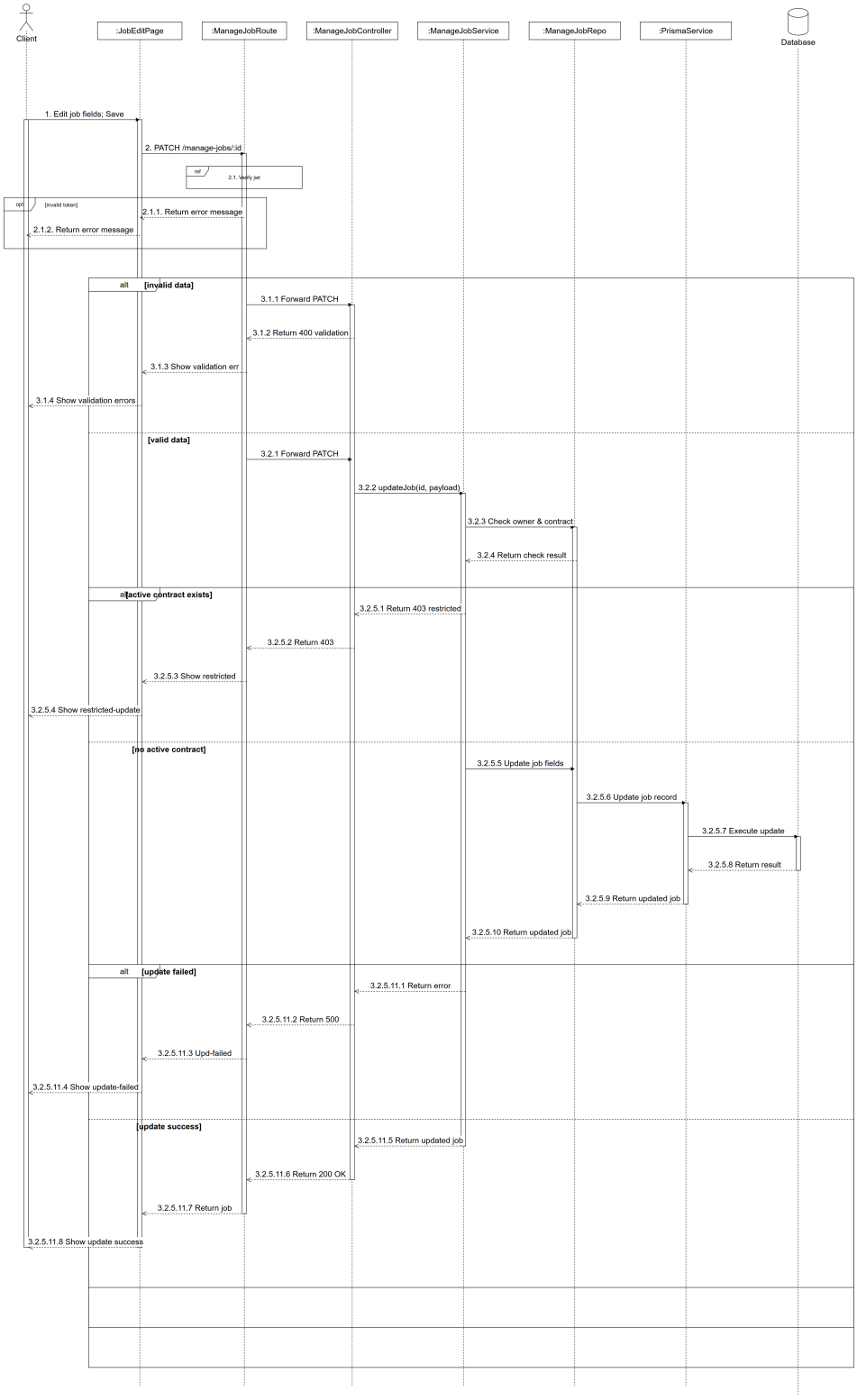

Figure 44: Sequence diagram of UC-11.04 Update Job

3.6.6 UC-11.05 – Delete Job

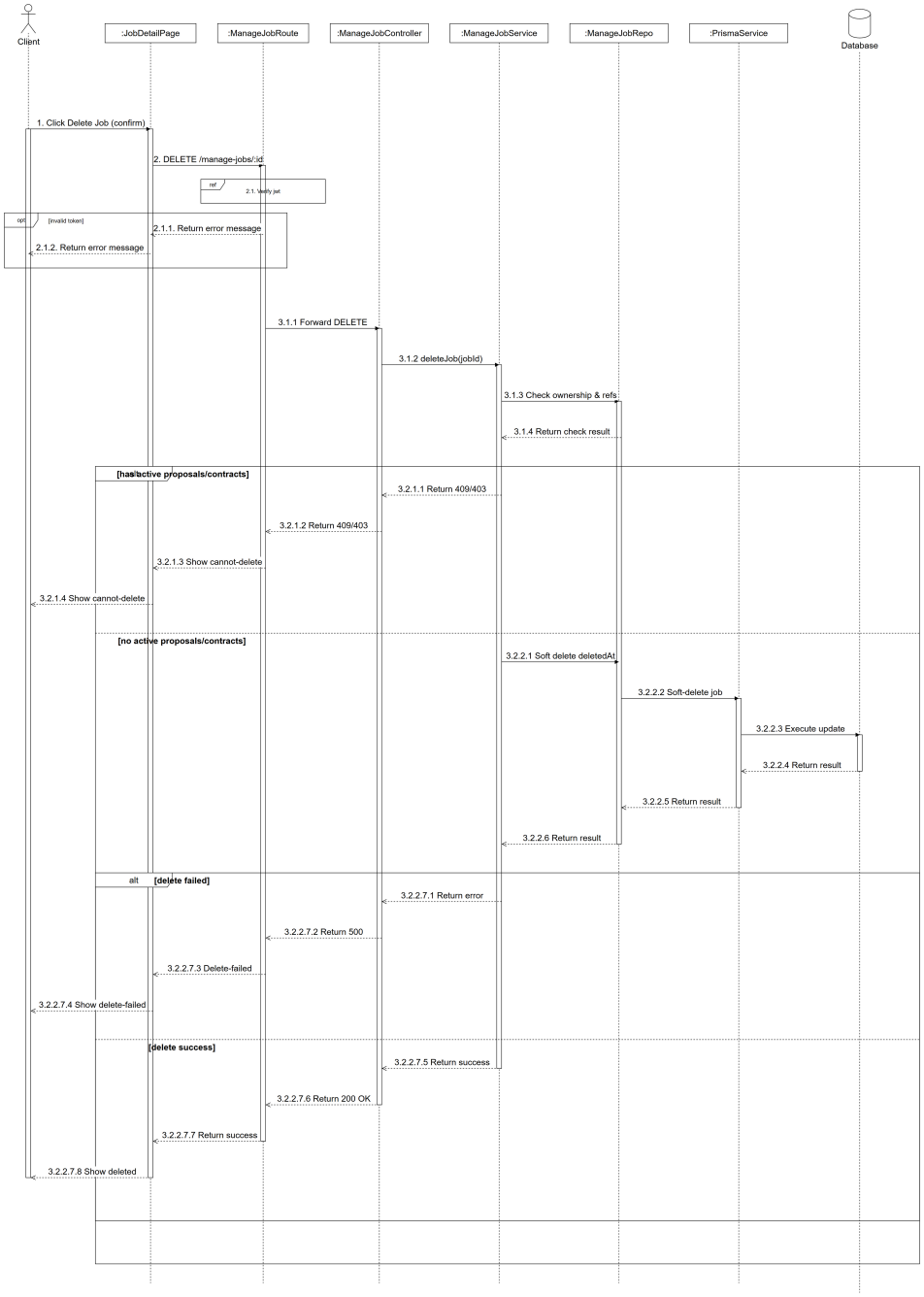

Figure 45: Sequence diagram of UC-11.05 Delete Job

3.6.7 UC-11.06 – Change Job Status

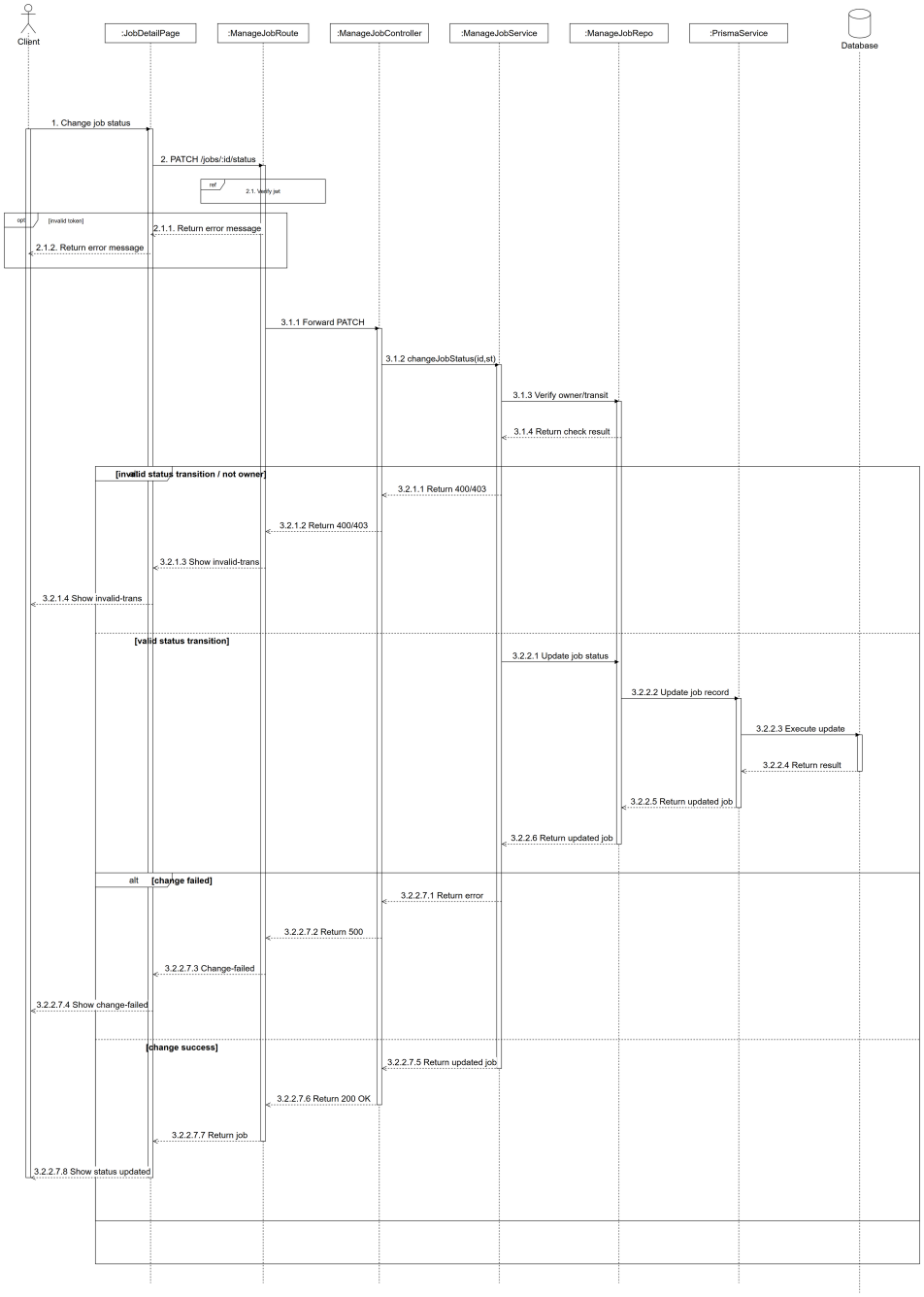

Figure 46: Sequence diagram of UC-11.06 Change Job Status

3.7 Bookmarked Jobs Management

Bookmarked Jobs Management allows freelancers to save interesting job postings for later review, inspect their list of bookmarked jobs, and remove bookmarks when no longer needed.

3.7.1 Class Diagram

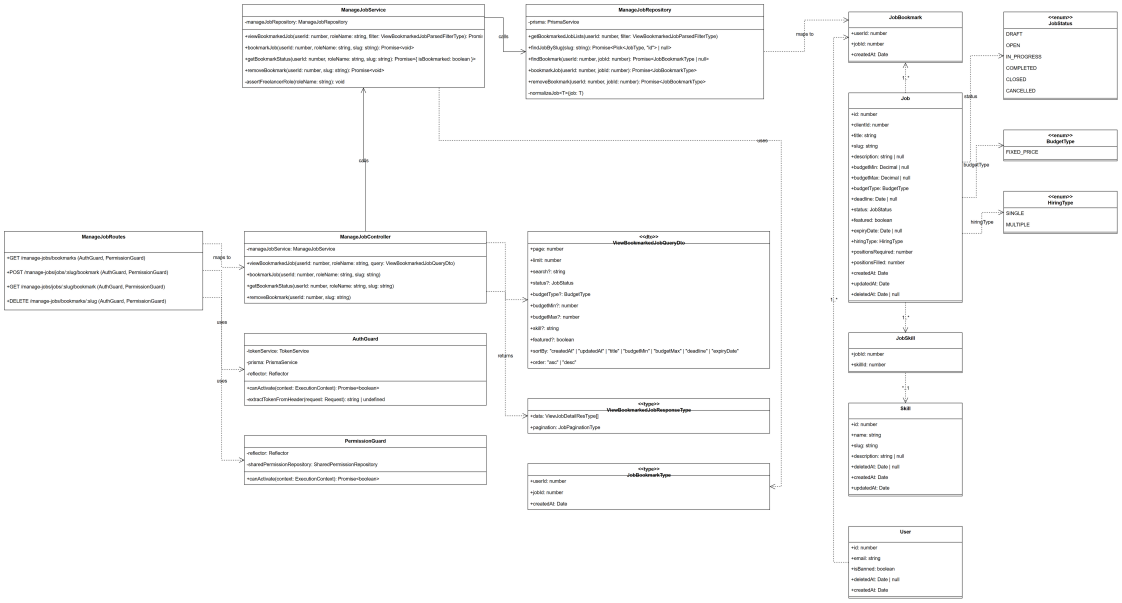

Figure 47: Class diagram of Bookmarked Jobs Management

3.7.2 UC-12.01 – View Bookmarked Job List

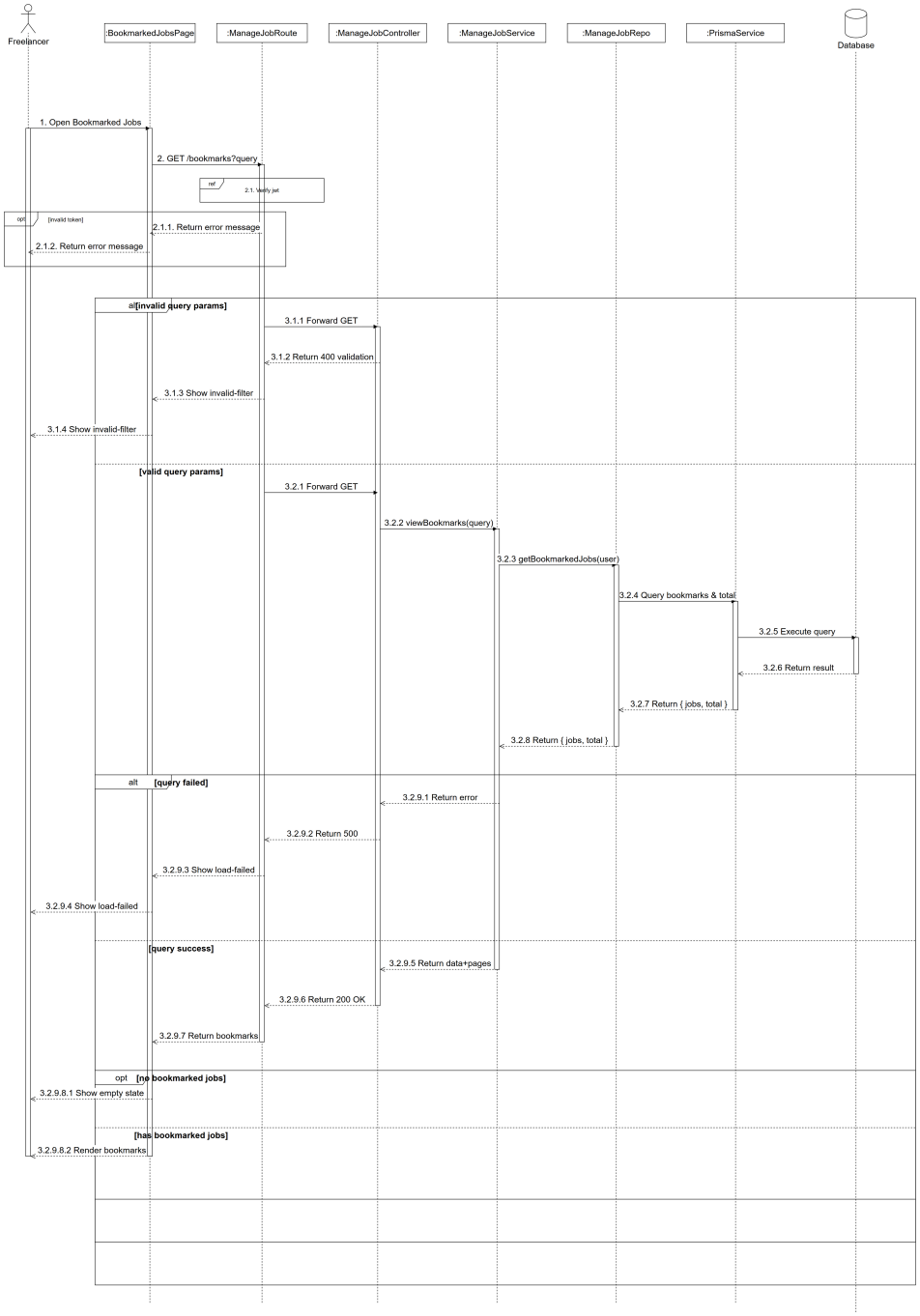

Figure 48: Sequence diagram of UC-12.01 View Bookmarked Job List

3.7.3 UC-12.02 – Bookmark Job

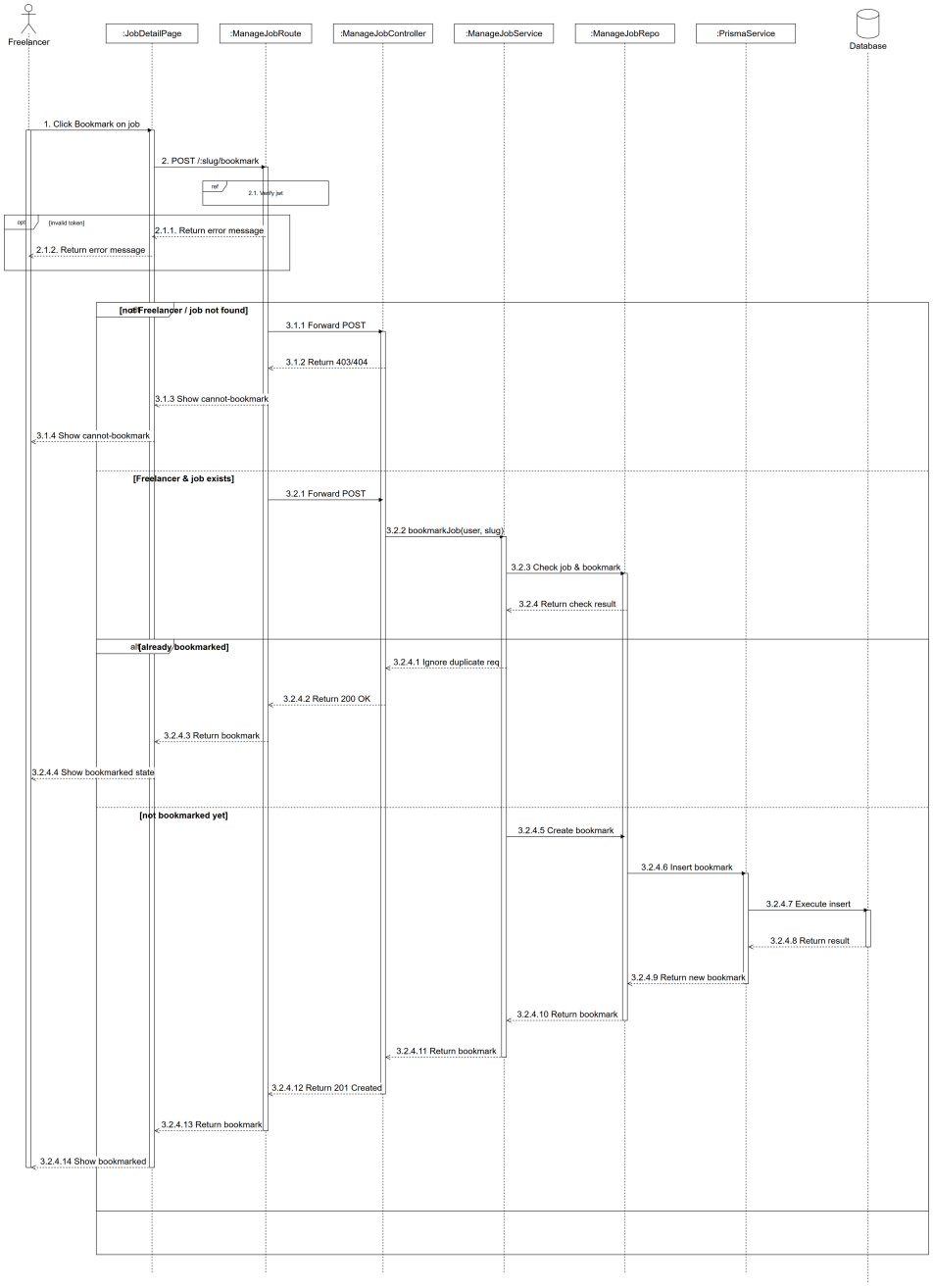

Figure 49: Sequence diagram of UC-12.02 Bookmark Job

3.7.4 UC-12.03 – Remove Bookmark

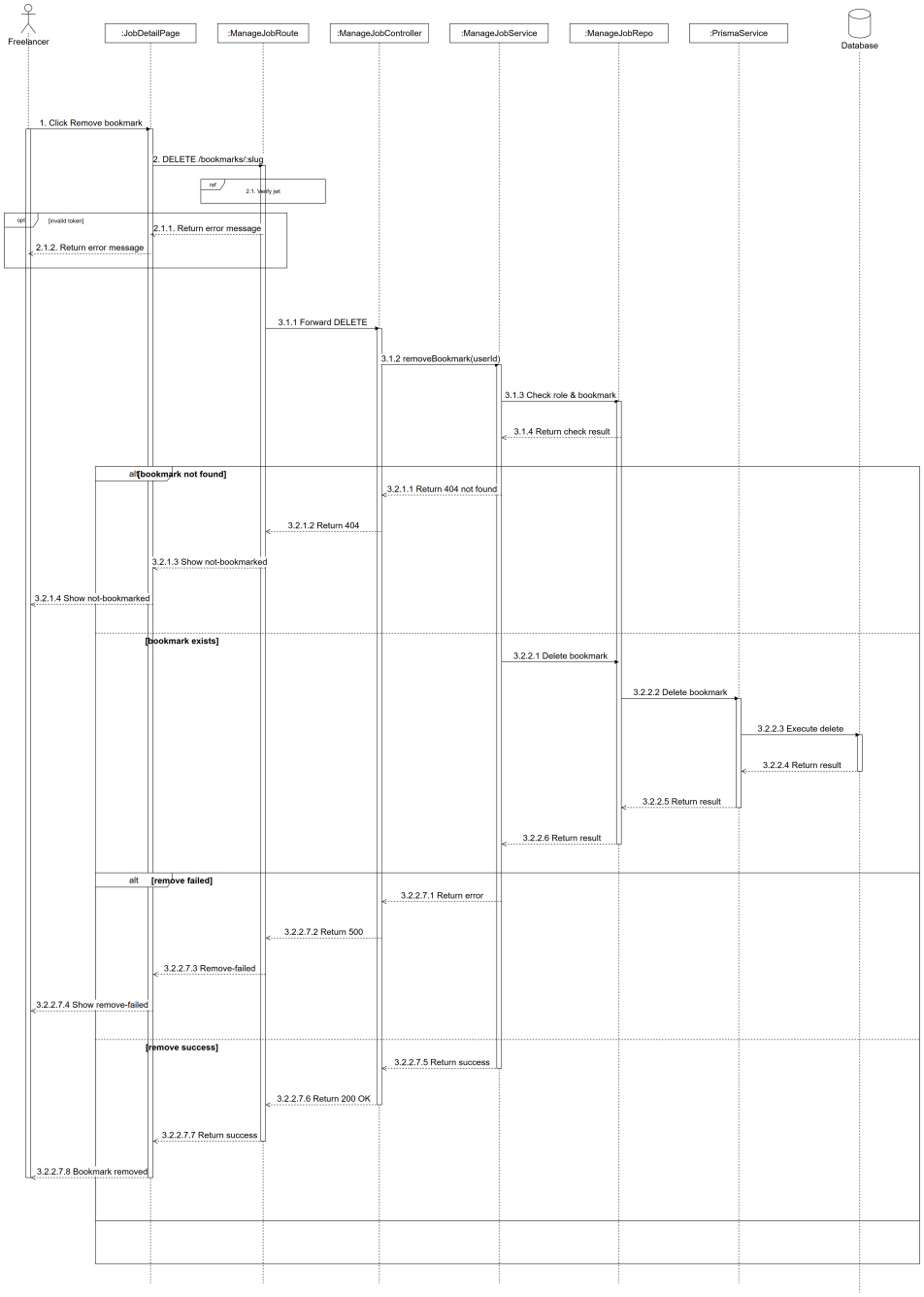

Figure 50: Sequence diagram of UC-12.03 Remove Bookmark

3.8 Verify jwt

Verify JWT is the shared security guard protecting API endpoints across all modules. Every incoming HTTP request with a Bearer token is intercepted, validated for cryptographic signature and expiration time, and verified against the database to guarantee the associated session remains active and has not been revoked.

a. Class Diagram

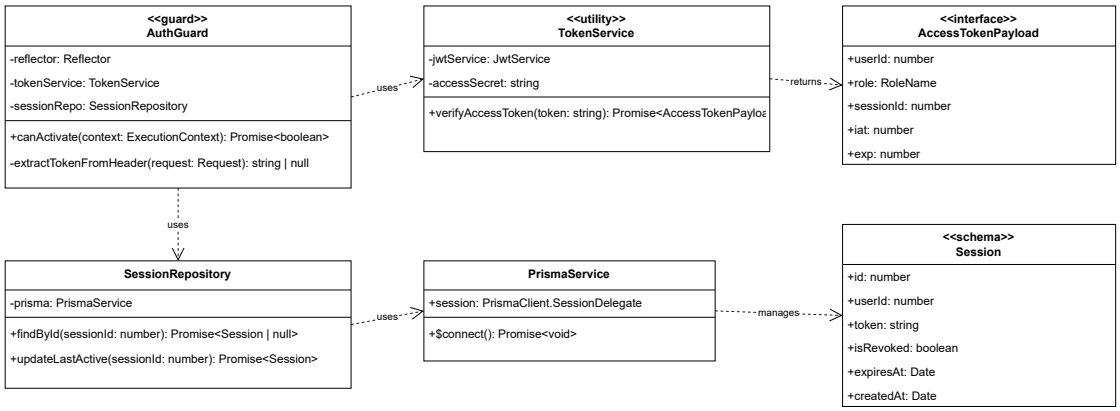

Figure 51: Class diagram of Verify JWT

b. Sequence Diagram(s)

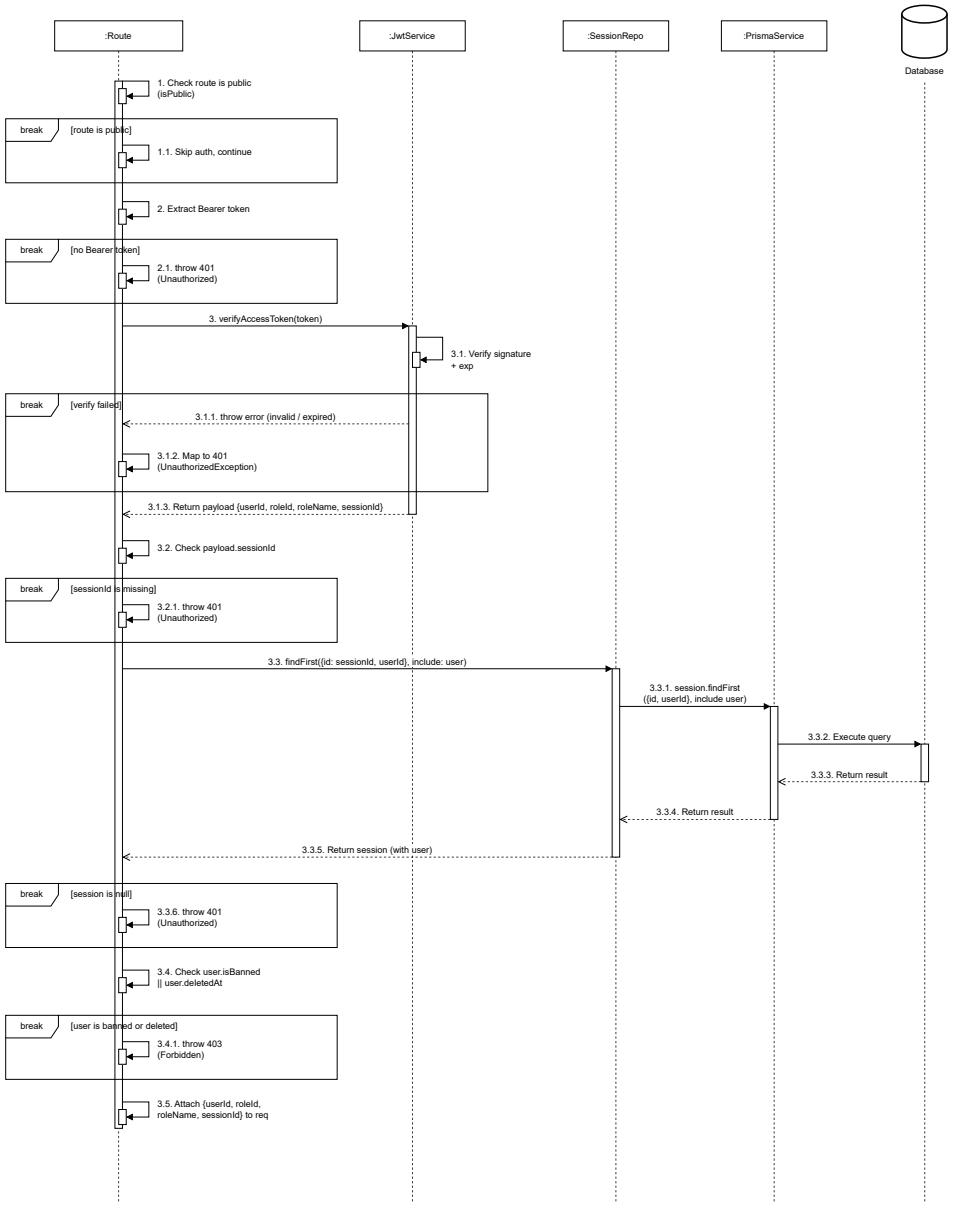

Figure 52: Sequence diagram of Verify JWT

3.9 Authentication

The authentication module covers account registration, email/password and Google OAuth login, logout, password recovery, and the JWT/session flow shared by all protected endpoints.

3.9.1 Class Diagram

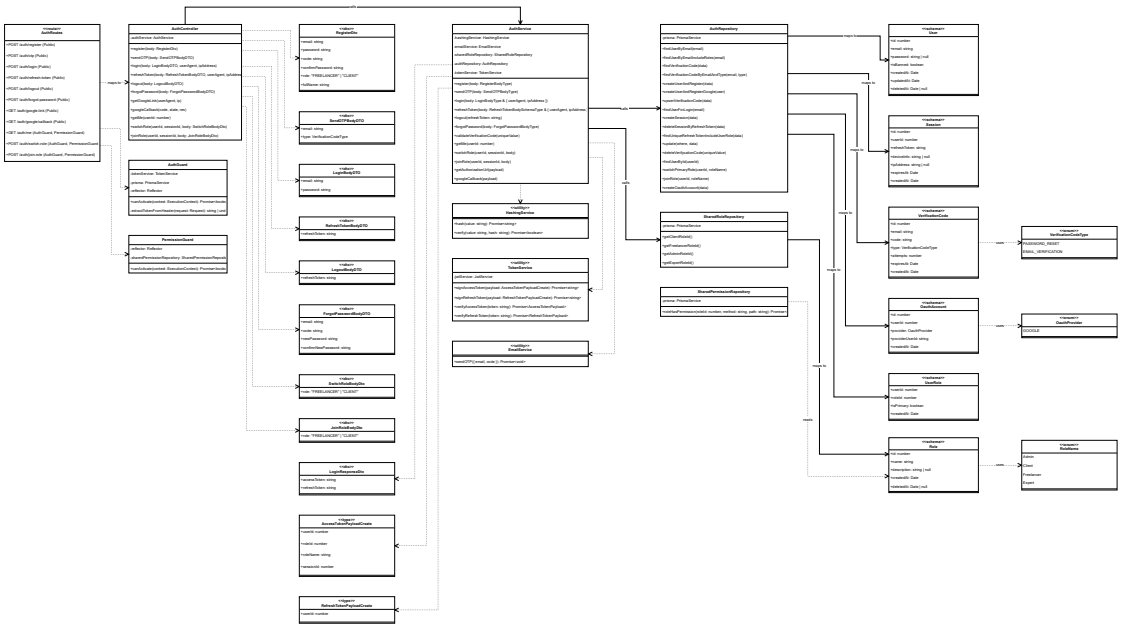

Figure 53: Class diagram of Authentication

3.9.2 UC-01 – Login with Google

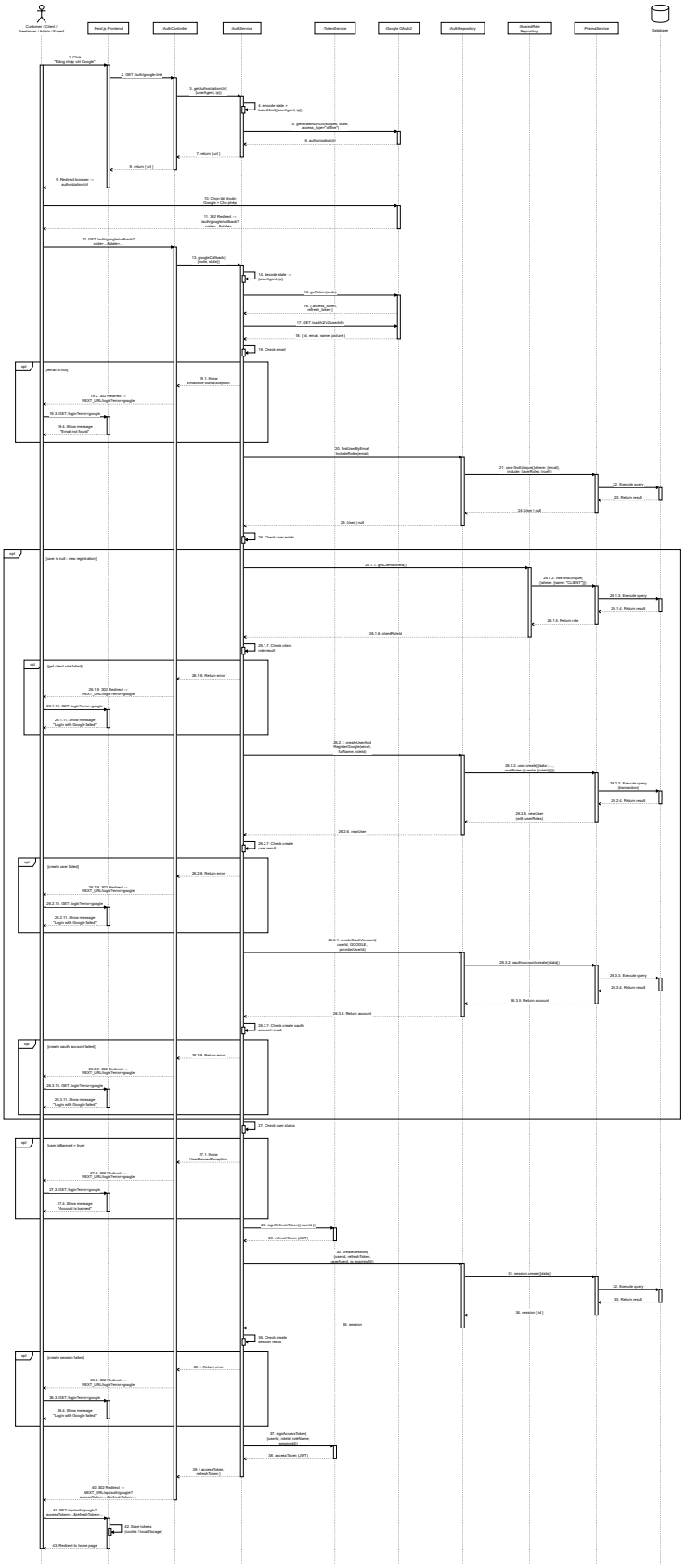

Figure 54: Sequence diagram of UC-01 Login with Google

3.9.3 UC-02 – Login

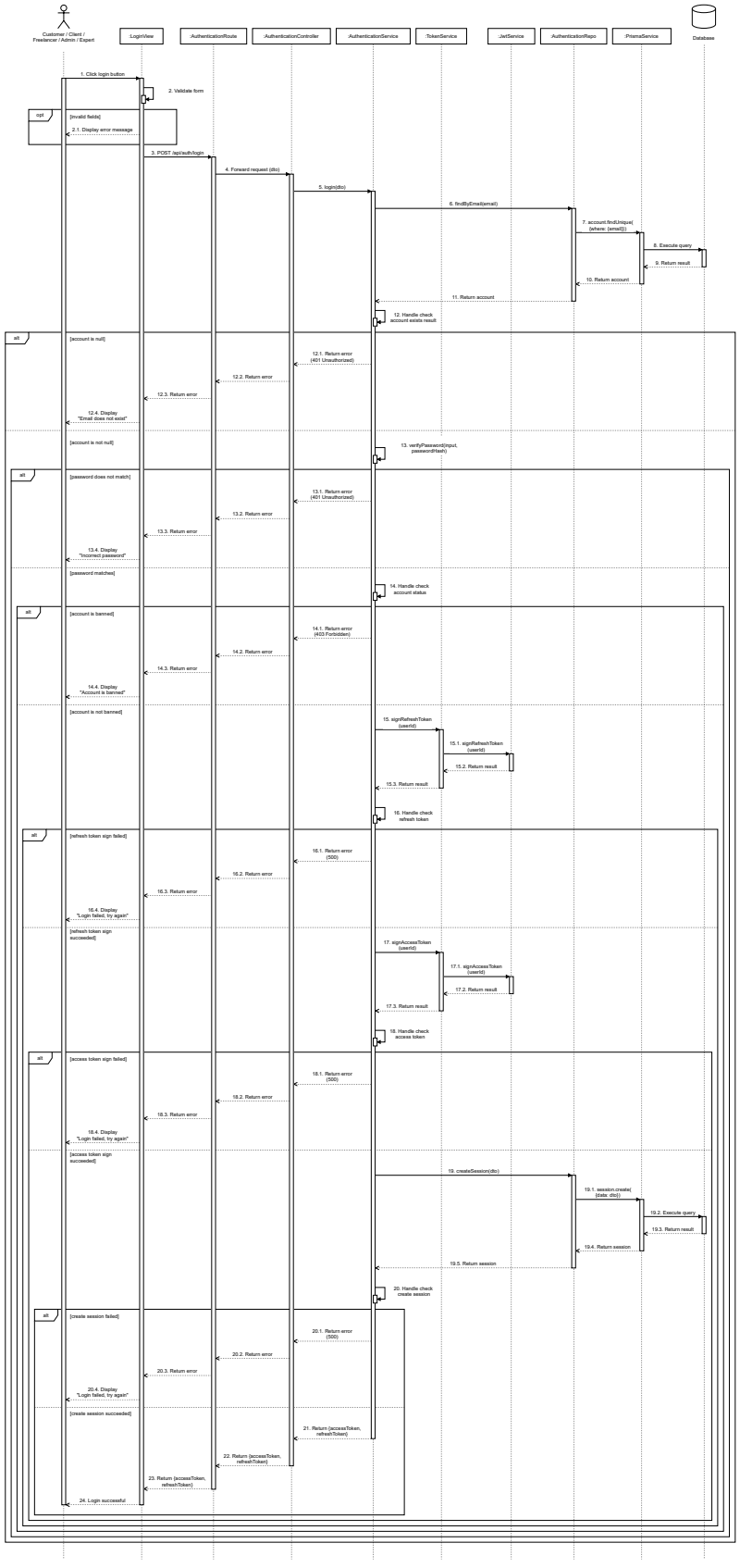

Figure 55: Sequence diagram of UC-02 Login

3.9.4 UC-03 – Logout

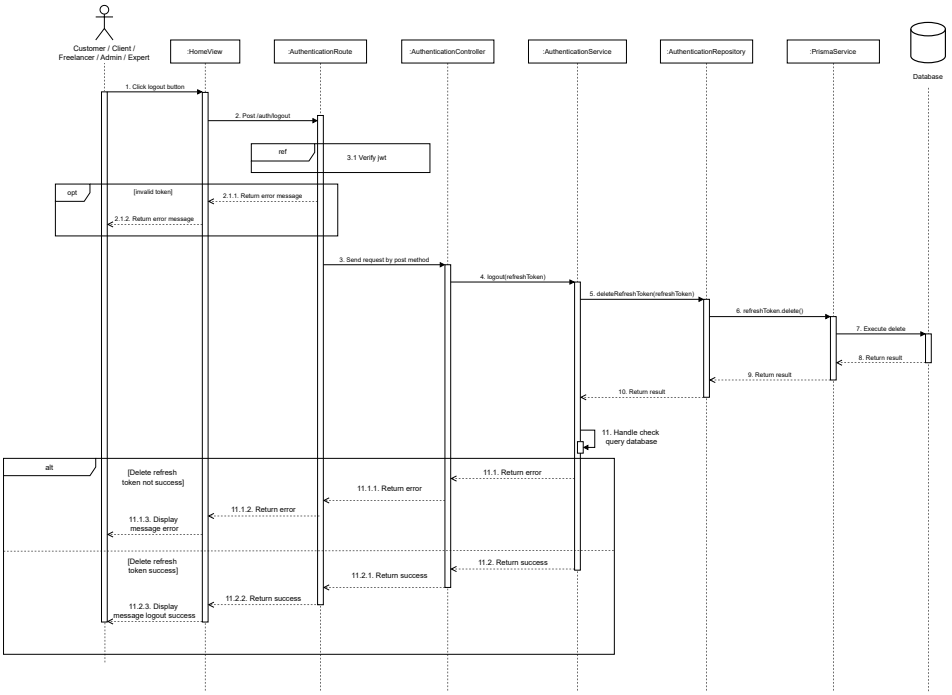

Figure 56: Sequence diagram of UC-03 Logout

3.9.5 UC-04 – Register

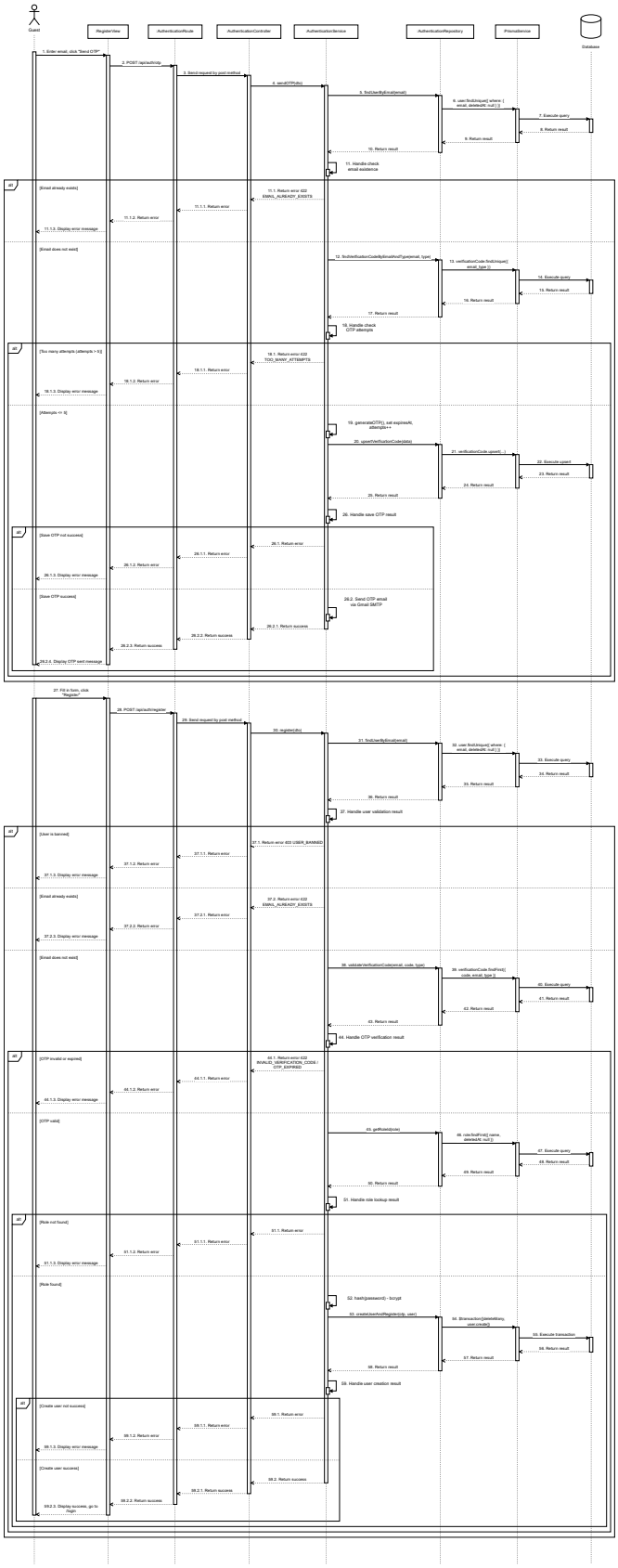

Figure 57: Sequence diagram of UC-04 Register

3.9.6 UC-05 – Forgot Password

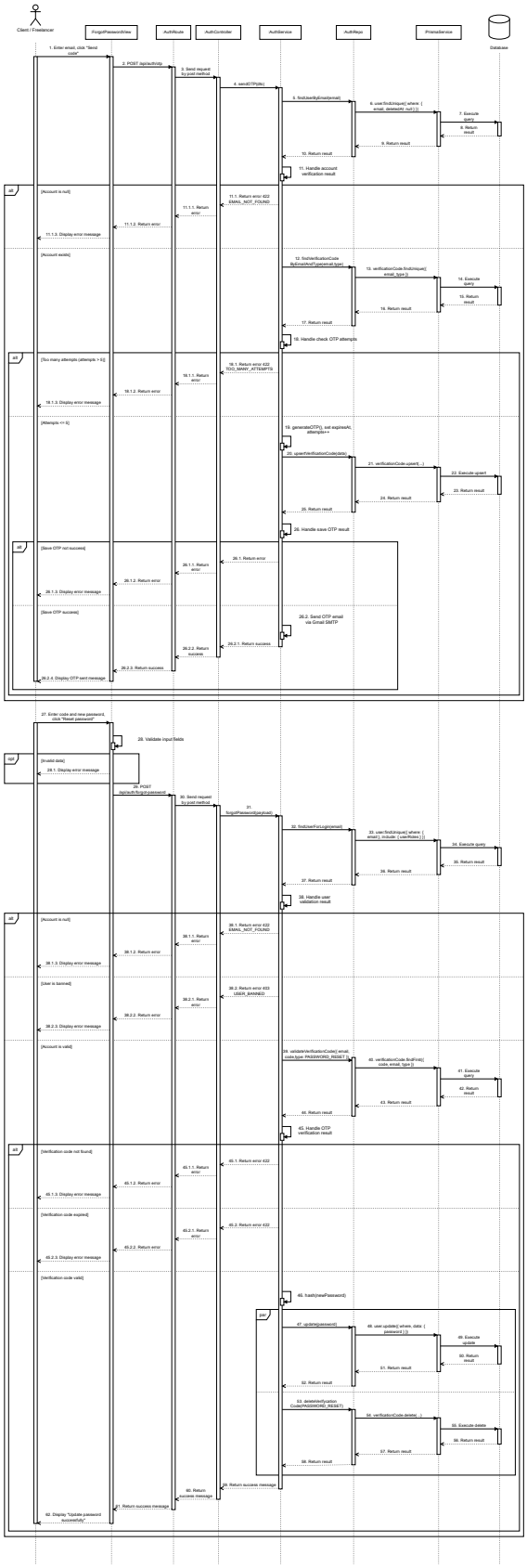

Figure 58: Sequence diagram of UC-05 Forgot Password

3.10 Role Management

Role Management lets an administrator inspect the role catalogue, maintain role metadata, and replace the permission set assigned to a role.

3.10.1 Class Diagram

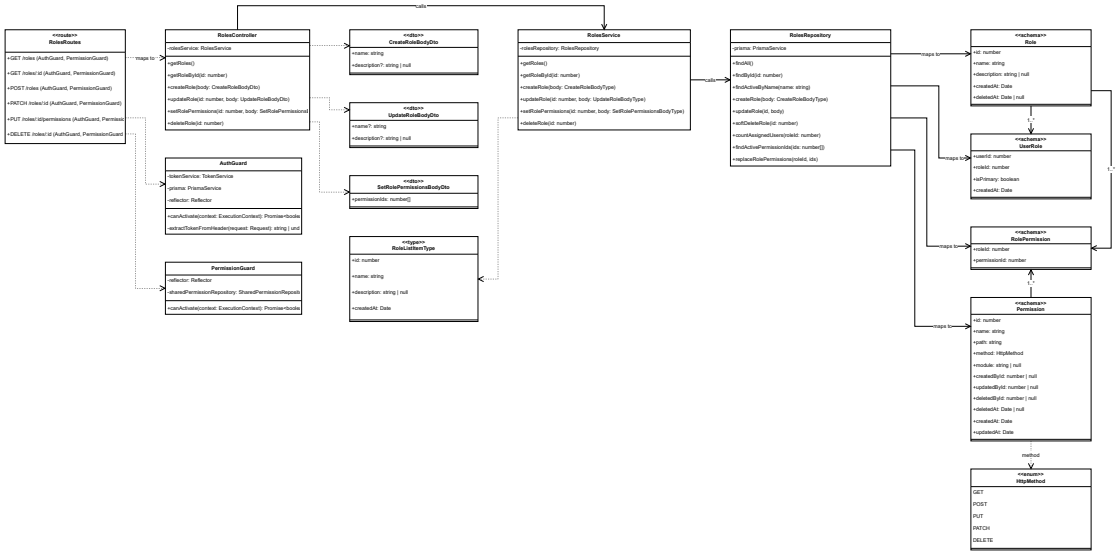

Figure 59: Class diagram of Role Management

3.10.2 UC-06.01 – View Role List

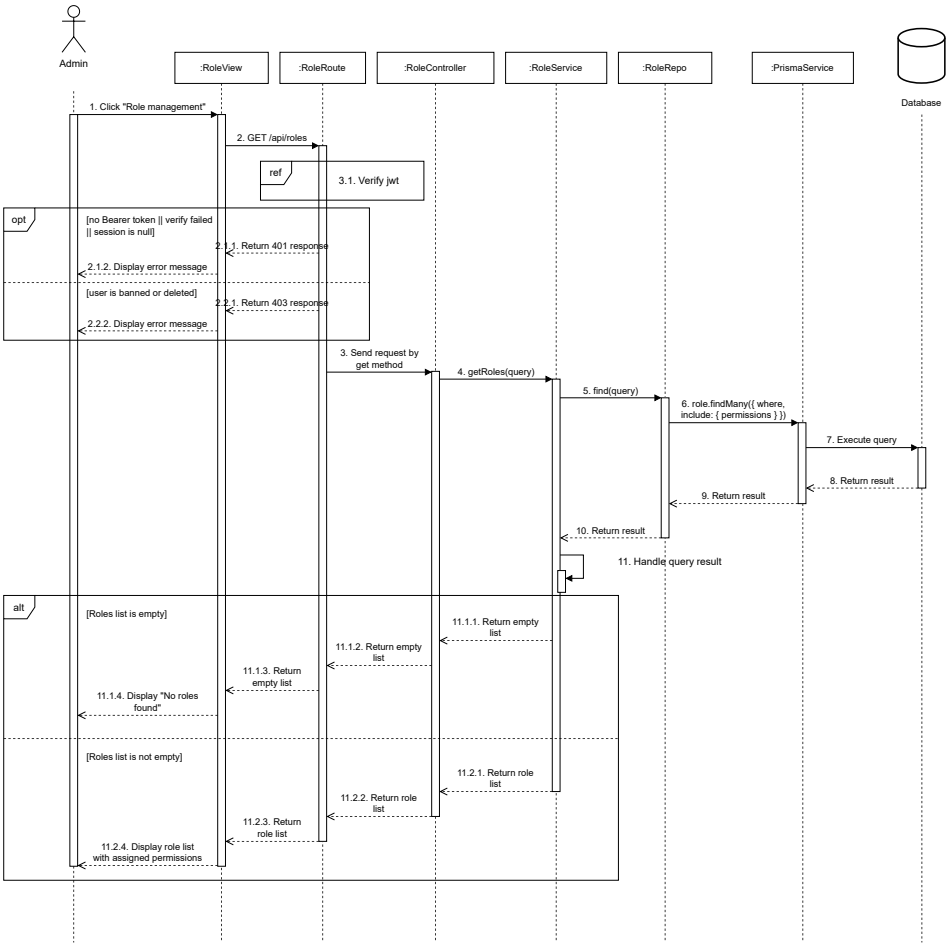

Figure 60: Sequence diagram of UC-06.01 View Role List

3.10.3 UC-06.02 – View Role Detail

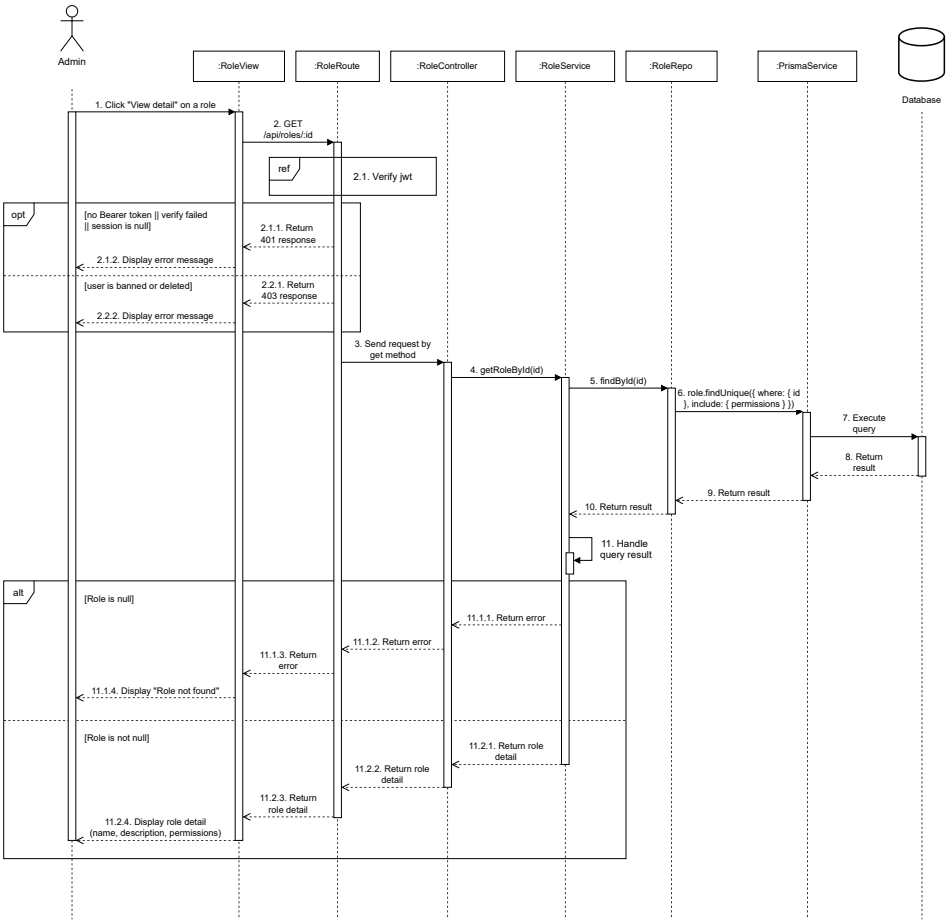

Figure 61: Sequence diagram of UC-06.02 View Role Detail

3.10.4 UC-06.03 – Create Role

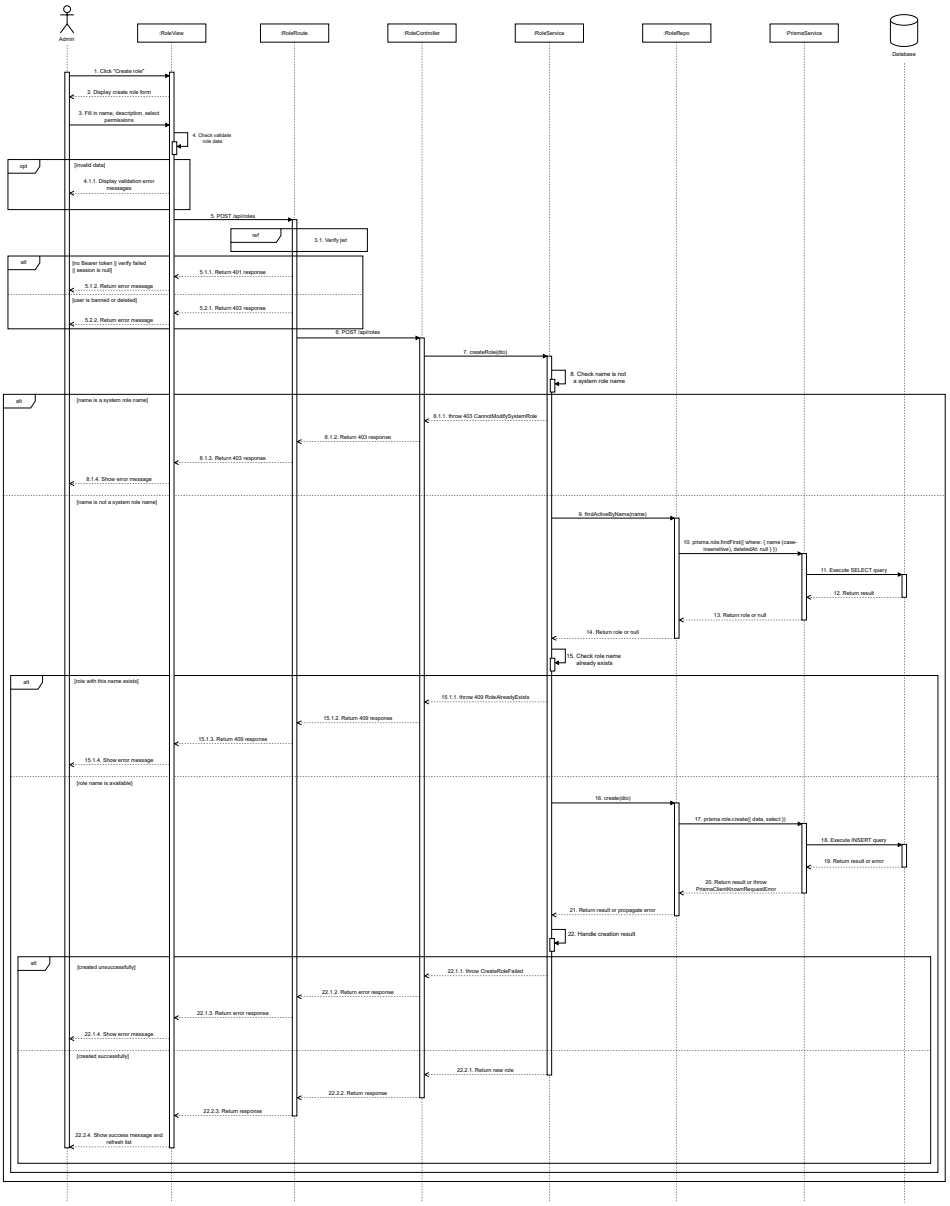

Figure 62: Sequence diagram of UC-06.03 Create Role

3.10.5 UC-06.04 – Update Role

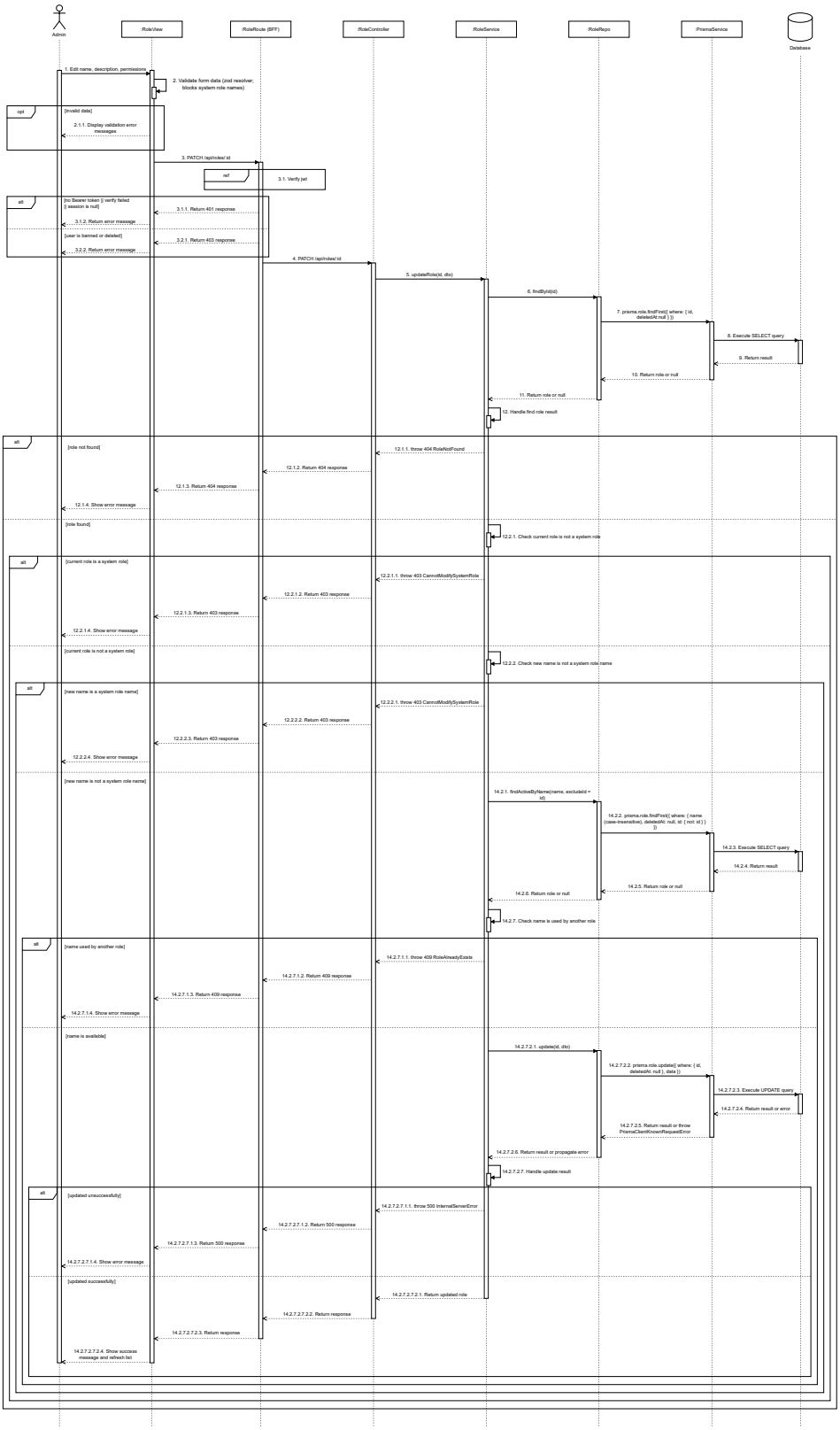

Figure 63: Sequence diagram of UC-06.04 Update Role

3.10.6 UC-06.05 – Delete Role

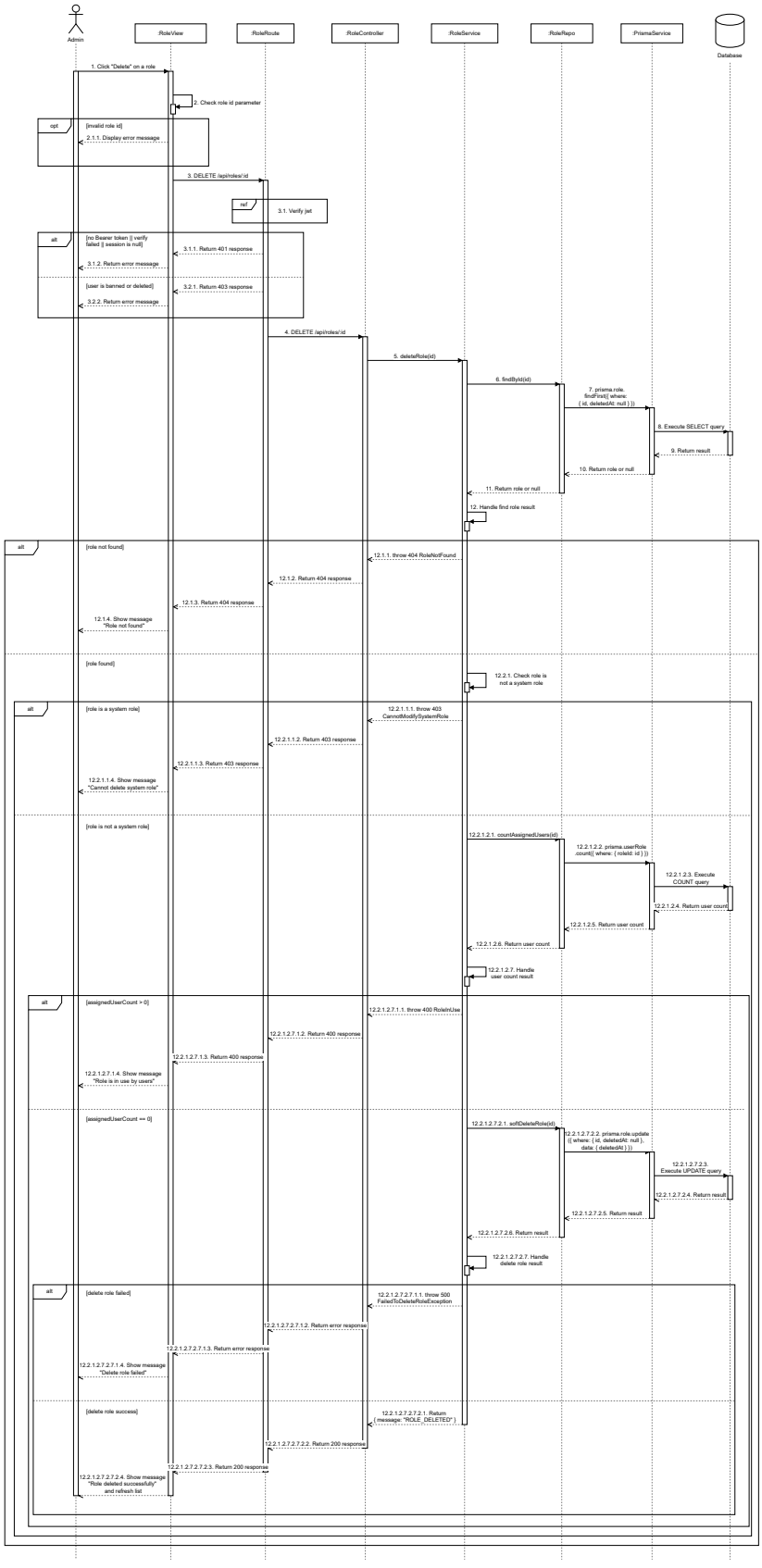

Figure 64: Sequence diagram of UC-06.05 Delete Role

3.10.7 UC-06.06 – Update Role Permissions

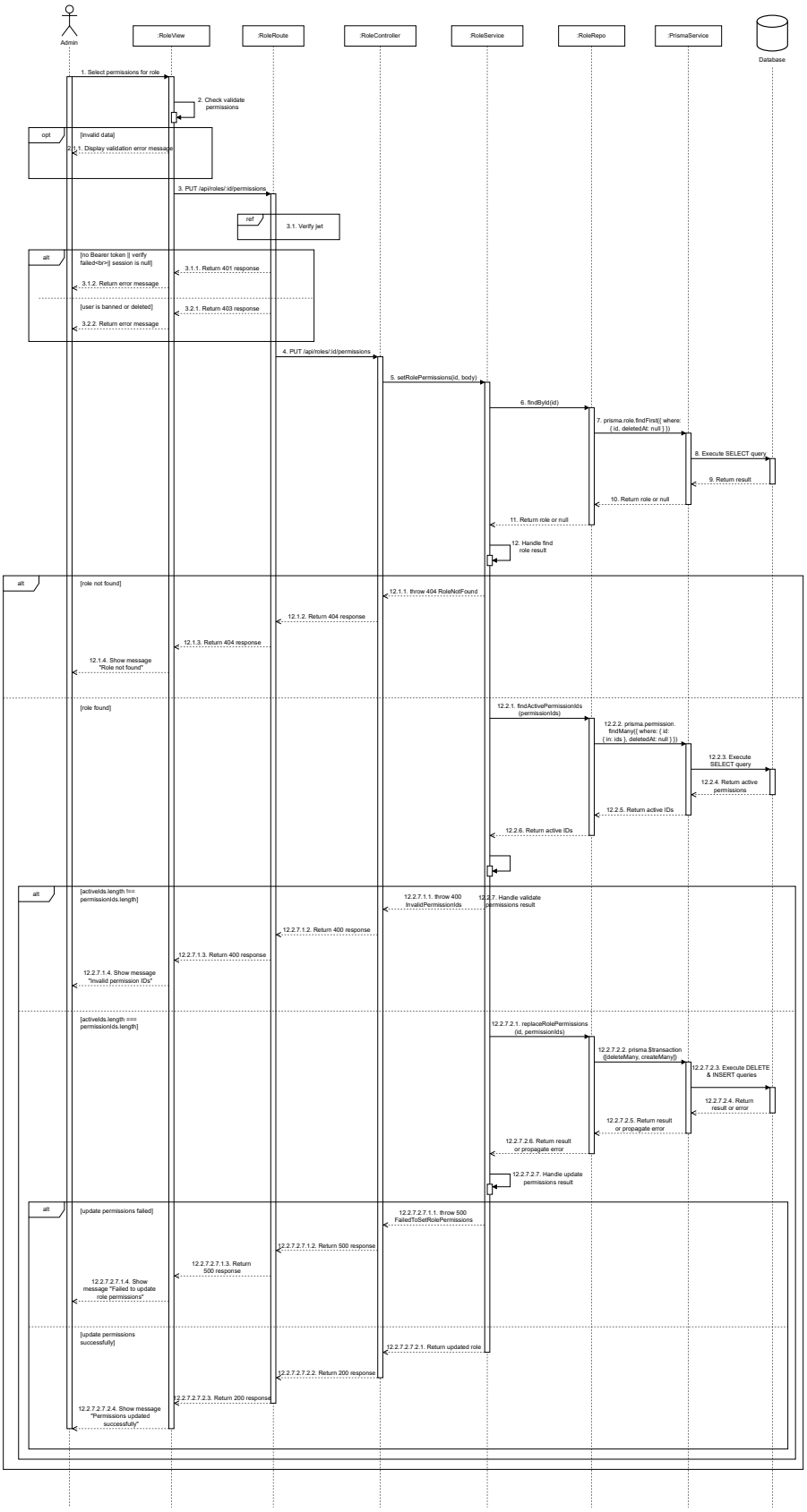

Figure 65: Sequence diagram of UC-06.06 Update Role Permissions

3.11 Manage Conversation

Manage Conversation covers the UC-27 private messaging module: listing and starting one-to-one conversations, opening a conversation, sending text messages and files, marking messages as read, recalling a message, and pinning or unpinning a conversation. Delivery is served both by the REST endpoints and by the realtime gateway in the same namespace.

3.11.1 Class Diagram

Figure 66: Class diagram of Manage Conversation

3.11.2 UC-27.01 – View Conversation List

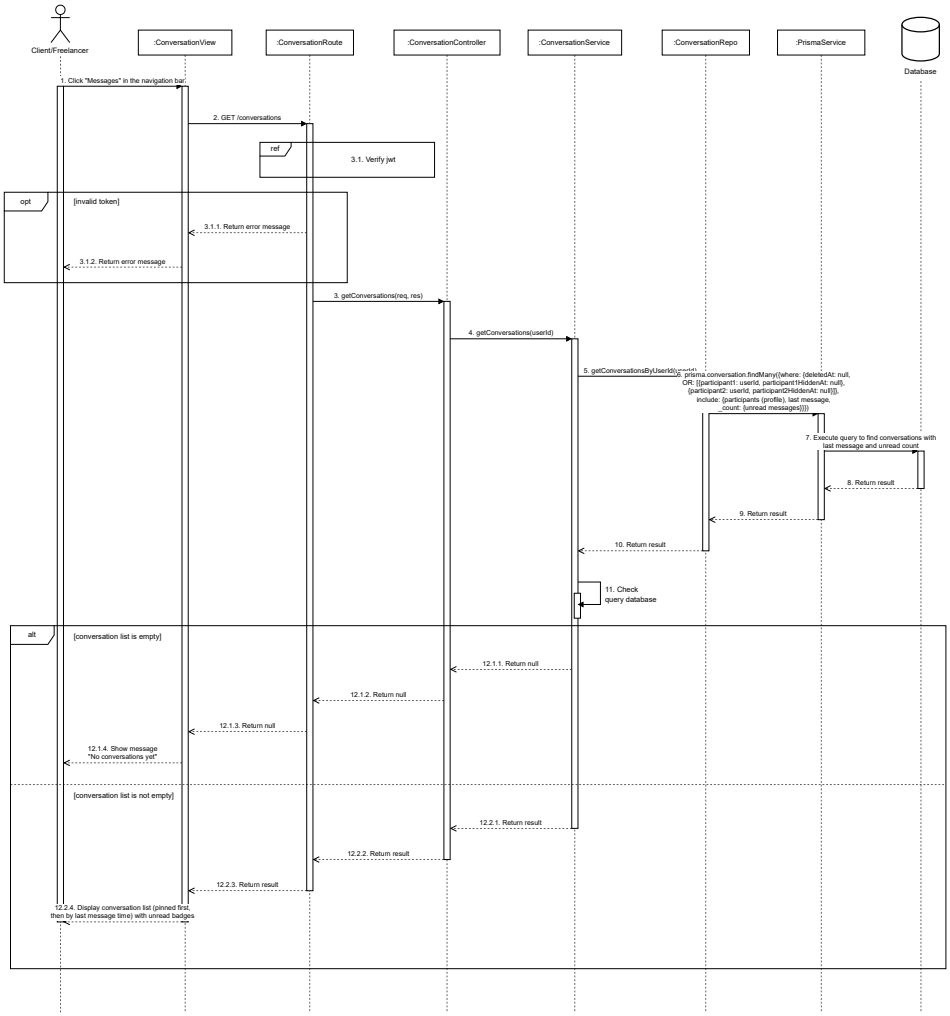

Figure 67: Sequence diagram of UC-27.01 View Conversation List

3.11.3 UC-27.02 – Start Conversation

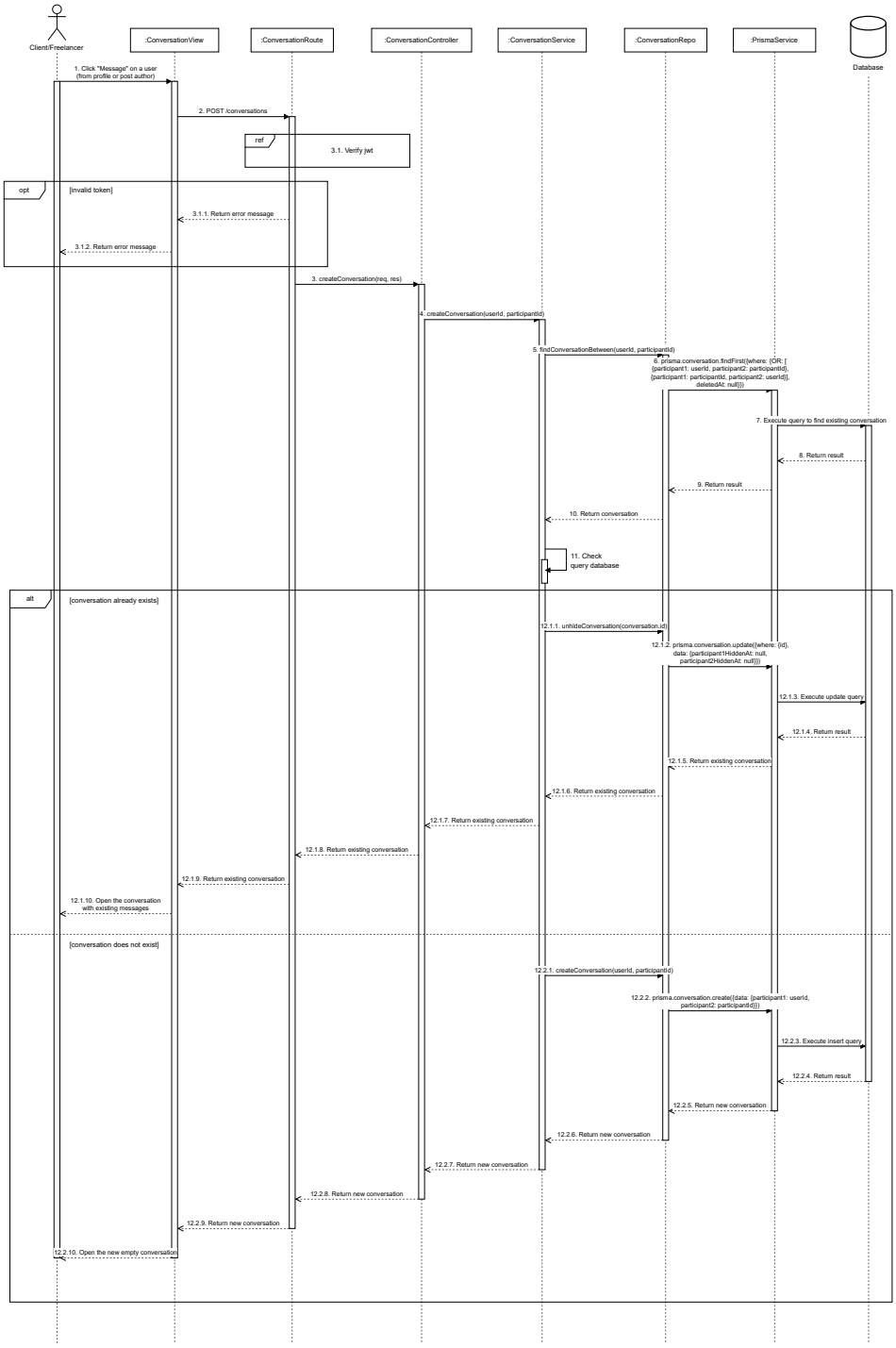

Figure 68: Sequence diagram of UC-27.02 Start Conversation

3.11.4 UC-27.03 – View Conversation Detail

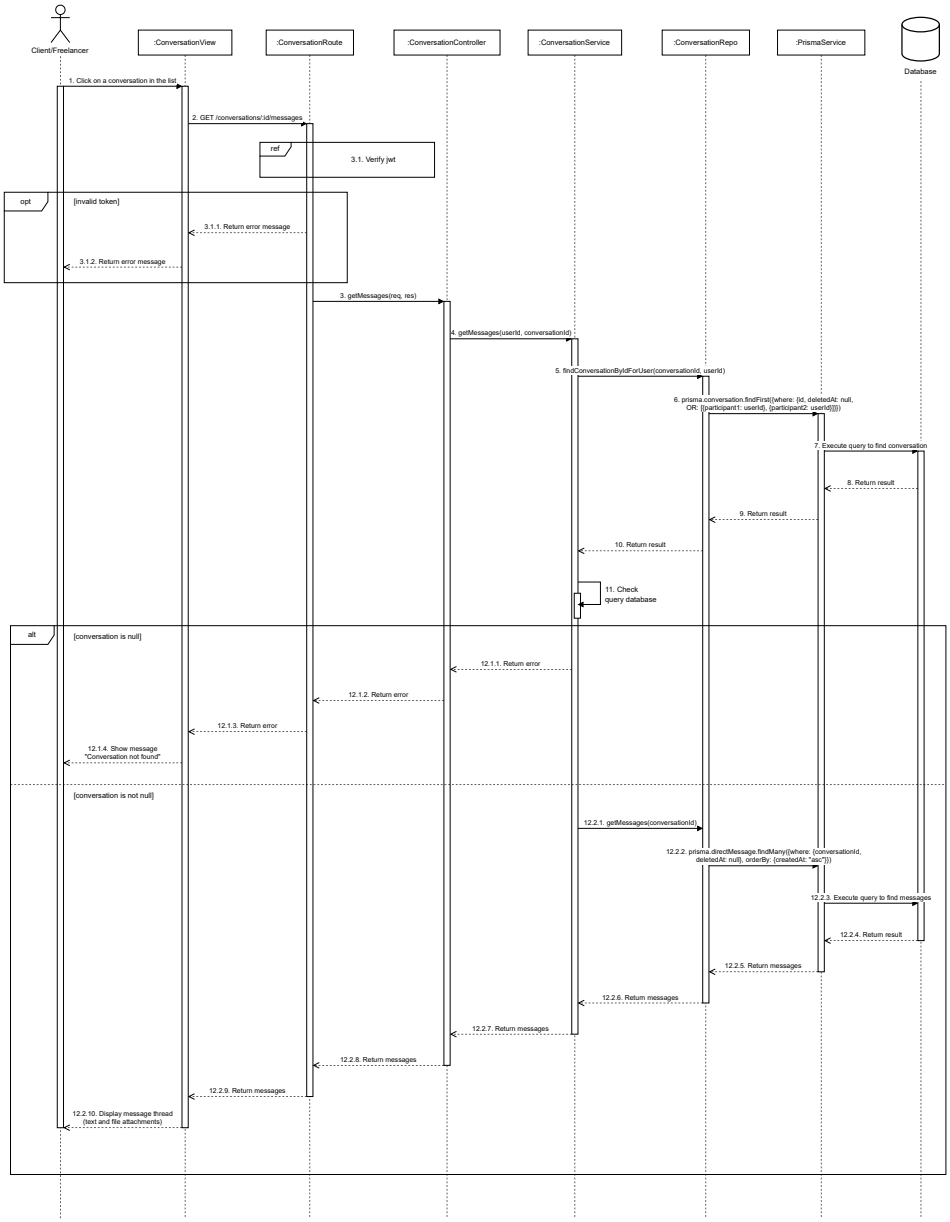

Figure 69: Sequence diagram of UC-27.03 View Conversation Detail

3.11.5 UC-27.04 – Send Message

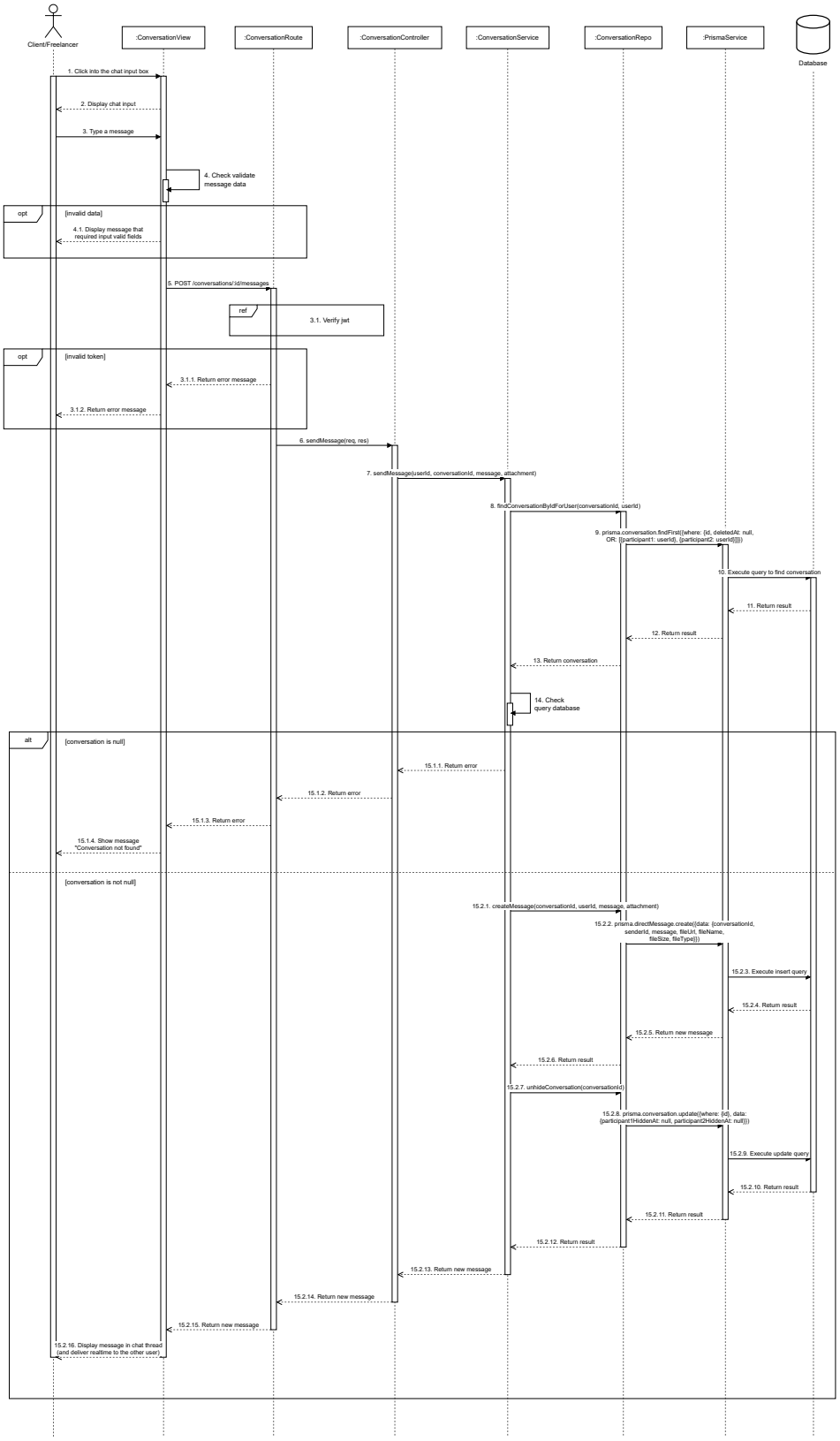

Figure 70: Sequence diagram of UC-27.04 Send Message

3.11.6 UC-27.05 – Mark Message as Read

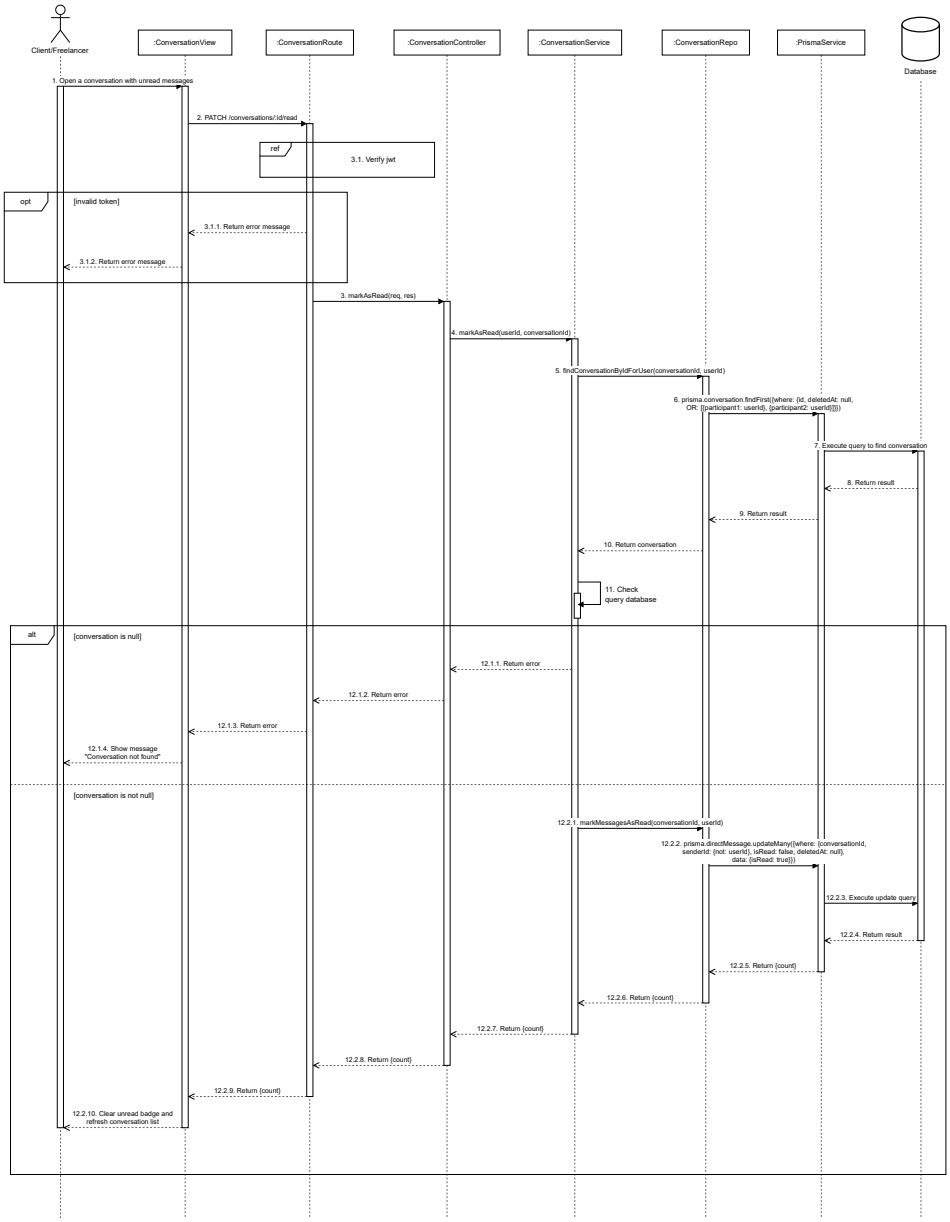

Figure 71: Sequence diagram of UC-27.05 Mark Message as Read

3.11.7 UC-27.06 – Delete/Recall Message

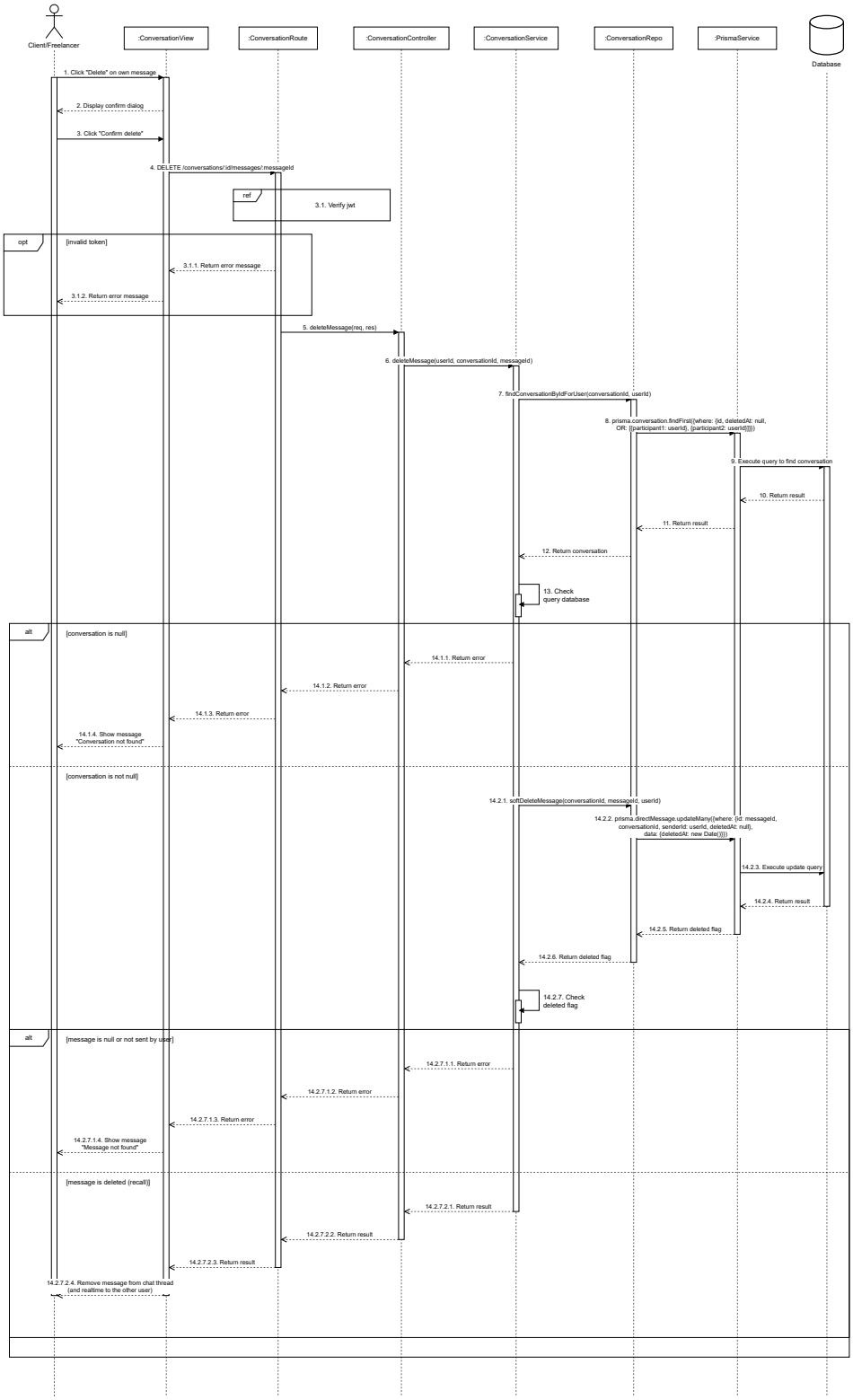

Figure 72: Sequence diagram of UC-27.06 Delete/Recall Message

3.11.8 UC-27.07 – Pin/Unpin Conversation

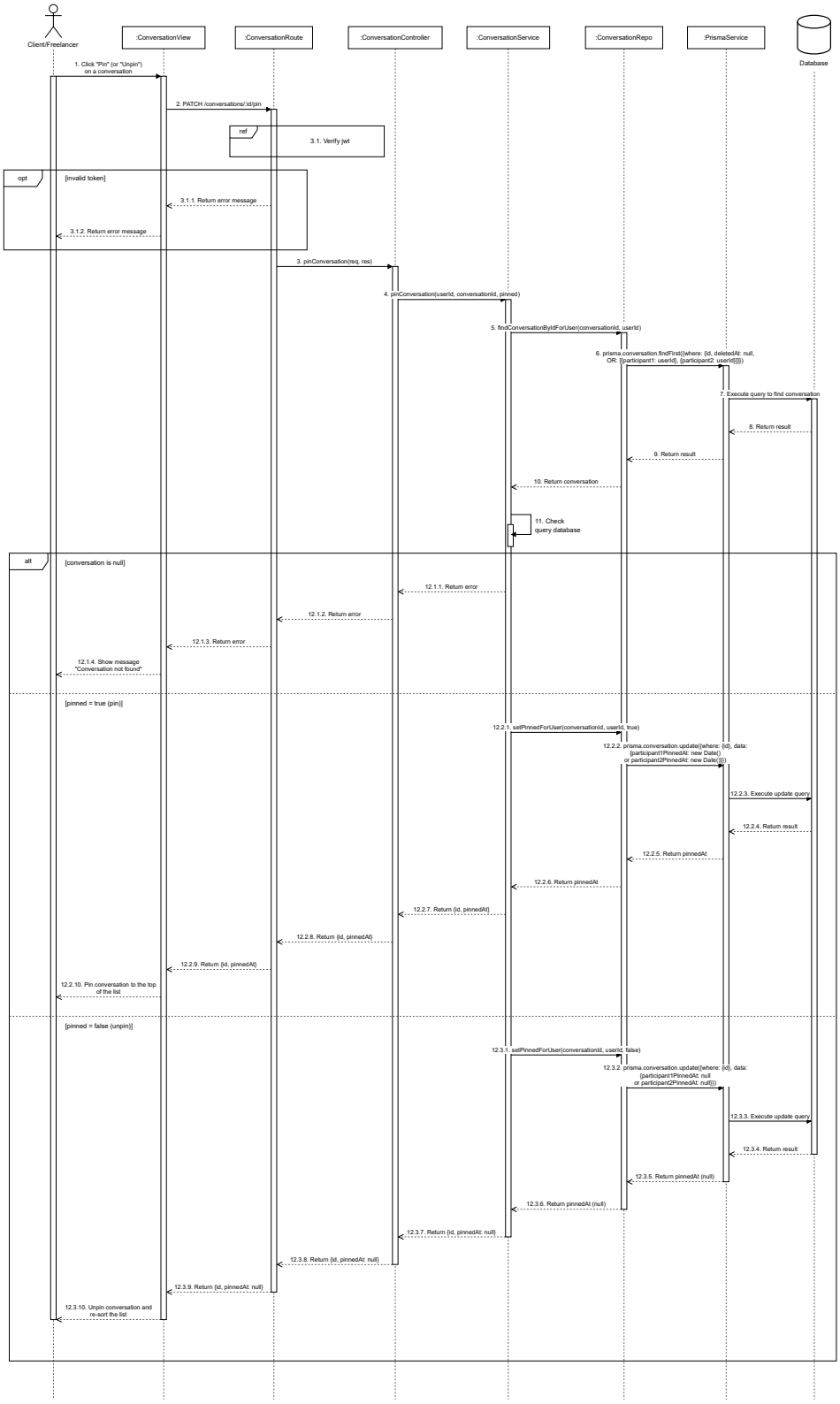

Figure 73: Sequence diagram of UC-27.07 Pin/Unpin Conversation

3.11.9 UC-27.08 – Send File

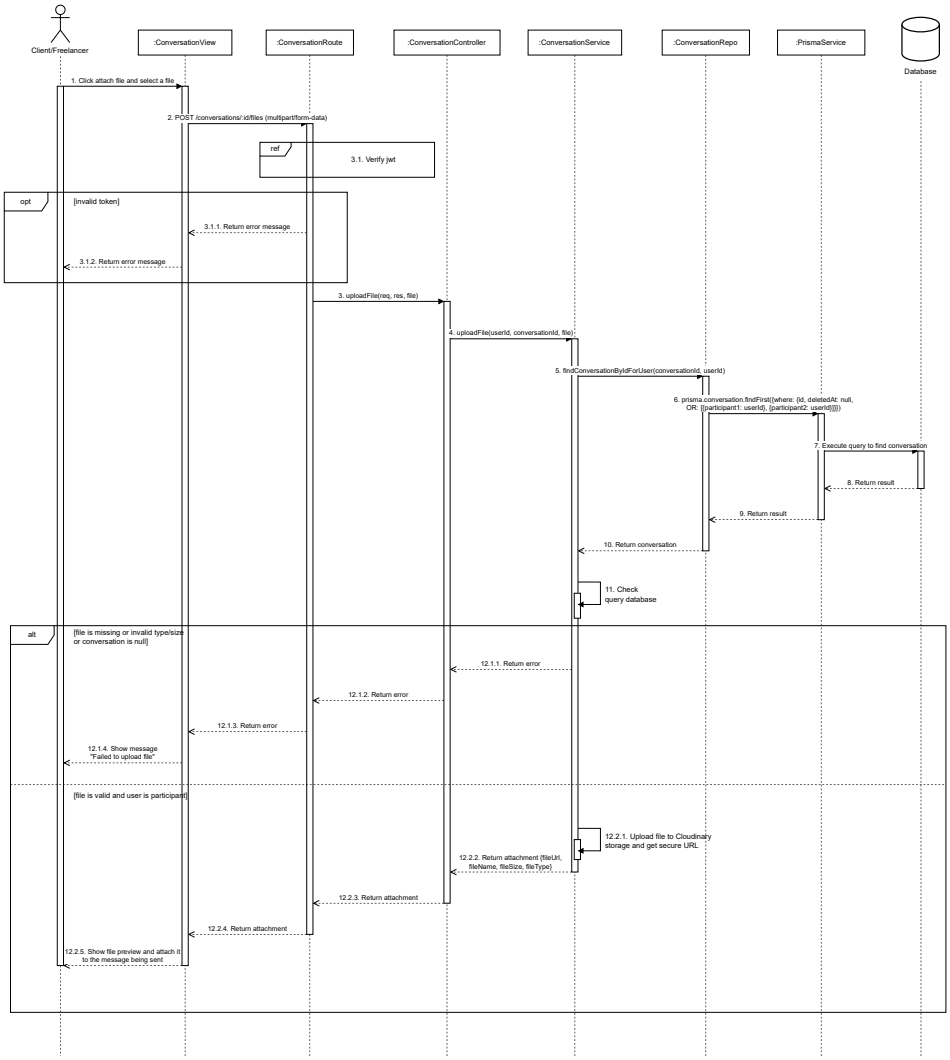

Figure 74: Sequence diagram of UC-27.08 Send File

3.12 Identity Verification

Freelancers upload identity documents, submit AI verification evidence, and inspect their verification status through the account-profile module.

3.12.1 Class Diagram

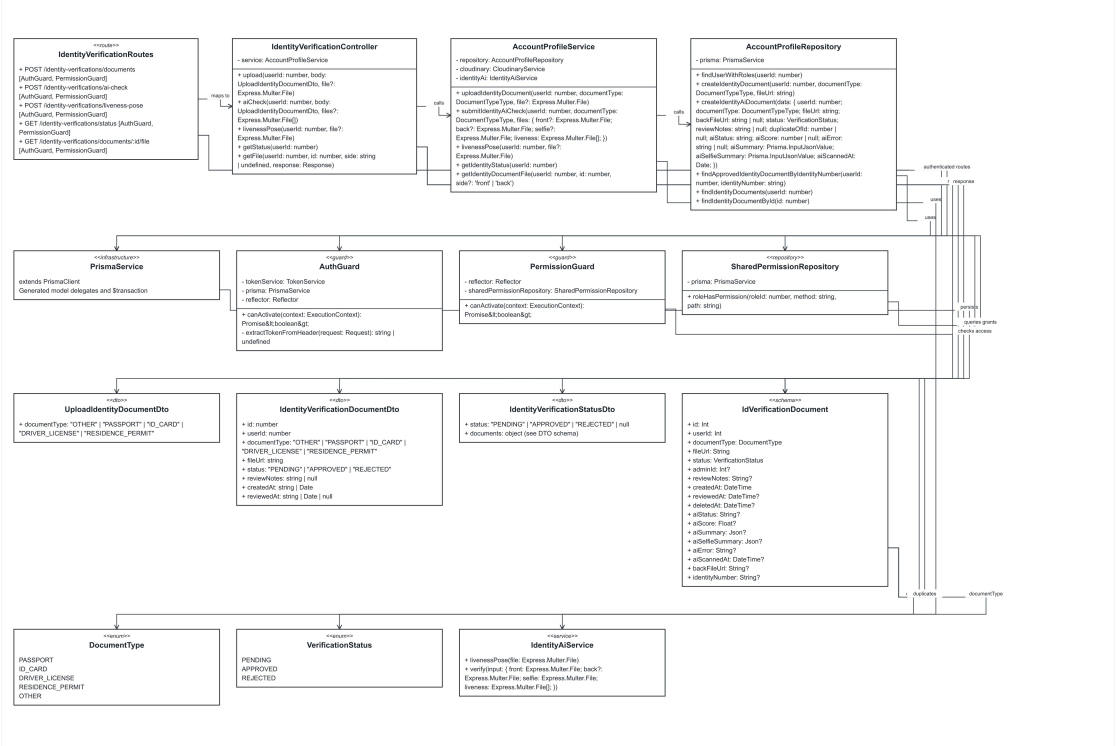

Figure 75: Class diagram of Identity Verification

3.12.2 UC-18.01 – Upload Identity Documents

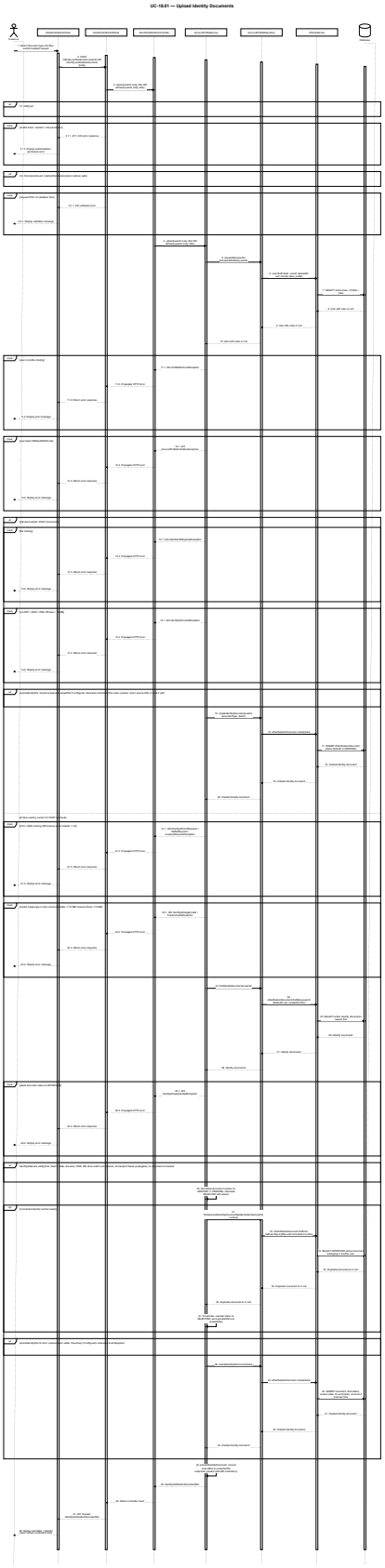

Figure 76: Sequence diagram of UC-18.01 Upload Identity Documents

3.12.3 UC-18.02 – View Verification Status

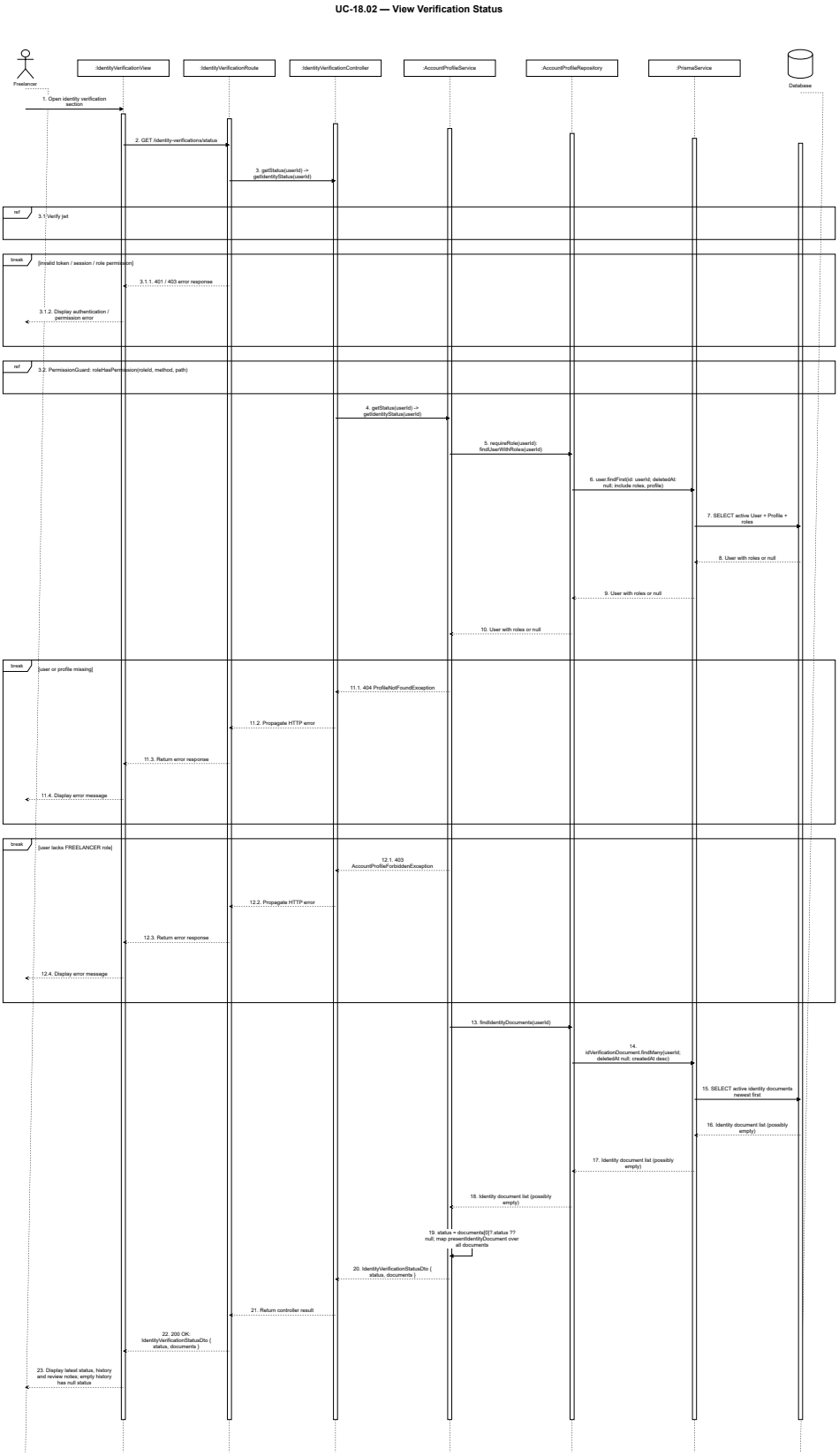

Figure 77: Sequence diagram of UC-18.02 View Verification Status

3.13 Client Profile Management

The public client profile is retrieved by user identifier. An authenticated client updates company information, subject to the profile revision policy.

3.13.1 Class Diagram

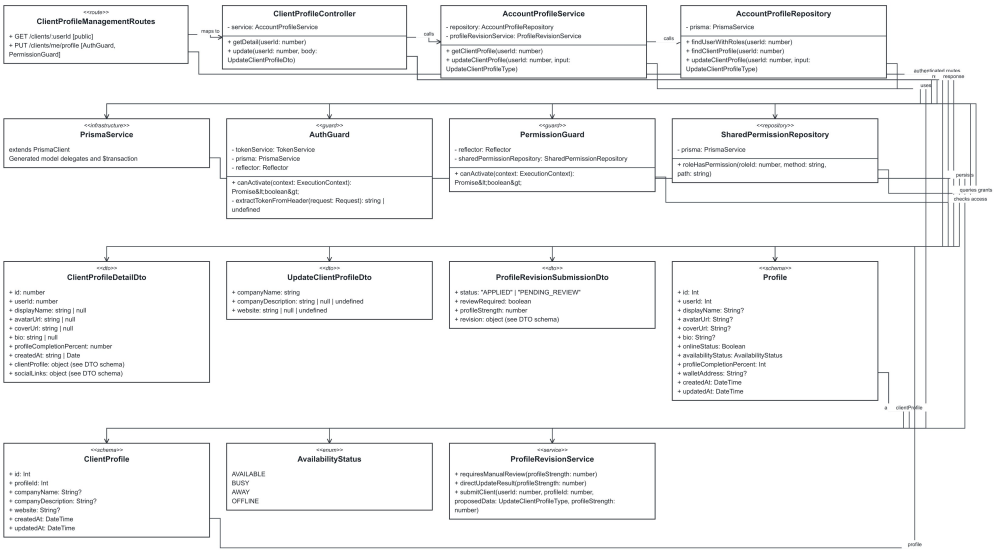

Figure 78: Class diagram of Client Profile Management

3.13.2 UC-19.01 – View Client Profile

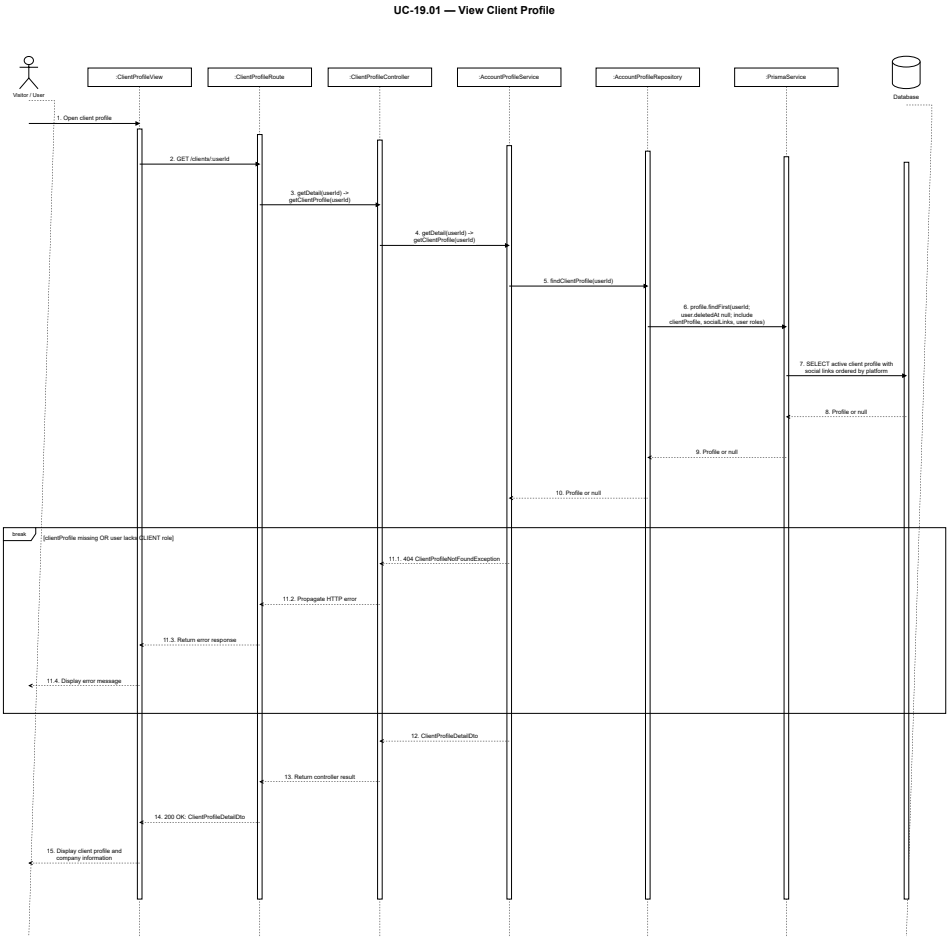

Figure 79: Sequence diagram of UC-19.01 View Client Profile

3.13.3 UC-19.02 – Update Company Information

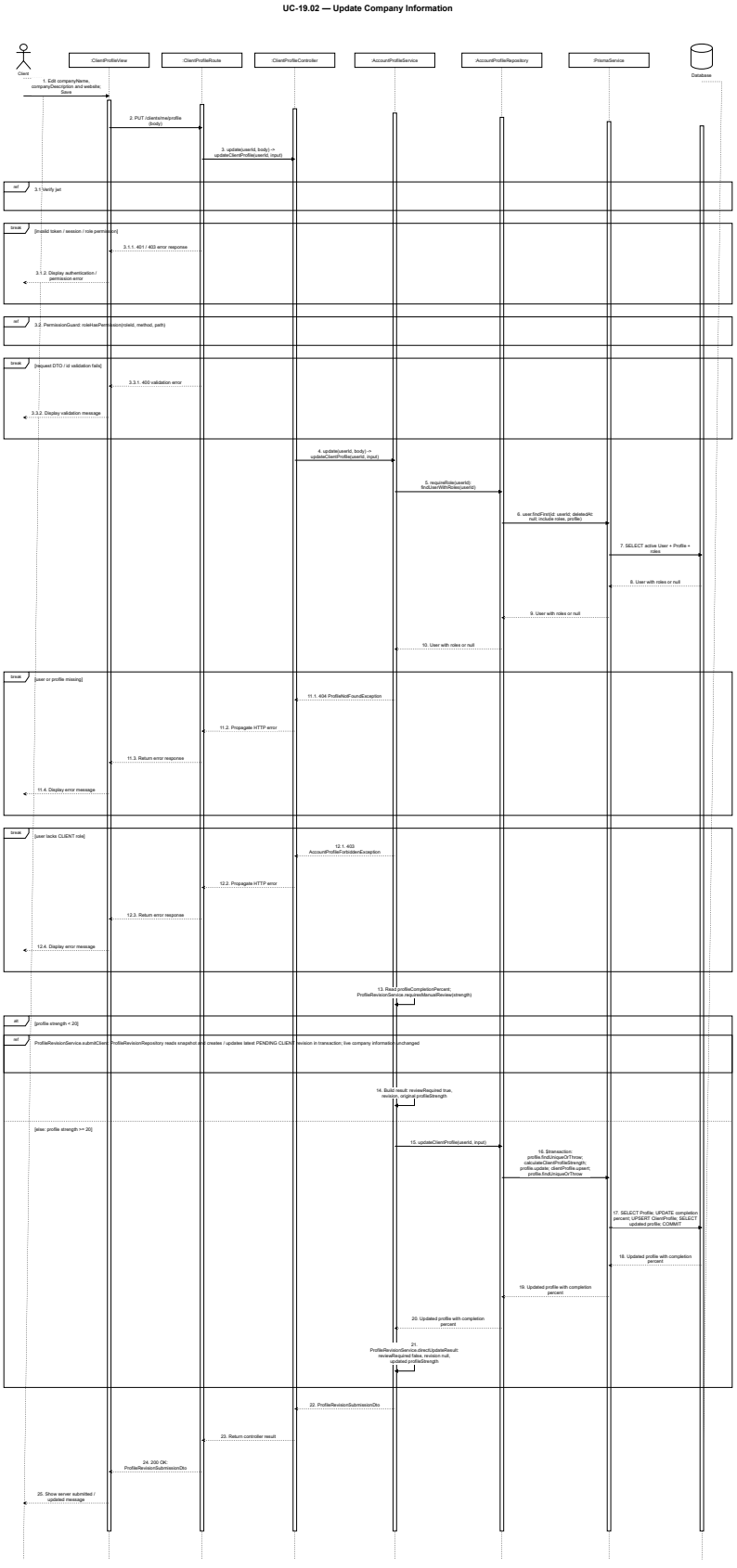

Figure 80: Sequence diagram of UC-19.02 Update Company Information

3.14 Social Link Management

Users view, add, and remove social links on their own profiles. The service checks profile existence, duplicate platforms, and link ownership.

3.14.1 Class Diagram

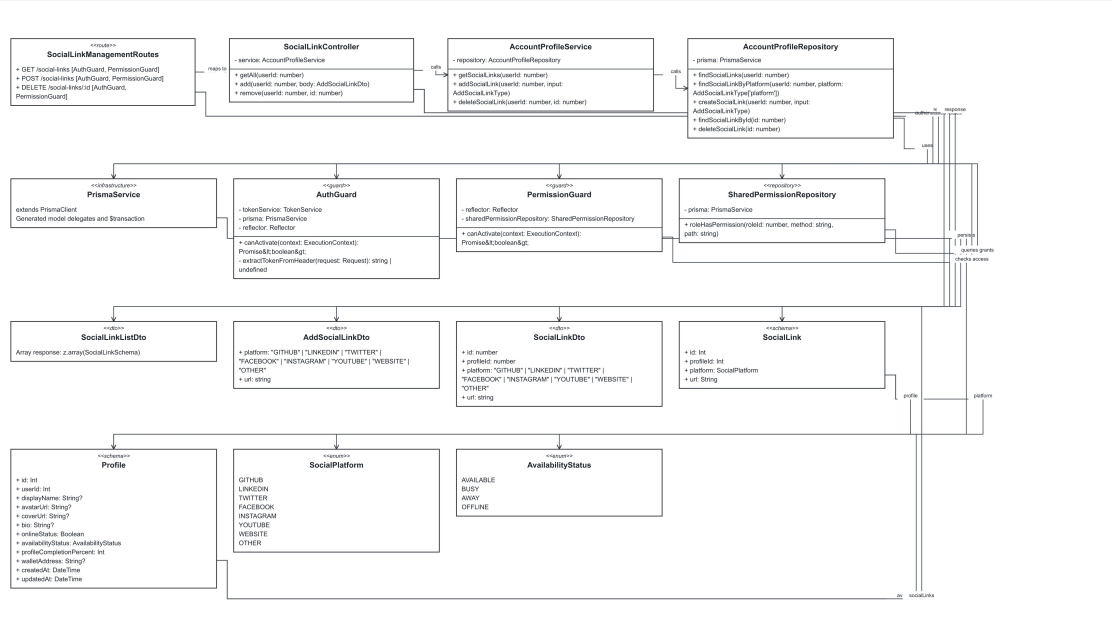

Figure 81: Class diagram of Social Link Management

3.14.2 UC-20.01 – View Social Link List

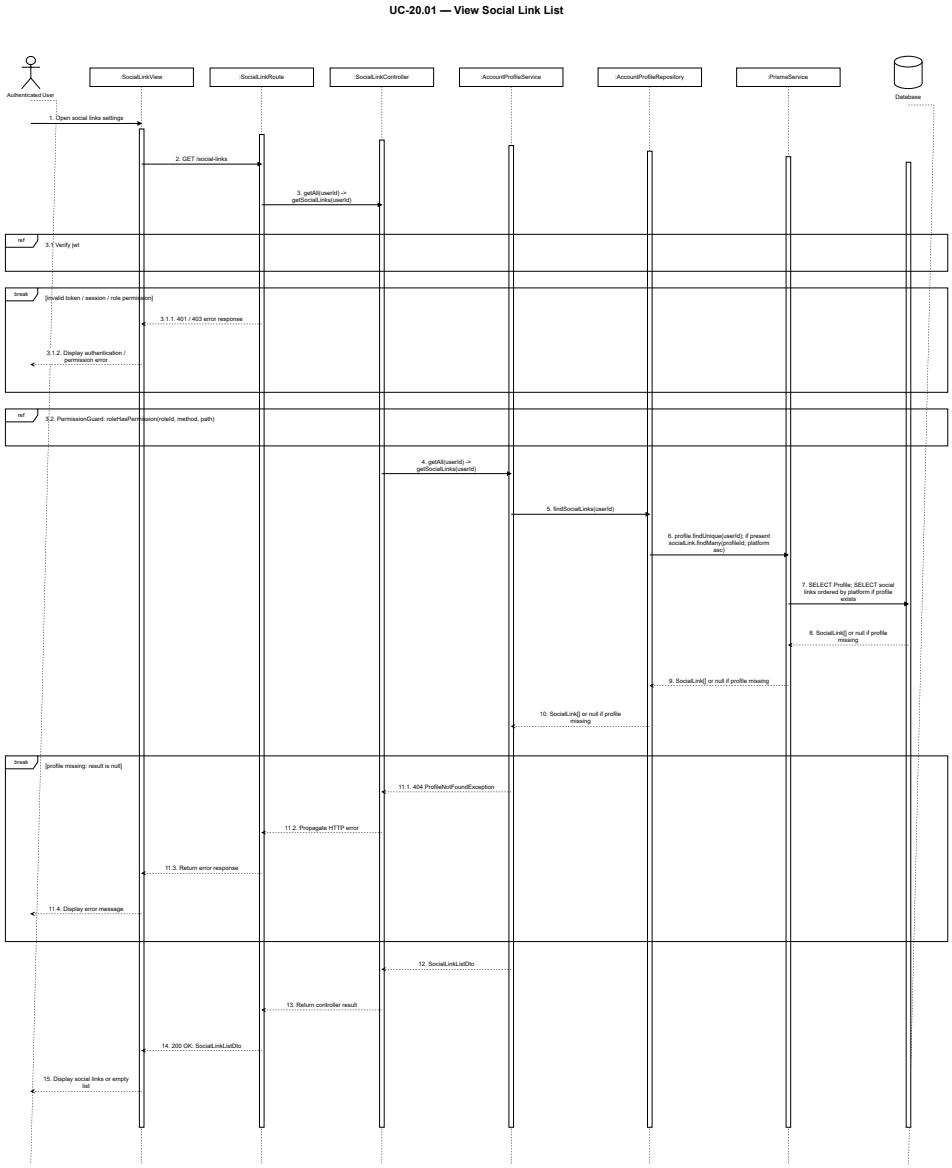

Figure 82: Sequence diagram of UC-20.01 View Social Link List

3.14.3 UC-20.02 – Add Social Link

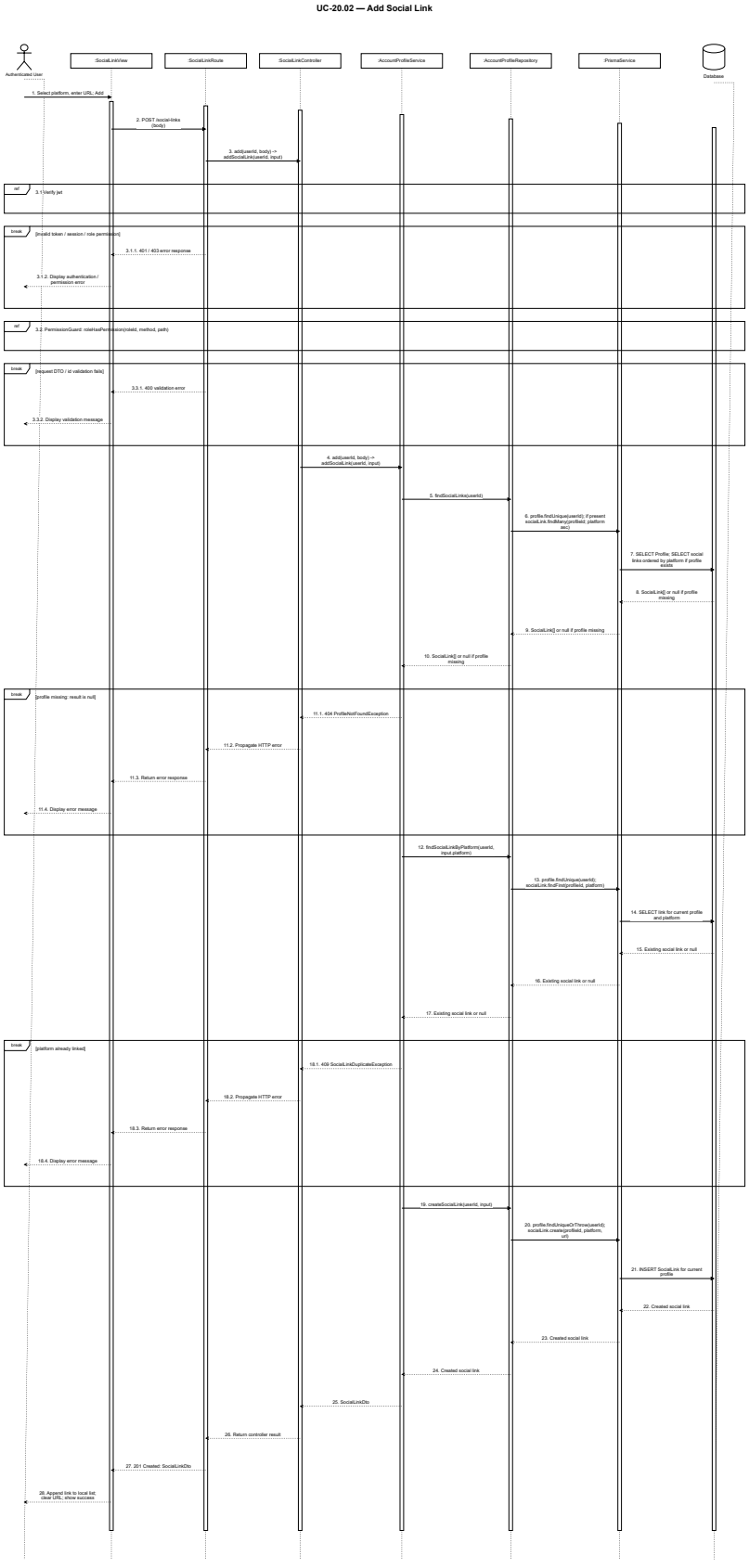

Figure 83: Sequence diagram of UC-20.02 Add Social Link

3.14.4 UC-20.03 – Delete Social Link

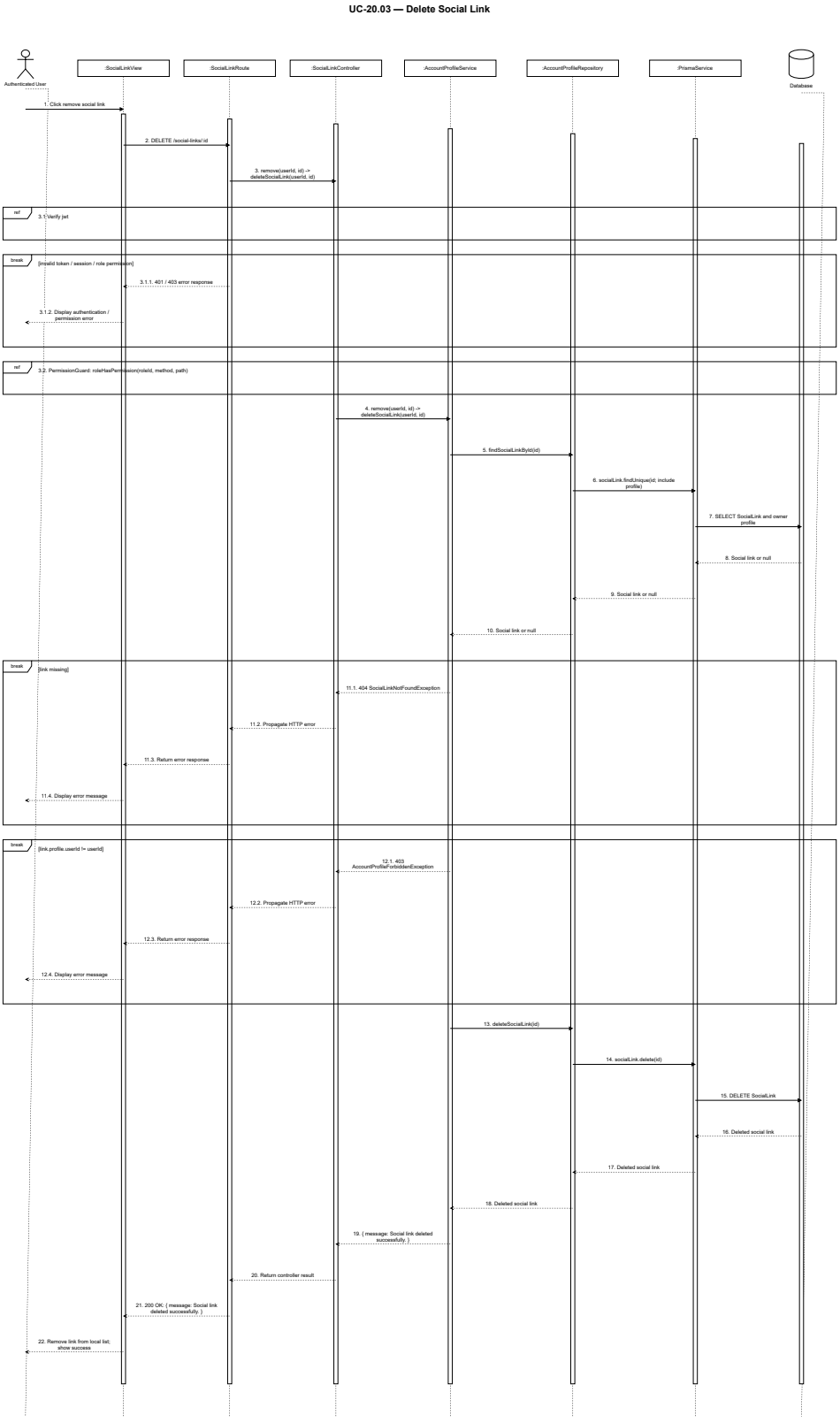

Figure 84: Sequence diagram of UC-20.03 Delete Social Link

3.15 Favorite Freelancer Management

Clients maintain a personal list of favourite freelancers. The service checks roles, duplicate relationships, and missing favourites.

3.15.1 Class Diagram

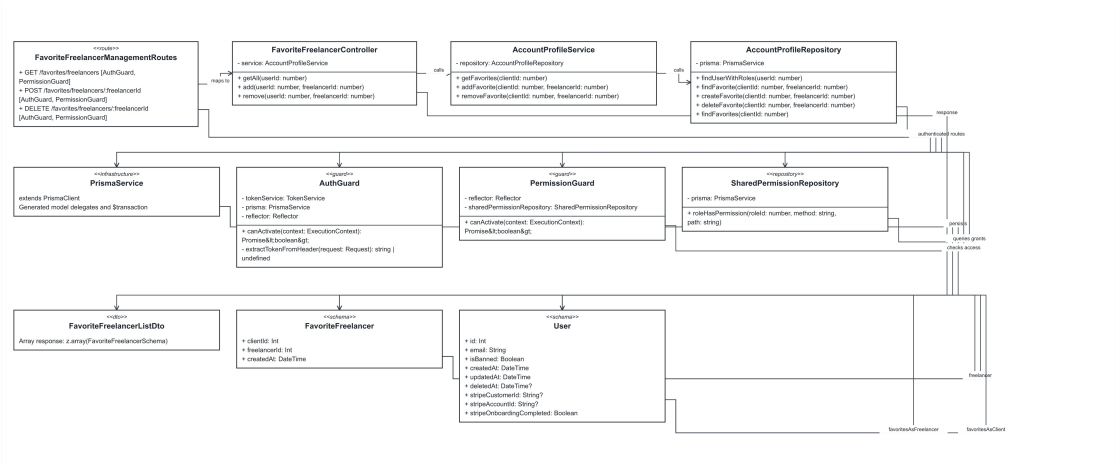

Figure 85: Class diagram of Favorite Freelancer Management

3.15.2 UC-21.01 – View Favorite Freelancer List

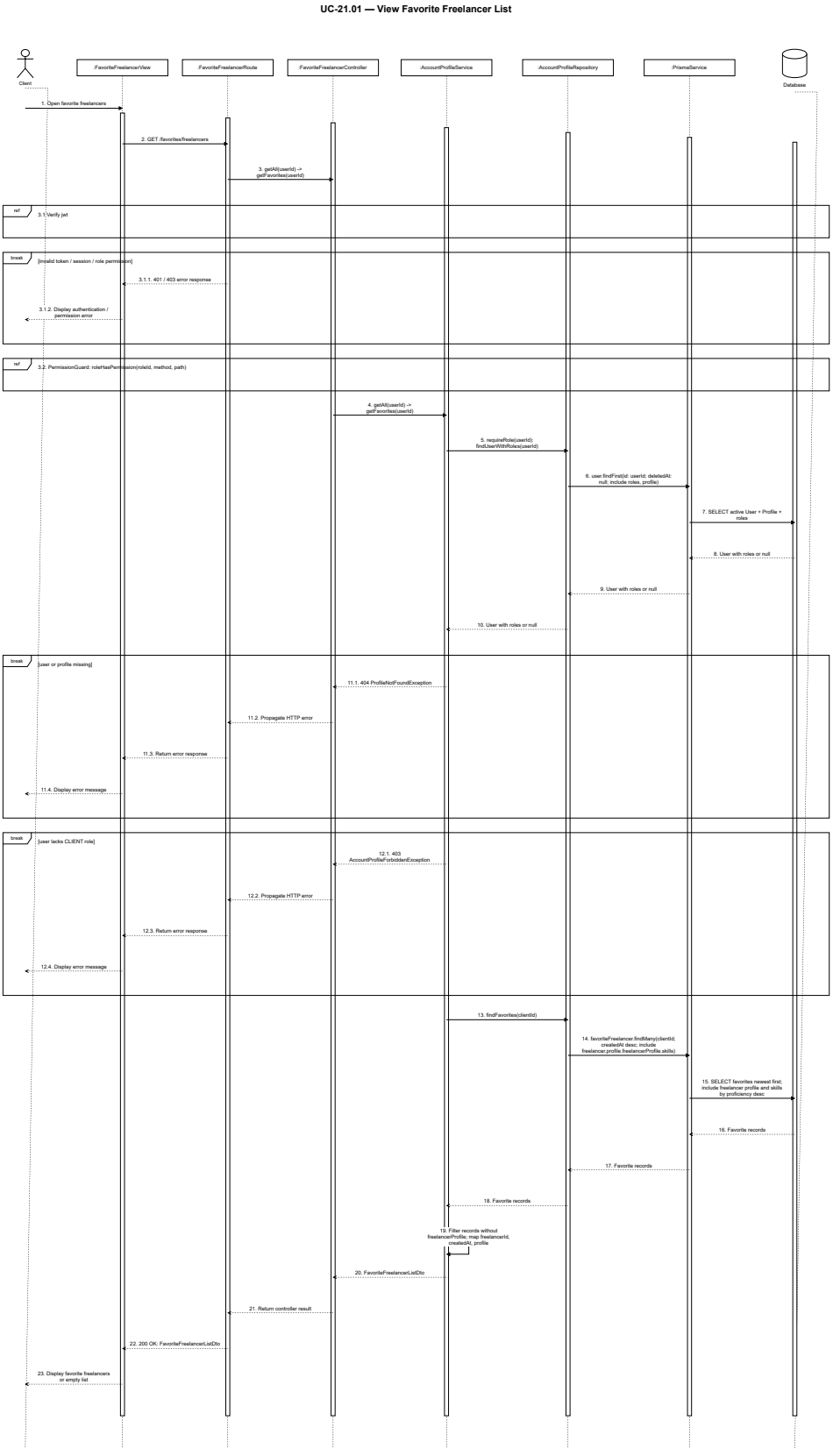

Figure 86: Sequence diagram of UC-21.01 View Favorite Freelancer List

3.15.3 UC-21.02 – Add Freelancer to Favorites

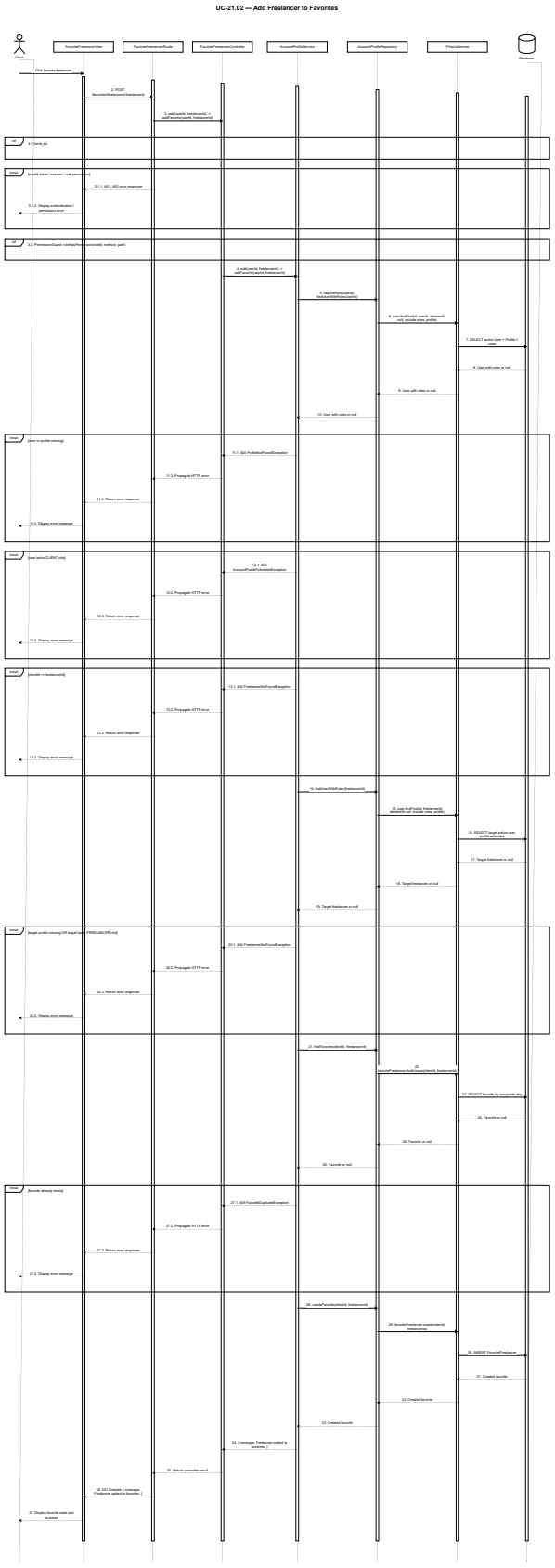

Figure 87: Sequence diagram of UC-21.02 Add Freelancer to Favorites

3.15.4 UC-21.03 – Remove Freelancer from Favorites

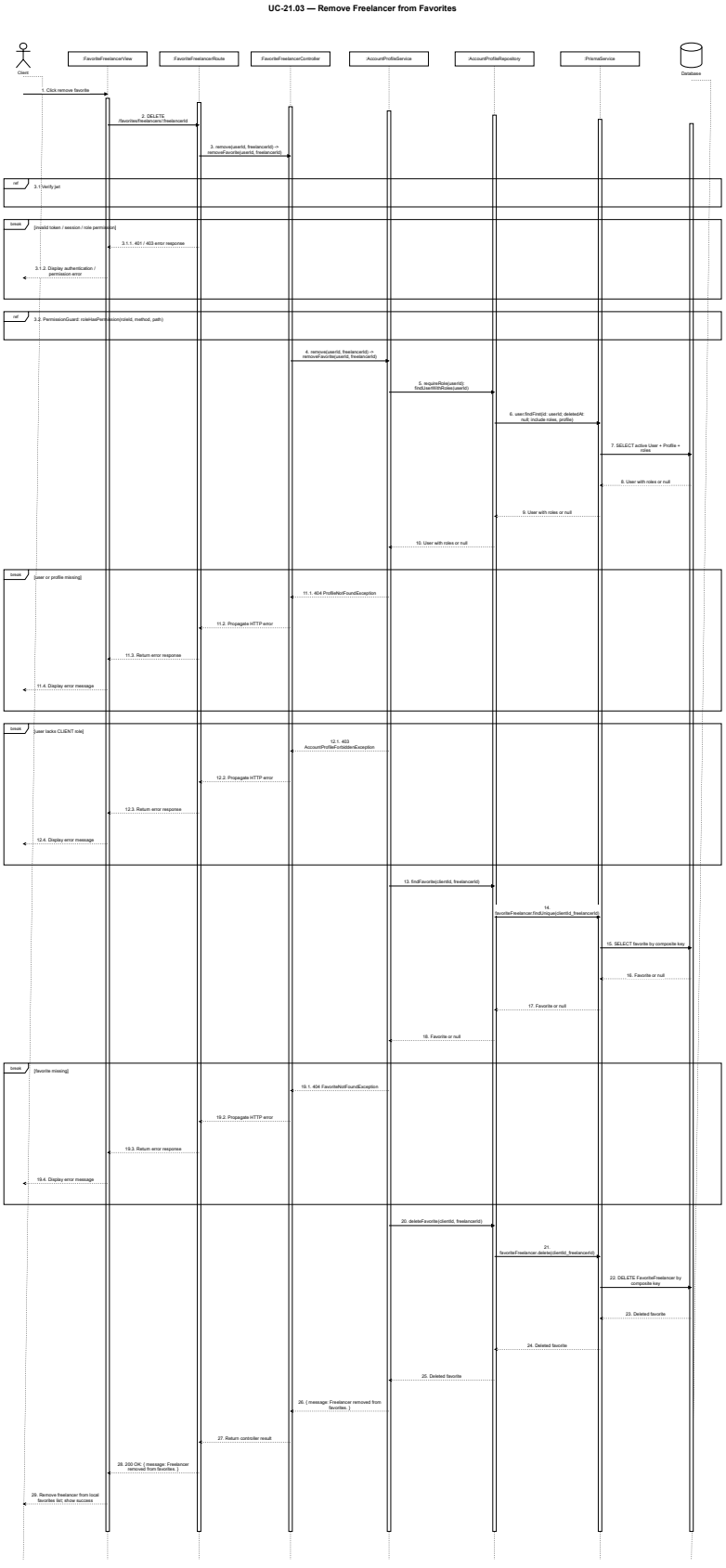

Figure 88: Sequence diagram of UC-21.03 Remove Freelancer from Favorites

3.16 Saved Search Management

Saved Search Management allows freelancers to save complex job filter criteria, view their list of saved searches, inspect search details, update filter parameters, and remove obsolete searches.

3.16.1 Class Diagram

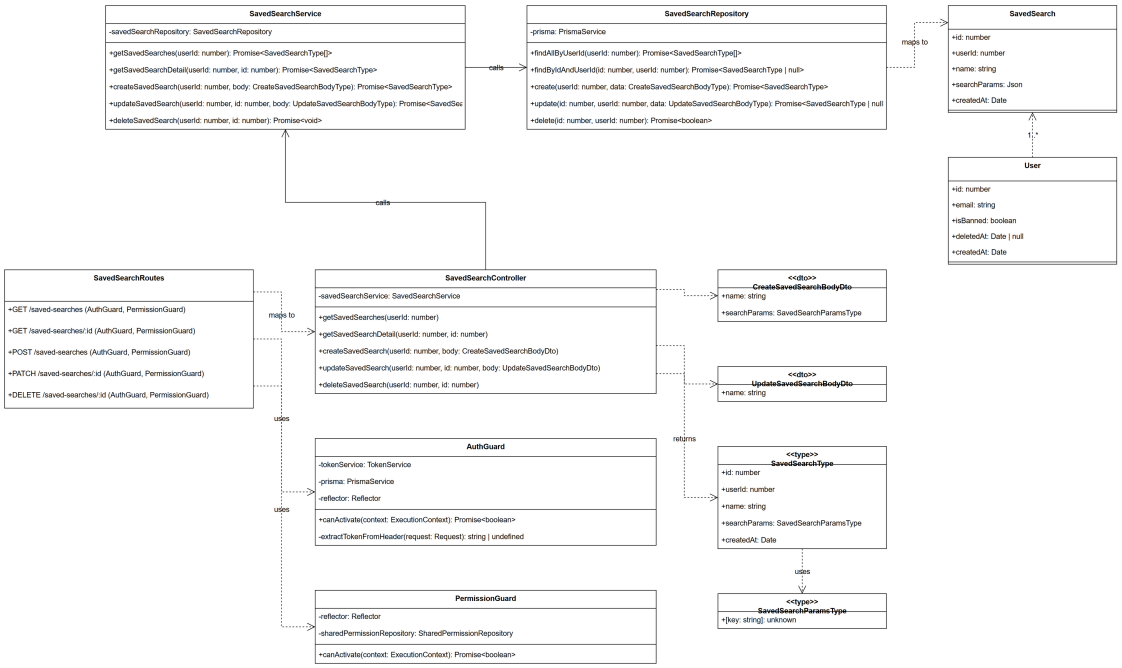

Figure 89: Class diagram of Saved Search Management

3.16.2 UC-22.01 – View Saved Search List

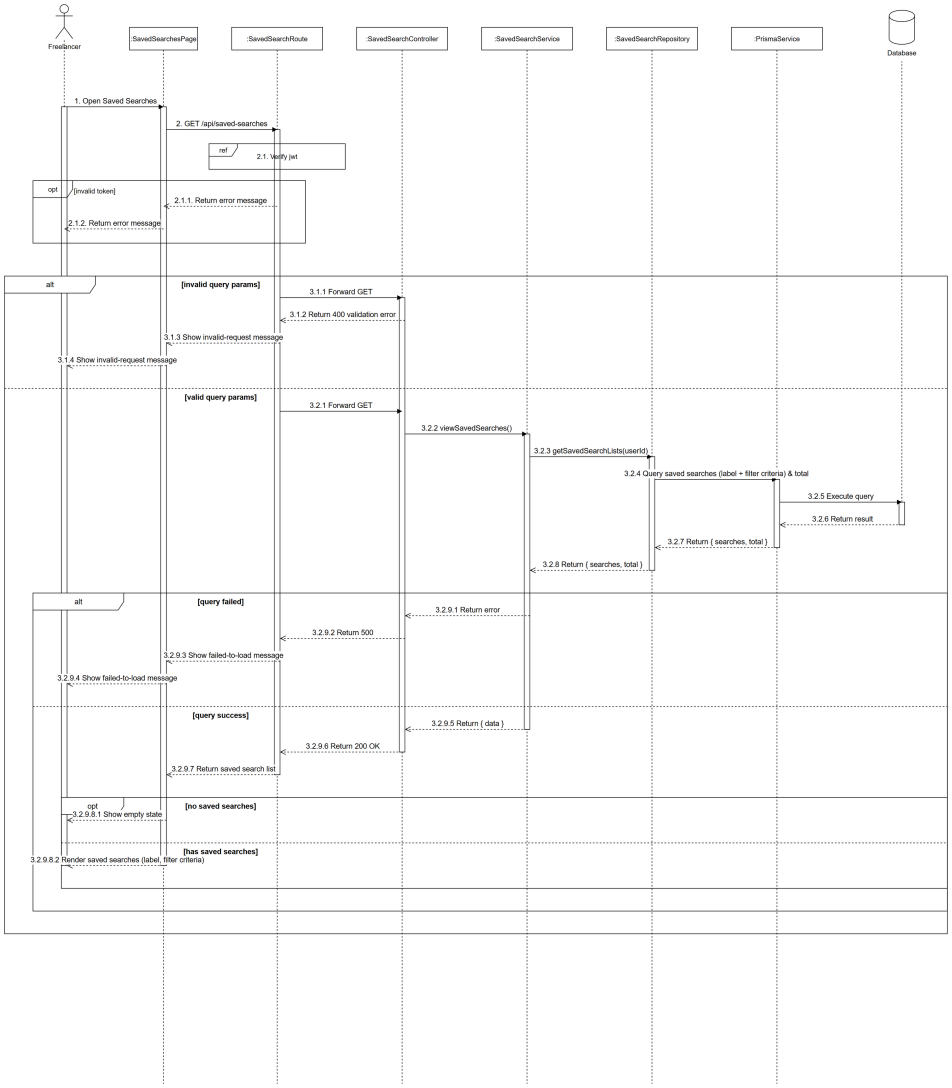

Figure 90: Sequence diagram of UC-22.01 View Saved Search List

3.16.3 UC-22.02 – View Saved Search Detail

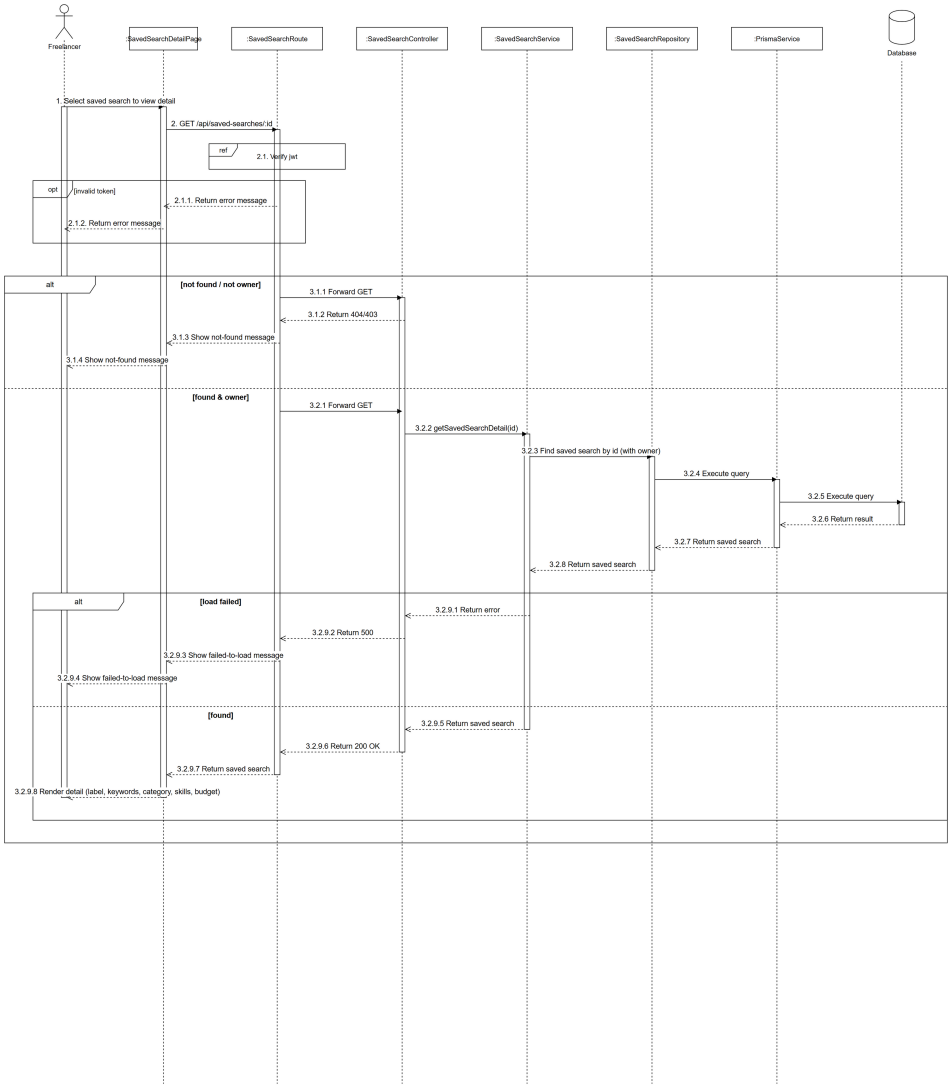

Figure 91: Sequence diagram of UC-22.02 View Saved Search Detail

3.16.4 UC-22.03 – Create Saved Search

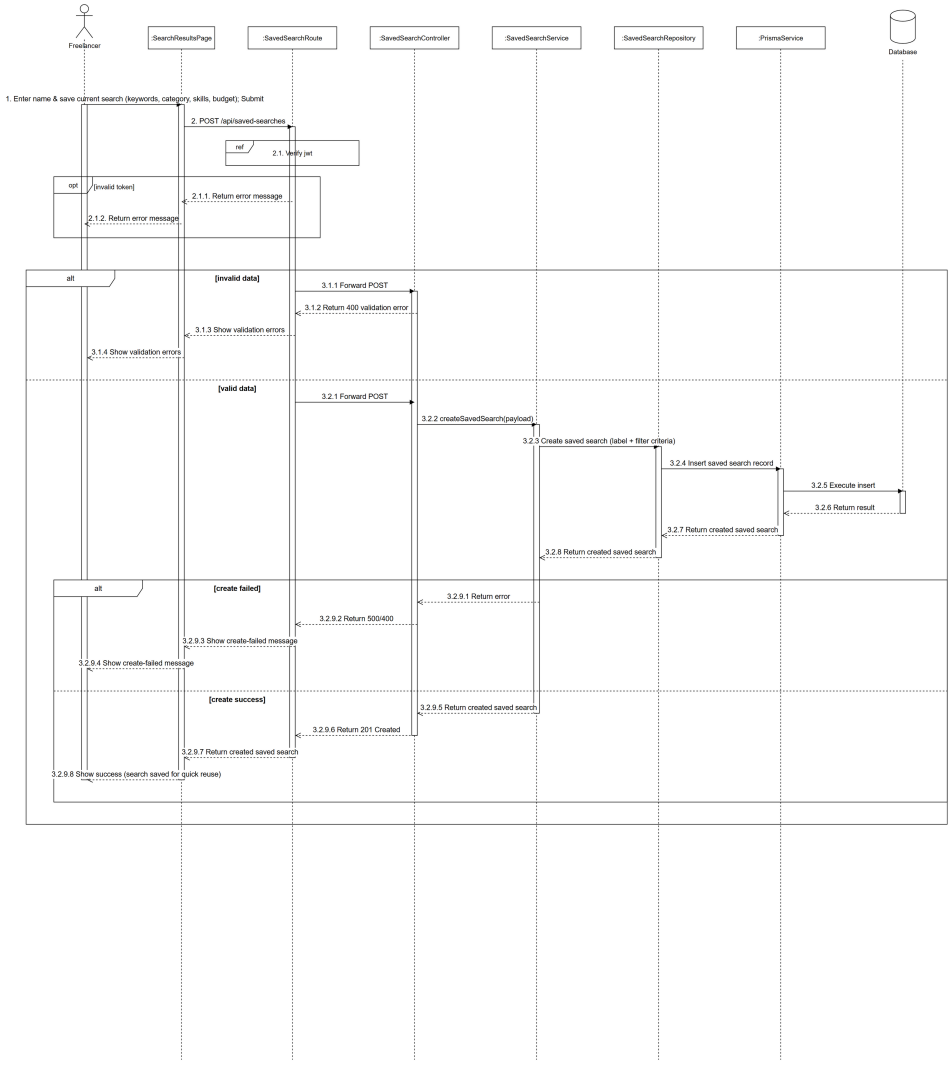

Figure 92: Sequence diagram of UC-22.03 Create Saved Search

3.16.5 UC-22.04 – Update Saved Search

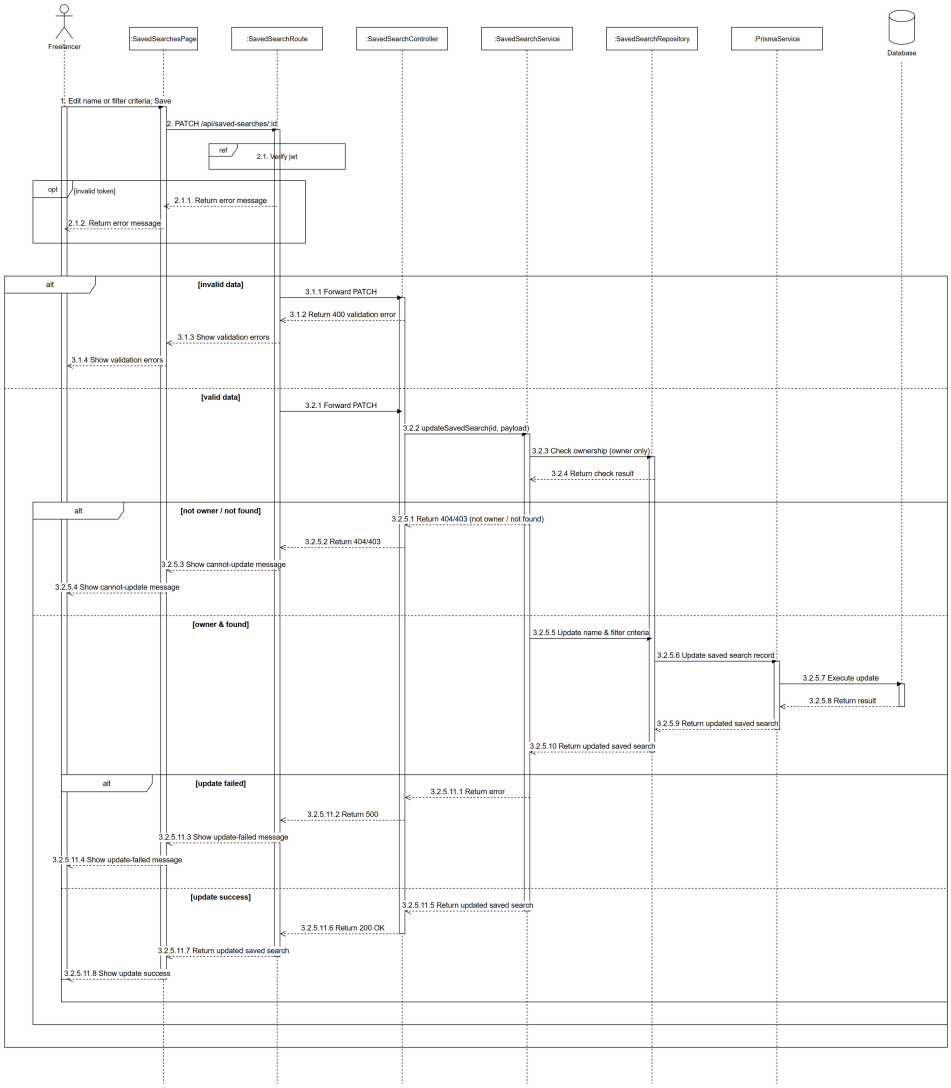

Figure 93: Sequence diagram of UC-22.04 Update Saved Search

3.16.6 UC-22.05 – Delete Saved Search

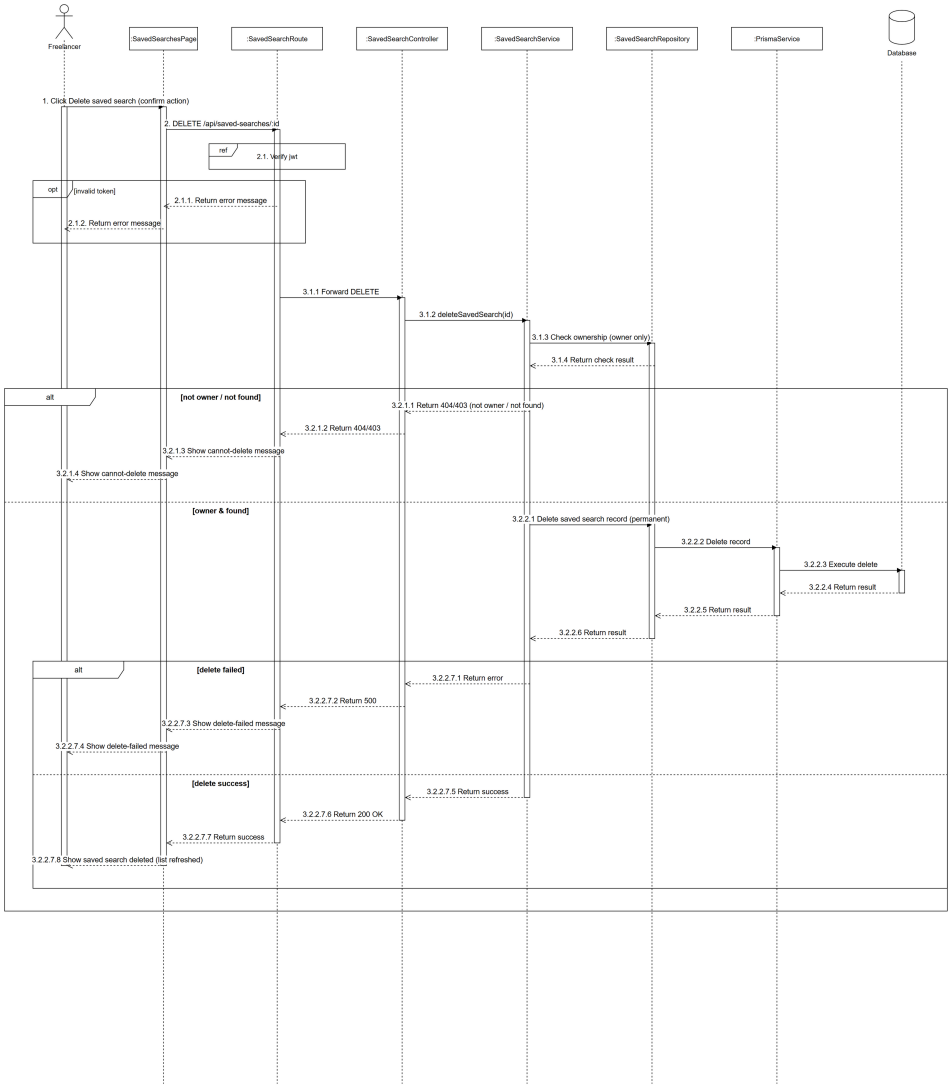

Figure 94: Sequence diagram of UC-22.05 Delete Saved Search

3.17 Job Alert Management

Job Alert Management enables freelancers to configure automated notifications based on job keywords, budget ranges, and category criteria, ensuring they receive prompt alerts when matching jobs are published.

3.17.1 Class Diagram

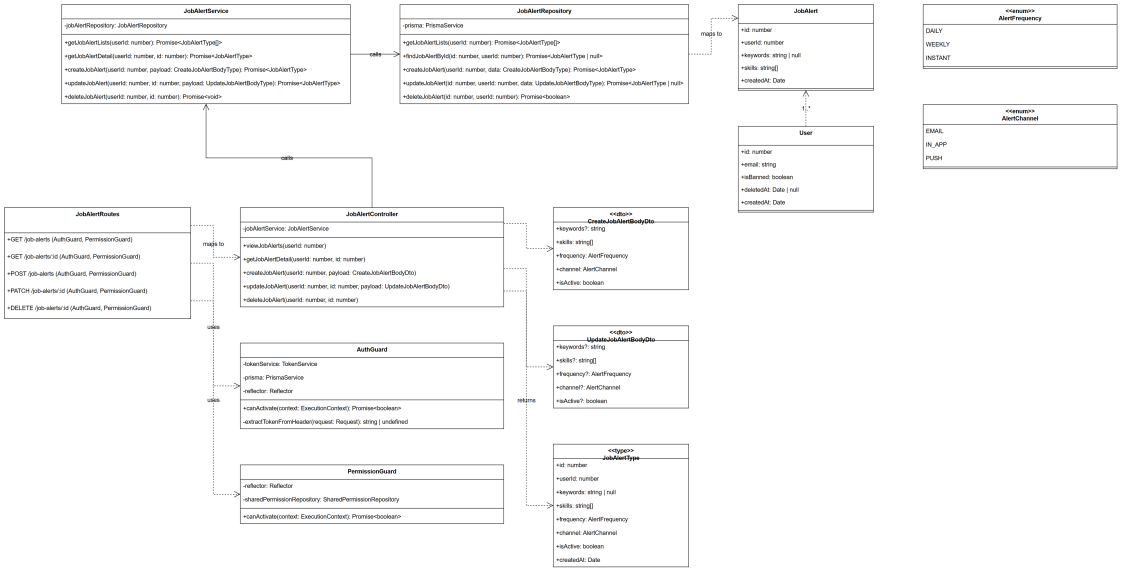

Figure 95: Class diagram of Job Alert Management

3.17.2 UC-23.01 – View Job Alert List

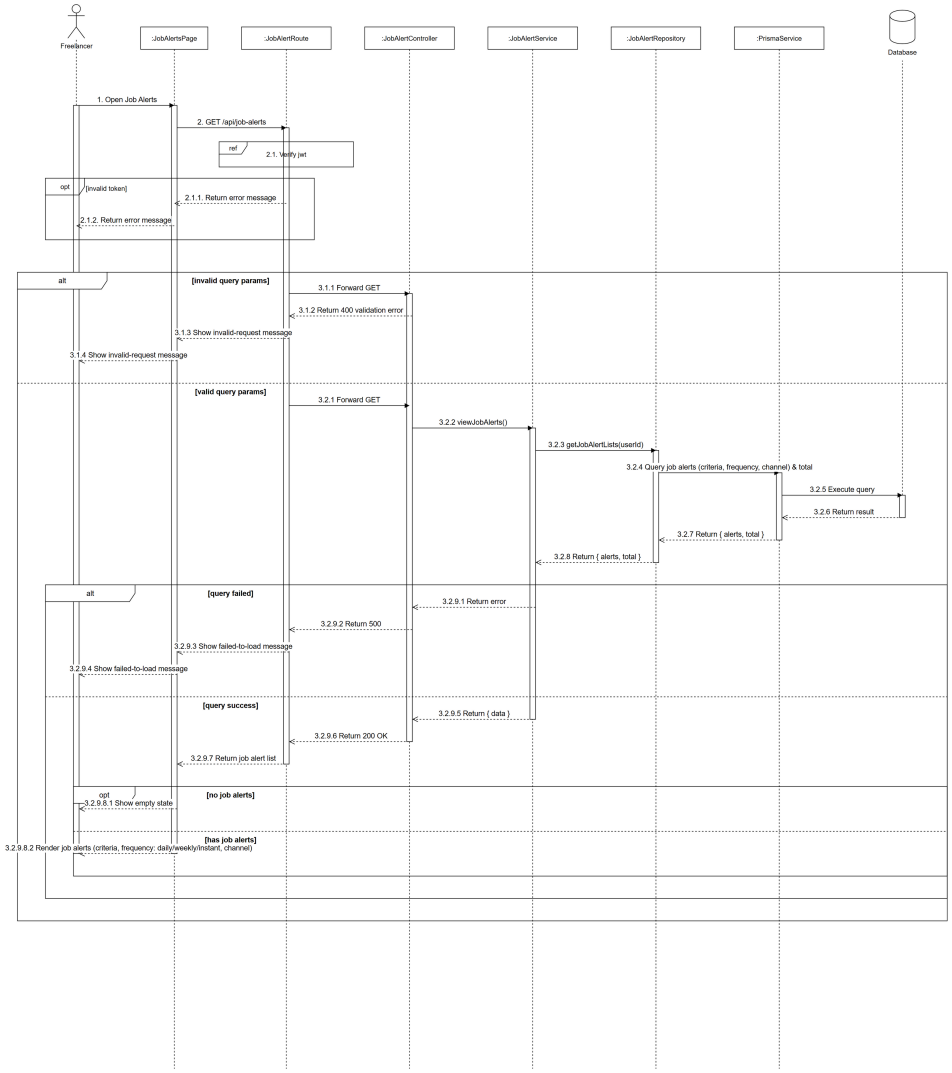

Figure 96: Sequence diagram of UC-23.01 View Job Alert List

3.17.3 UC-23.02 – View Job Alert Detail

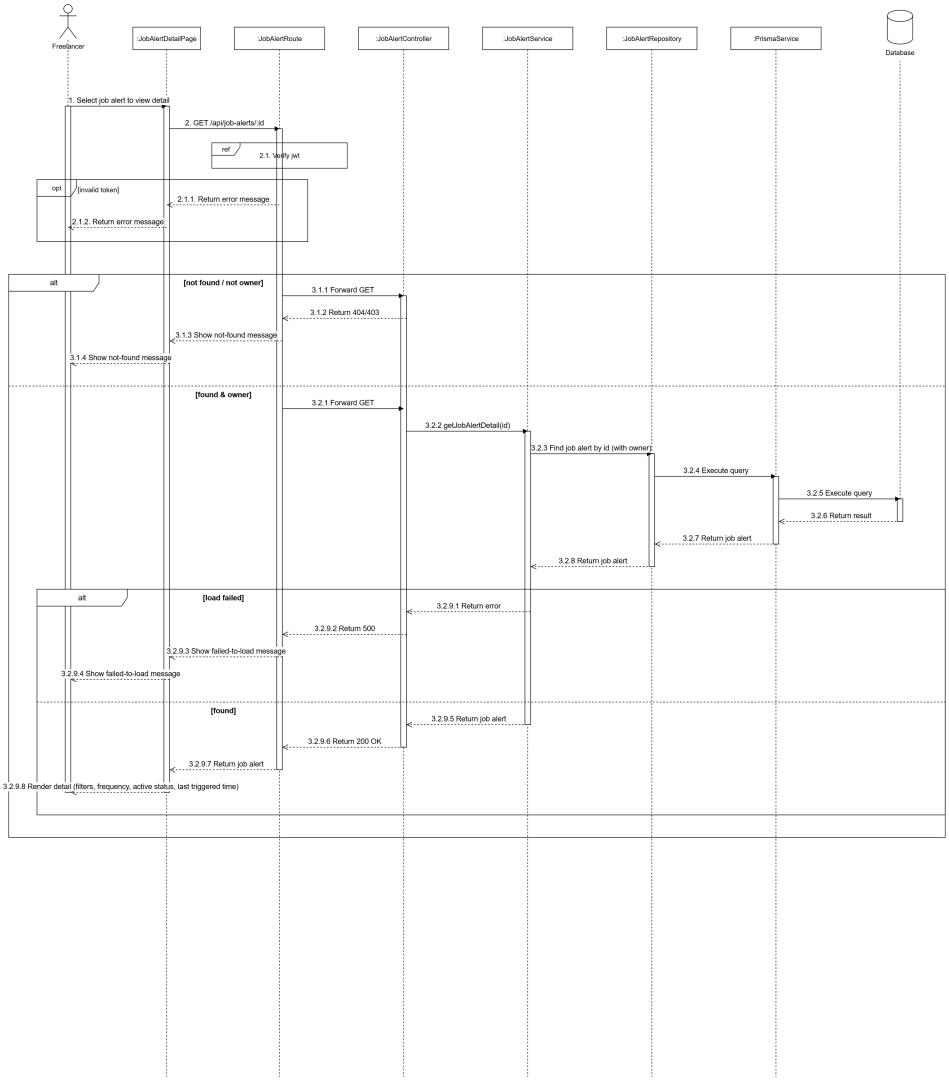

Figure 97: Sequence diagram of UC-23.02 View Job Alert Detail

3.17.4 UC-23.03 – Create Job Alert

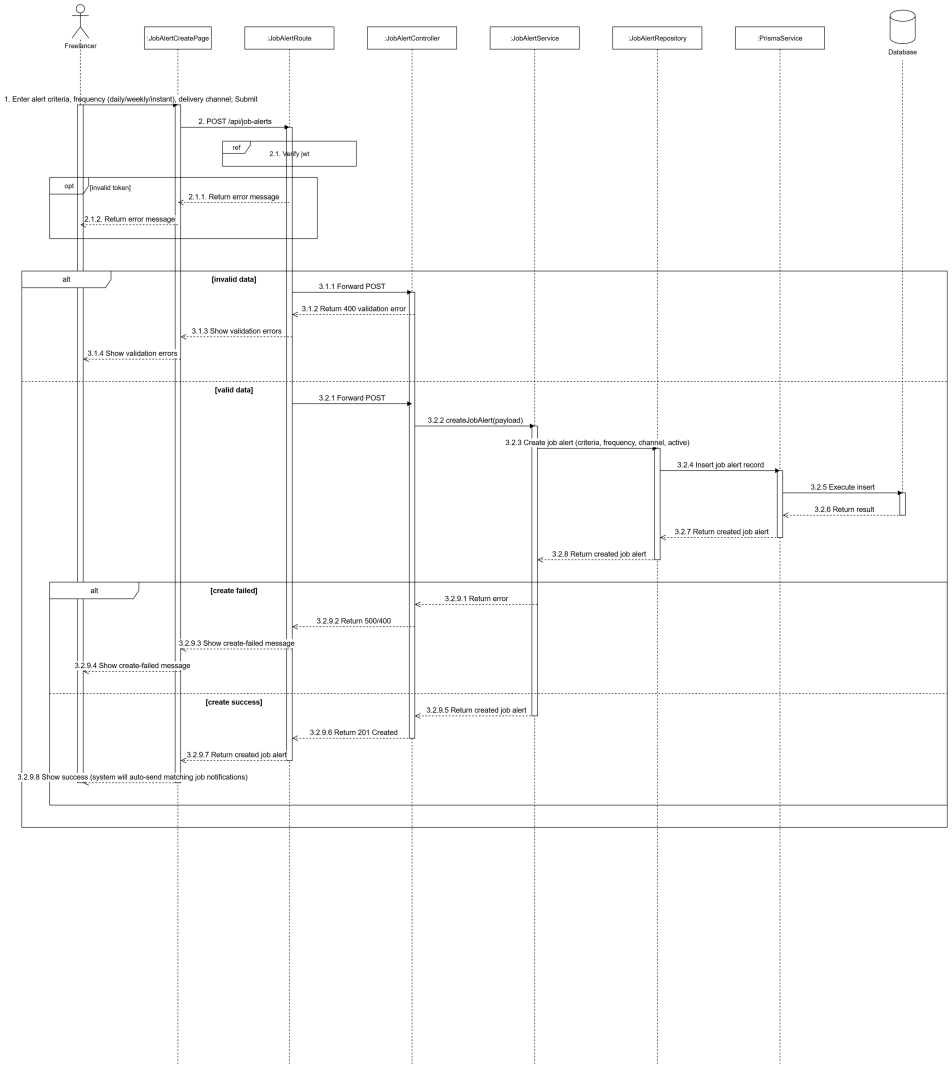

Figure 98: Sequence diagram of UC-23.03 Create Job Alert

3.17.5 UC-23.04 – Update Job Alert

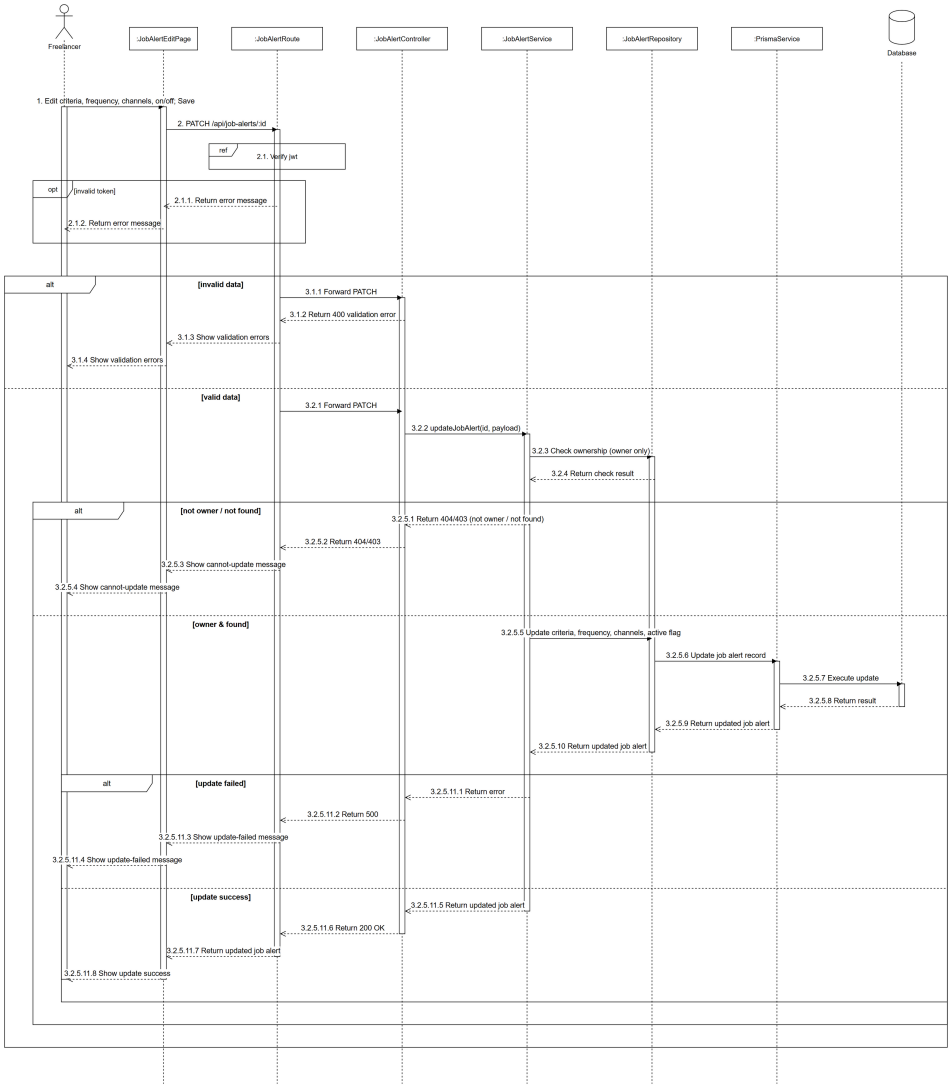

Figure 99: Sequence diagram of UC-23.04 Update Job Alert

3.17.6 UC-23.05 – Delete Job Alert

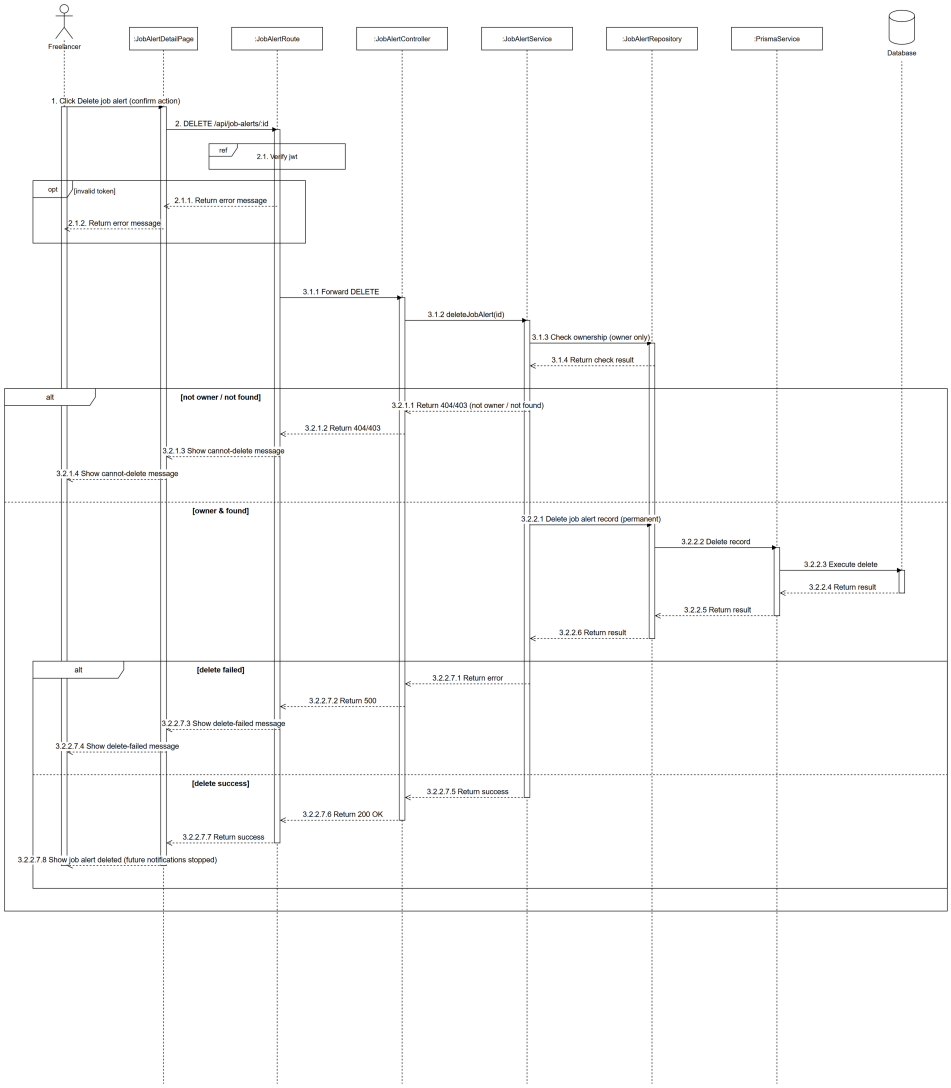

Figure 100: Sequence diagram of UC-23.05 Delete Job Alert

3.18 Permission Management

Permission Management exposes the route-level permission catalogue so that an administrator can inspect the permissions available for role assignment.

3.18.1 Class Diagram

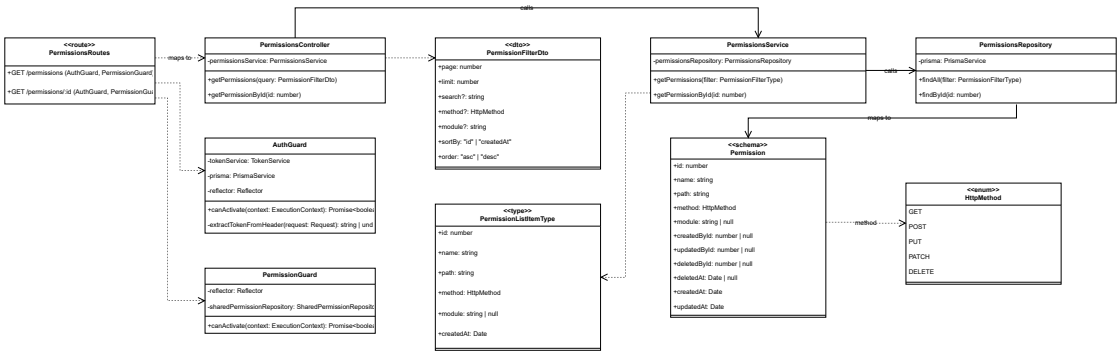

Figure 101: Class diagram of Permission Management

3.18.2 UC-15.01 – View Permission List

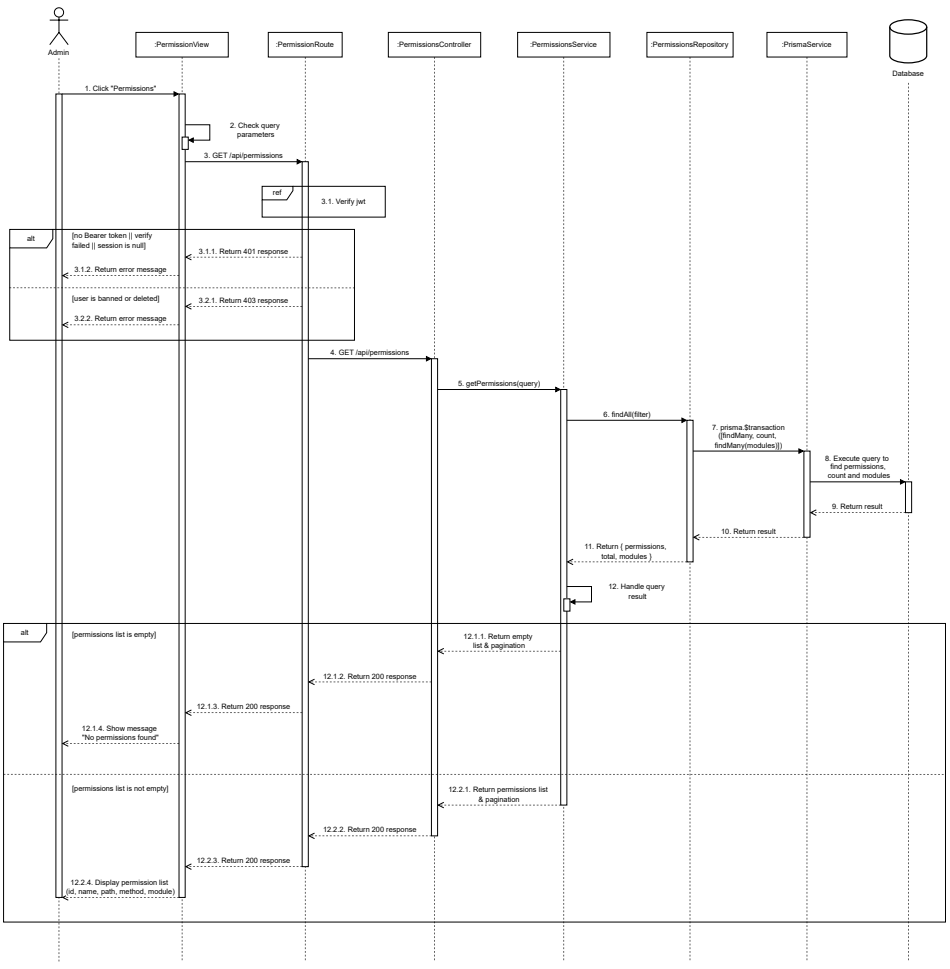

Figure 102: Sequence diagram of UC-15.01 View Permission List

3.18.3 UC-15.02 – View Permission Detail

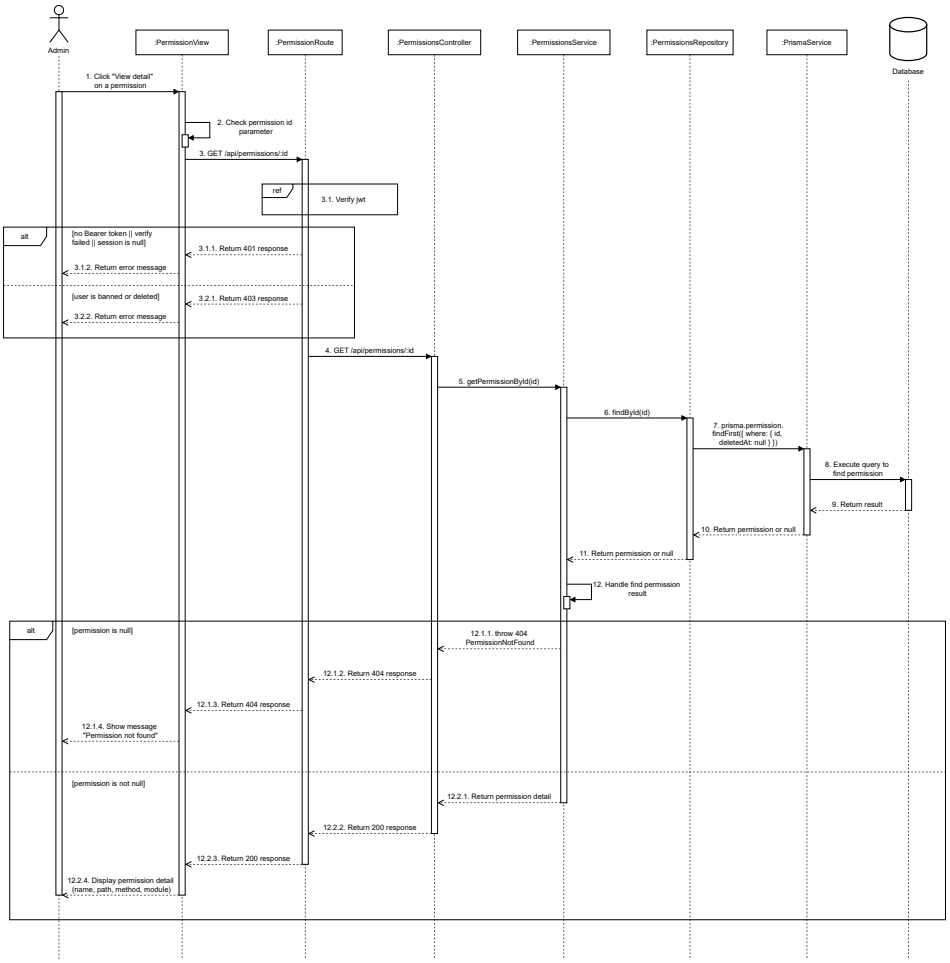

Figure 103: Sequence diagram of UC-15.02 View Permission Detail

3.19 Session Management

Session Management gives an authenticated user visibility over the devices that currently hold a refresh-token session and allows a session to be revoked remotely.

3.19.1 Class Diagram

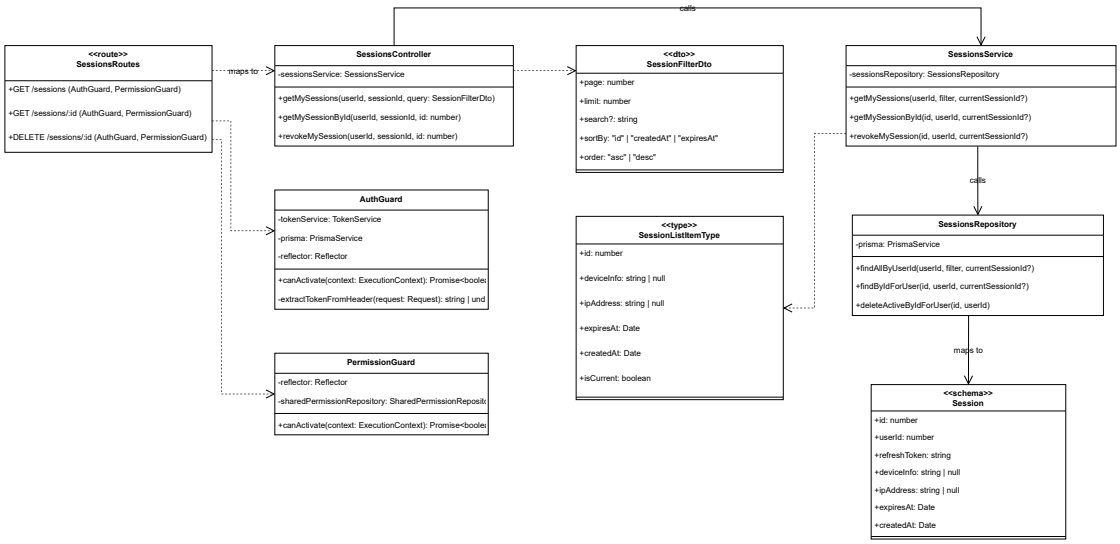

Figure 104: Class diagram of Session Management

3.19.2 UC-16.01 – View Session List

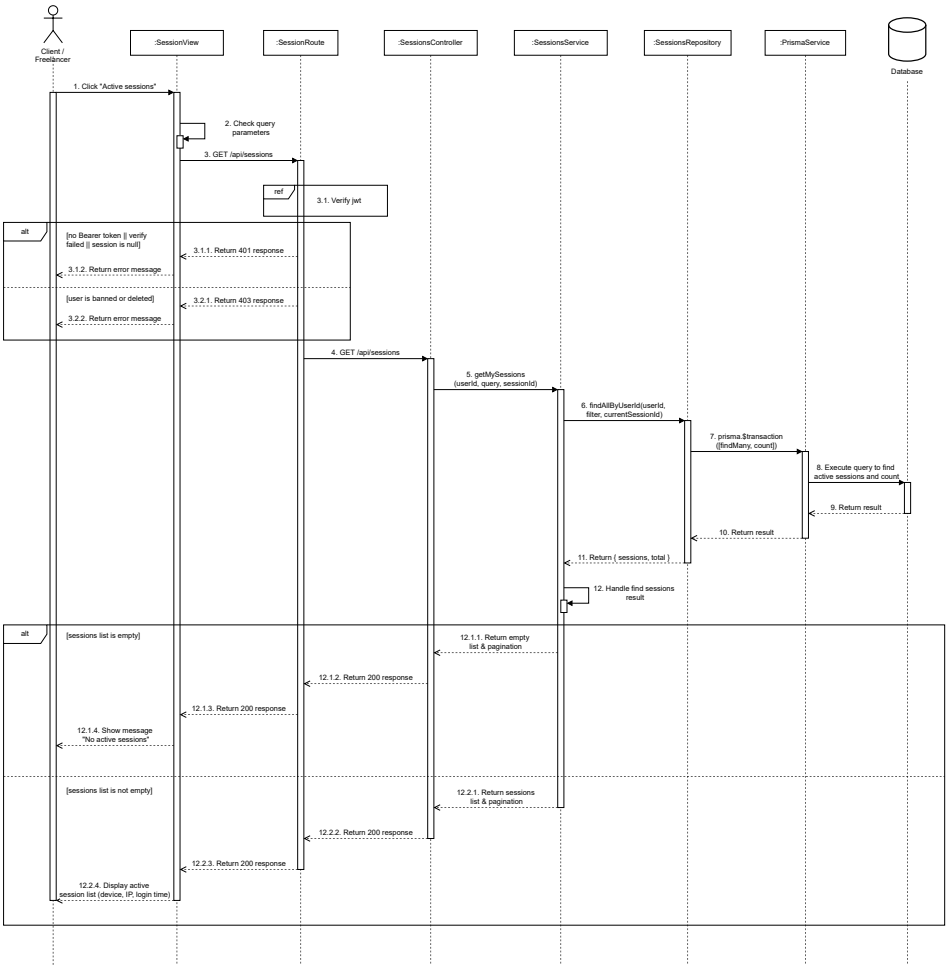

Figure 105: Sequence diagram of UC-16.01 View Session List

3.19.3 UC-16.02 – View Session Detail

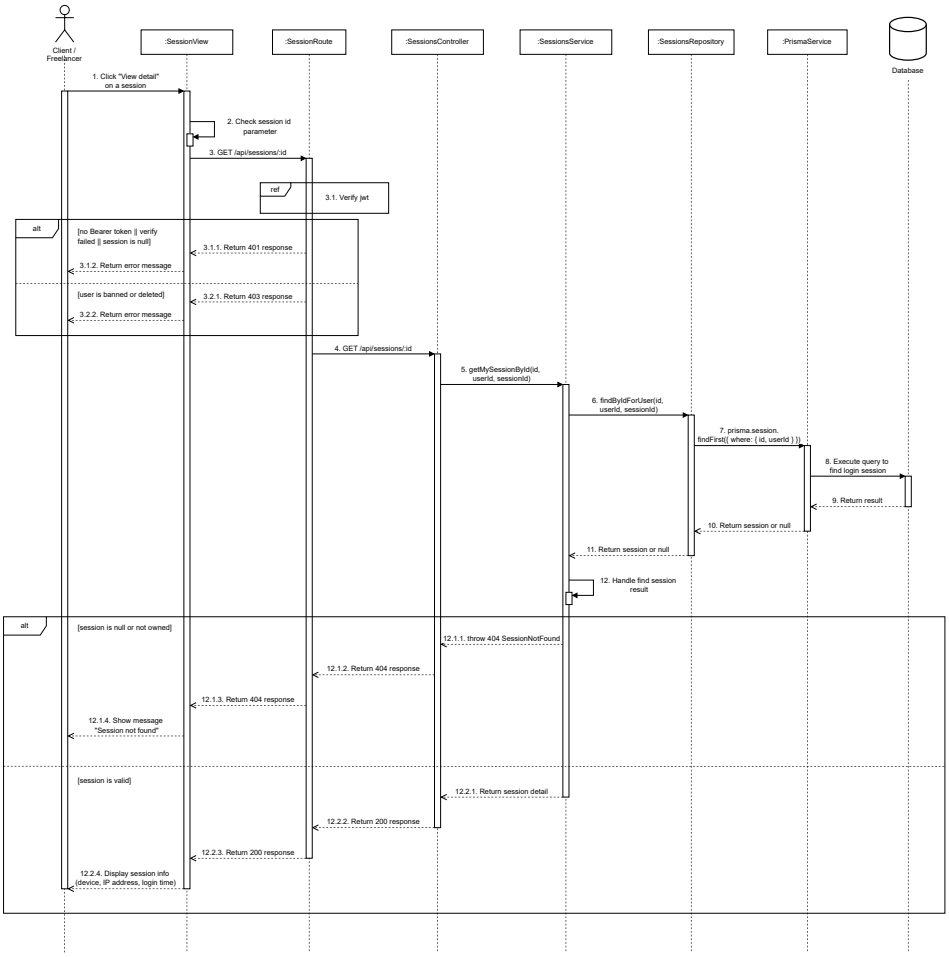

Figure 106: Sequence diagram of UC-16.02 View Session Detail

3.19.4 UC-16.03 – Revoke Session

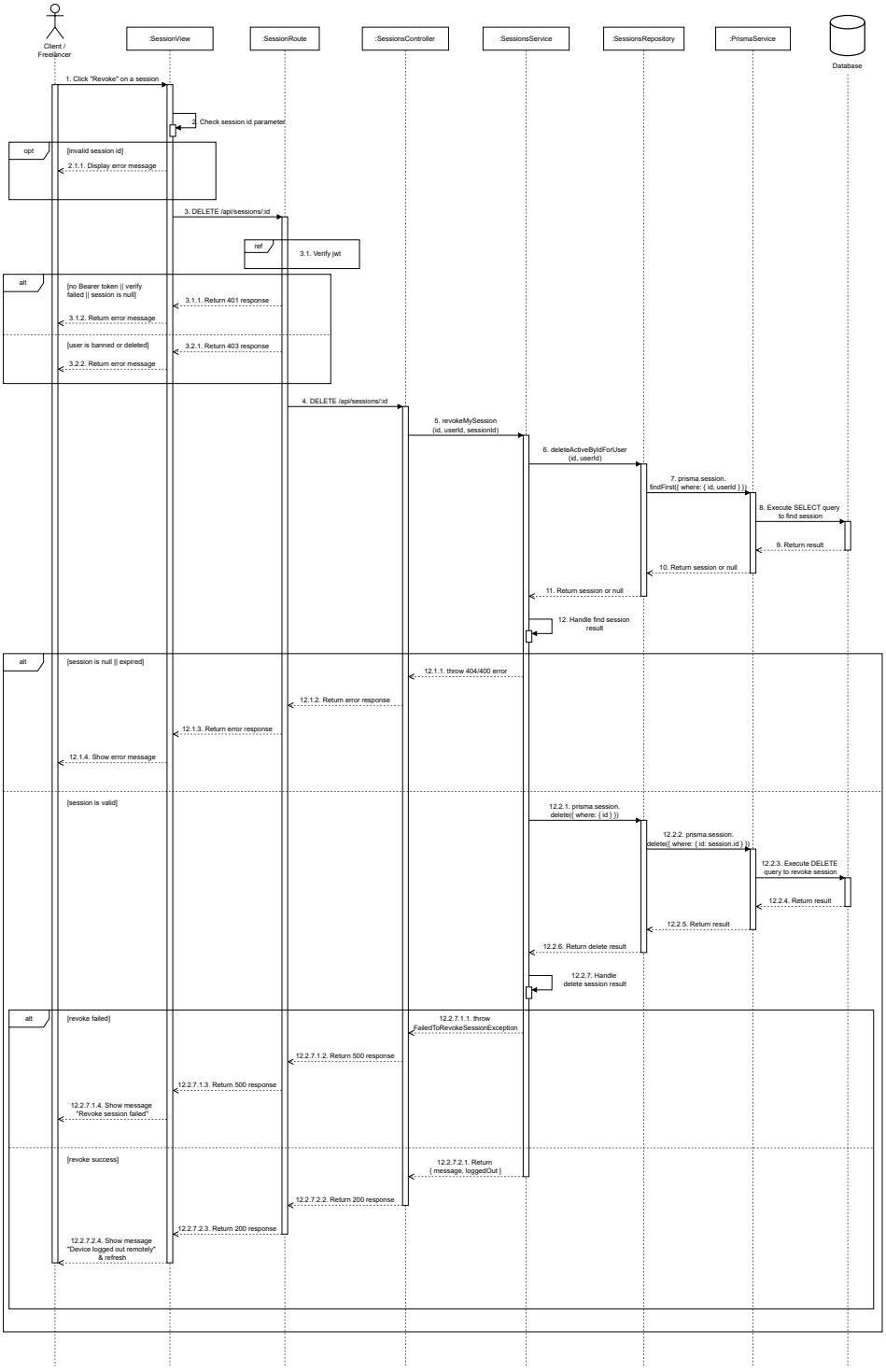

Figure 107: Sequence diagram of UC-16.03 Revoke Session

3.20 Forum Category Management

Forum Category Management covers the administrator-side category catalogue and the public category views used by the forum. Deletion is a soft delete, so a category can be restored later.

3.20.1 Class Diagram

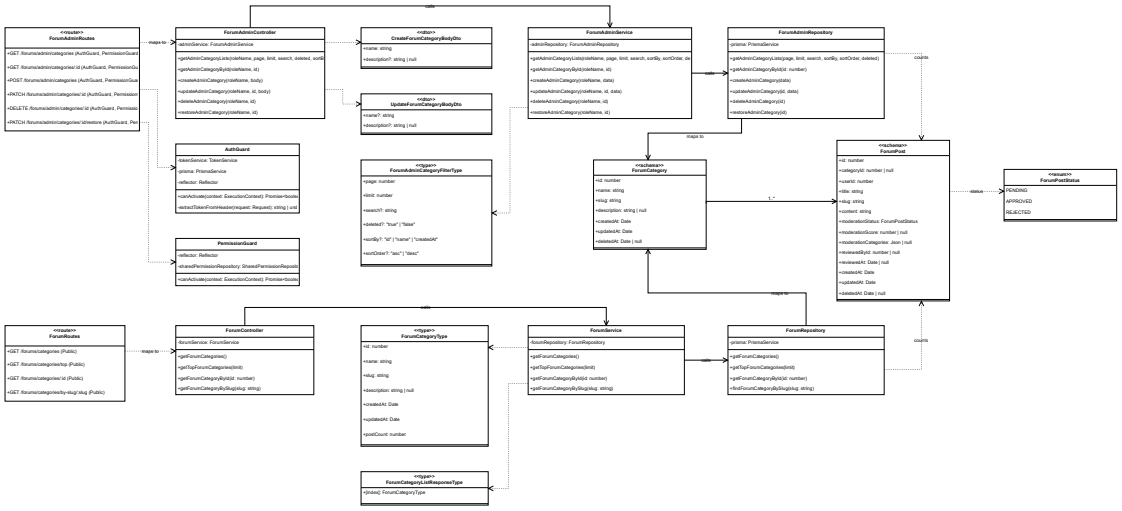

Figure 108: Class diagram of Forum Category Management

3.20.2 UC-17.01 – Create Forum Category

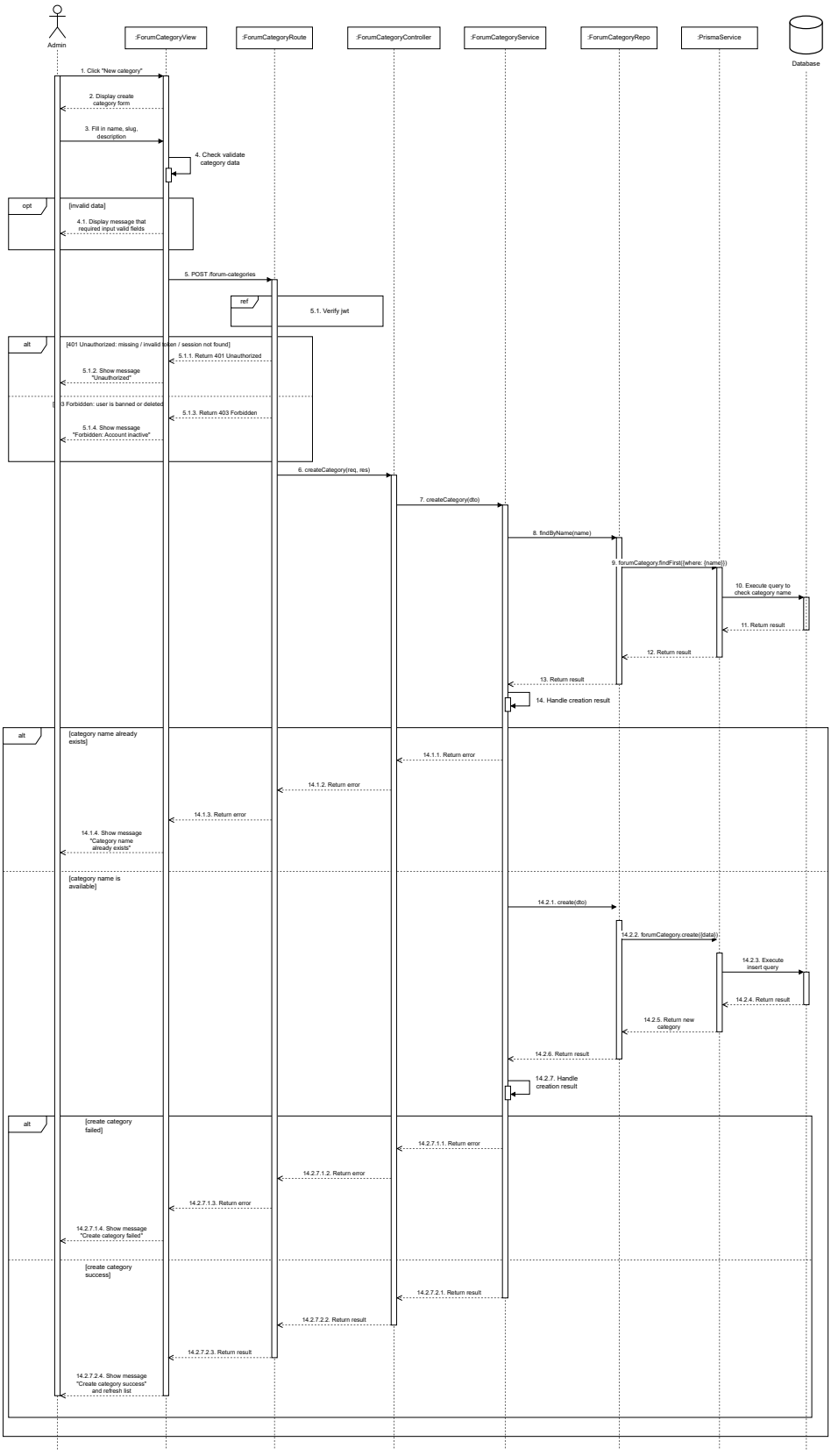

Figure 109: Sequence diagram of UC-17.01 Create Forum Category

3.20.3 UC-17.02 – Update Forum Category

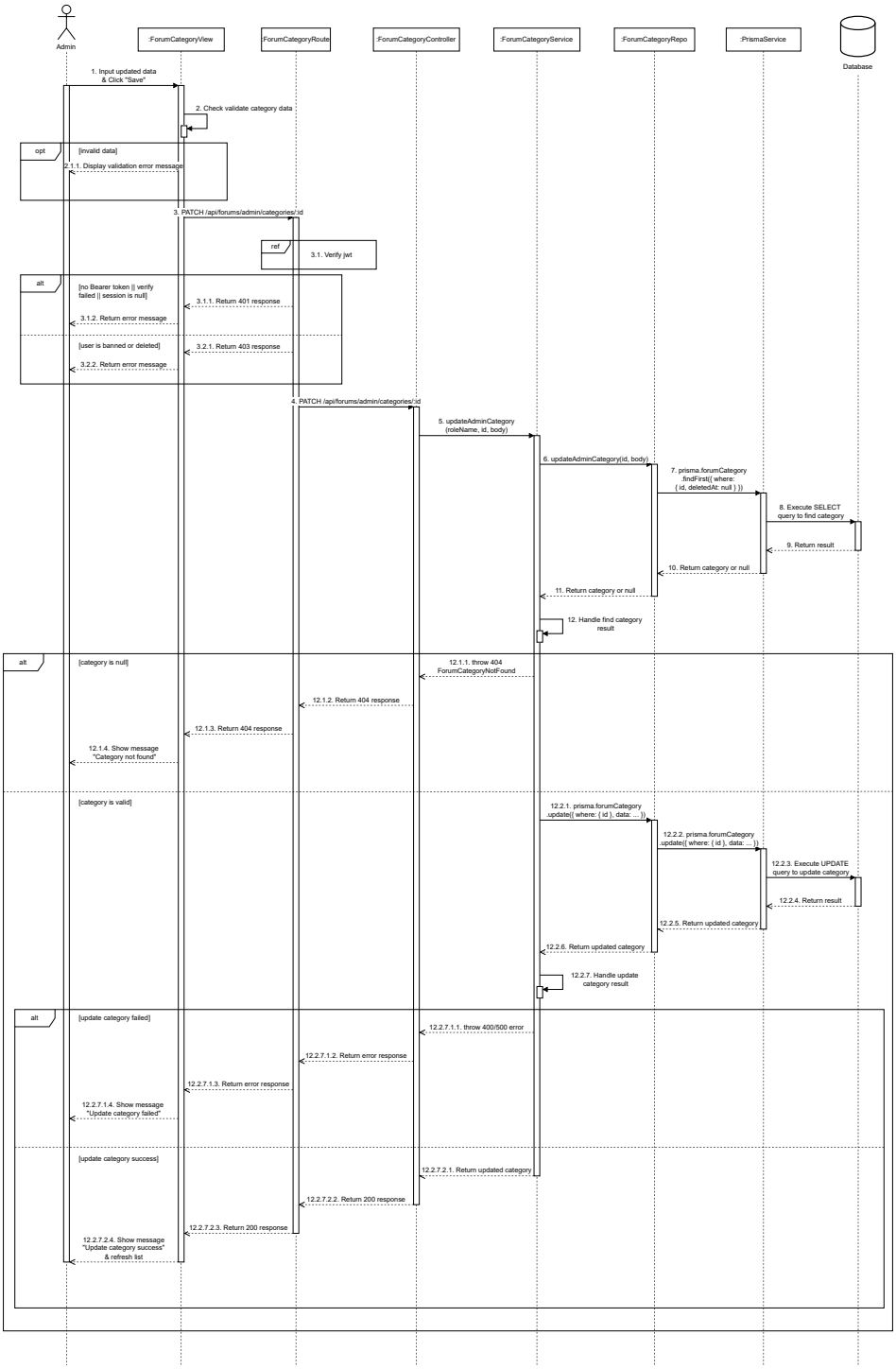

Figure 110: Sequence diagram of UC-17.02 Update Forum Category

3.20.4 UC-17.03 – Delete Forum Category

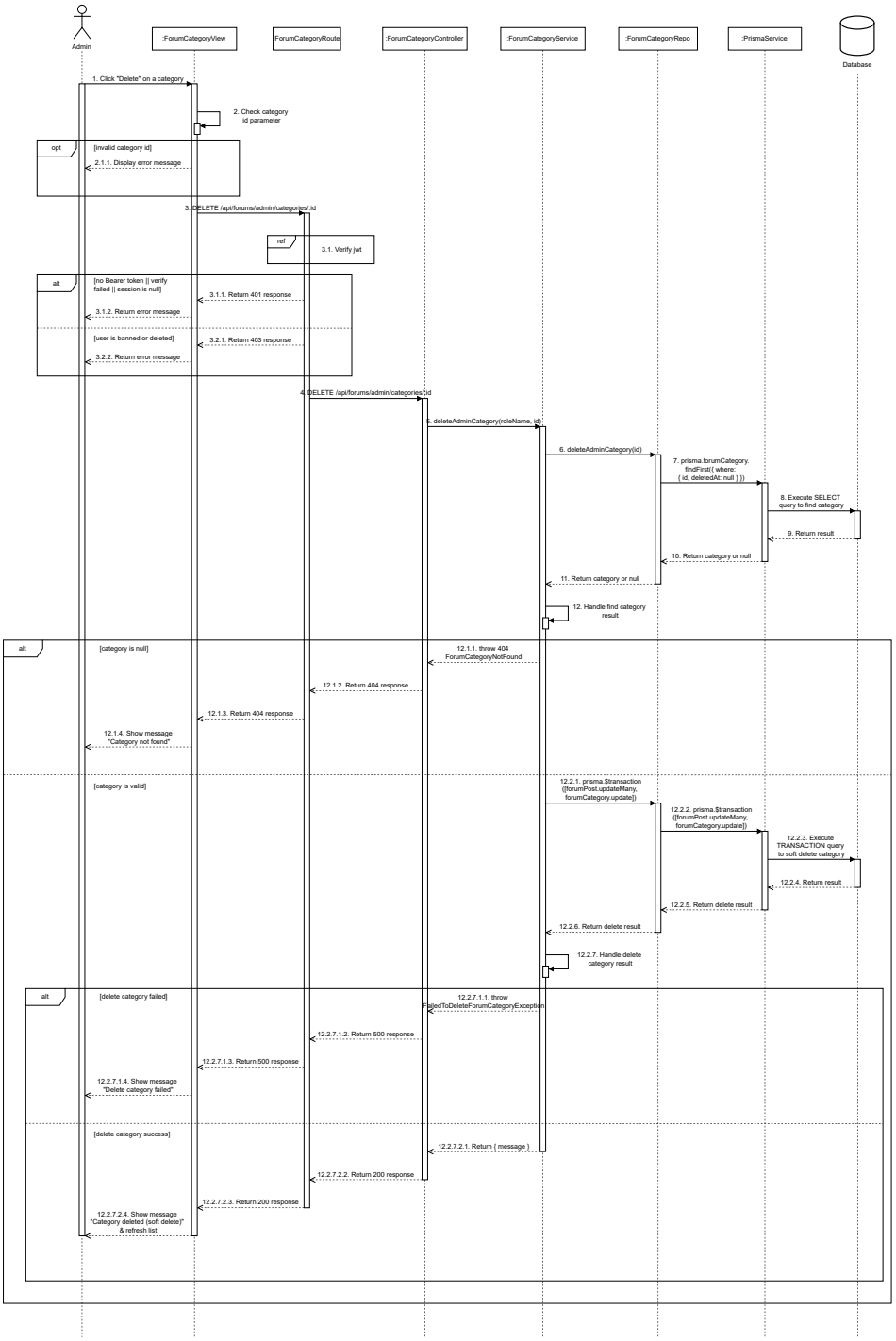

Figure 111: Sequence diagram of UC-17.03 Delete Forum Category

3.20.5 UC-17.06 – Restore Forum Category

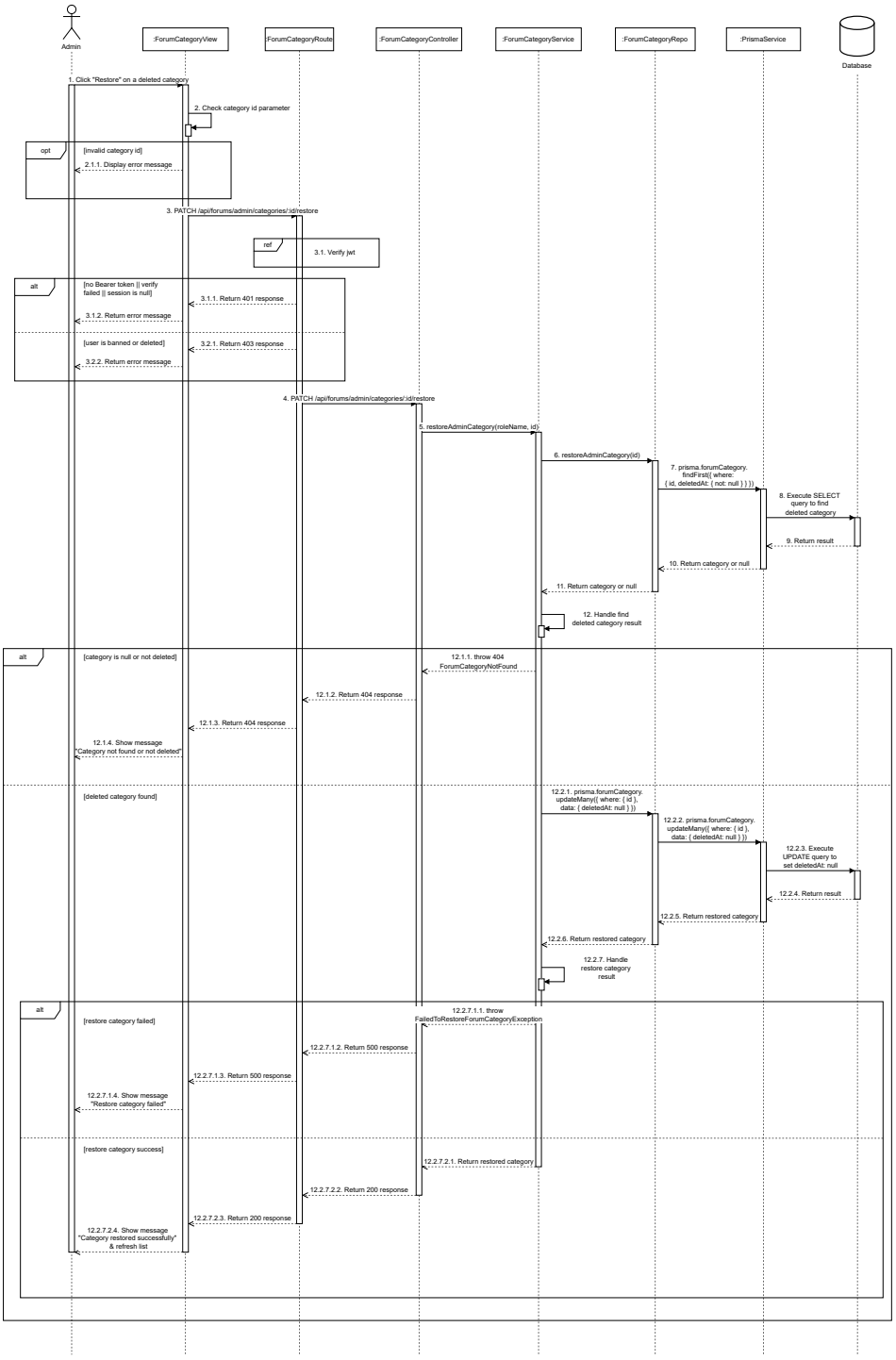

Figure 112: Sequence diagram of UC-17.06 Restore Forum Category

3.20.6 UC-14.01 – View Forum Category List

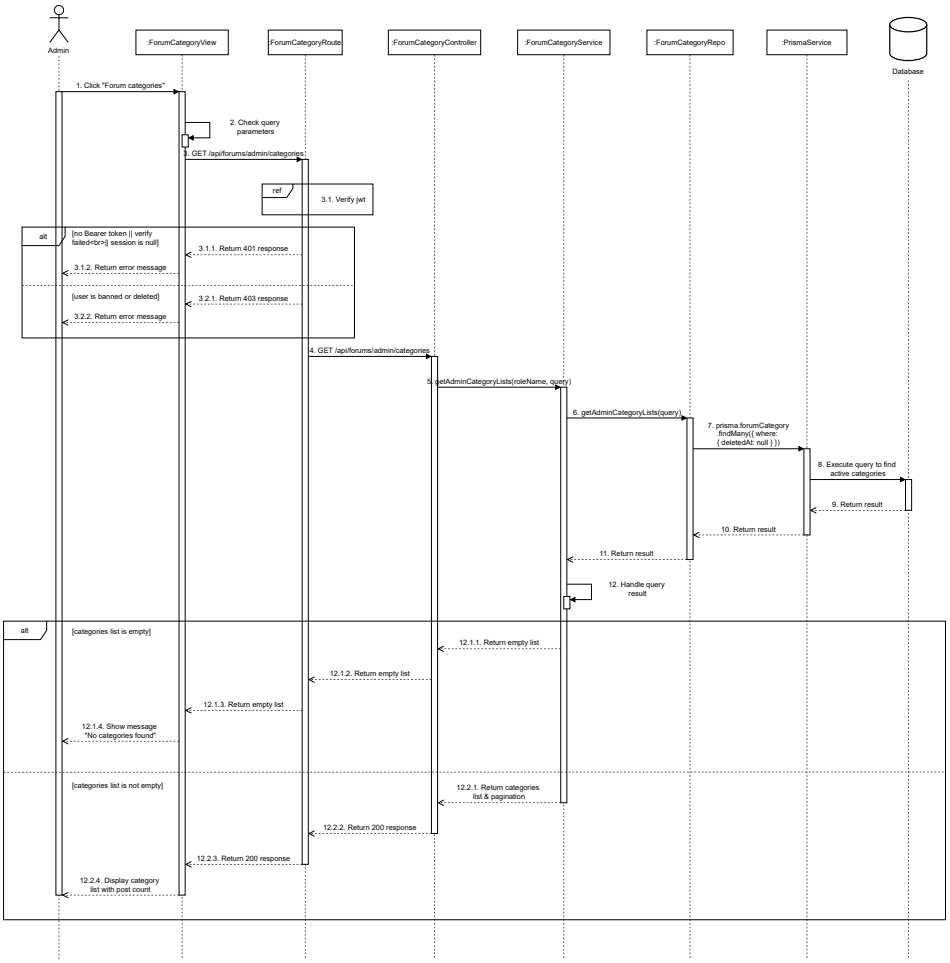

Figure 113: Sequence diagram of UC-14.01 View Forum Category List

3.20.7 UC-14.02 – View Forum Category Detail

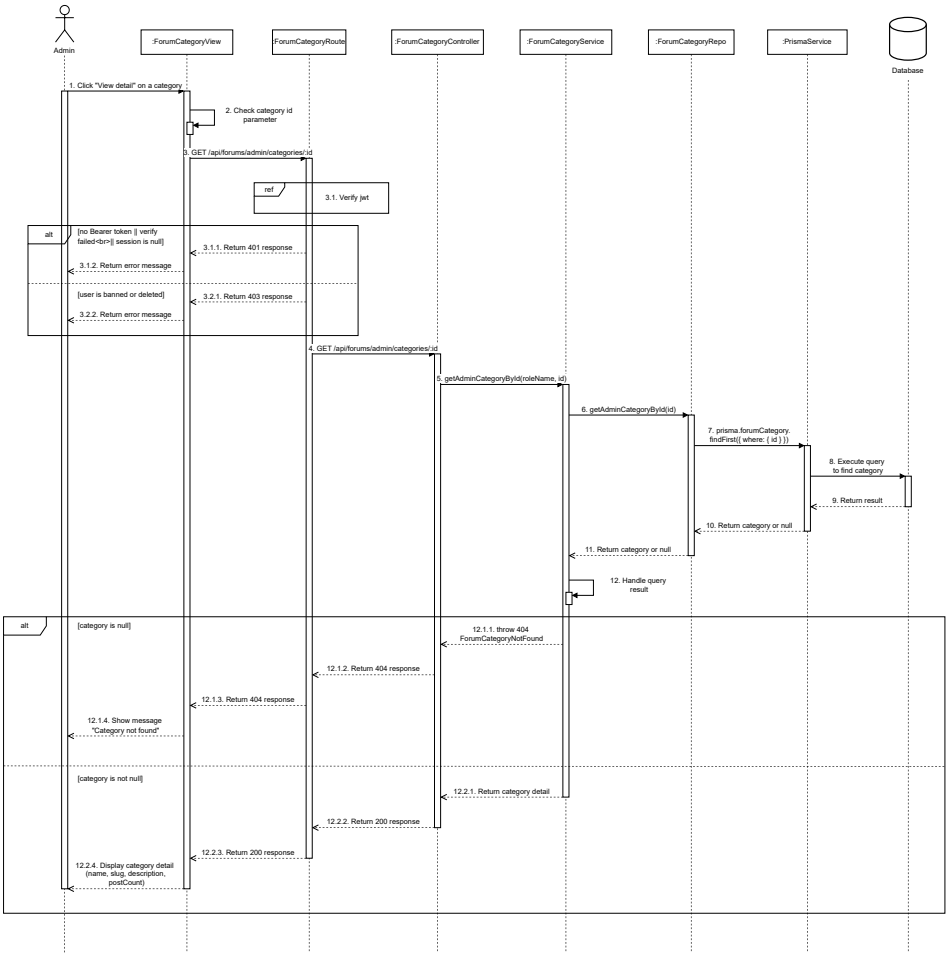

Figure 114: Sequence diagram of UC-14.02 View Forum Category Detail

3.21 Manage Moderation

Manage Moderation covers the UC-34 review workflow of the administration console: listing the posts waiting for review, opening a flagged post and approving or rejecting it. A review writes the moderation status, score and reviewer back to the forum post.

3.21.1 Class Diagram

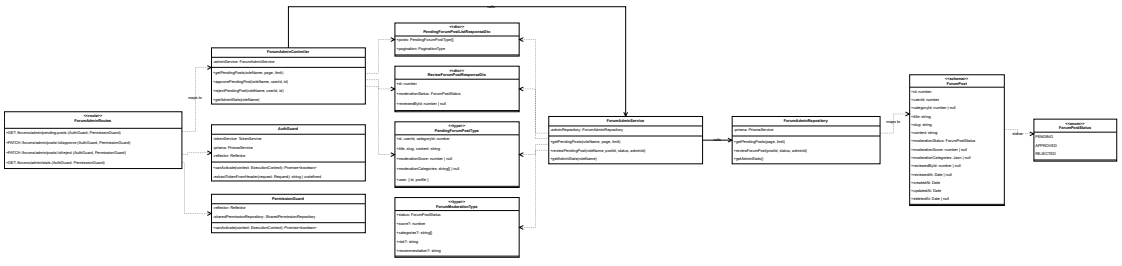

Figure 115: Class diagram of Manage Moderation

3.21.2 UC-34.01 – View Moderation Queue

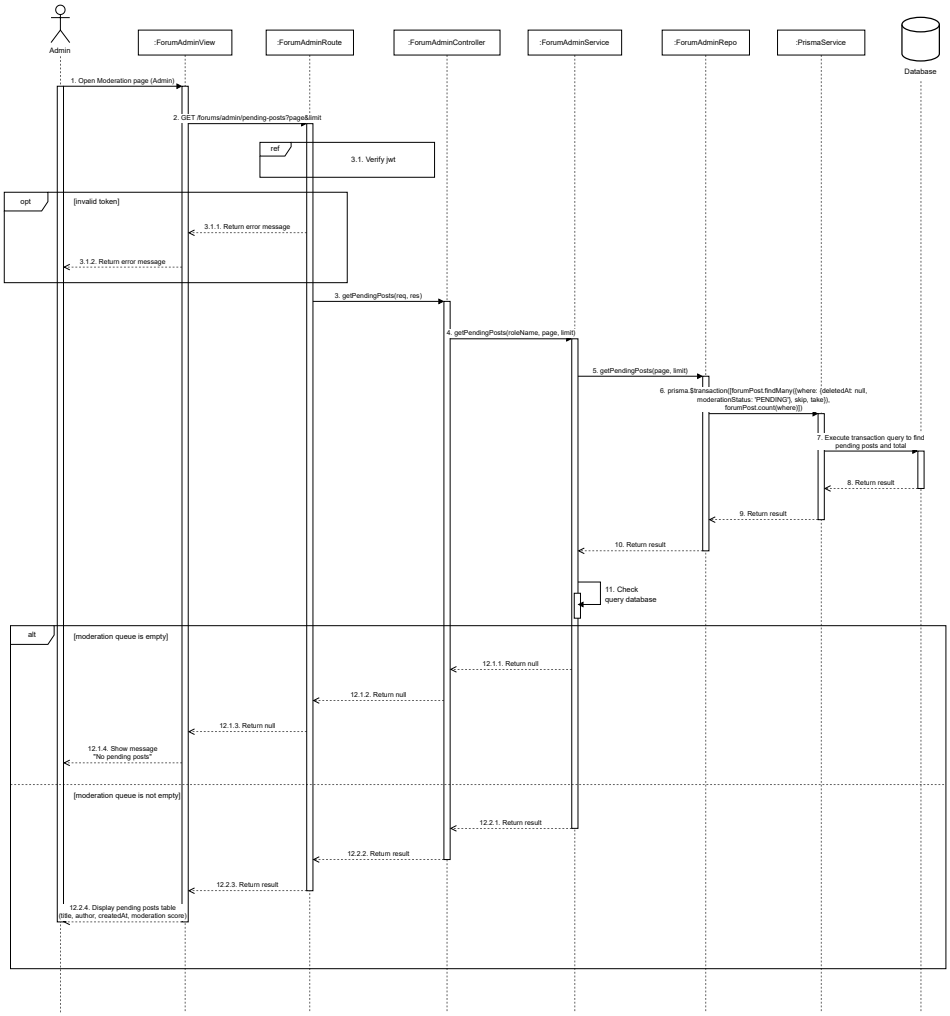

Figure 116: Sequence diagram of UC-34.01 View Moderation Queue

3.21.3 UC-34.02 – View Moderation Item Detail

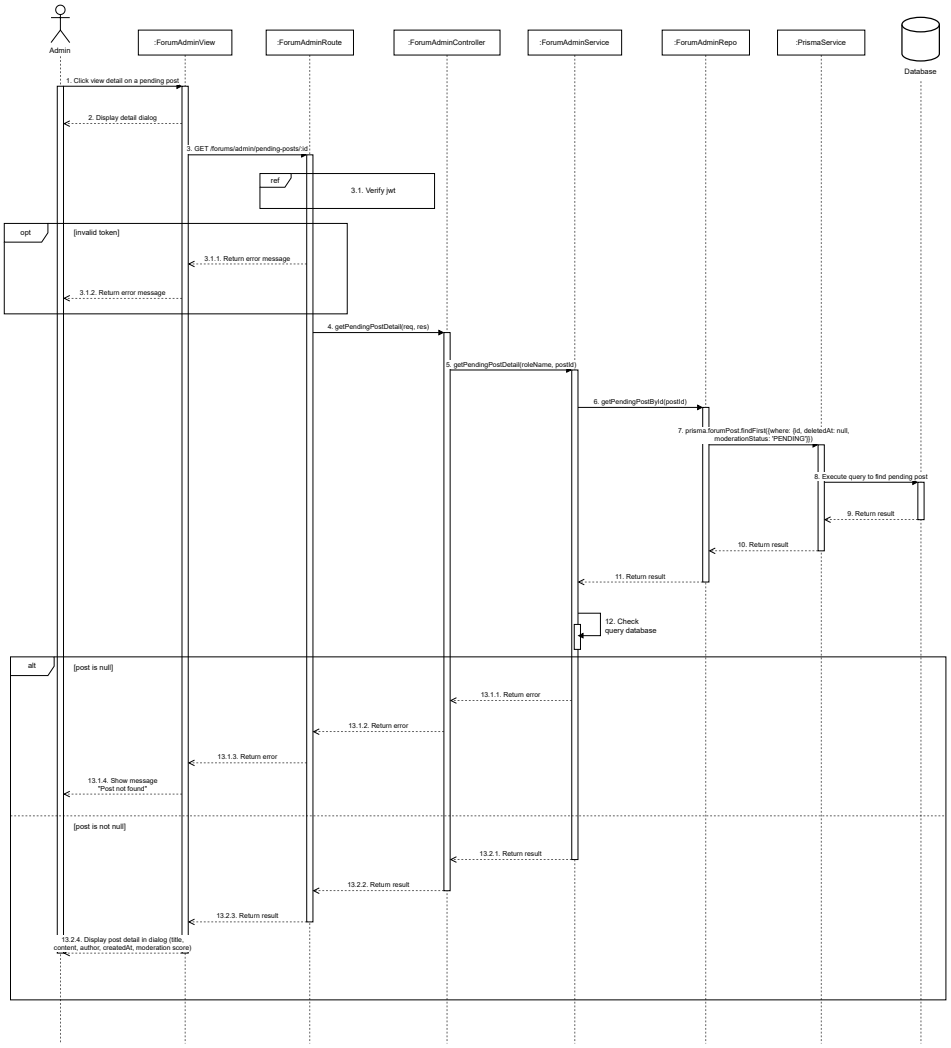

Figure 117: Sequence diagram of UC-34.02 View Moderation Item Detail

3.21.4 UC-34.03 – Resolve Moderation Item

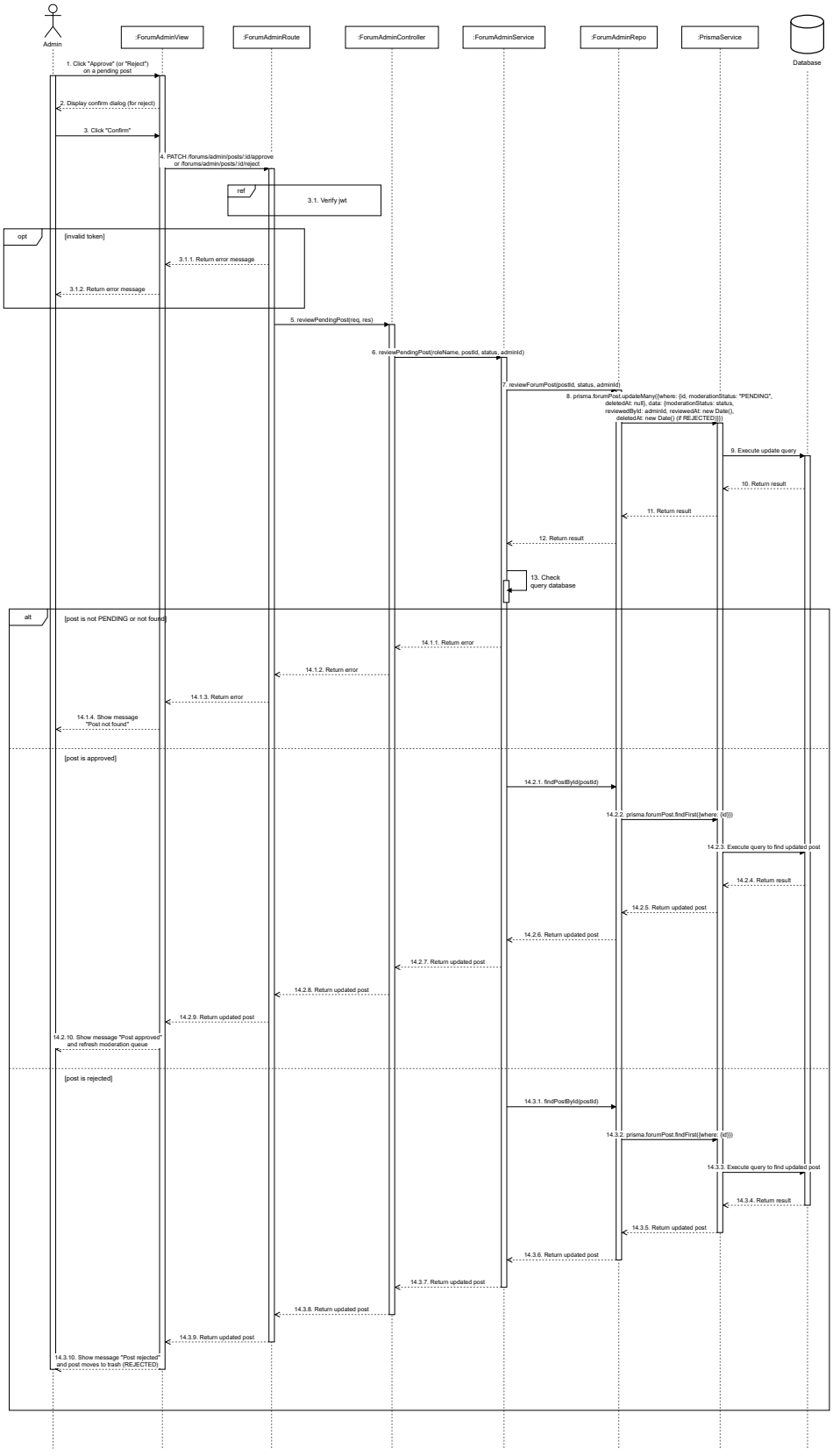

Figure 118: Sequence diagram of UC-34.03 Resolve Moderation Item

3.22 Manage ID Verification

Manage ID Verification covers the UC-35 identity review workflow: listing the submitted identity documents, opening a request with its uploaded file, and approving or rejecting the request. A decision updates the verification status and the reviewer of the identity document.

3.22.1 Class Diagram

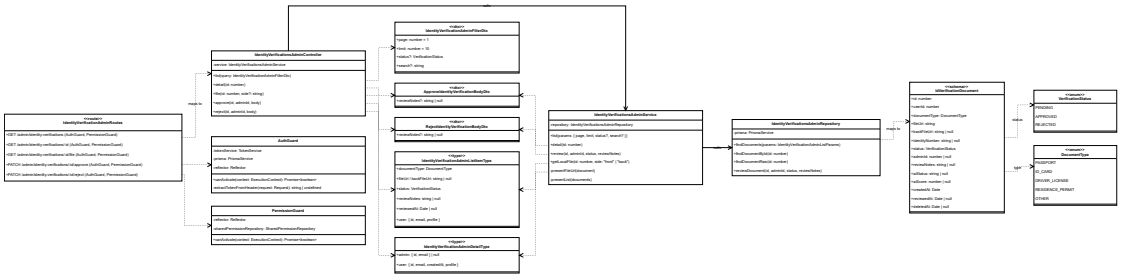

Figure 119: Class diagram of Manage ID Verification

3.22.2 UC-35.01 – View ID Verification Request List

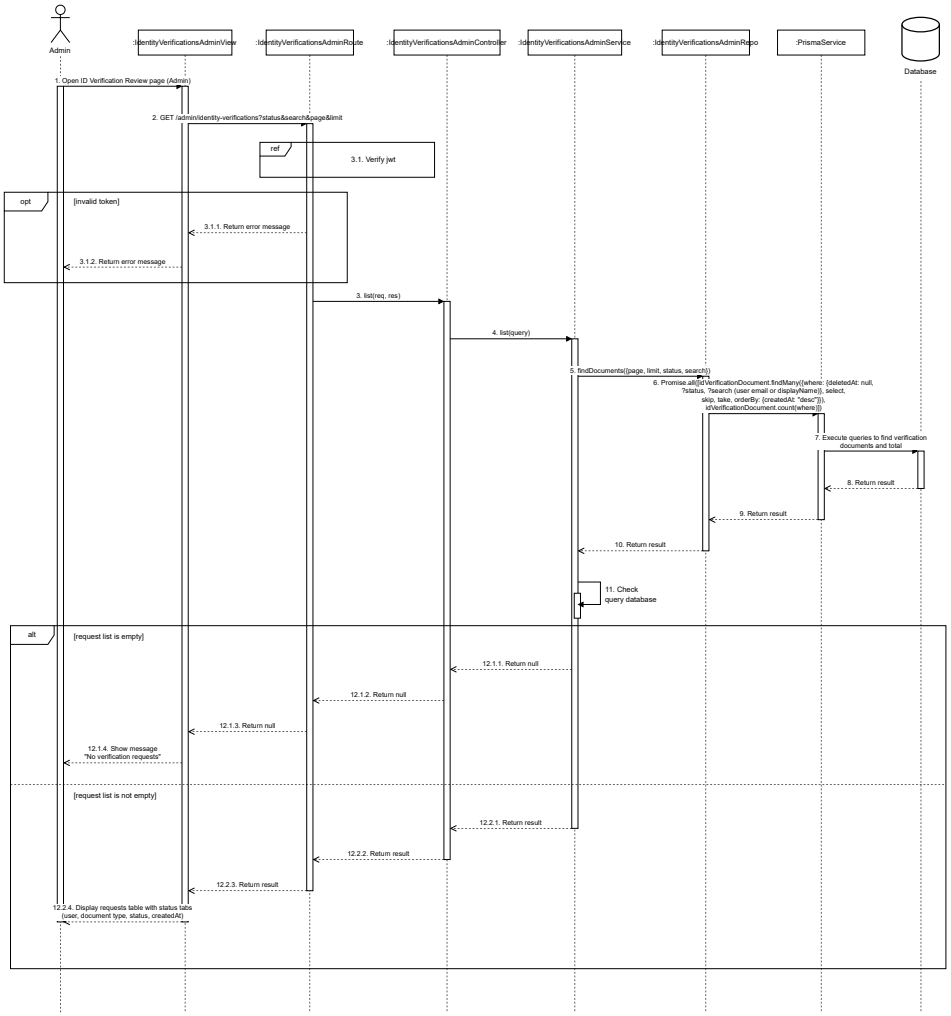

Figure 120: Sequence diagram of UC-35.01 View ID Verification Request List

3.22.3 UC-35.02 – View ID Verification Request Detail

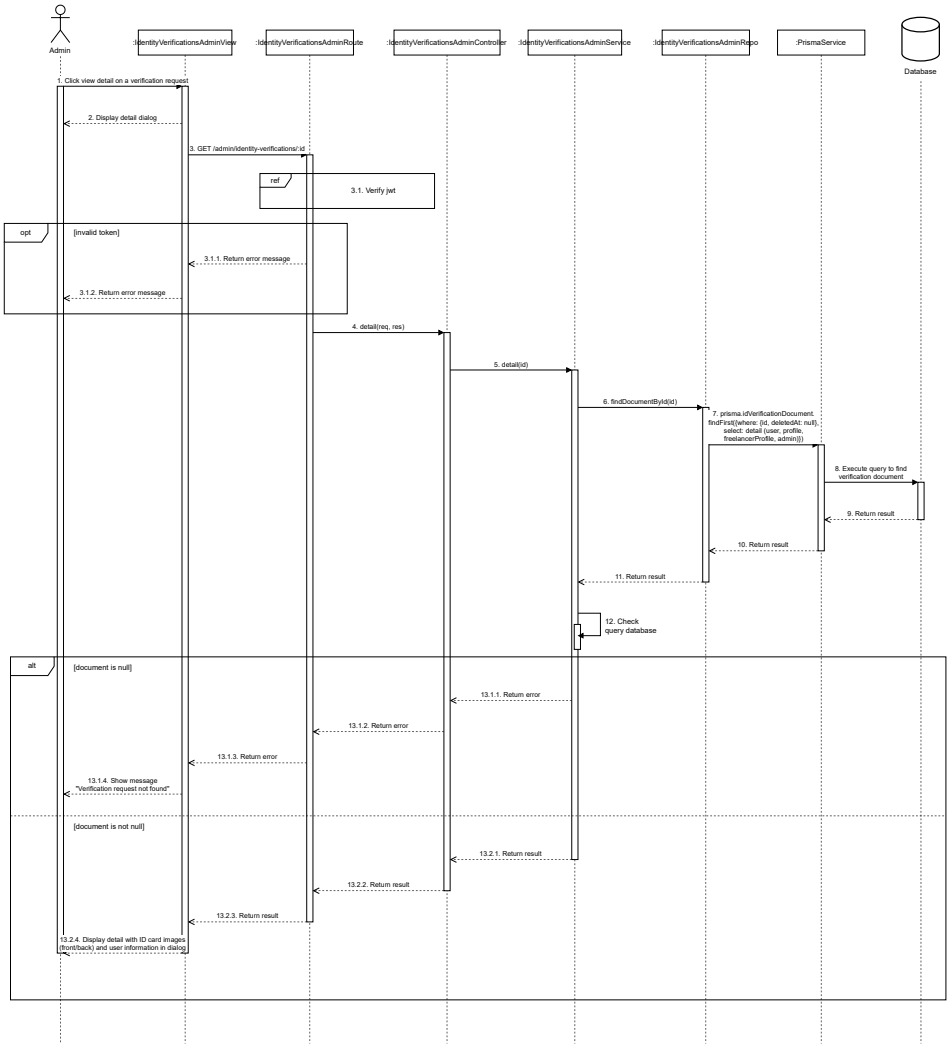

Figure 121: Sequence diagram of UC-35.02 View ID Verification Request Detail

3.22.4 UC-35.03 – Approve ID Verification

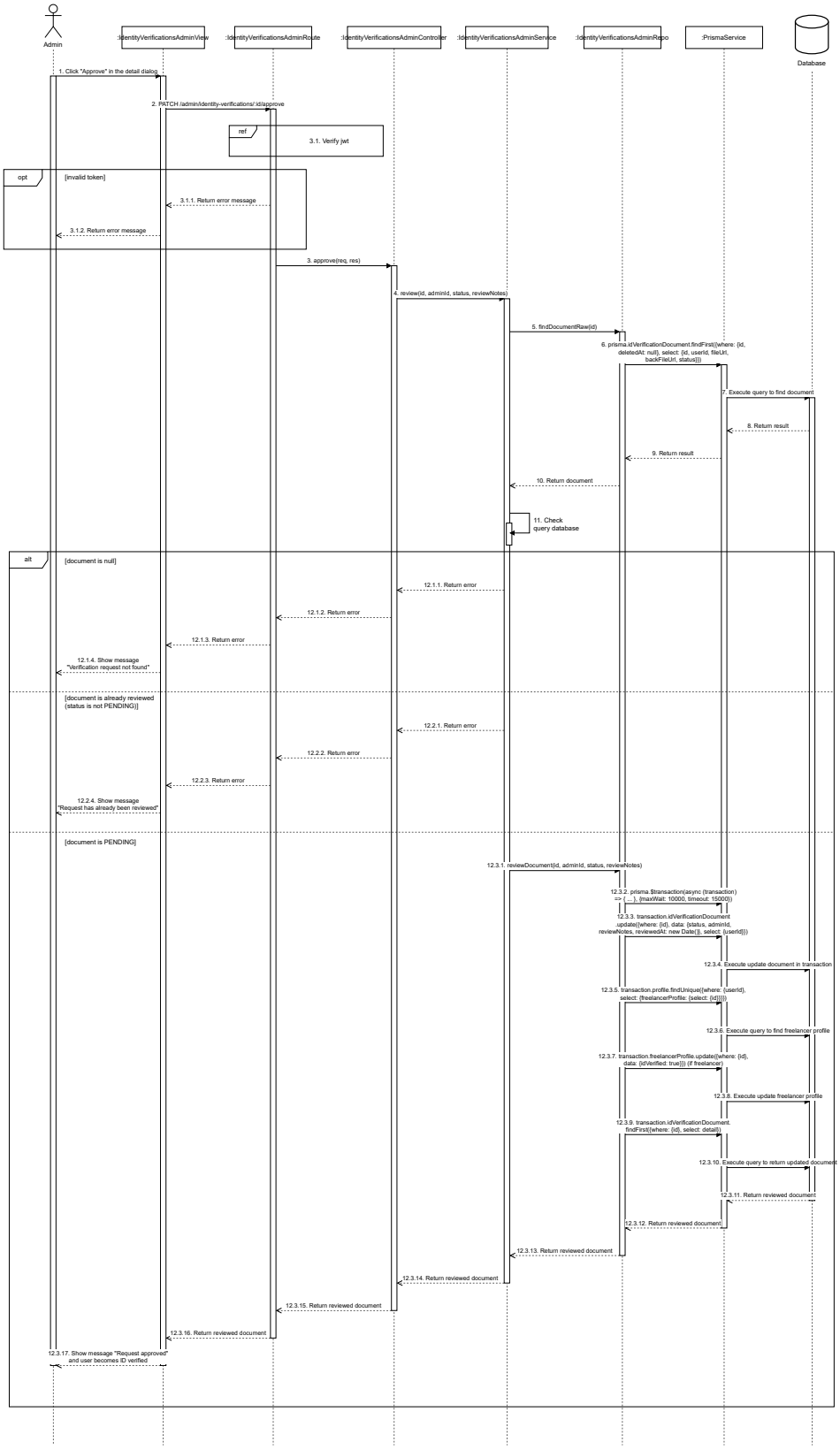

Figure 122: Sequence diagram of UC-35.03 Approve ID Verification

3.22.5 UC-35.04 – Reject ID Verification

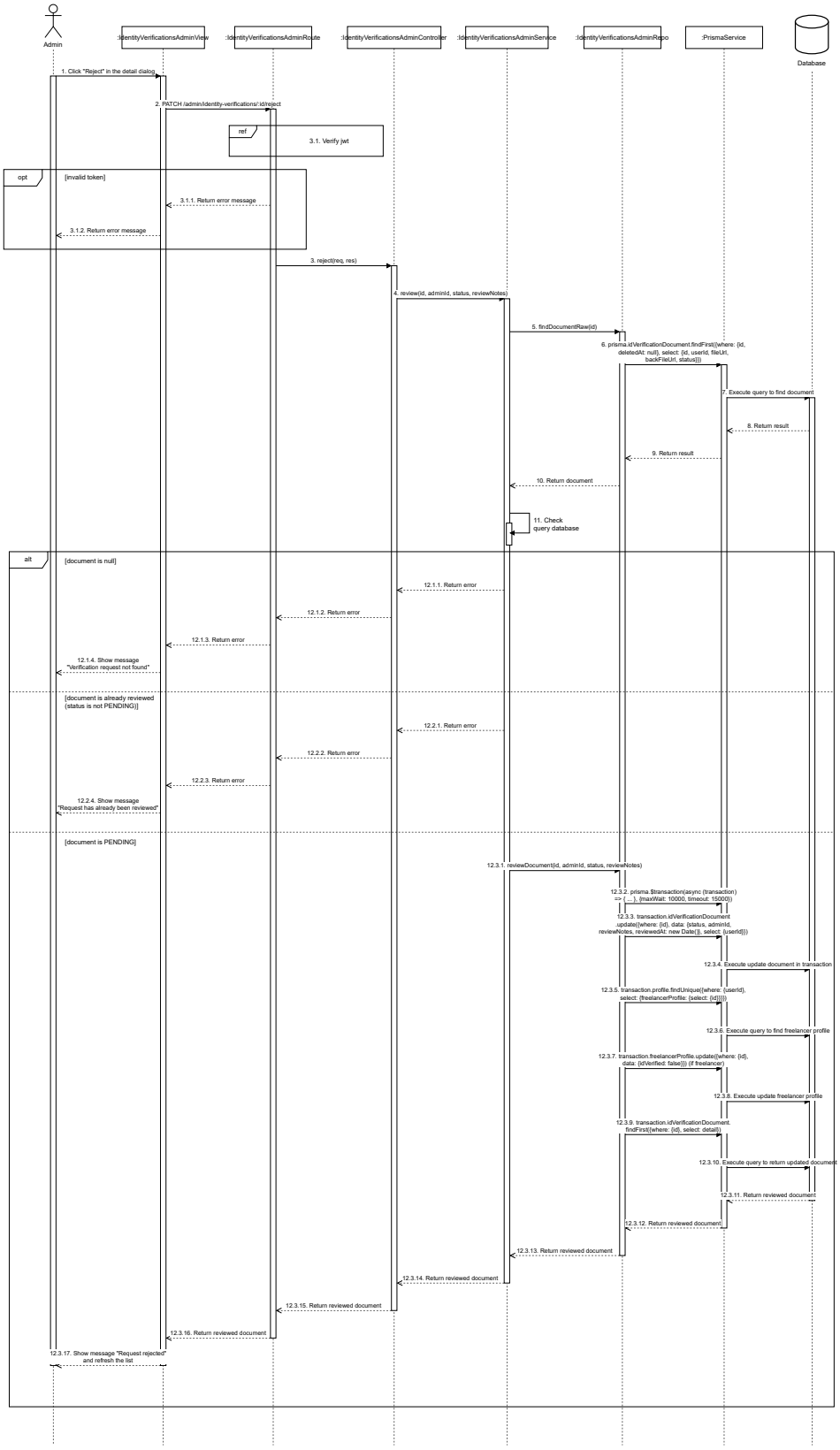

Figure 123: Sequence diagram of UC-35.04 Reject ID Verification

3.23 Manage Forum Trash

Manage Forum Trash covers the UC-36 recovery workflow: listing the soft-deleted posts and comments kept for review, opening one deleted item, and restoring it so that it becomes visible in the forum again.

3.23.1 Class Diagram

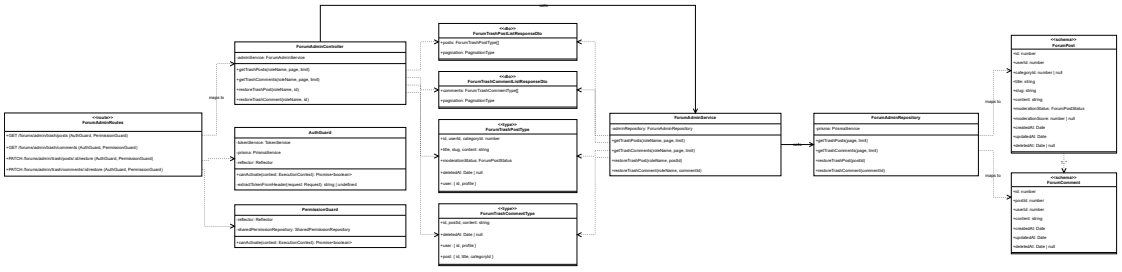

Figure 124: Class diagram of Manage Forum Trash

3.23.2 UC-36.01 – View Deleted Posts/Comments List

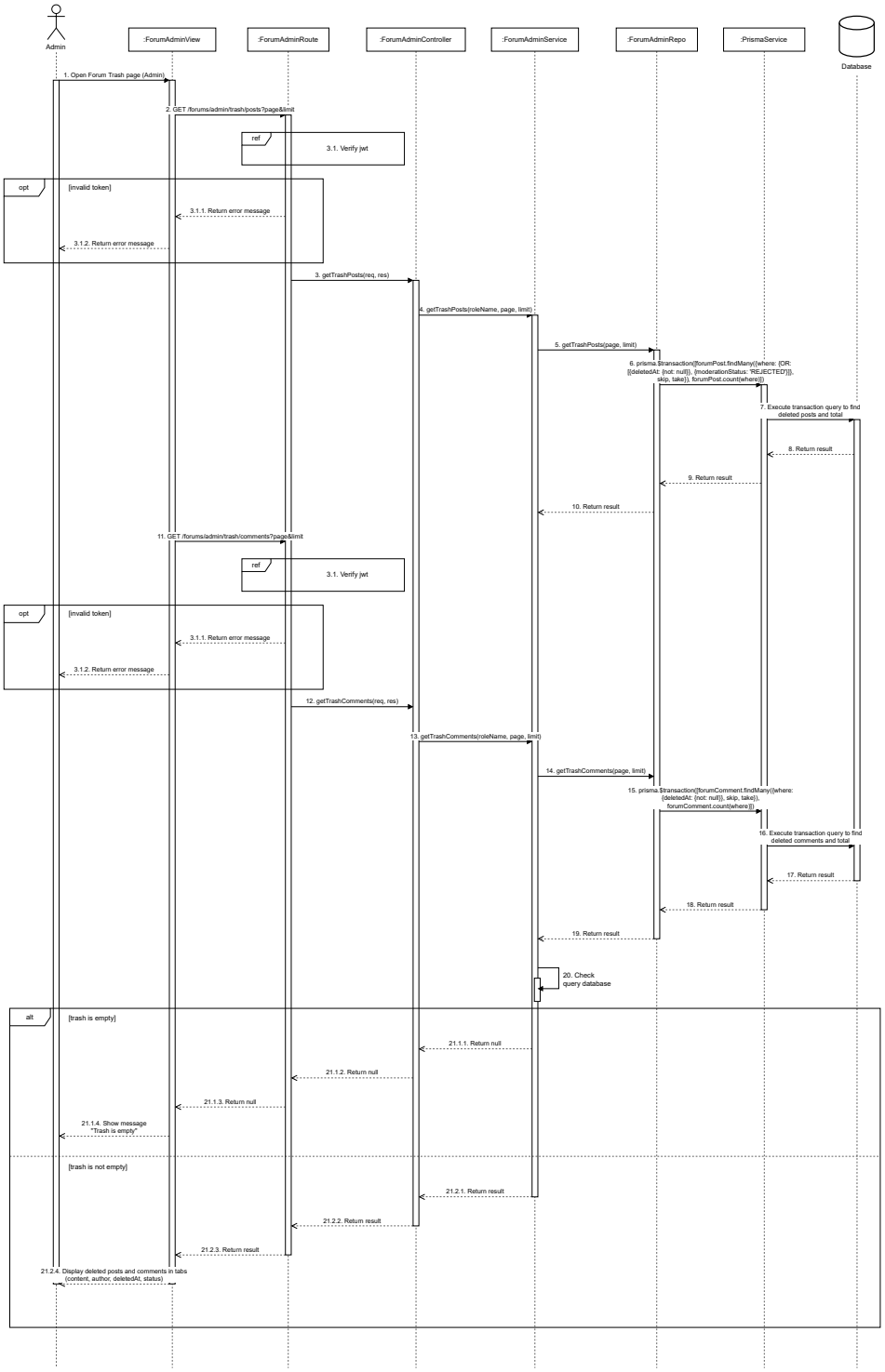

Figure 125: Sequence diagram of UC-36.01 View Deleted Posts/Comments List

3.23.3 UC-36.02 – View Deleted Posts/Comments Detail

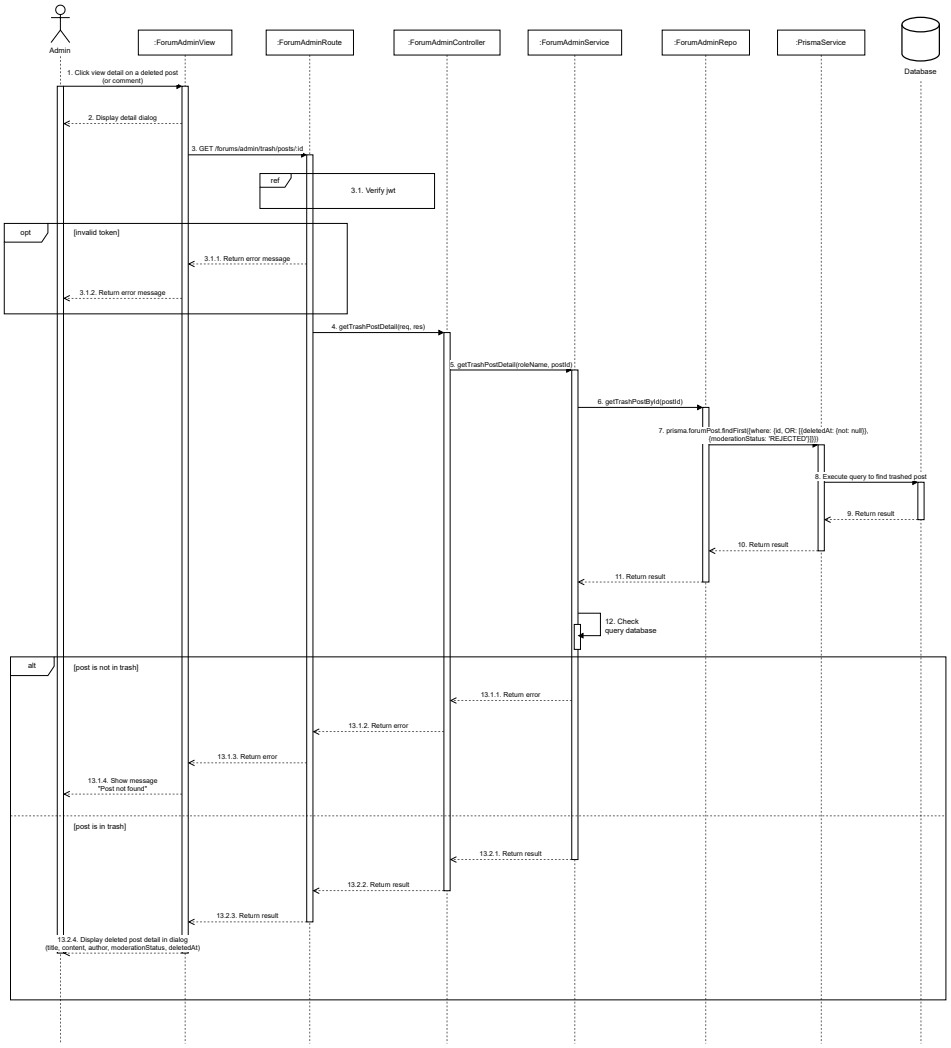

Figure 126: Sequence diagram of UC-36.02 View Deleted Posts/Comments Detail

3.23.4 UC-36.03 – Restore Deleted Posts/Comments

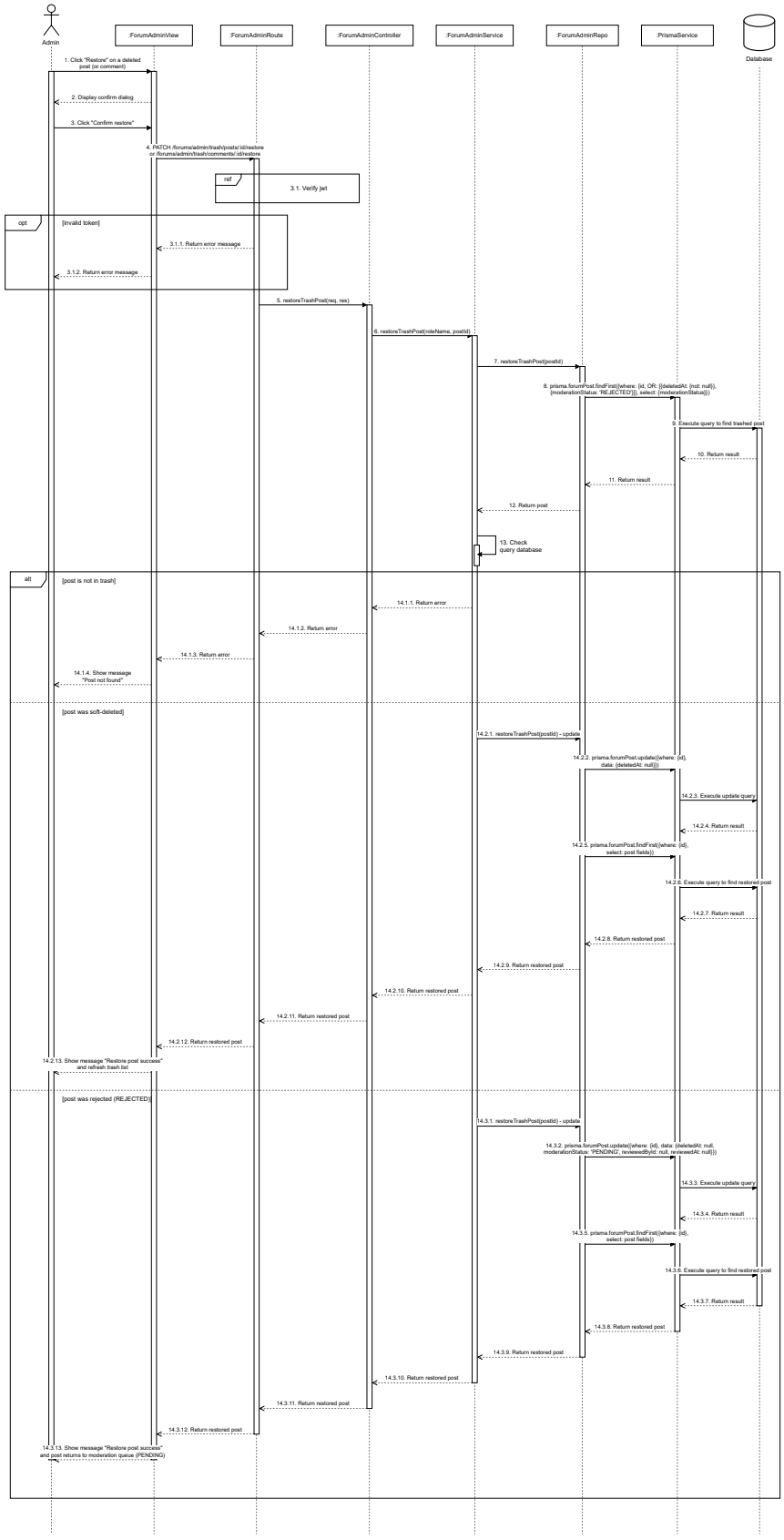

Figure 127: Sequence diagram of UC-36.03 Restore Deleted Posts/Comments

3.24 Following Management

Clients retrieve followed and discoverable freelancers, follow a freelancer, or remove an existing following relationship. New follows also create a notification.

3.24.1 Class Diagram

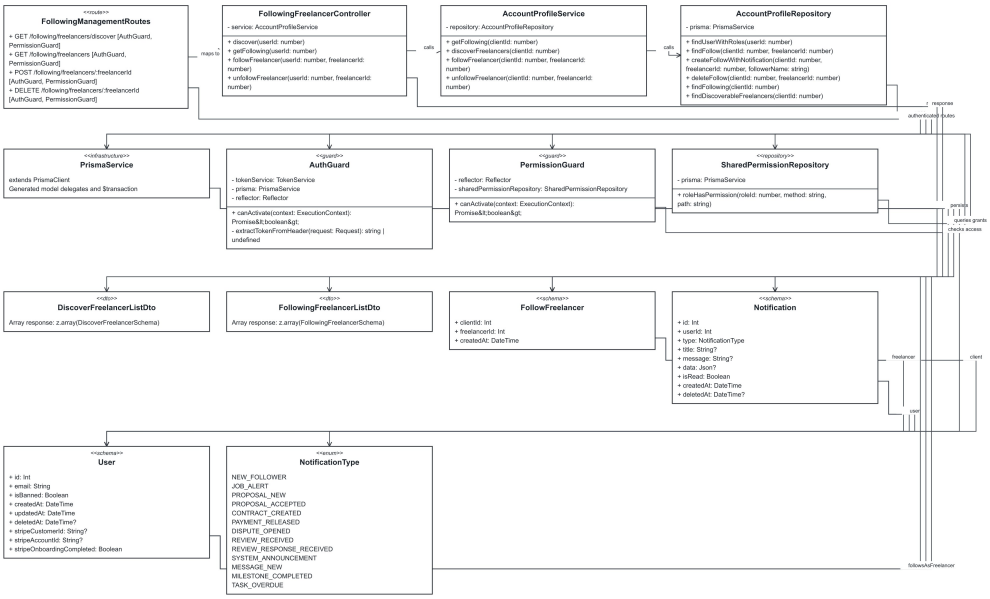

Figure 128: Class diagram of Following Management

3.24.2 UC-29.01 – View Following List

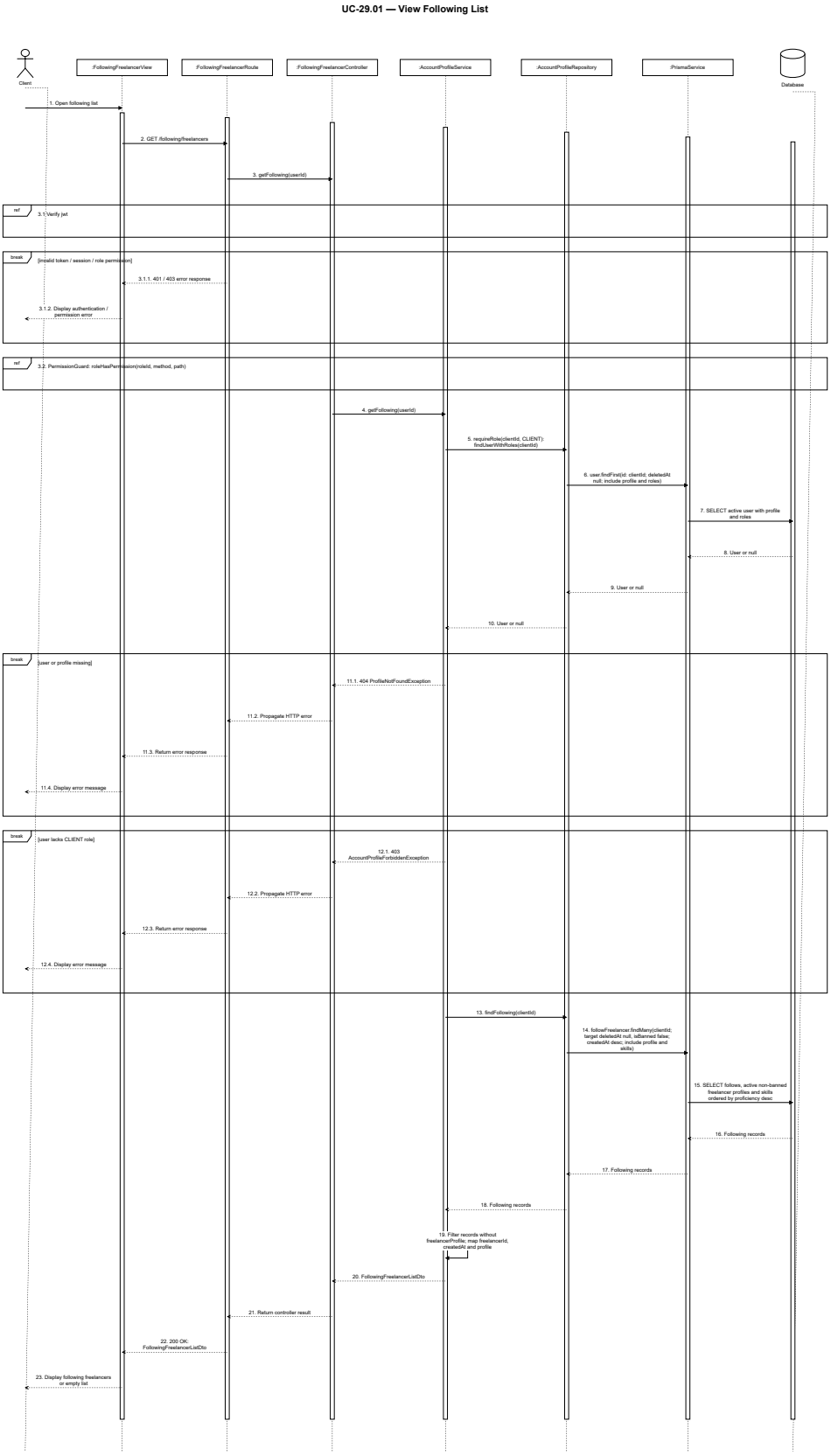

Figure 129: Sequence diagram of UC-29.01 View Following List

3.24.3 UC-29.02 – Follow Freelancer

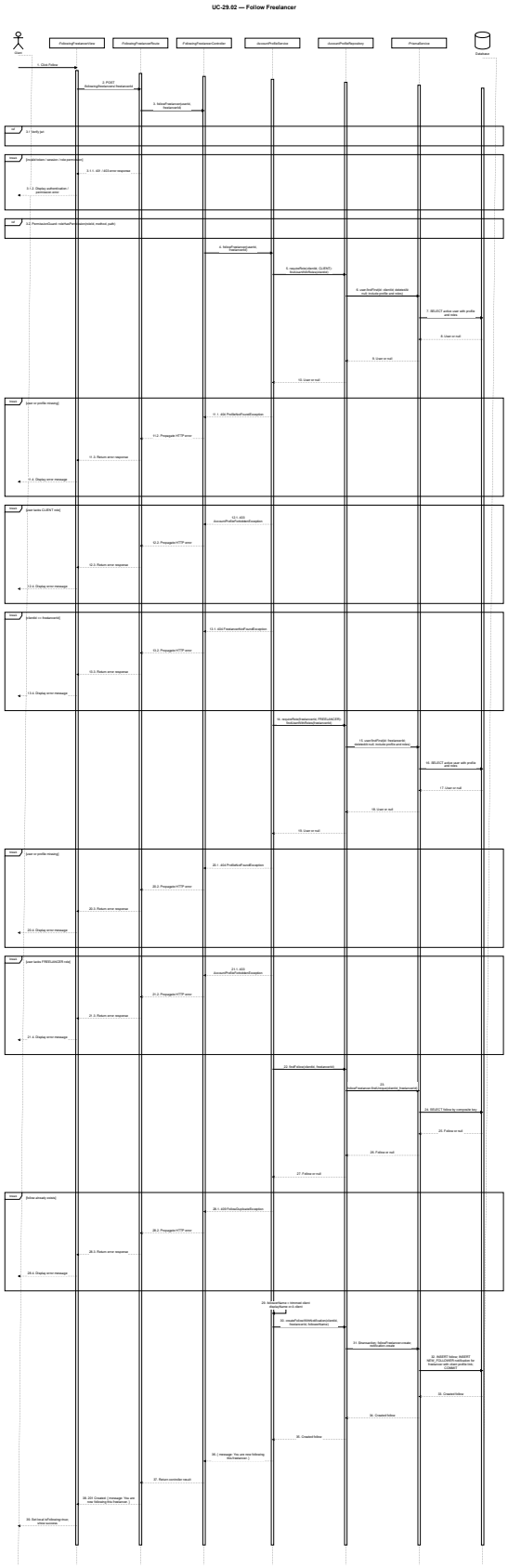

Figure 130: Sequence diagram of UC-29.02 Follow Freelancer

3.24.4 UC-29.03 – Unfollow Freelancer

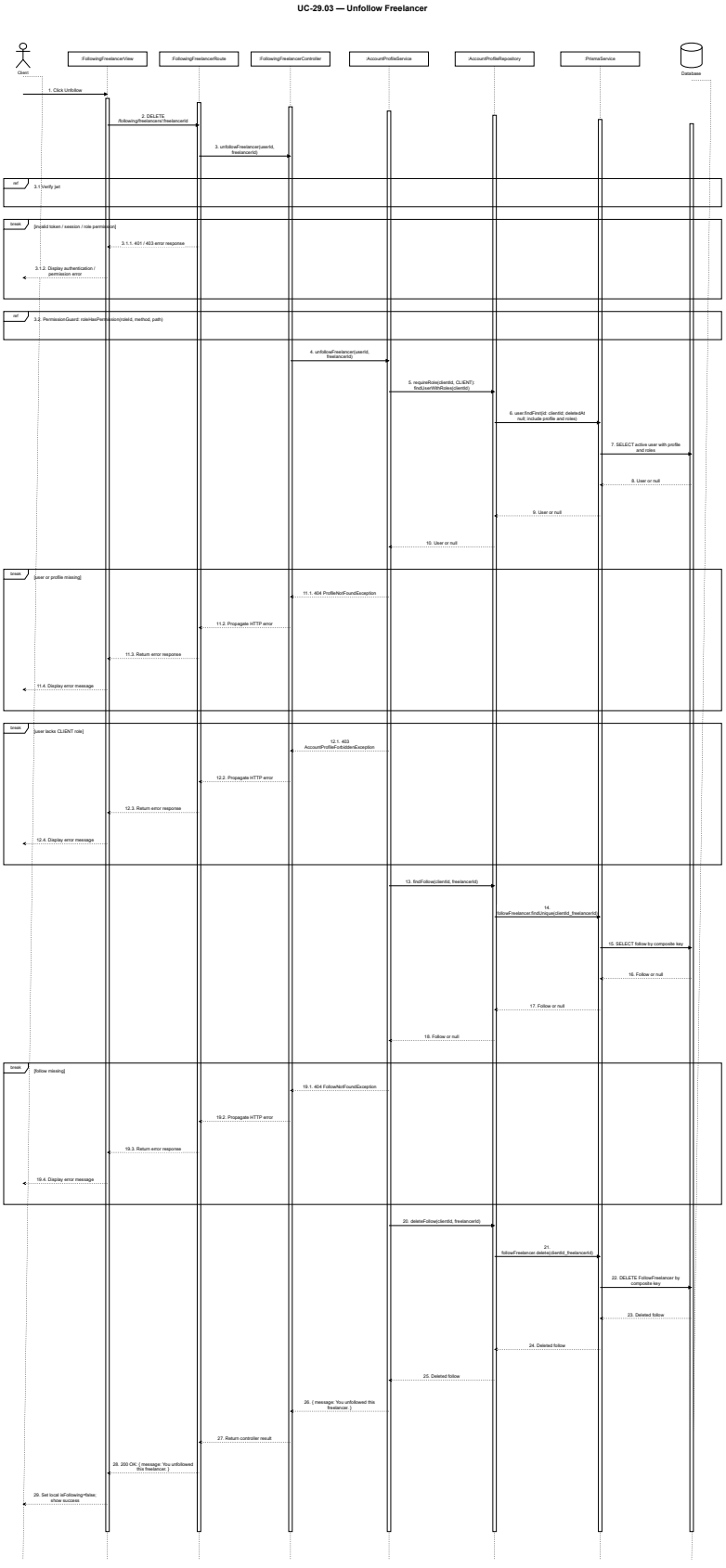

Figure 131: Sequence diagram of UC-29.03 Unfollow Freelancer

3.25 Notification Management

Users retrieve their notifications, mark one or all as read, and delete an owned notification. Deletion is persisted as a soft delete.

3.25.1 Class Diagram

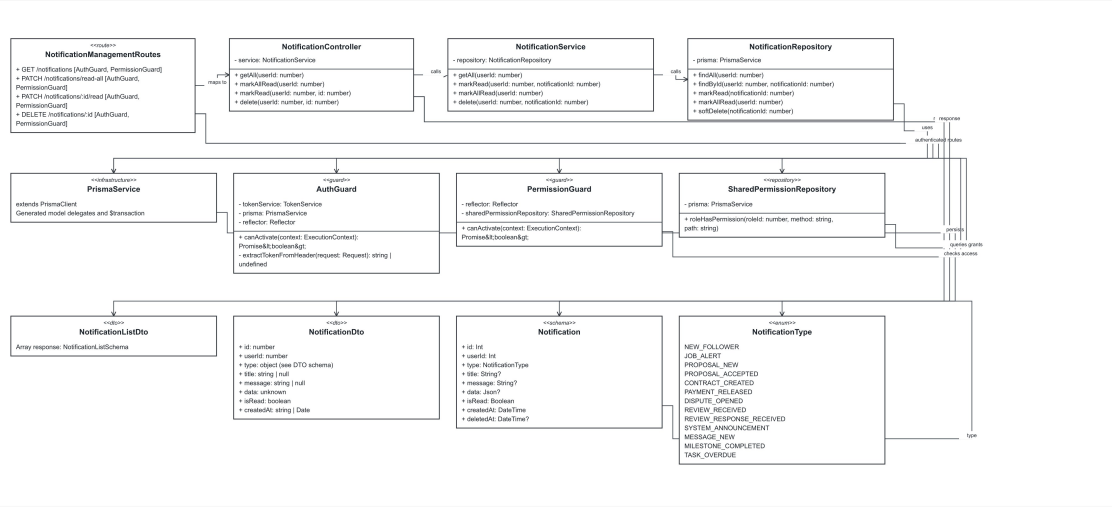

Figure 132: Class diagram of Notification Management

3.25.2 UC-30.01 – View Notification List

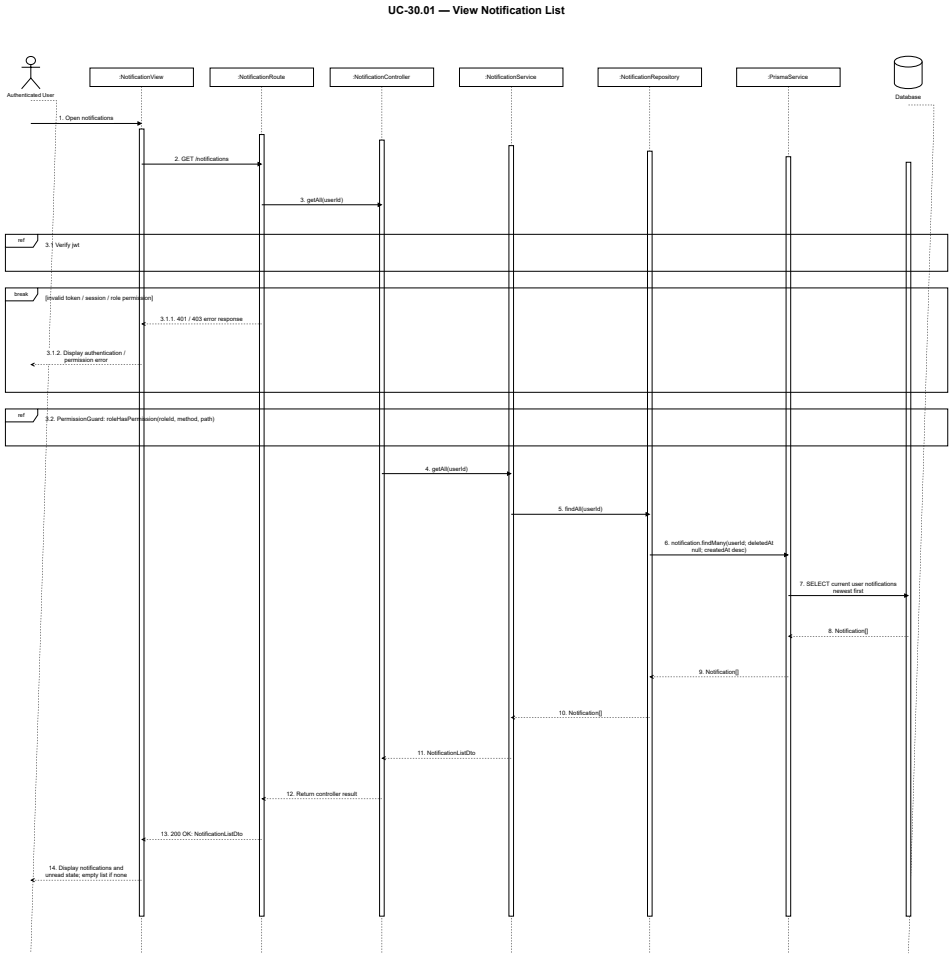

Figure 133: Sequence diagram of UC-30.01 View Notification List

3.25.3 UC-30.02 – Mark Notification as Read

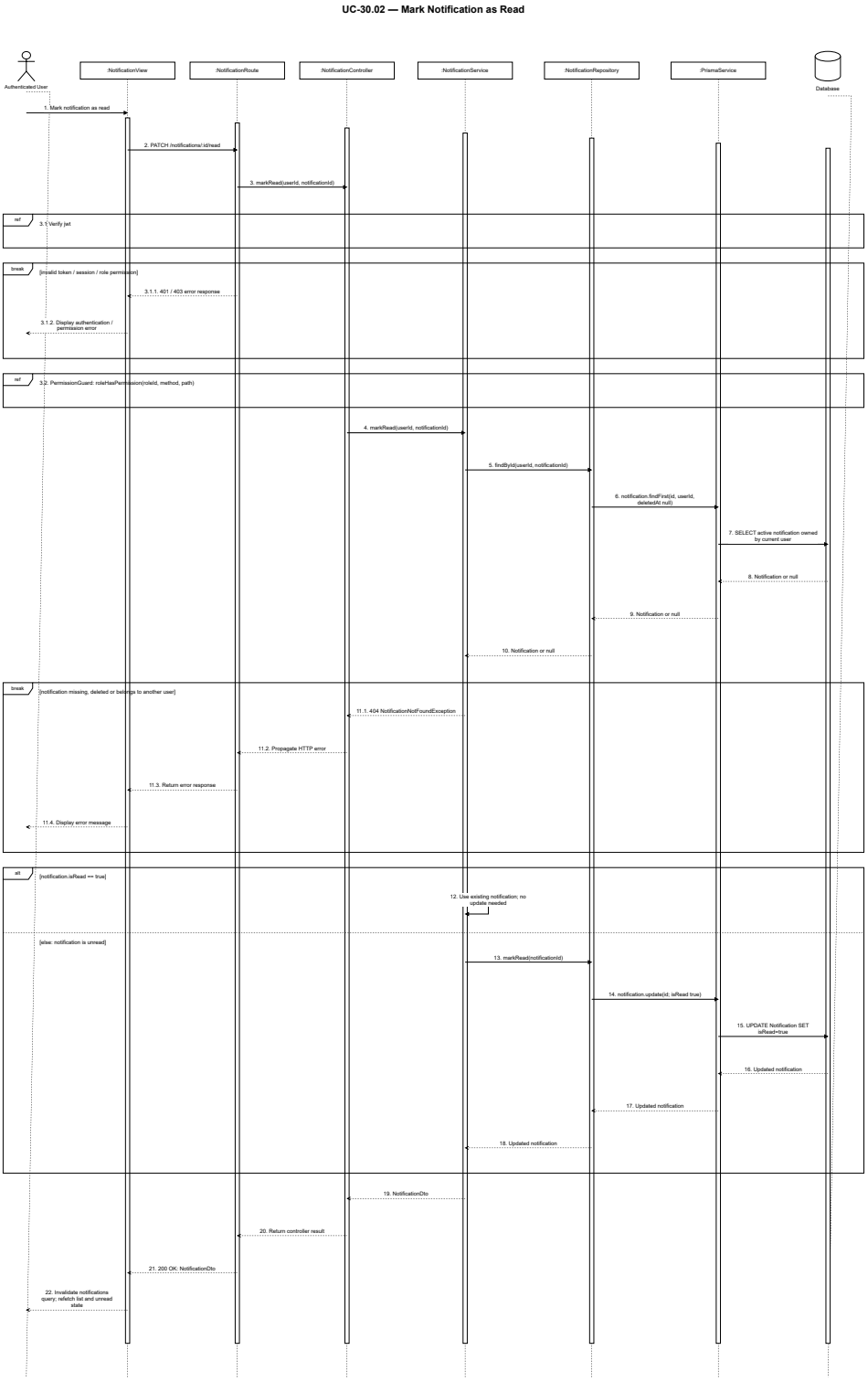

Figure 134: Sequence diagram of UC-30.02 Mark Notification as Read

3.25.4 UC-30.03 – Mark All Notifications as Read

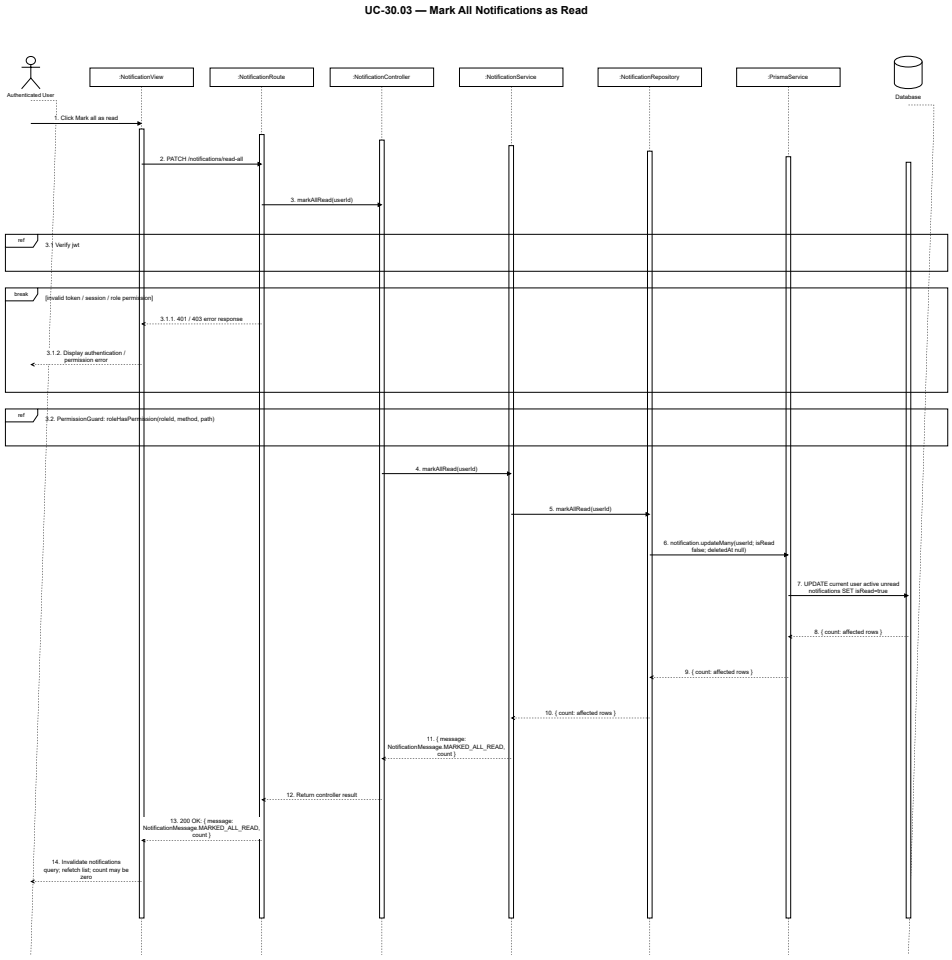

Figure 135: Sequence diagram of UC-30.03 Mark All Notifications as Read

3.25.5 UC-30.04 – Delete Notification

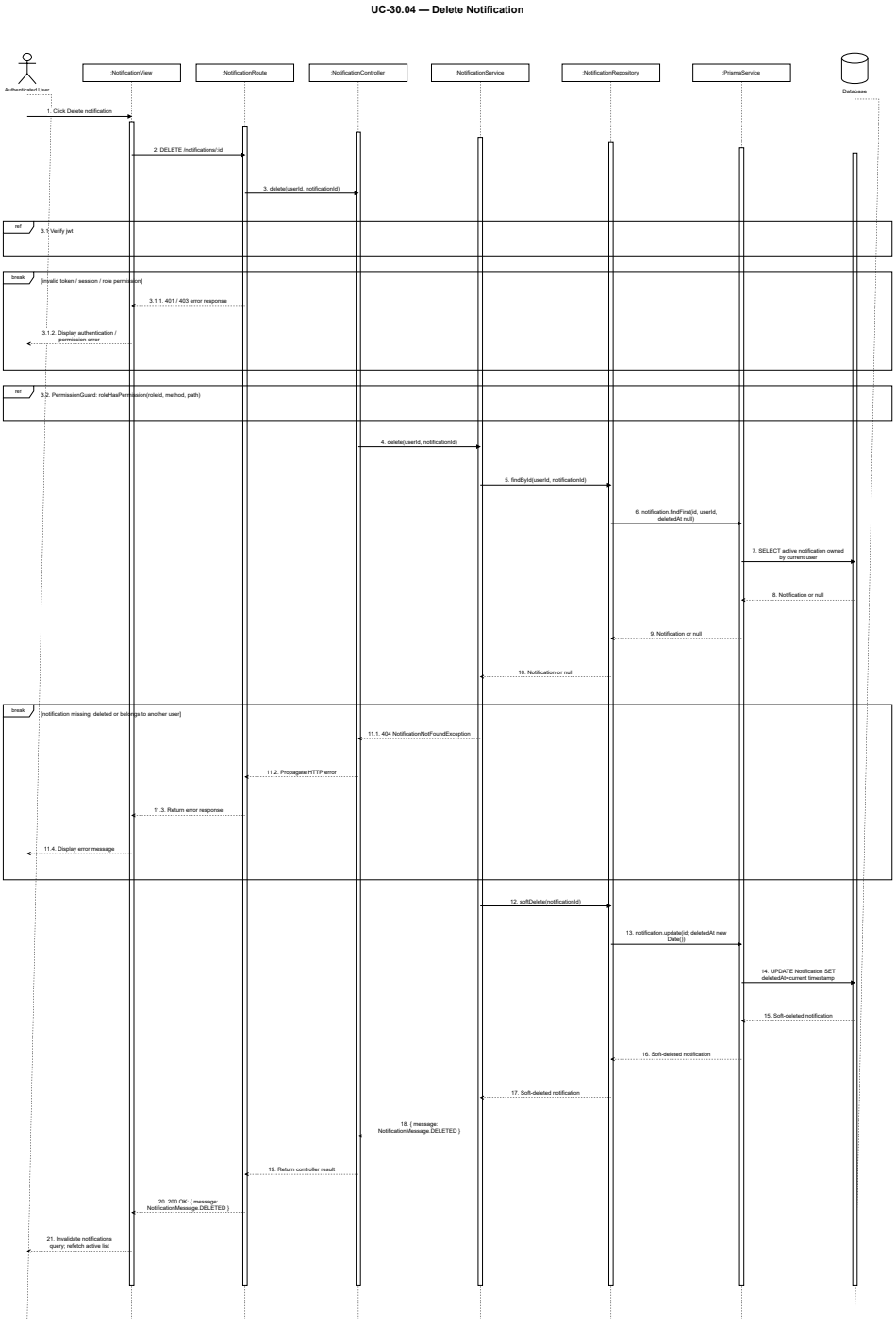

Figure 136: Sequence diagram of UC-30.04 Delete Notification

3.26 Role Switching

Authenticated users switch to an assigned role or join a new supported role. The role assignment is updated and a new access token is issued for the existing session.

3.26.1 Class Diagram

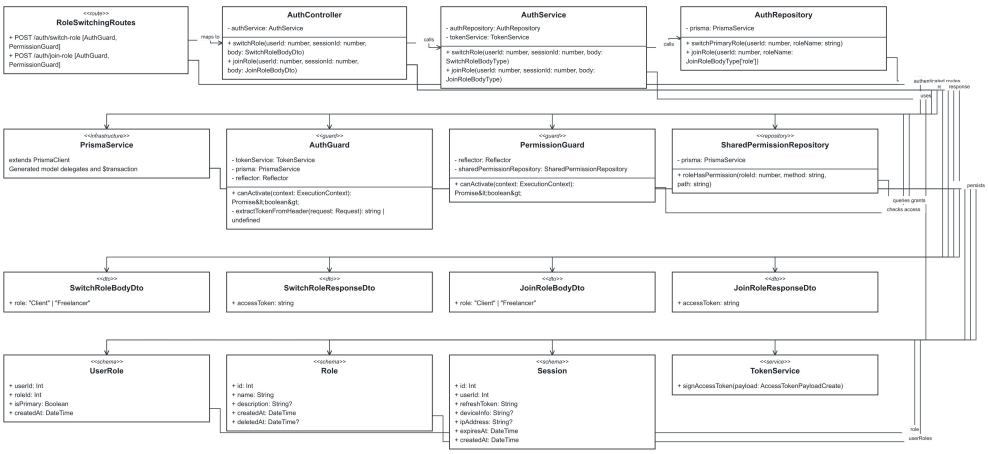

Figure 137: Class diagram of Role Switching

3.26.2 UC-31.01 – Switch Role

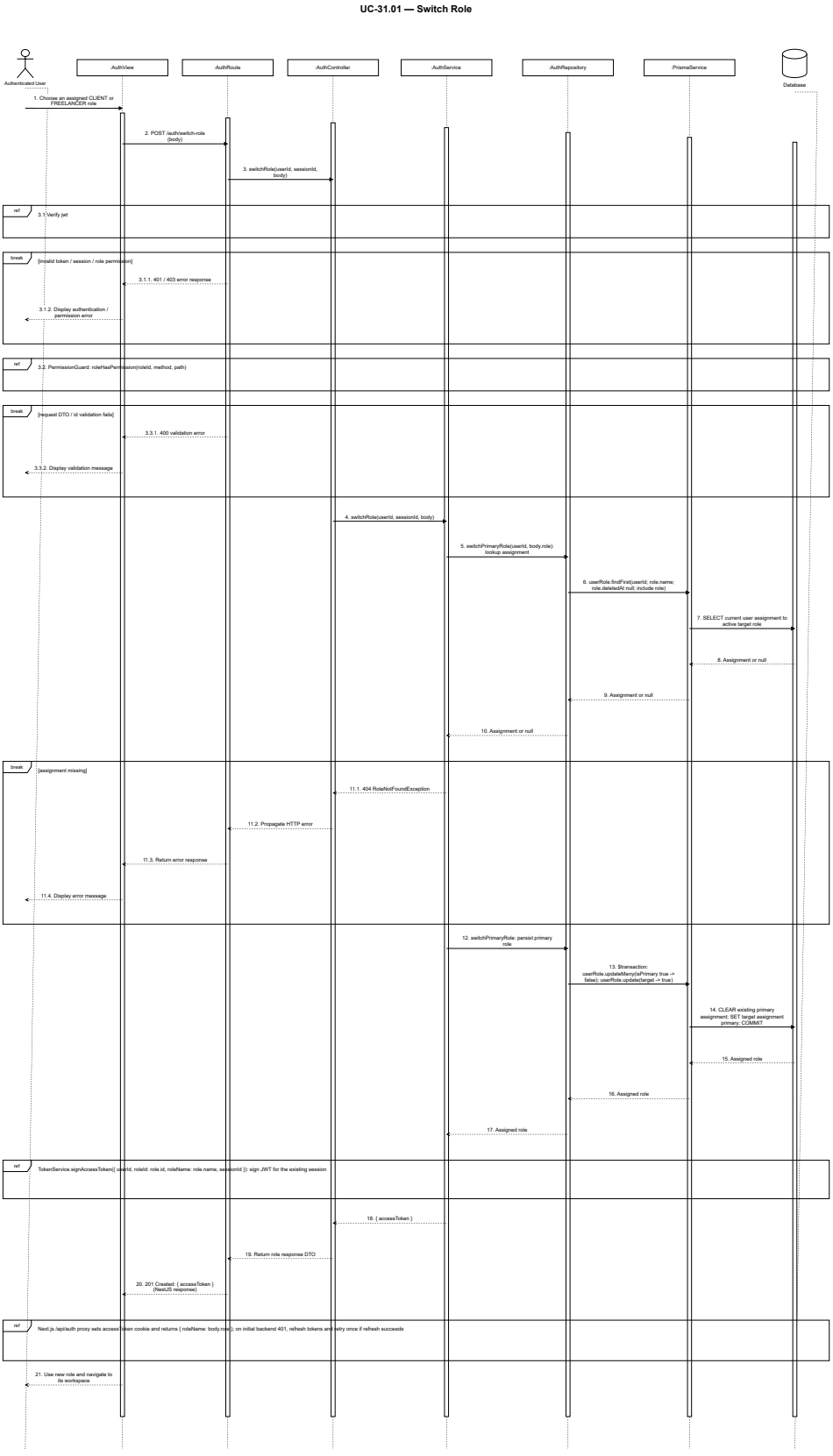

Figure 138: Sequence diagram of UC-31.01 Switch Role

3.26.3 UC-31.02 – Join New Role

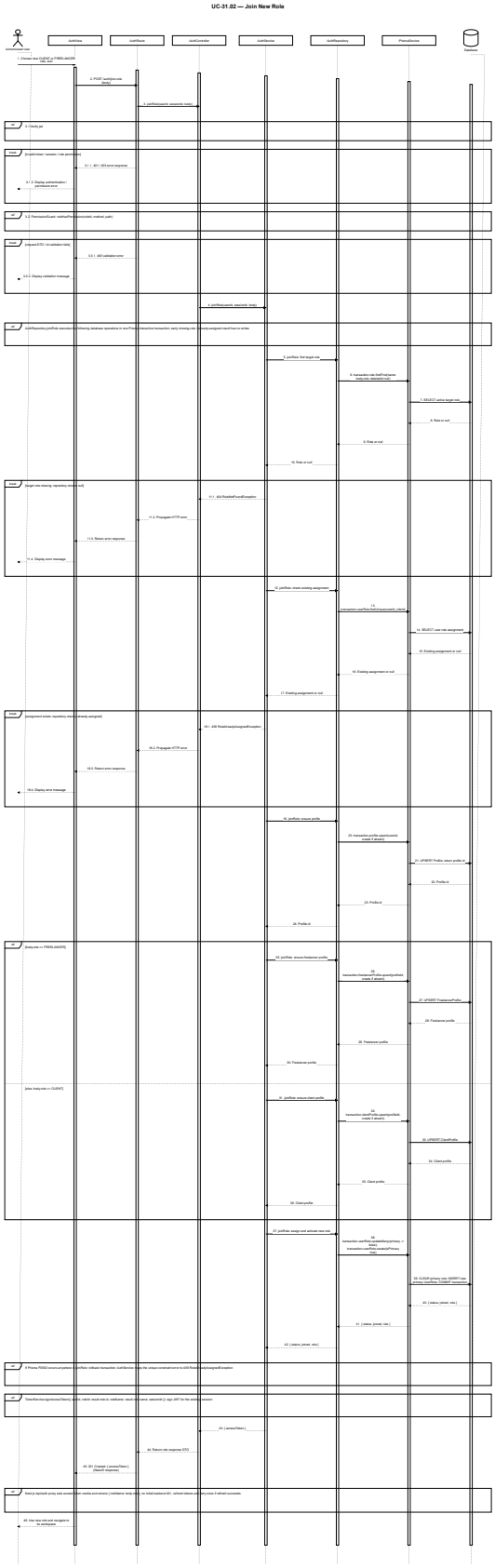

Figure 139: Sequence diagram of UC-31.02 Join New Role

3.27 Proposal Management

Proposal Management covers the full proposal workflow between freelancers and clients. Freelancers can submit proposals with cover letters, bid amounts, and delivery estimates, view their submitted proposals, modify pending proposals, or withdraw them. Clients can view all proposals submitted for their jobs, inspect detailed freelancer bids, and accept or reject proposals to initiate contracting.

3.27.1 Class Diagram

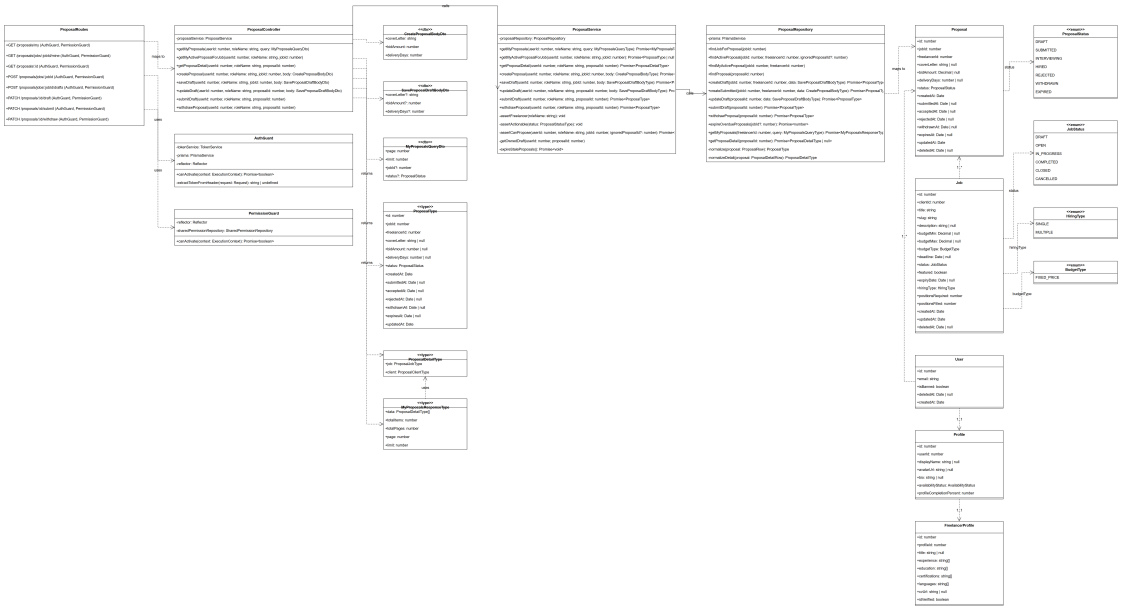

Figure 140: Class diagram of Freelancer Proposal Management

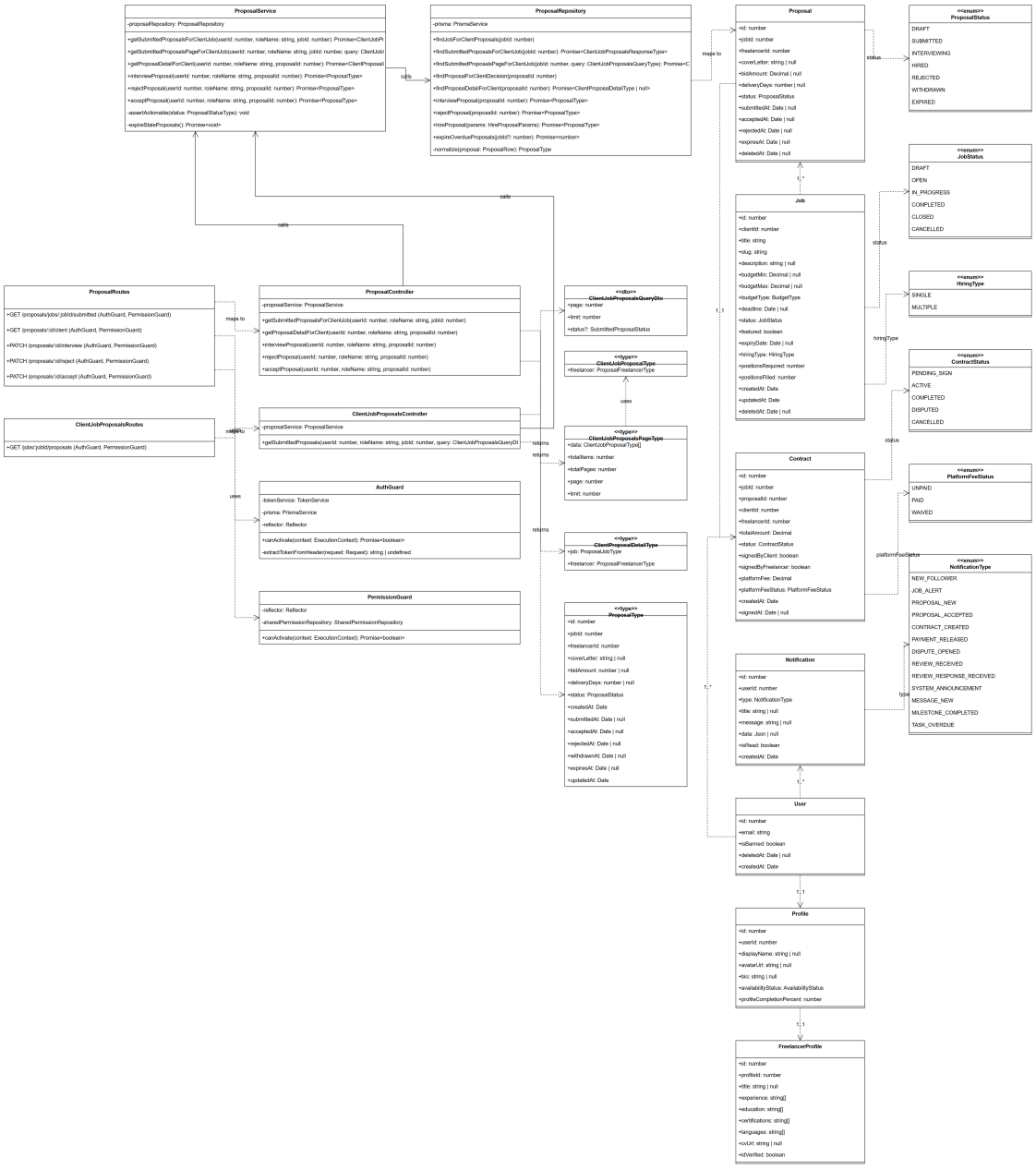

Figure 141: Class diagram of Client Proposal Management

3.27.2 UC-32.01 – View My Proposal List

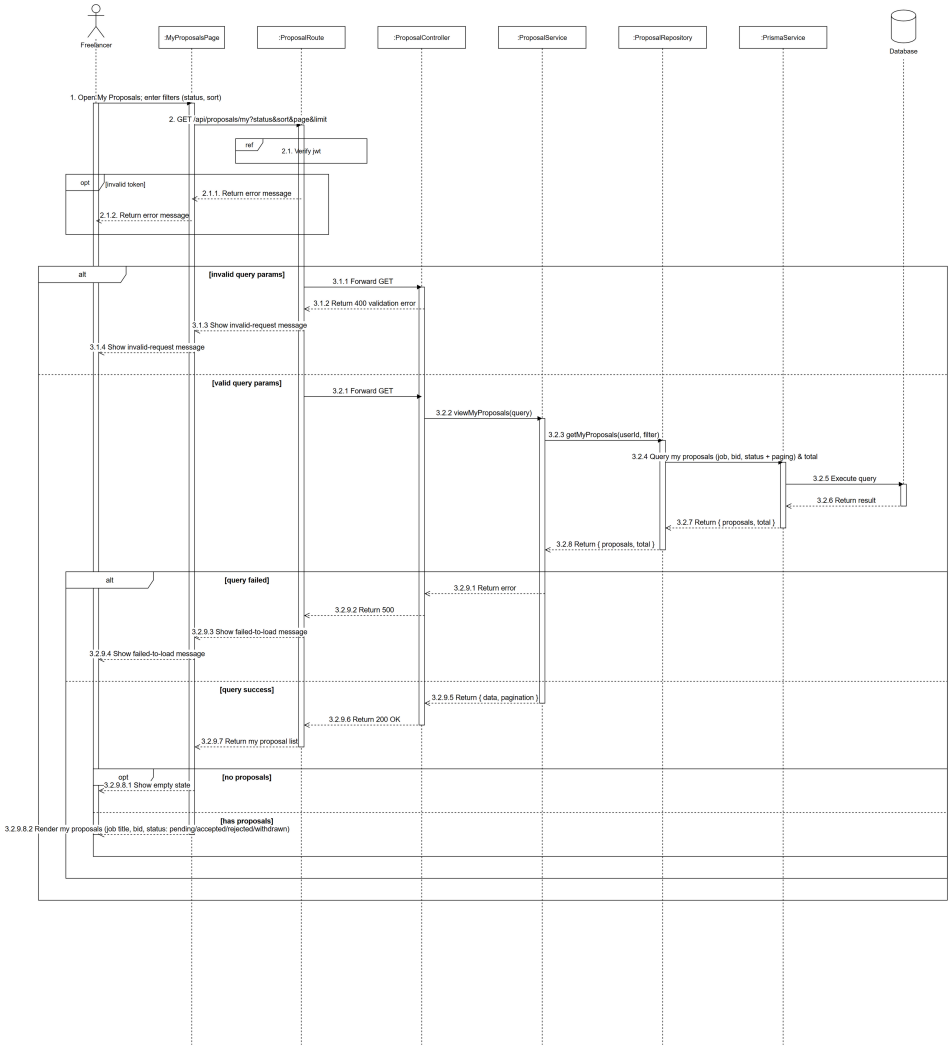

Figure 142: Sequence diagram of UC-32.01 View My Proposal List

3.27.3 UC-32.02 – View My Proposal Detail

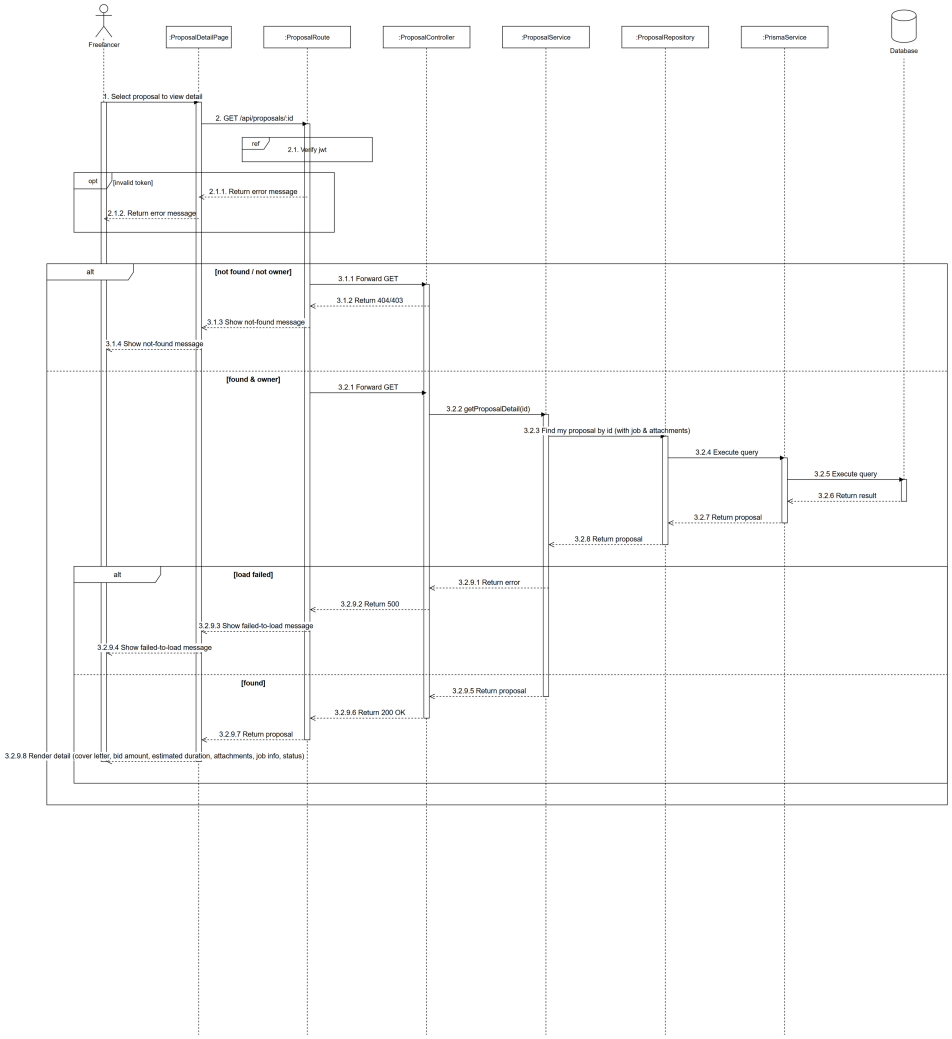

Figure 143: Sequence diagram of UC-32.02 View My Proposal Detail

3.27.4 UC-32.03 – Submit Proposal

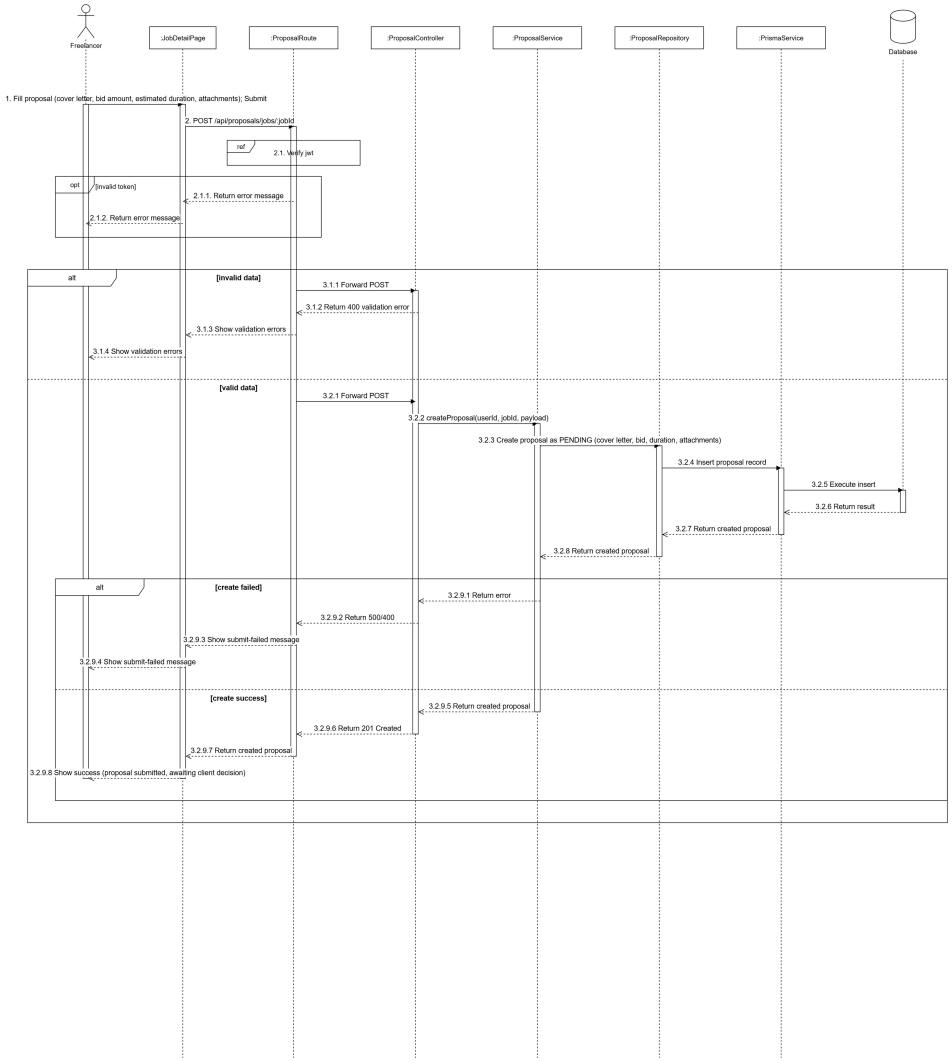

Figure 144: Sequence diagram of UC-32.03 Submit Proposal

3.27.5 UC-32.04 – Update Proposal

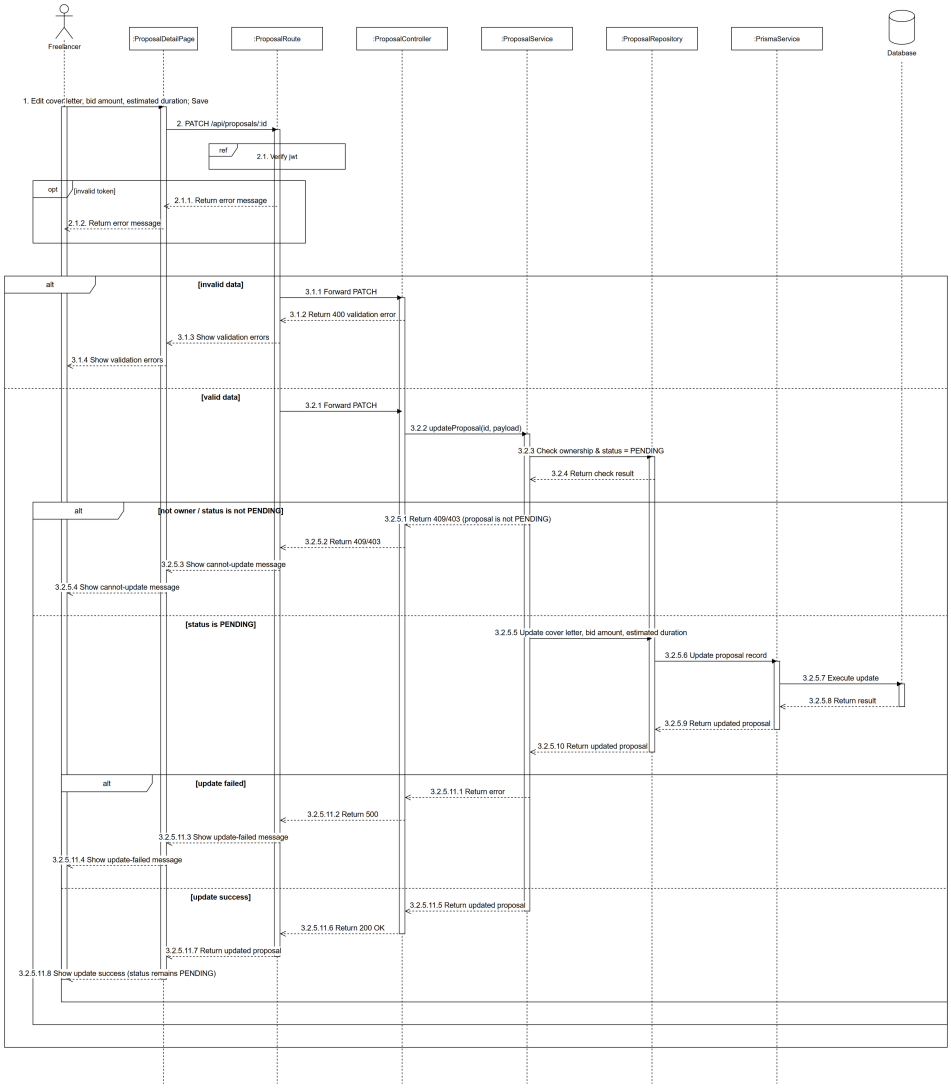

Figure 145: Sequence diagram of UC-32.04 Update Proposal

3.27.6 UC-32.05 – Withdraw Proposal

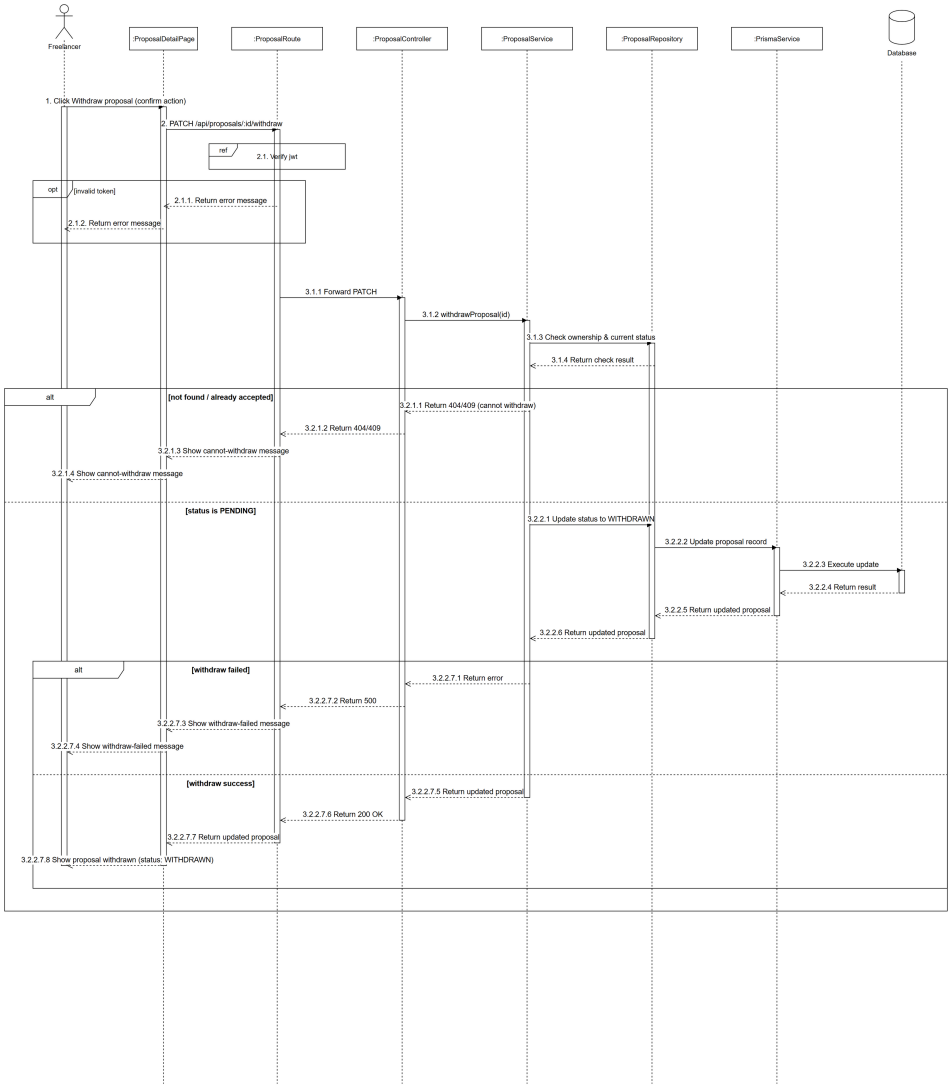

Figure 146: Sequence diagram of UC-32.05 Withdraw Proposal

3.27.7 UC-32.06 – Accept Proposal

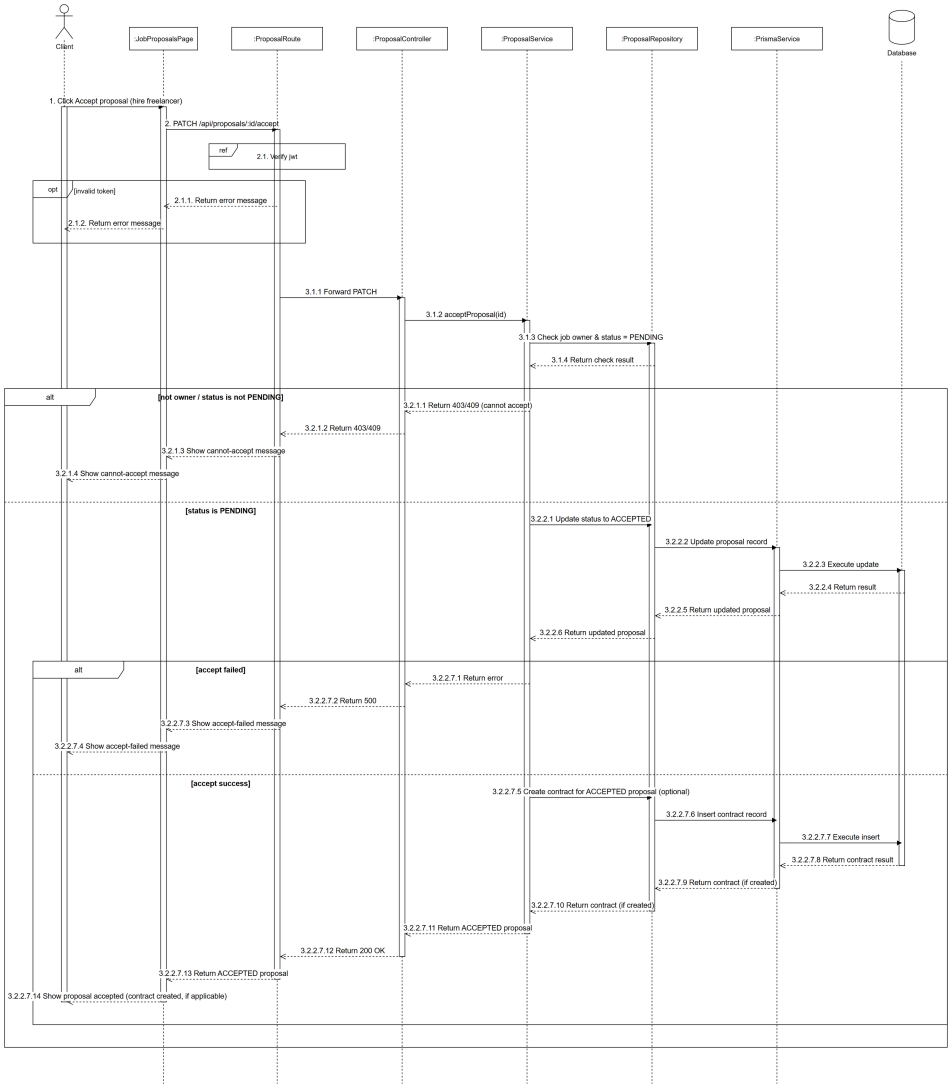

Figure 147: Sequence diagram of UC-32.06 Accept Proposal

3.27.8 UC-32.07 – Reject Proposal

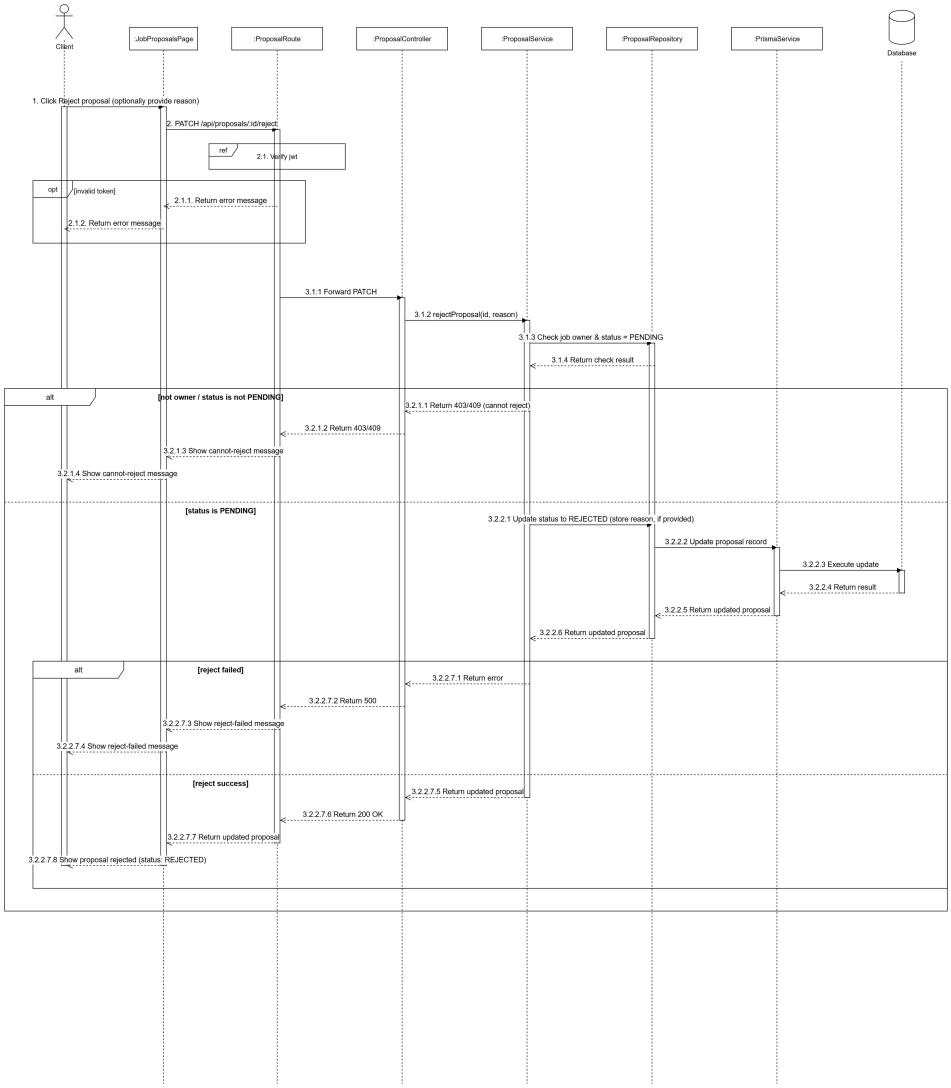

Figure 148: Sequence diagram of UC-32.07 Reject Proposal

3.27.9 UC-32.08 – View Job Proposal List

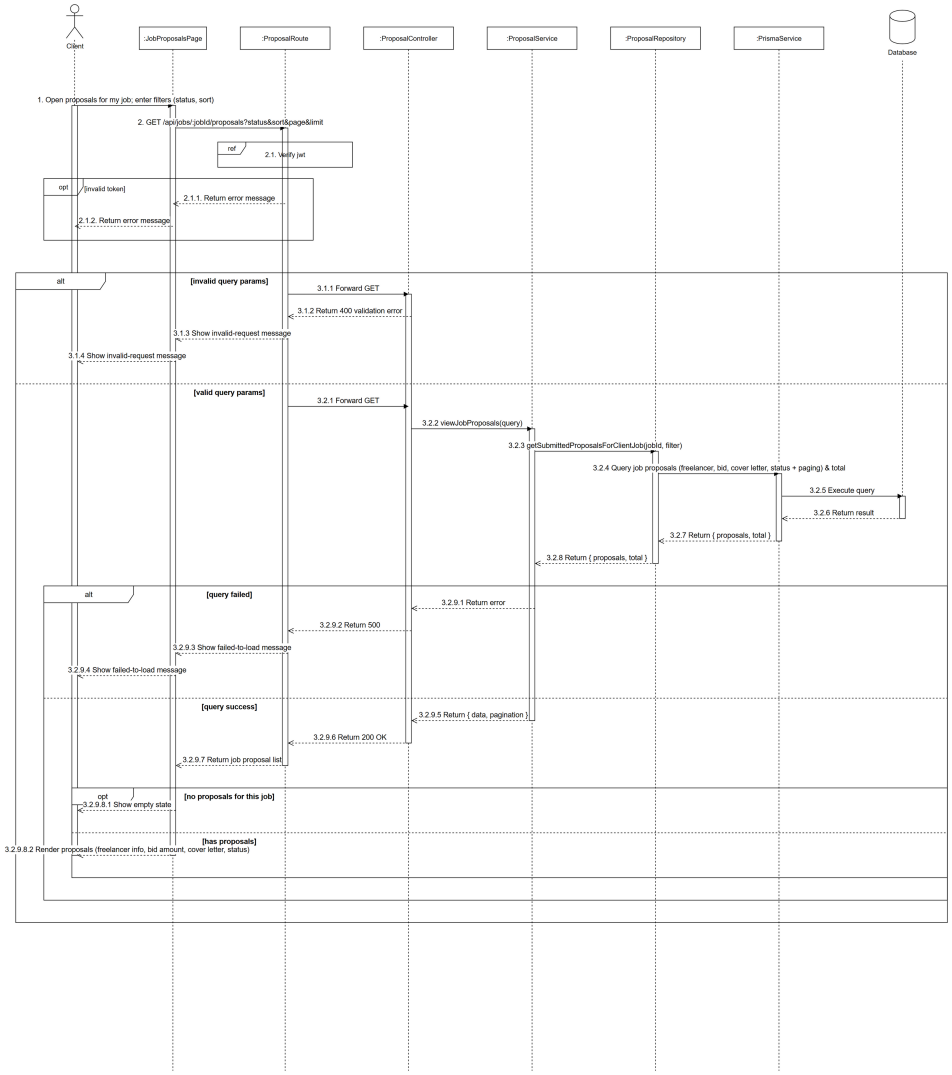

Figure 149: Sequence diagram of UC-32.08 View Job Proposal List

3.27.10 UC-32.09 – View Proposal Detail

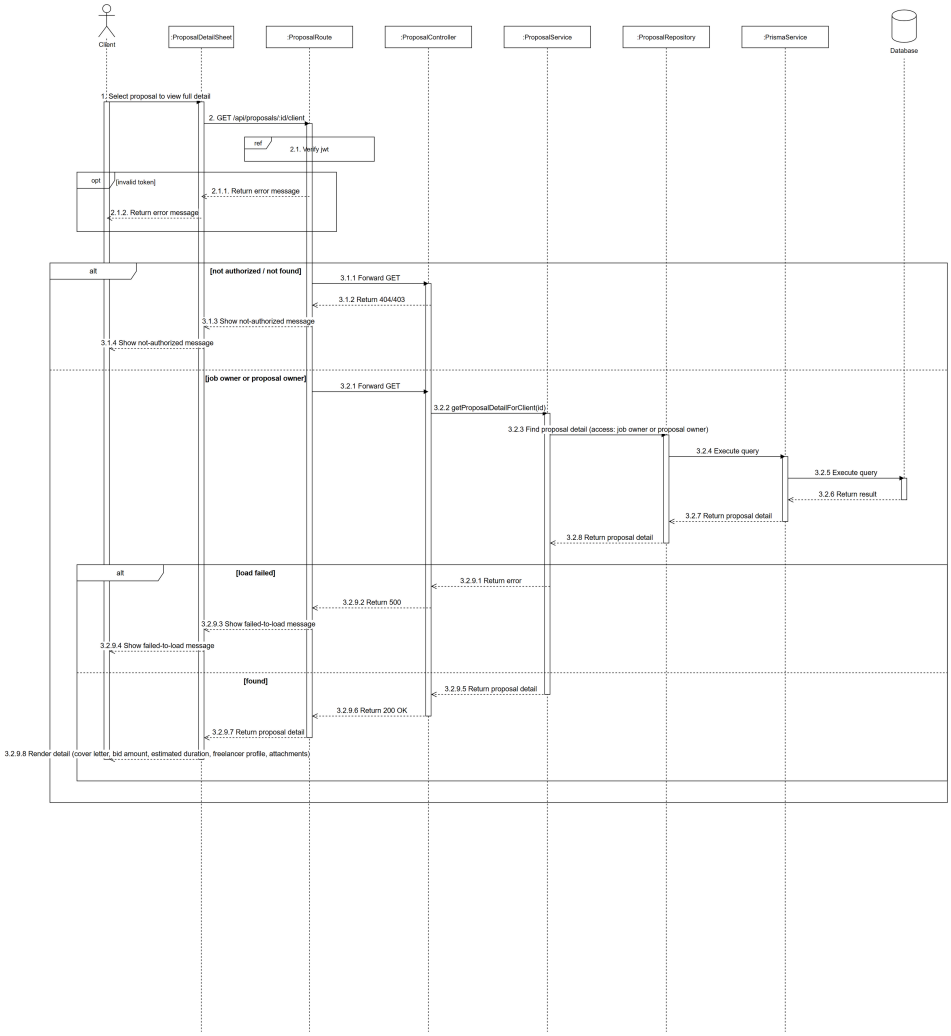

Figure 150: Sequence diagram of UC-32.09 View Proposal Detail

3.28 User Management

User Management provides the administrator with a searchable user catalogue, a consolidated account detail view, and dedicated editors for the Client and Freelancer profiles attached to an account.

3.28.1 Class Diagram

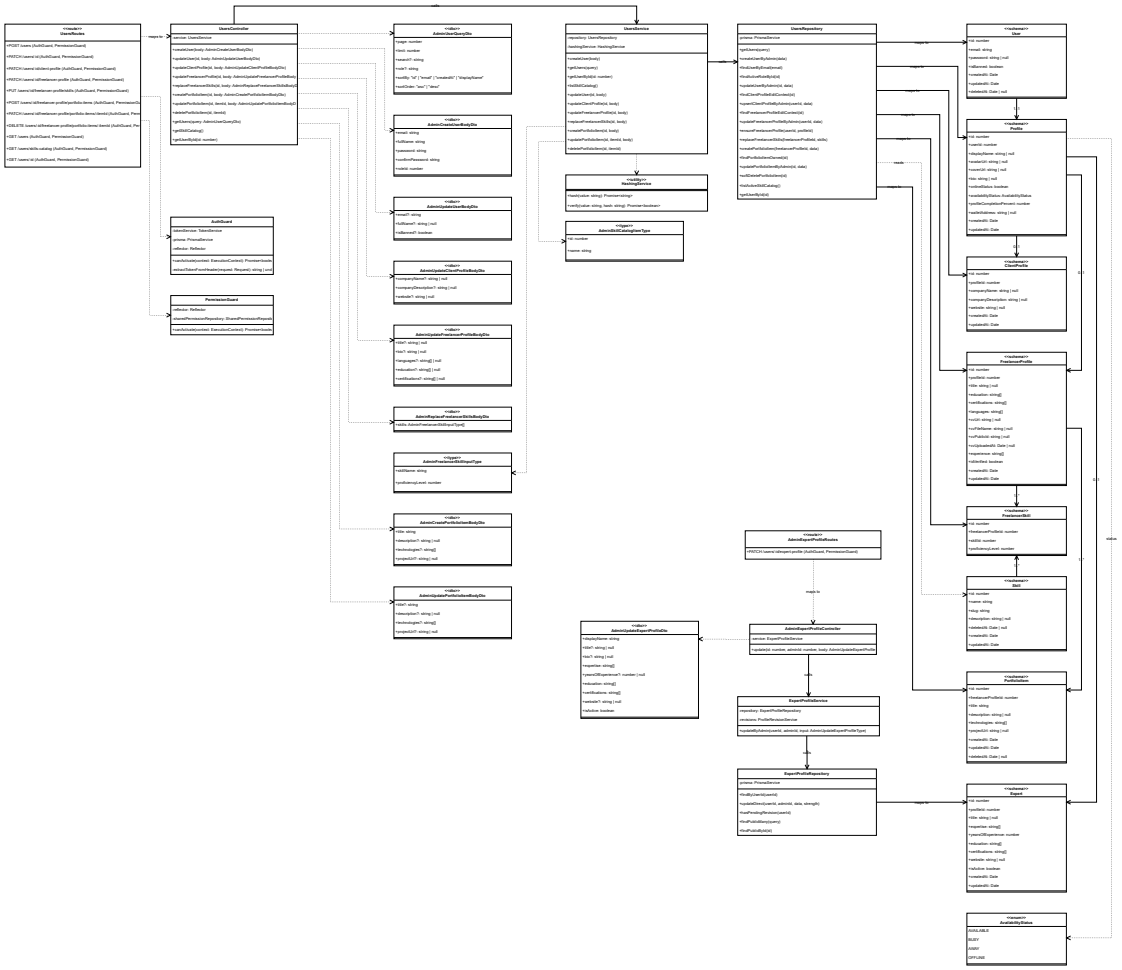

Figure 151: Class diagram of User Management

3.28.2 UC-28.01 – View User List

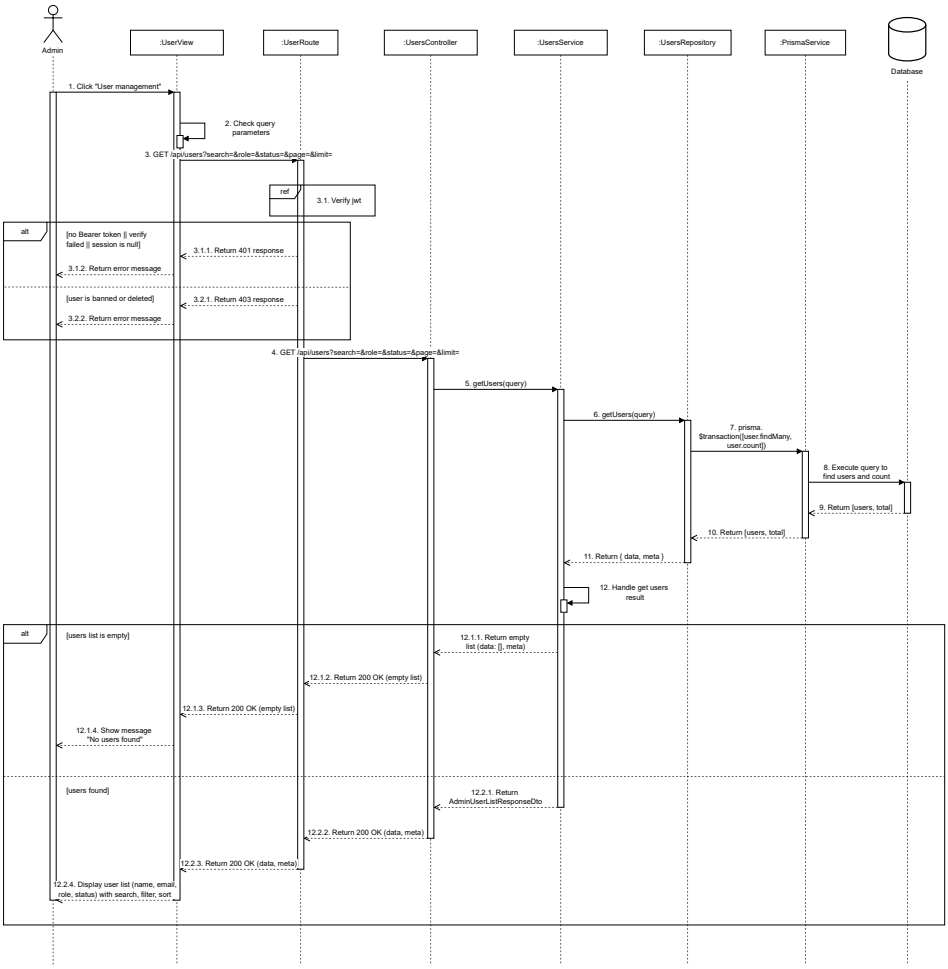

Figure 152: Sequence diagram of UC-28.01 View User List

3.28.3 UC-28.02 – View User Detail

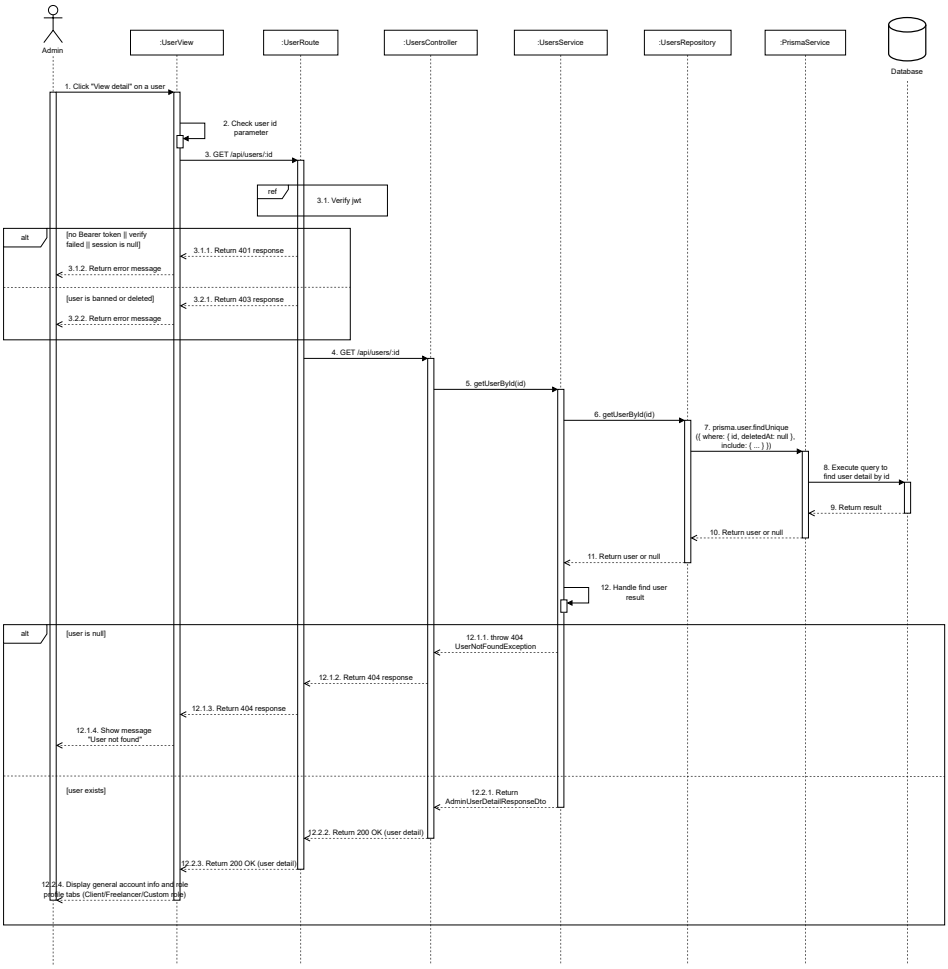

Figure 153: Sequence diagram of UC-28.02 View User Detail

3.28.4 UC-28.03 – Create User

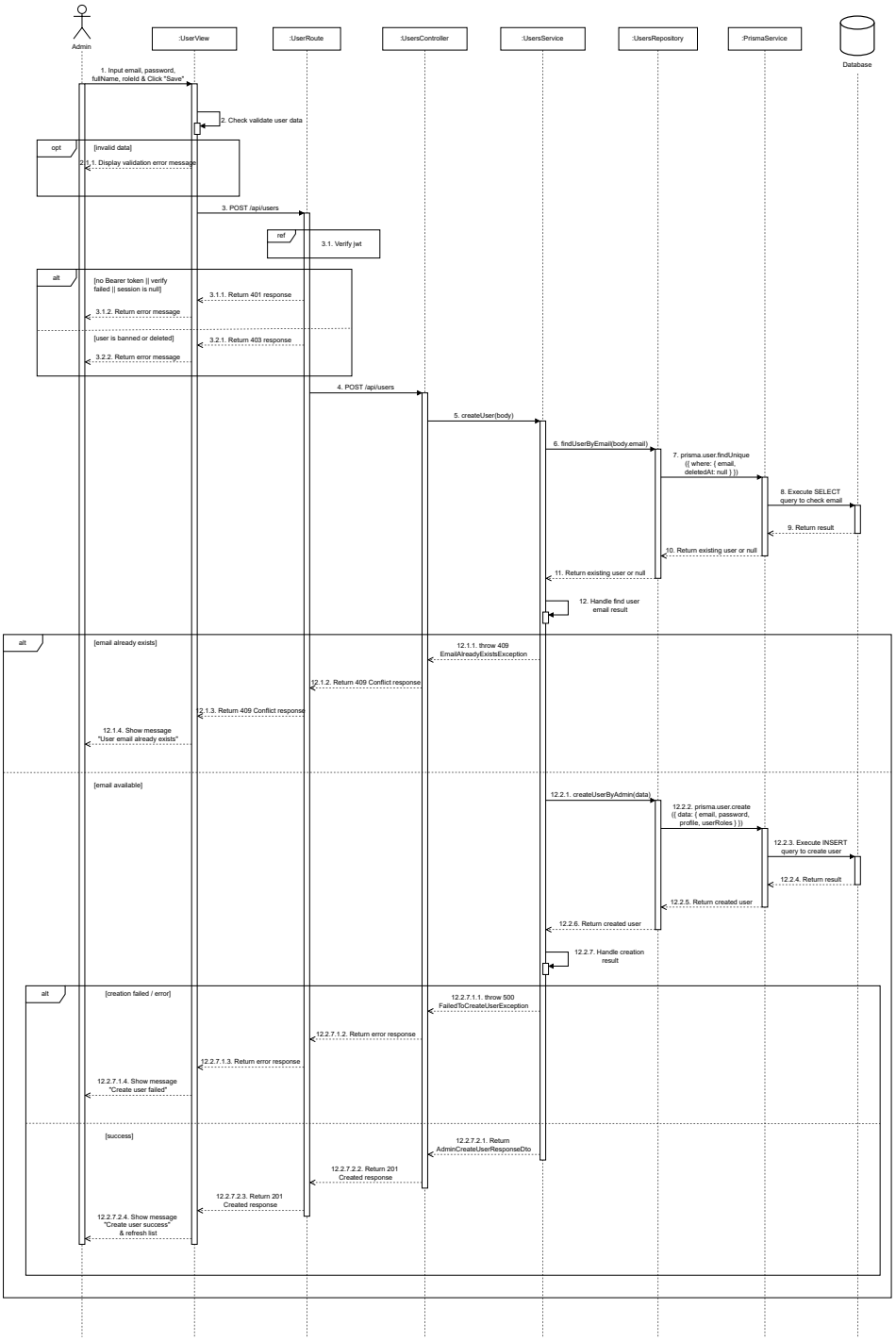

Figure 154: Sequence diagram of UC-28.03 Create User

3.28.5 UC-28.04 – Update User

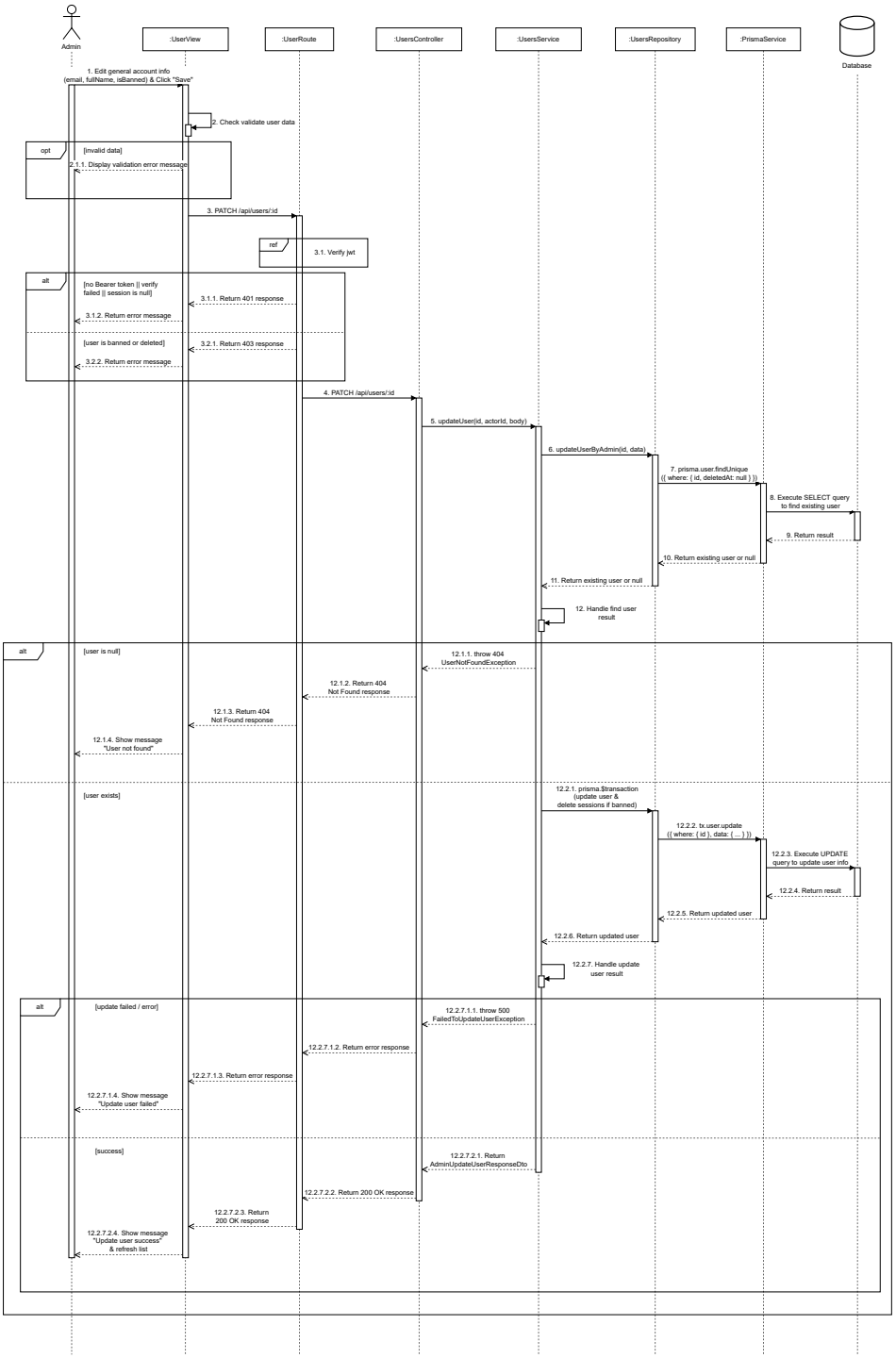

Figure 155: Sequence diagram of UC-28.04 Update User

3.28.6 UC-28.05 – Update Client Profile

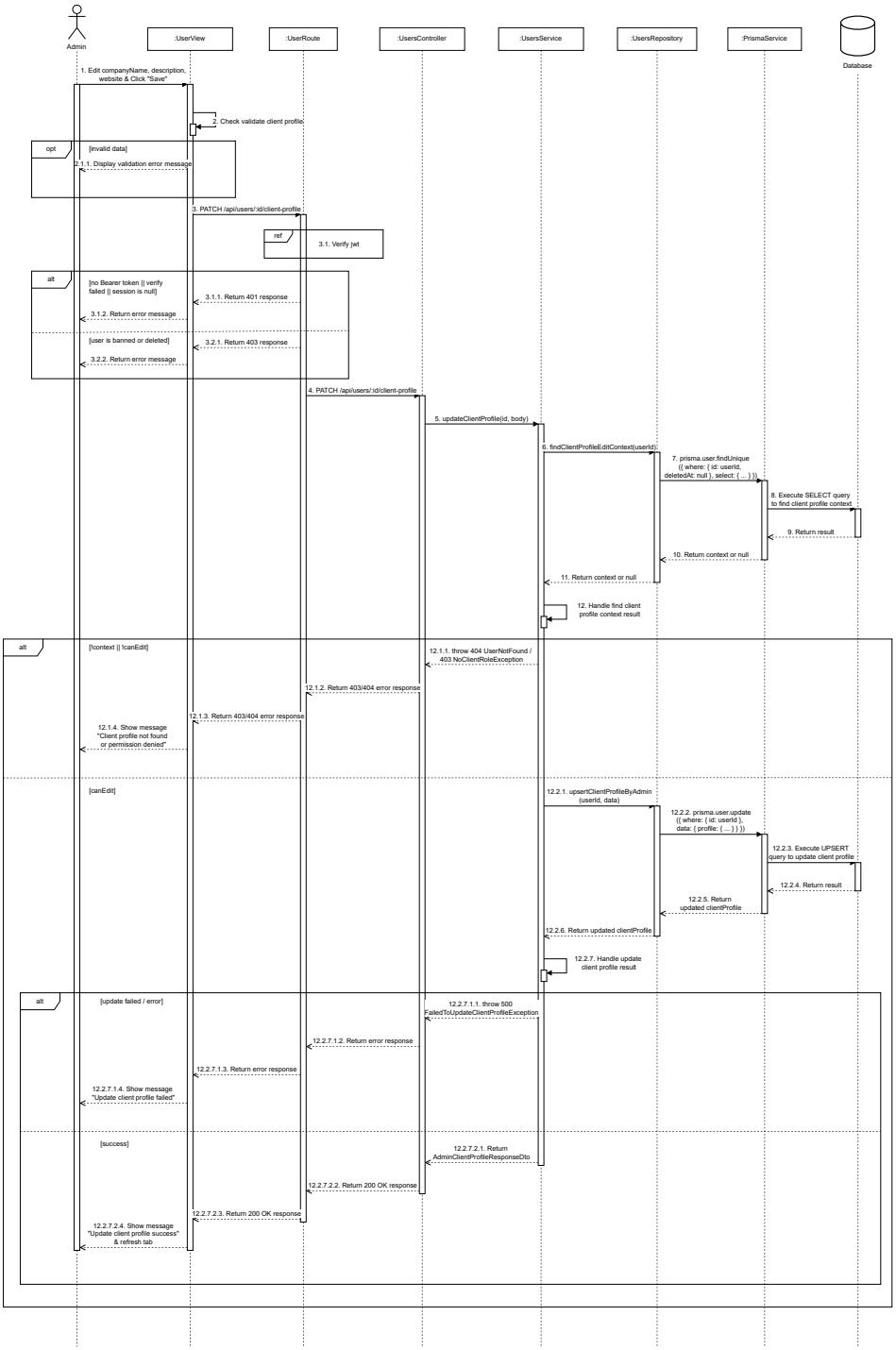

Figure 156: Sequence diagram of UC-28.05 Update Client Profile

3.28.7 UC-28.06 – Update Freelancer Basic Info

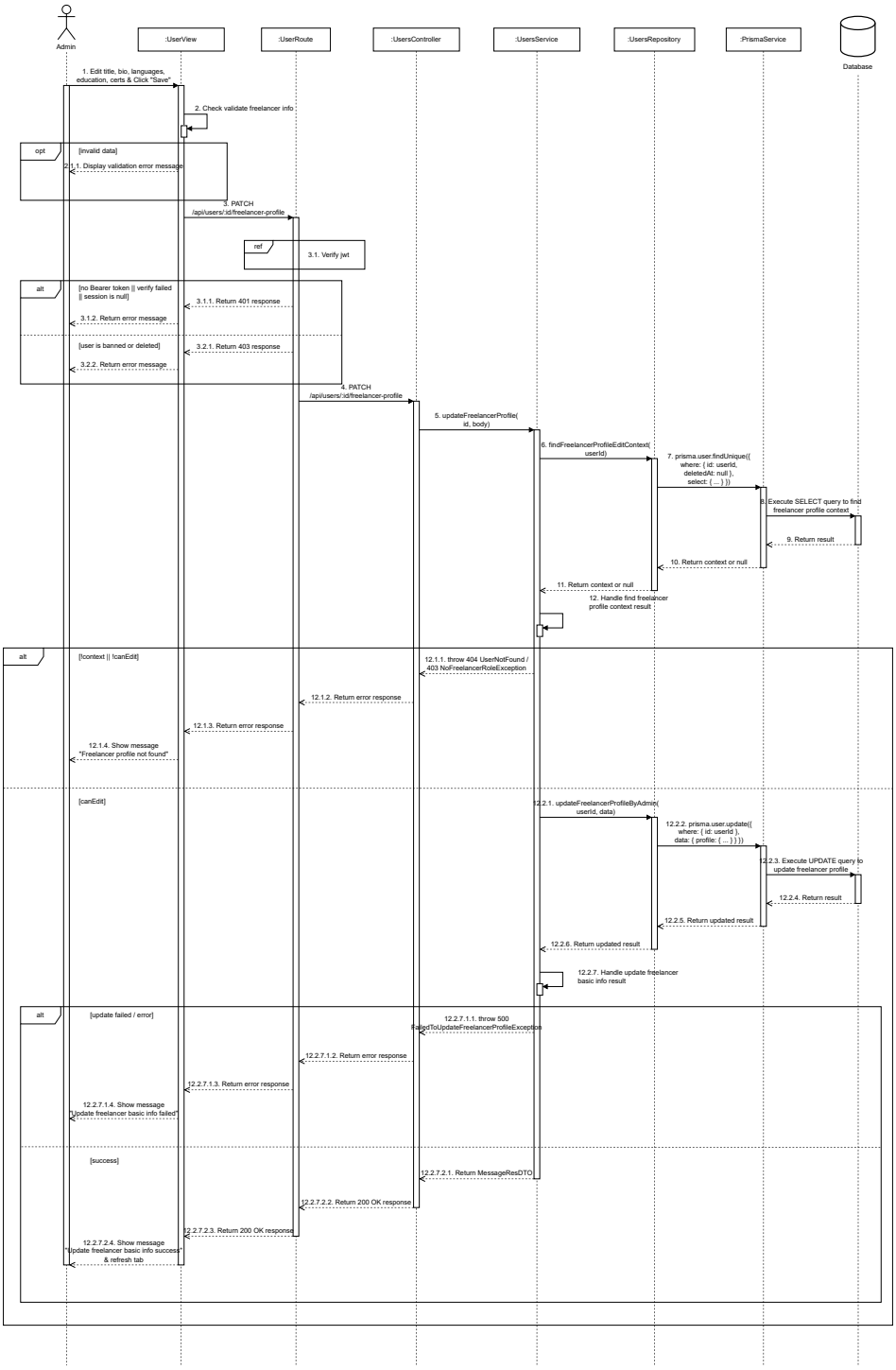

Figure 157: Sequence diagram of UC-28.06 Update Freelancer Basic Info

3.28.8 UC-28.07 – Update Freelancer Skills

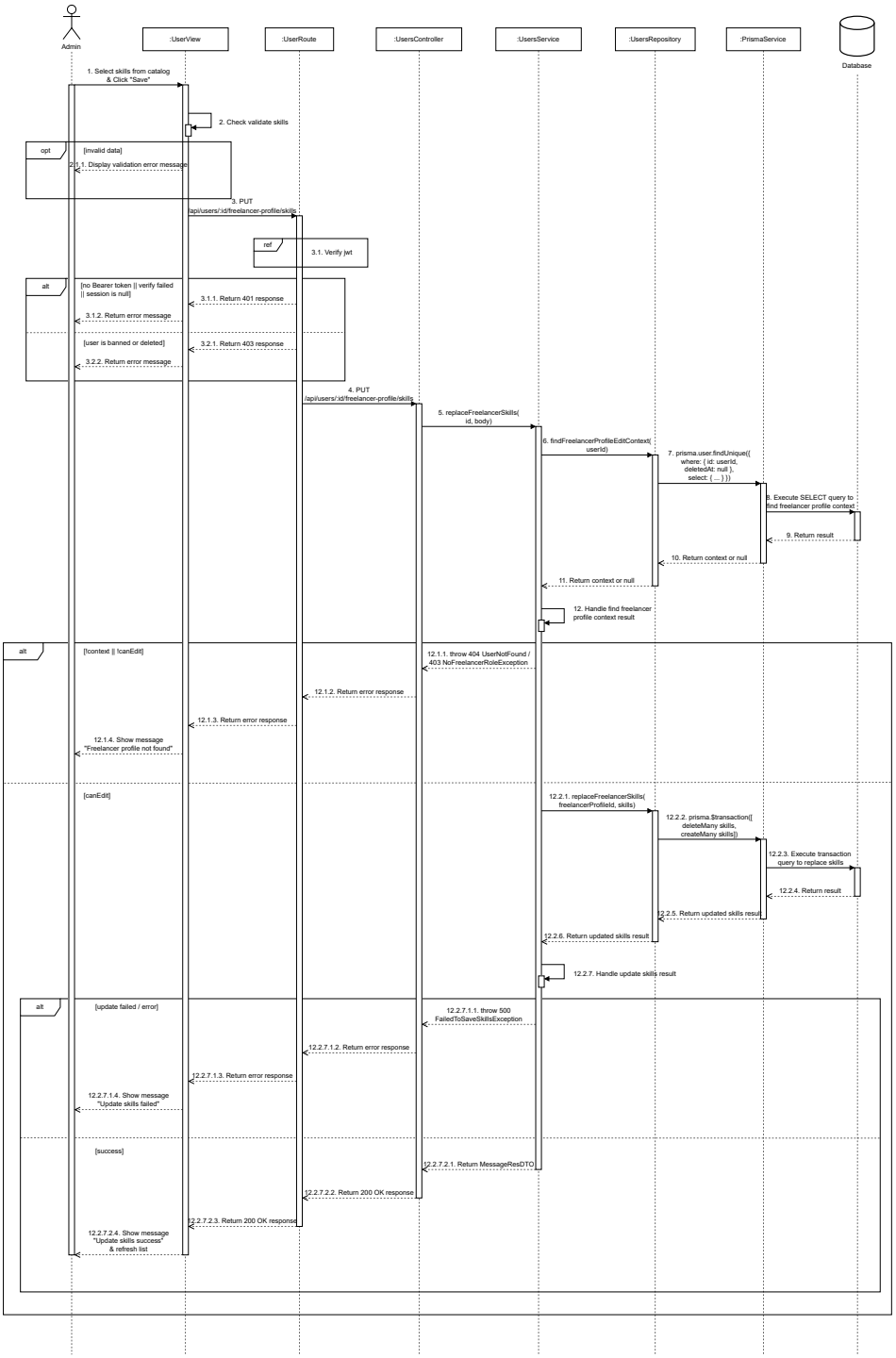

Figure 158: Sequence diagram of UC-28.07 Update Freelancer Skills

3.28.9 UC-28.08 – Update Freelancer Portfolio

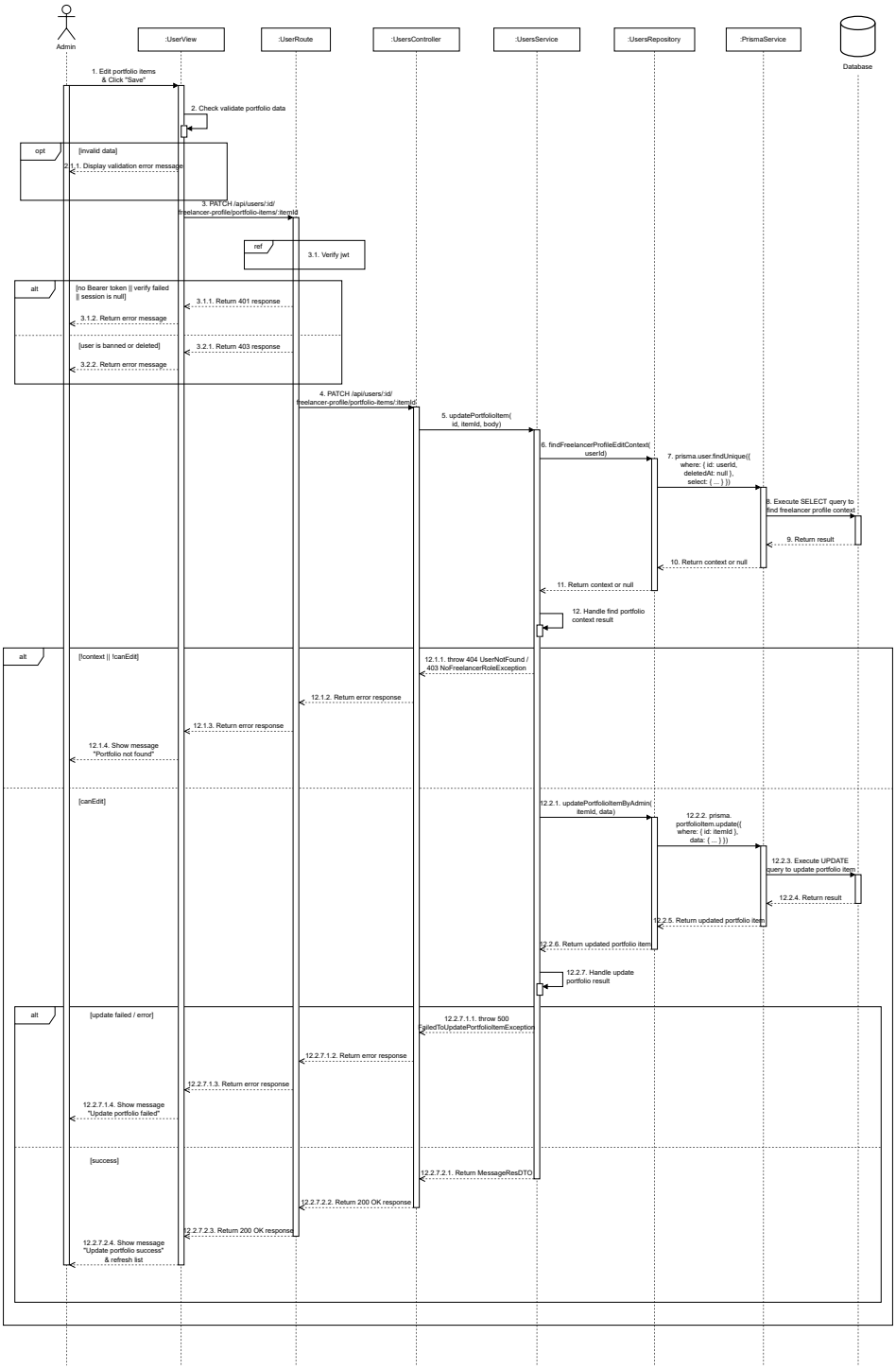

Figure 159: Sequence diagram of UC-28.08 Update Freelancer Portfolio

3.29 Review Management

Contract participants view reviews and create, update, or delete their own review according to contract eligibility and authorship checks.

3.29.1 Class Diagram

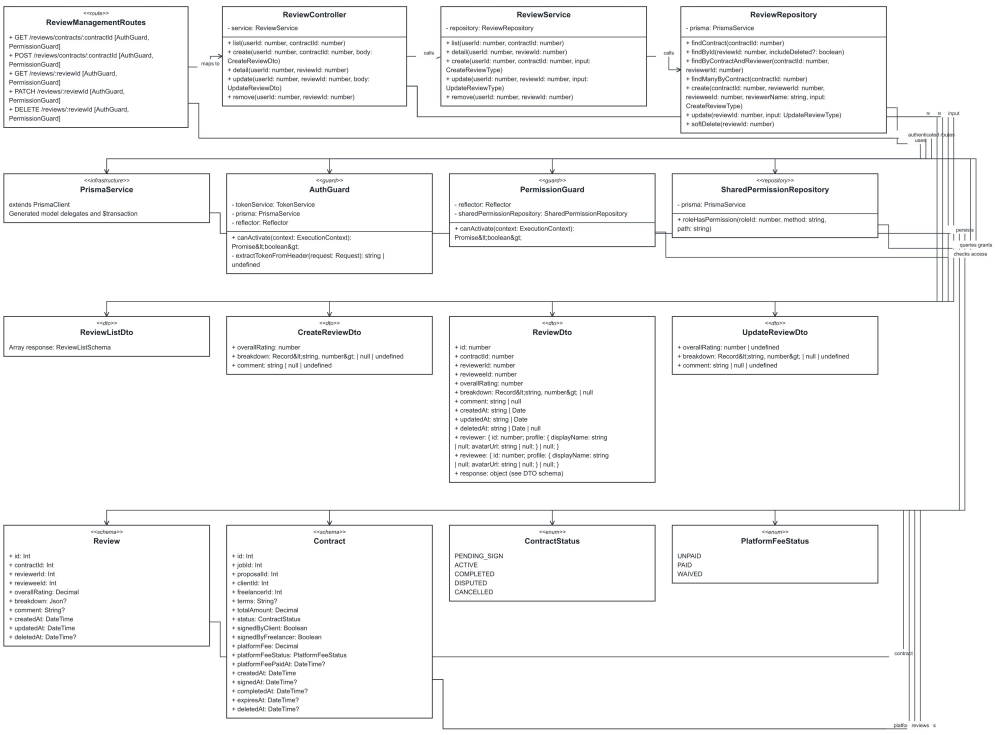

Figure 160: Class diagram of Review Management

3.29.2 UC-39.01 – View Review List

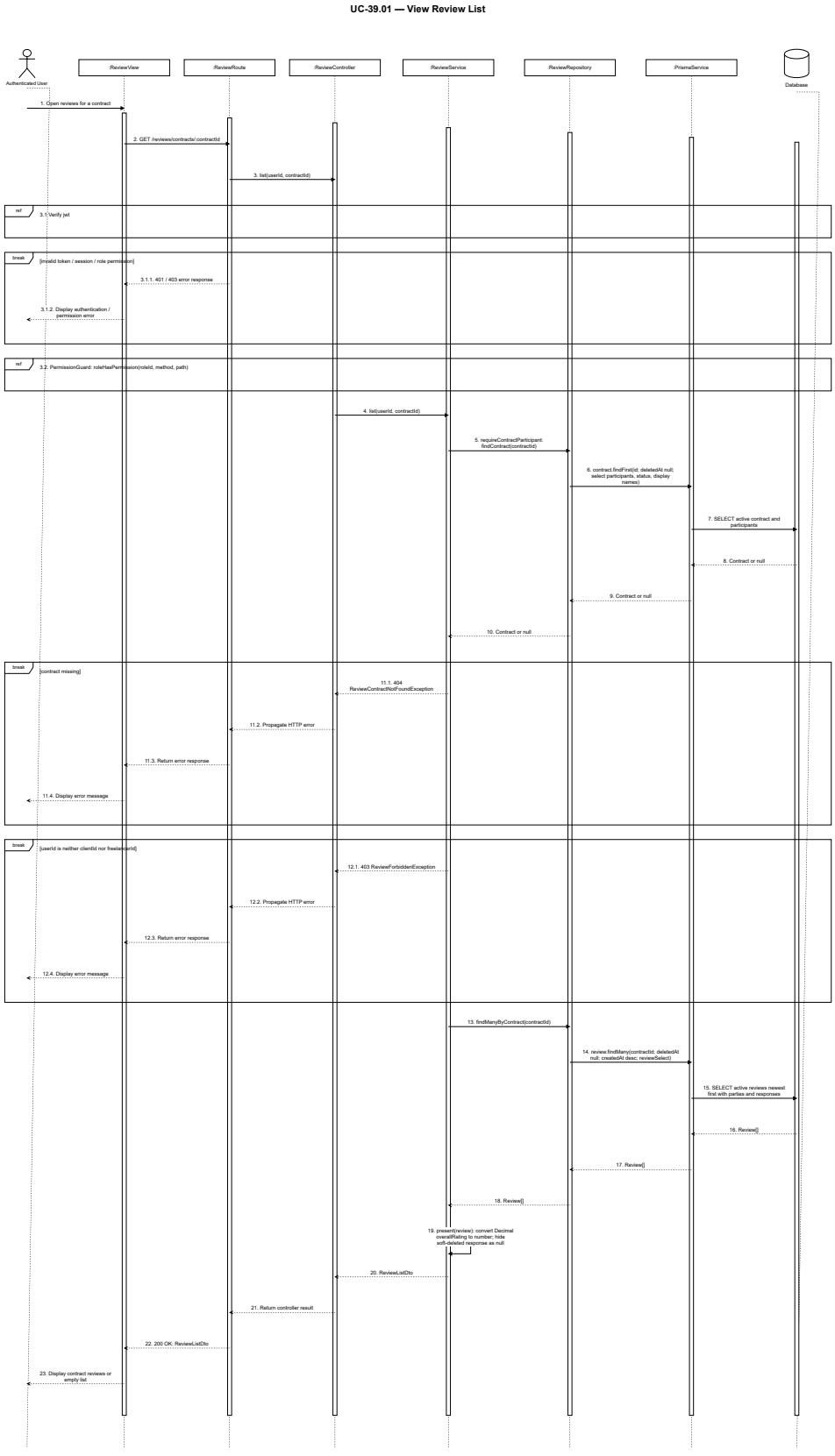

Figure 161: Sequence diagram of UC-39.01 View Review List

3.29.3 UC-39.02 – View Review Detail

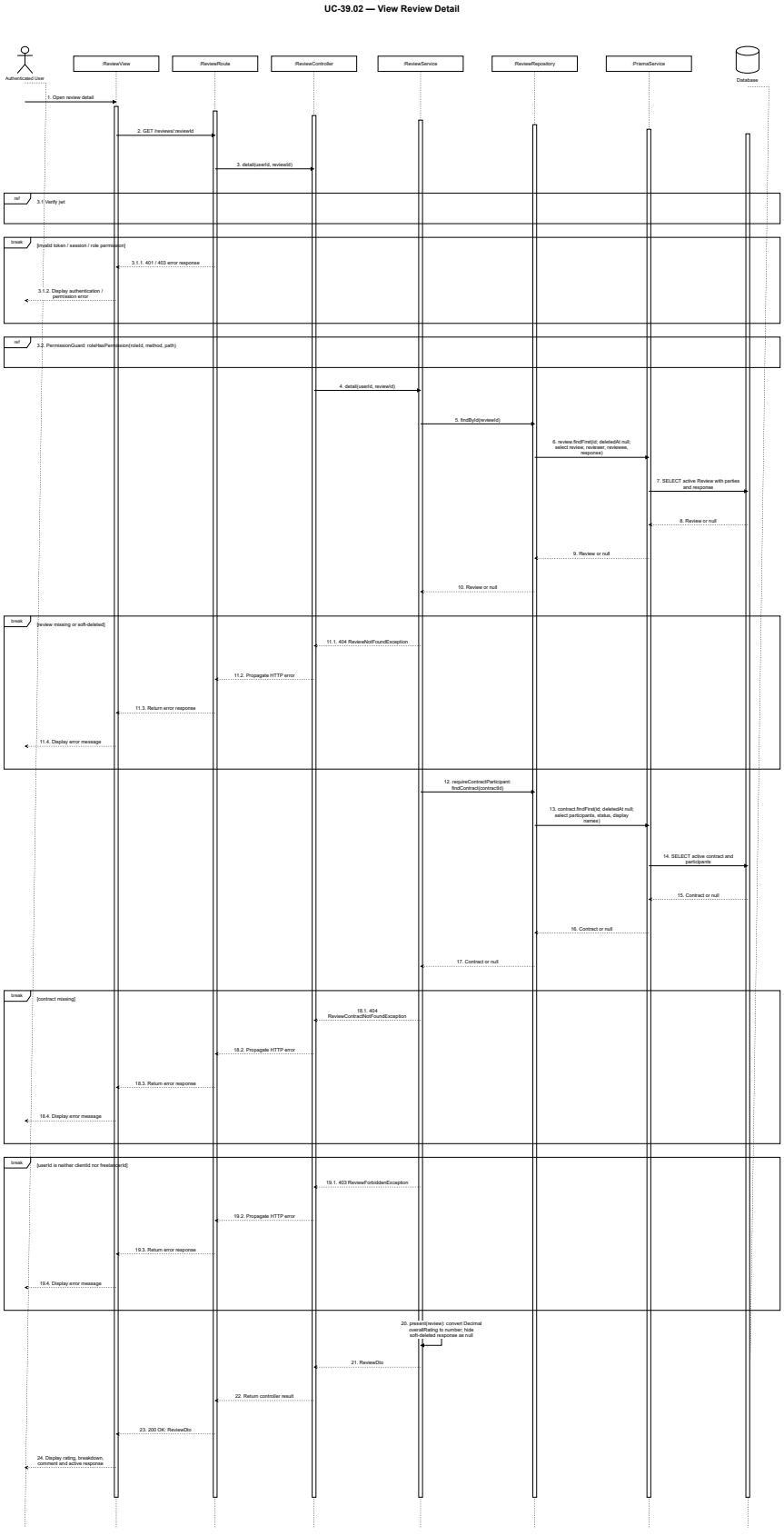

Figure 162: Sequence diagram of UC-39.02 View Review Detail

3.29.4 UC-39.03 – Write Review

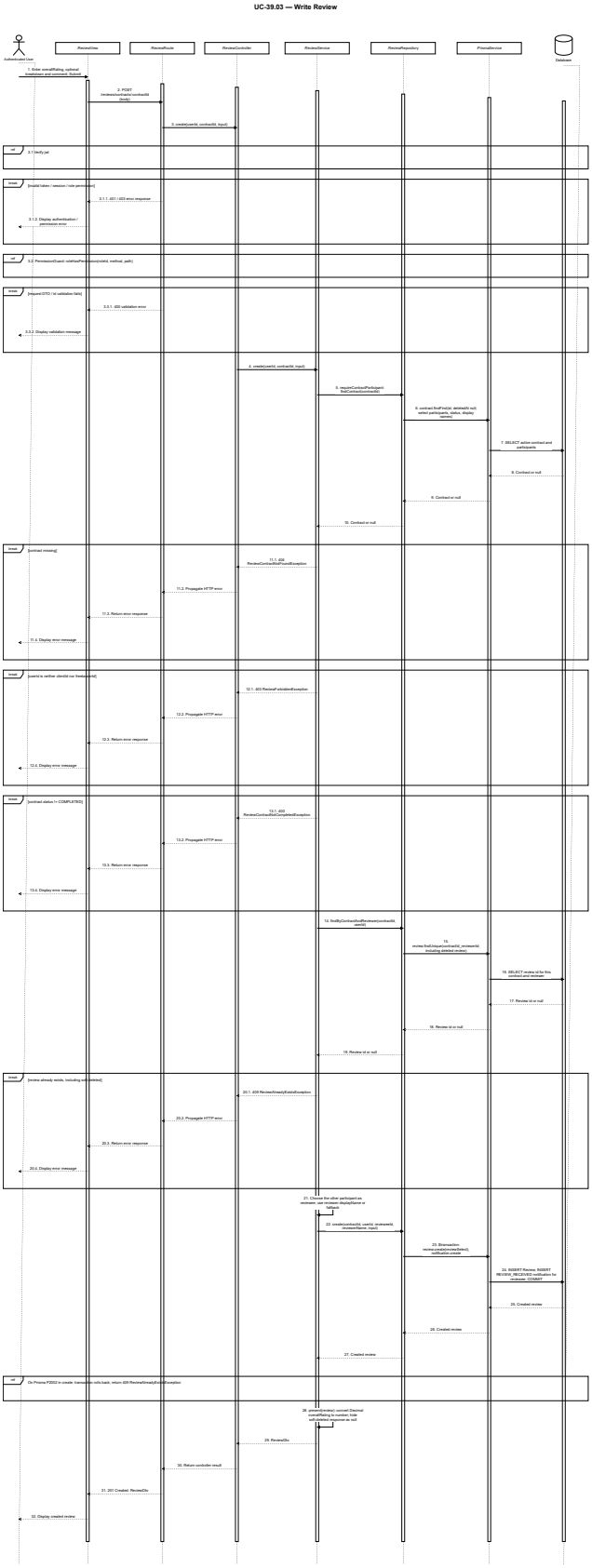

Figure 163: Sequence diagram of UC-39.03 Write Review

3.29.5 UC-39.04 – Update Review

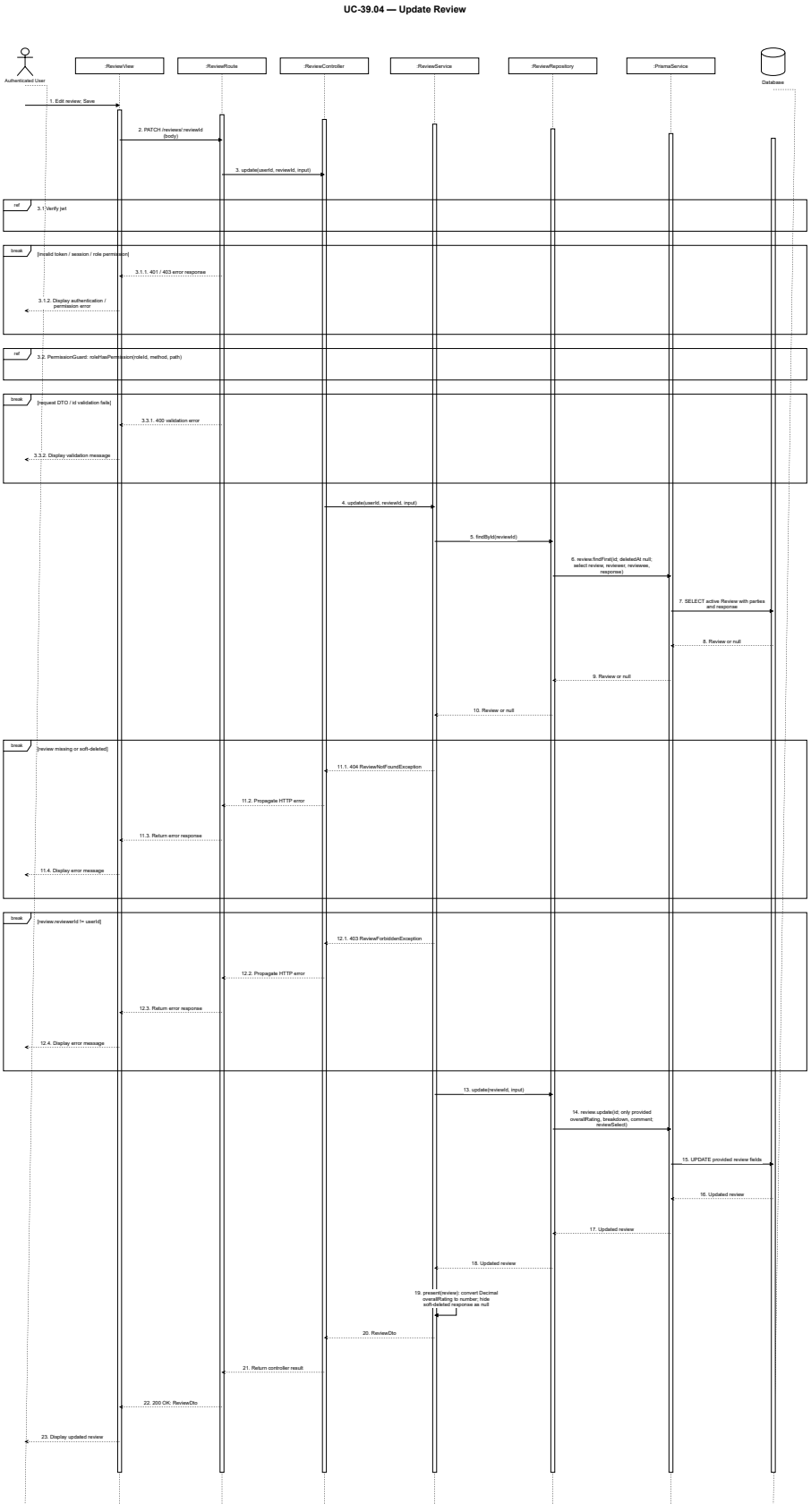

Figure 164: Sequence diagram of UC-39.04 Update Review

3.29.6 UC-39.05 – Delete Review

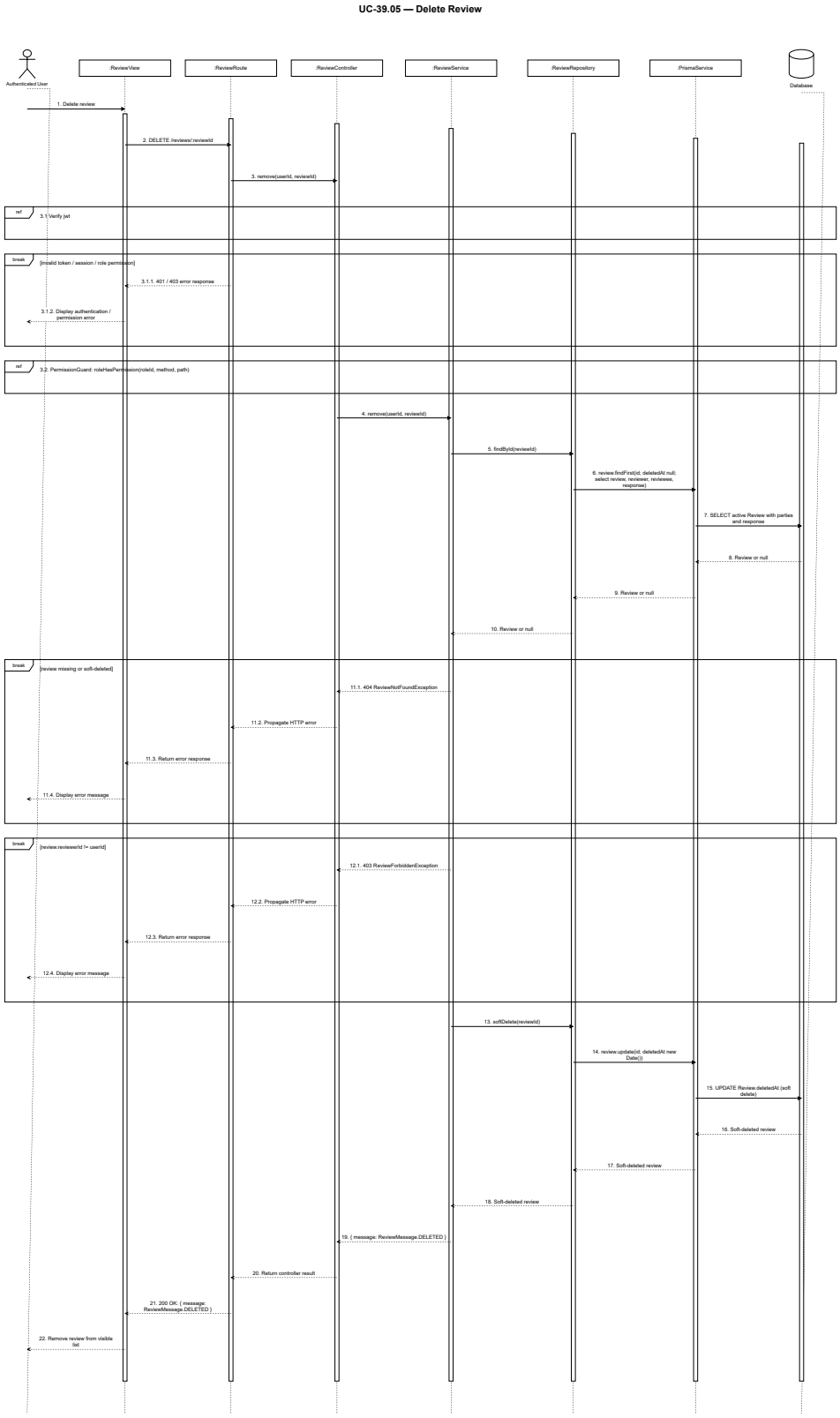

Figure 165: Sequence diagram of UC-39.05 Delete Review

3.30 Review Response Management

The recipient of a review can respond and maintain their own response. Responses can notify the original reviewer.

3.30.1 Class Diagram

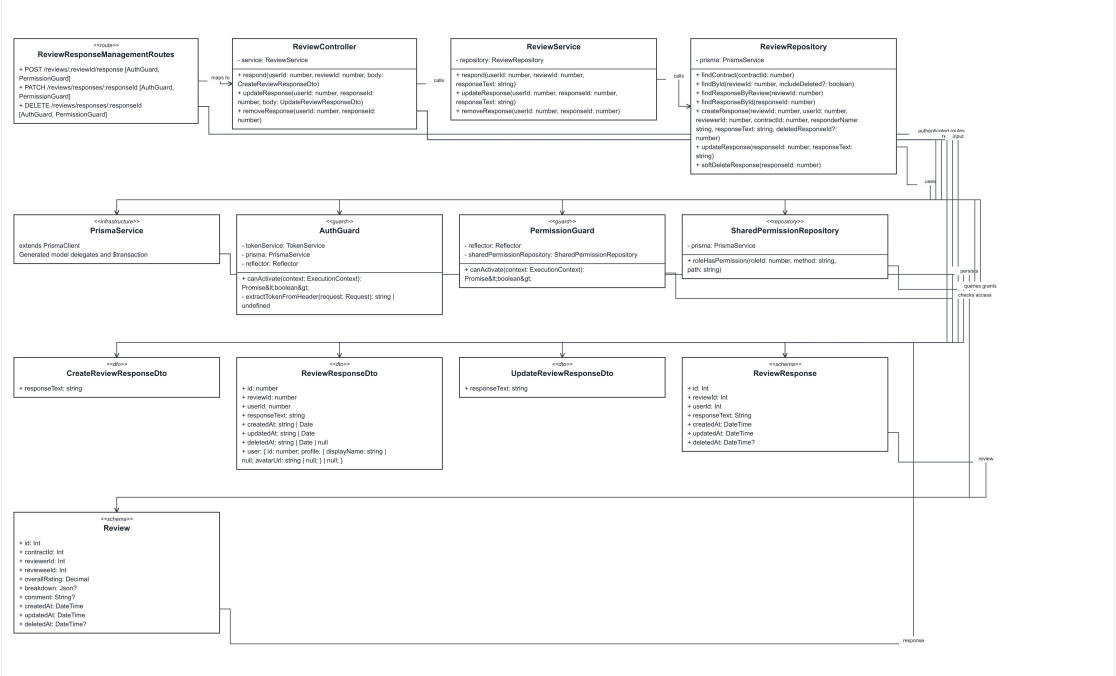

Figure 166: Class diagram of Review Response Management

3.30.2 UC-40.01 – Respond to Review

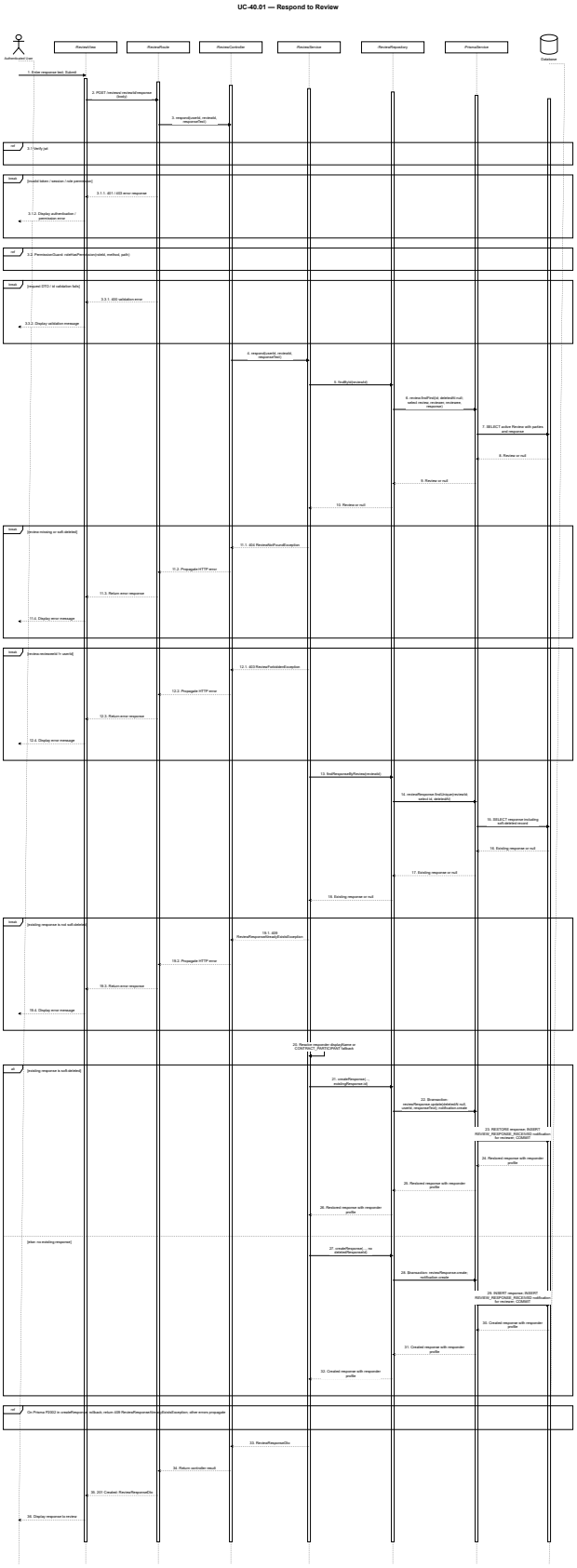

Figure 167: Sequence diagram of UC-40.01 Respond to Review

3.30.3 UC-40.02 – Update Review Response

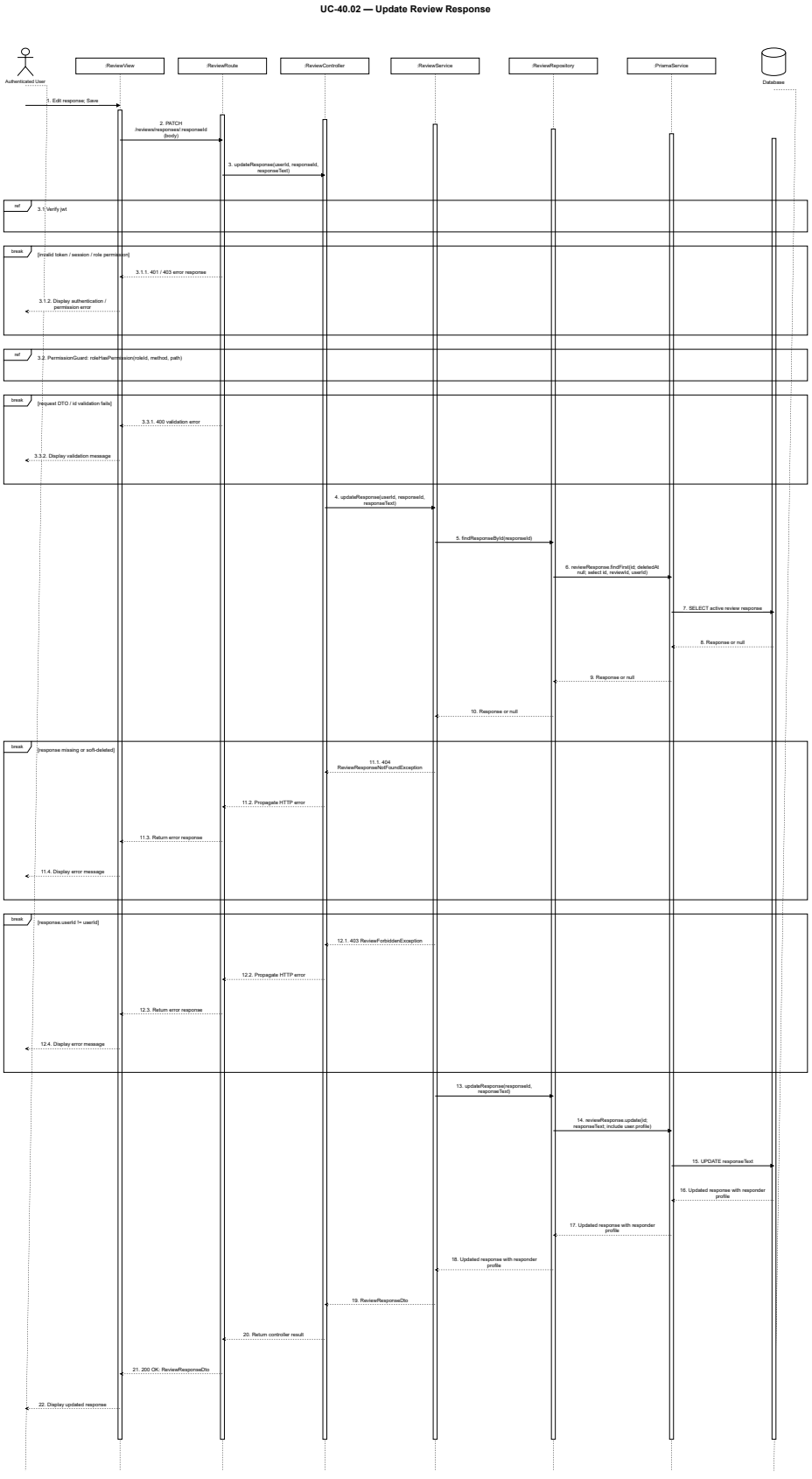

Figure 168: Sequence diagram of UC-40.02 Update Review Response

3.30.4 UC-40.03 – Delete Review Response

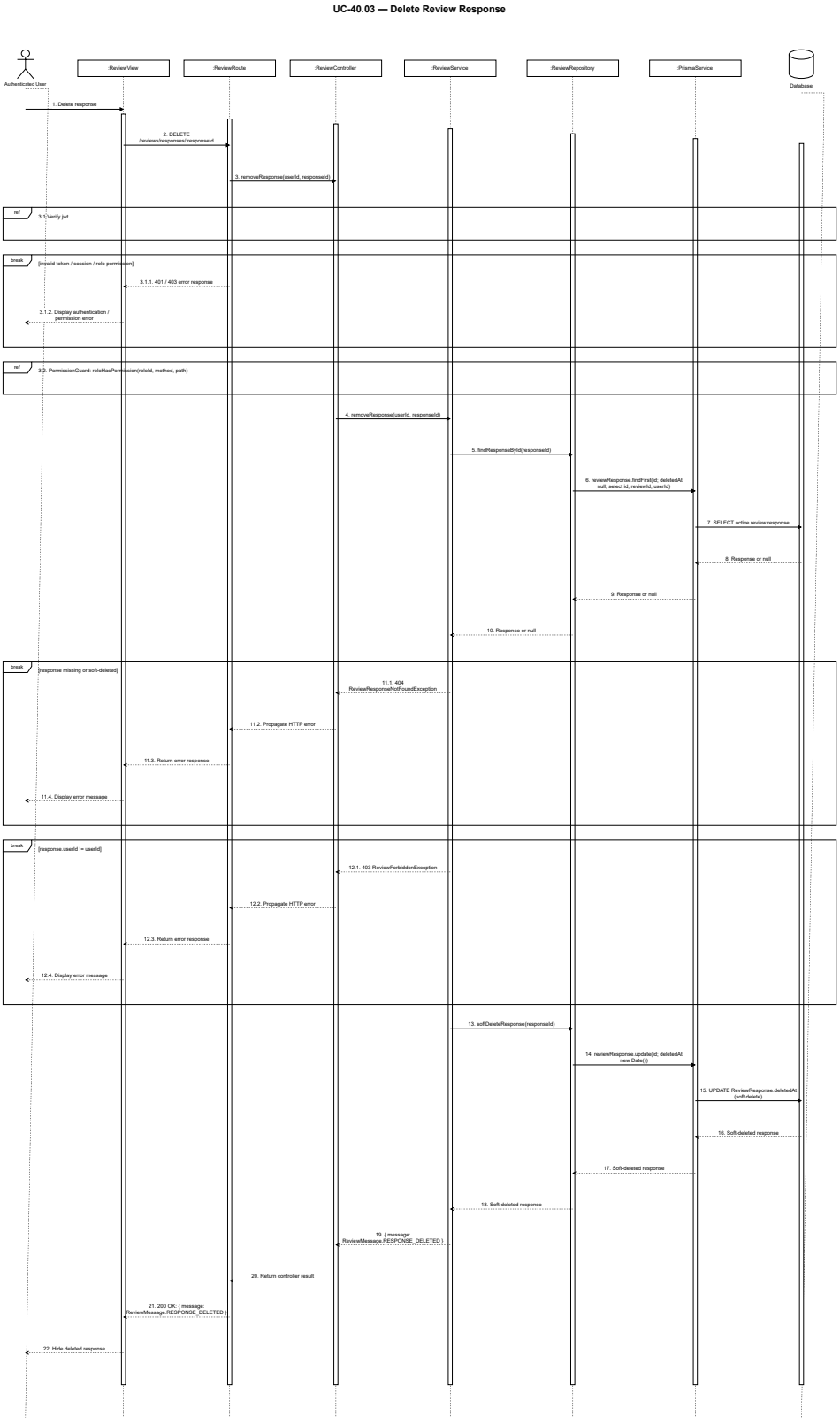

Figure 169: Sequence diagram of UC-40.03 Delete Review Response

3.31 Account Profile Management

Authenticated users view general account information, submit profile edits, change their password, and upload or retrieve their avatar.

3.31.1 Class Diagram

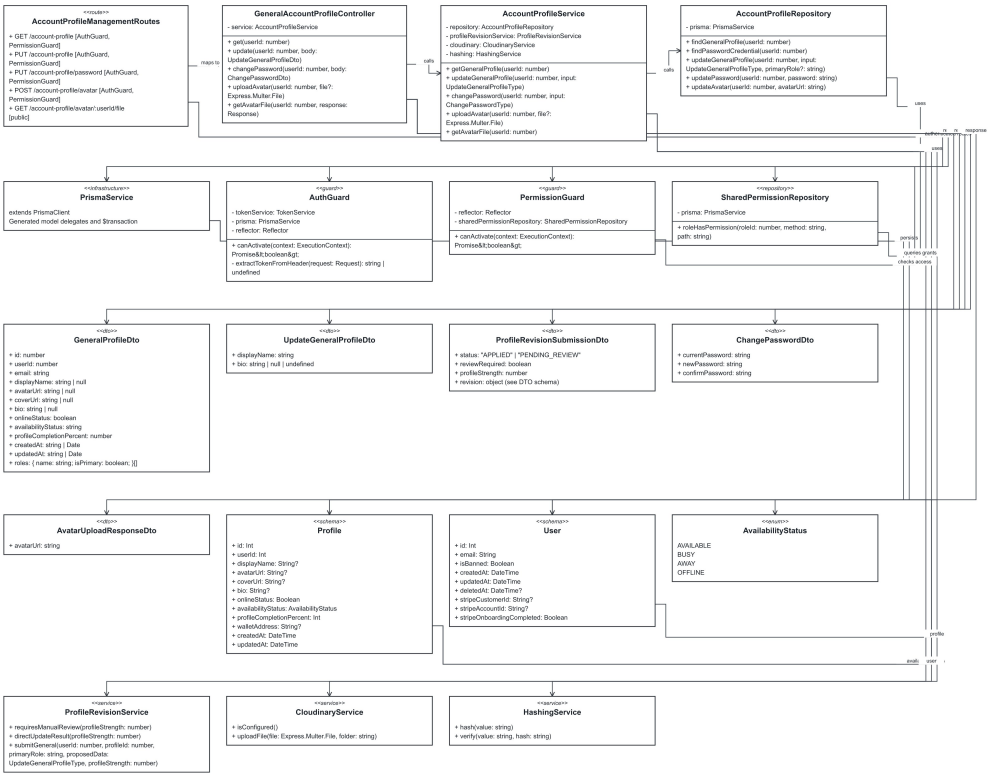

Figure 170: Class diagram of Account Profile Management

3.31.2 UC-41.01 – View Profile

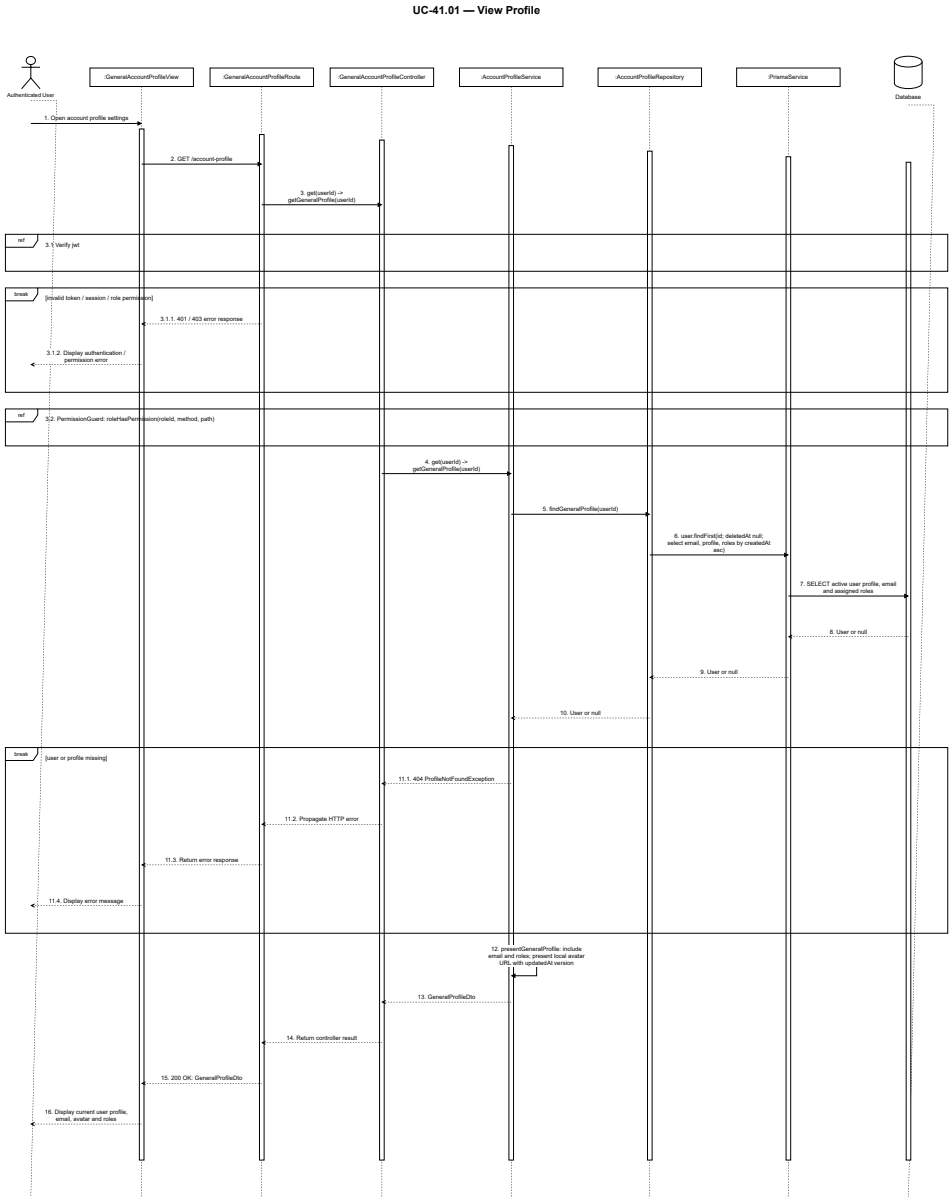

Figure 171: Sequence diagram of UC-41.01 View Profile

3.31.3 UC-41.02 – Update Profile

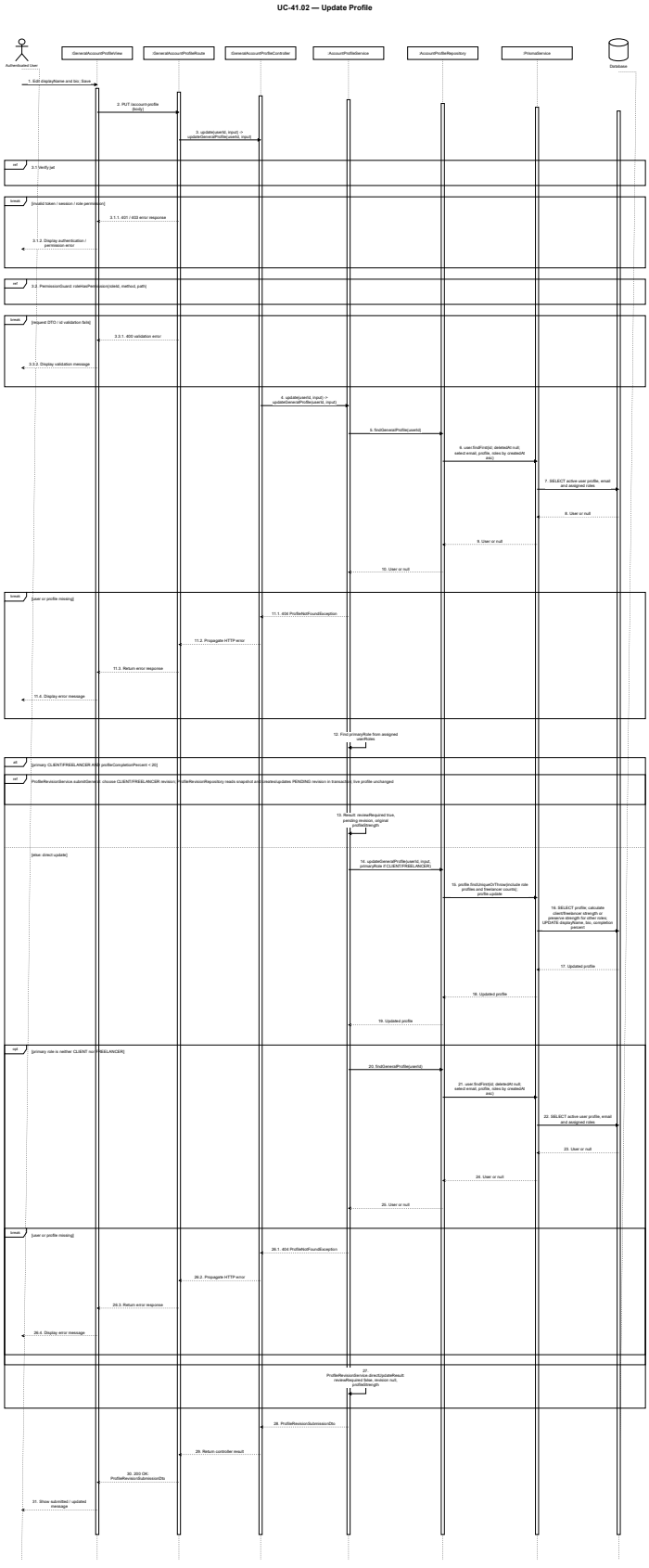

Figure 172: Sequence diagram of UC-41.02 Update Profile

3.31.4 UC-41.03 – Change Password

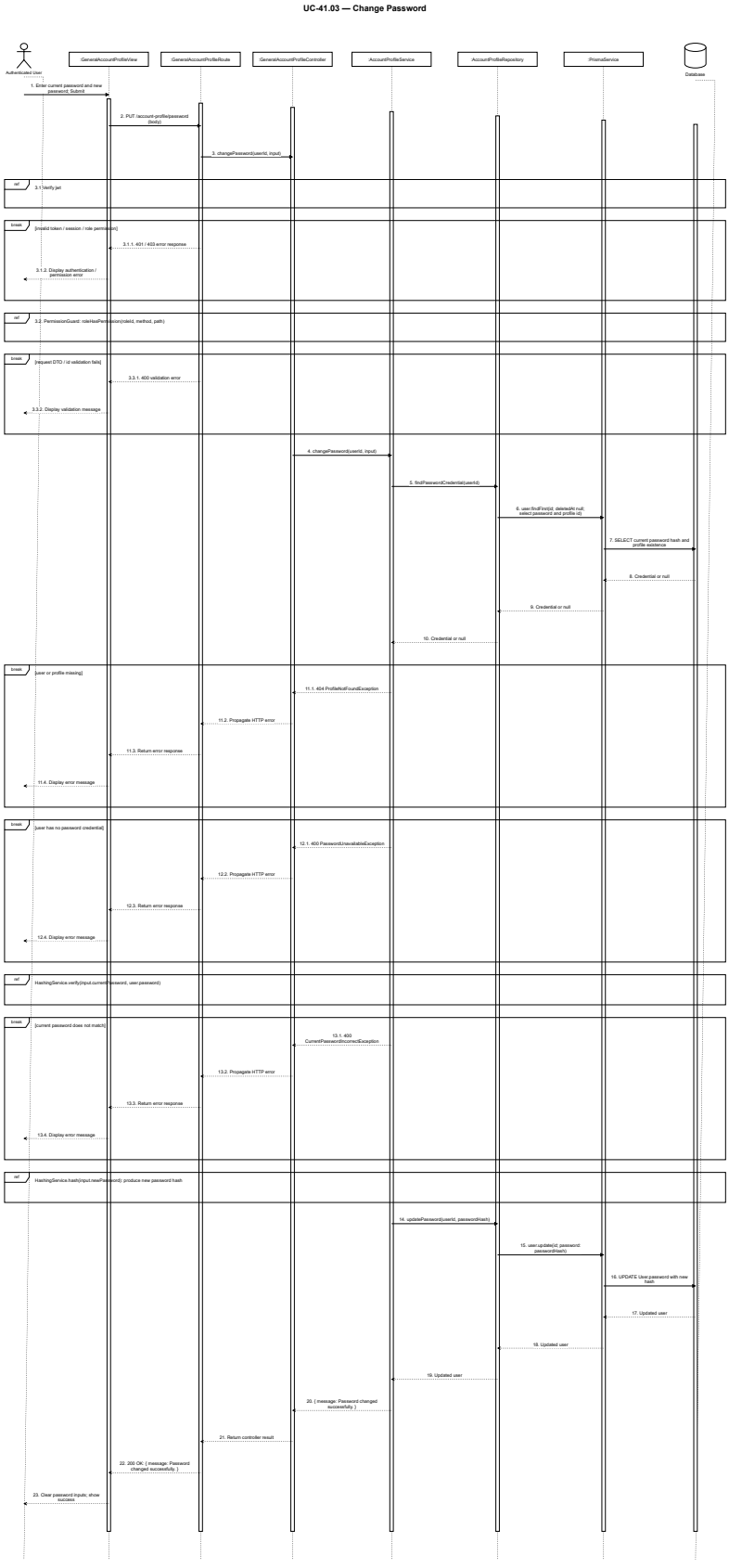

Figure 173: Sequence diagram of UC-41.03 Change Password

3.31.5 UC-41.04 – Upload Avatar

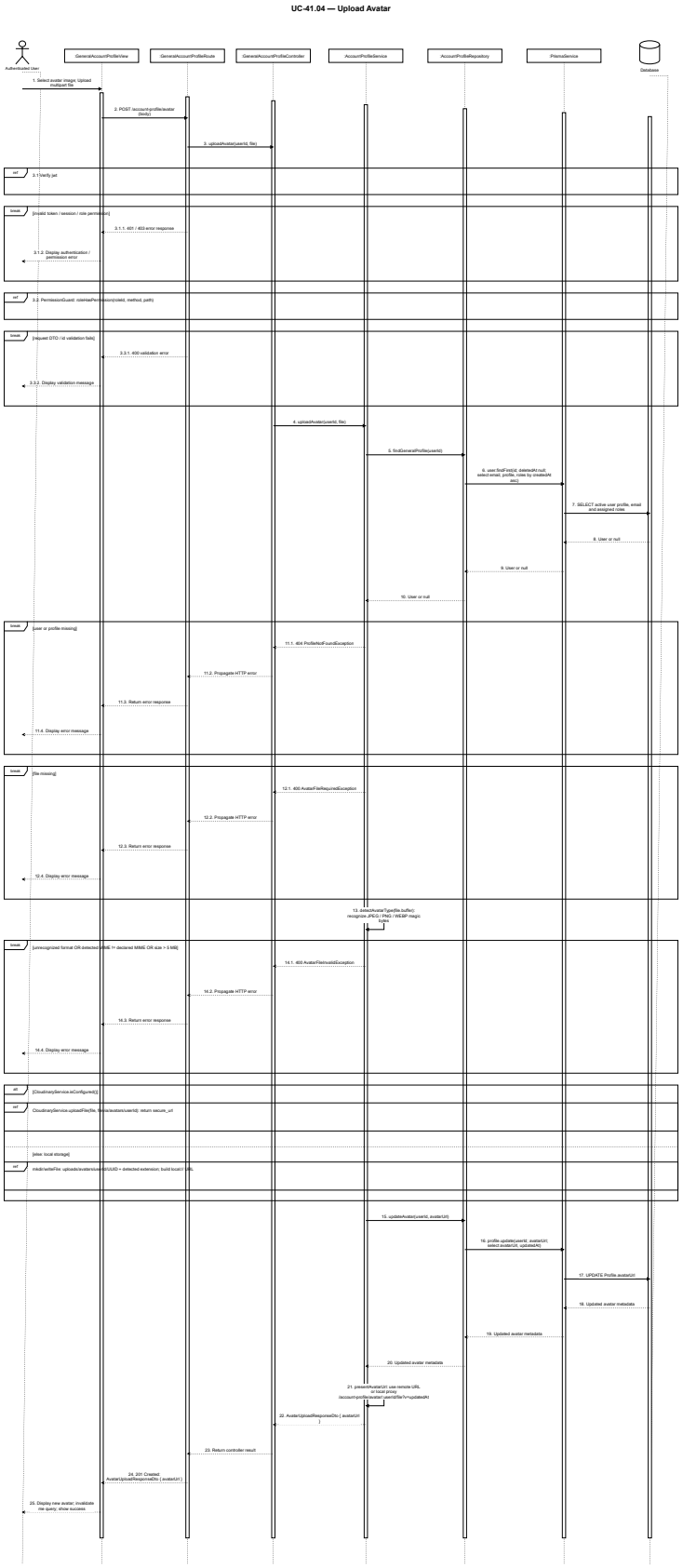

Figure 174: Sequence diagram of UC-41.04 Upload Avatar

3.32 Banner Management

Banner Management maintains the advertisement banners shown in configured page positions. The module supports image upload, lifecycle dates, activation state, and hard deletion.

3.32.1 Class Diagram

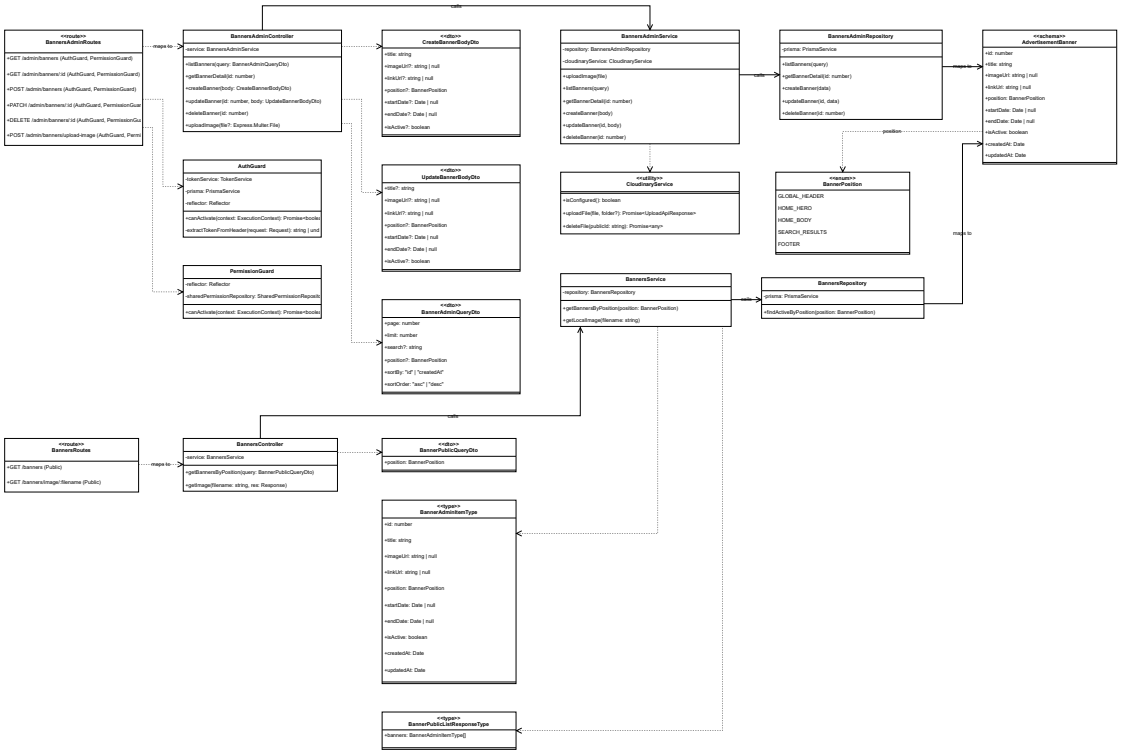

Figure 175: Class diagram of Banner Management

3.32.2 UC-37.01 – View Banner List

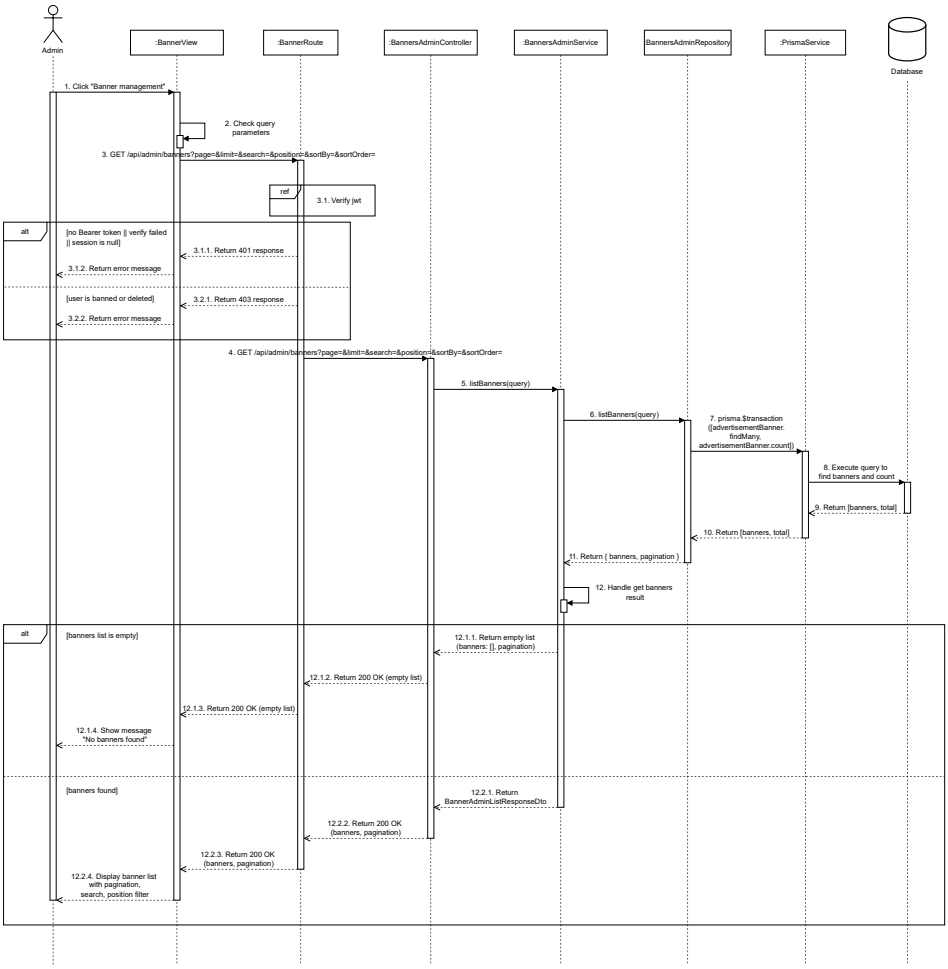

Figure 176: Sequence diagram of UC-37.01 View Banner List

3.32.3 UC-37.02 – View Banner Detail

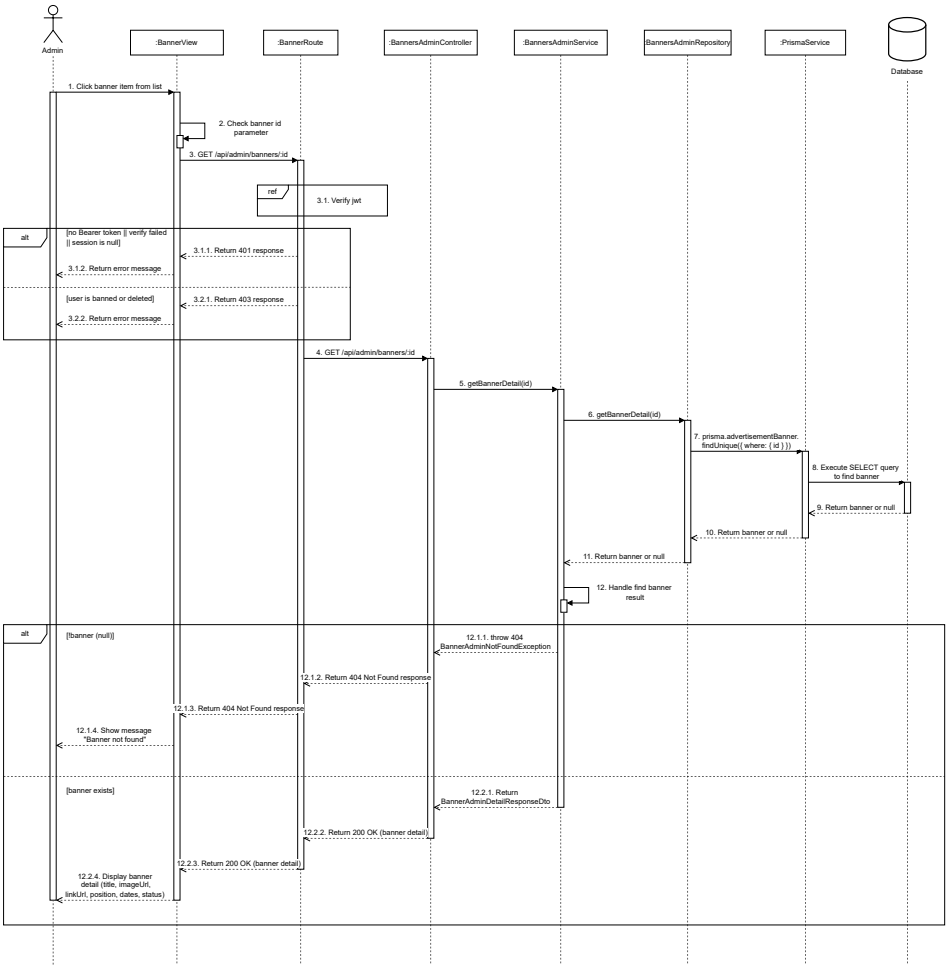

Figure 177: Sequence diagram of UC-37.02 View Banner Detail

3.32.4 UC-37.03 – Create Banner

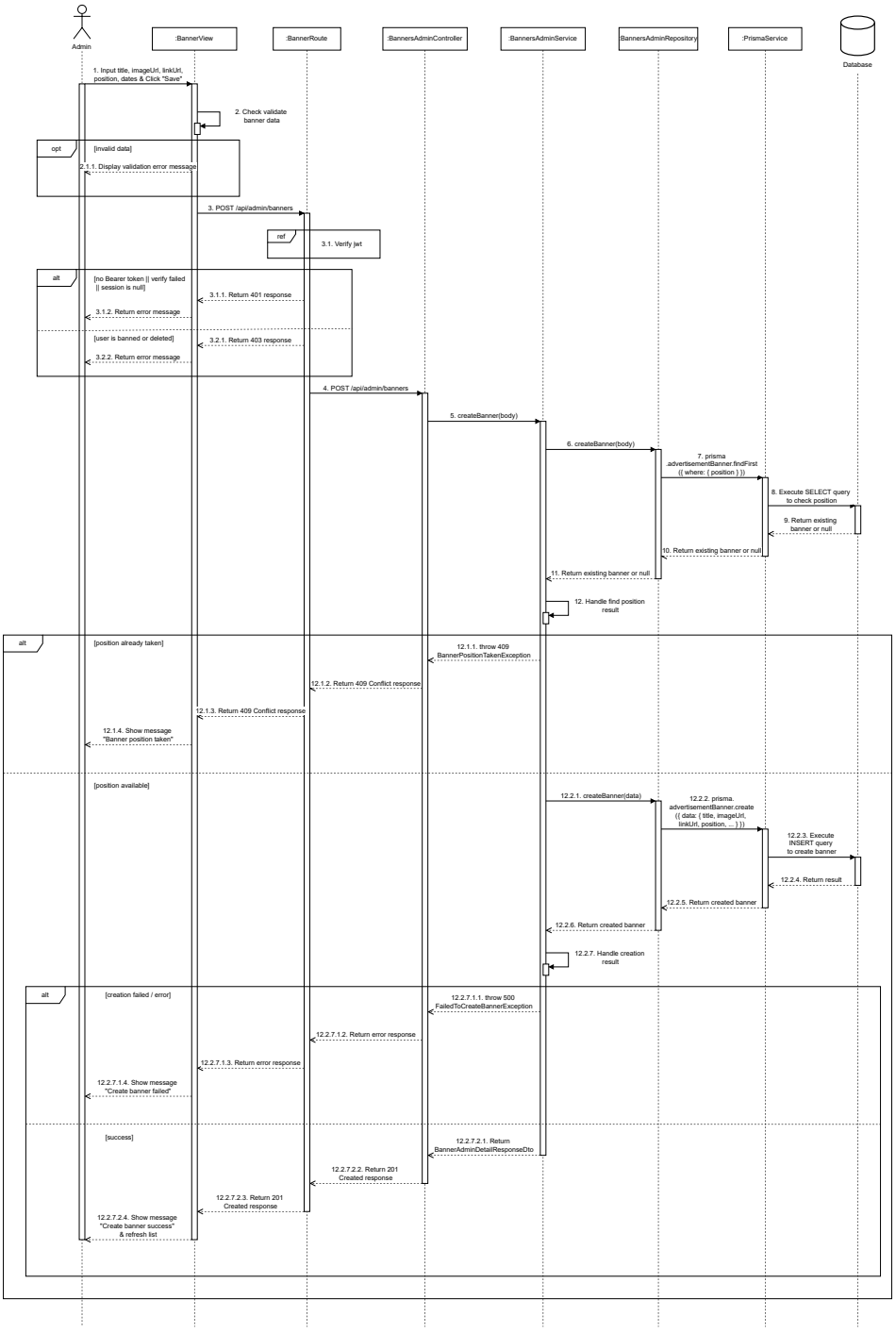

Figure 178: Sequence diagram of UC-37.03 Create Banner

3.32.5 UC-37.04 – Update Banner

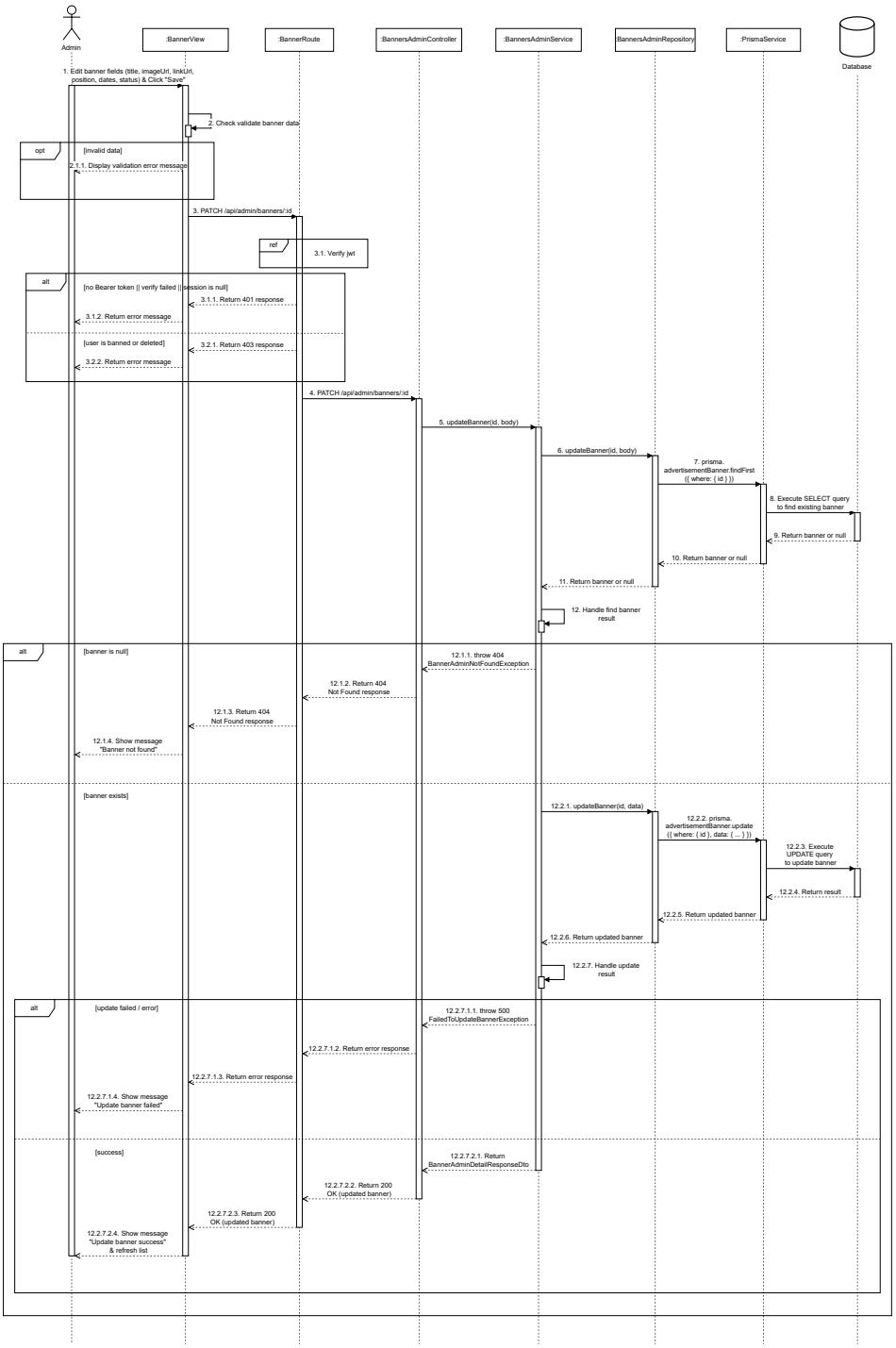

Figure 179: Sequence diagram of UC-37.04 Update Banner

3.32.6 UC-37.05 – Delete Banner

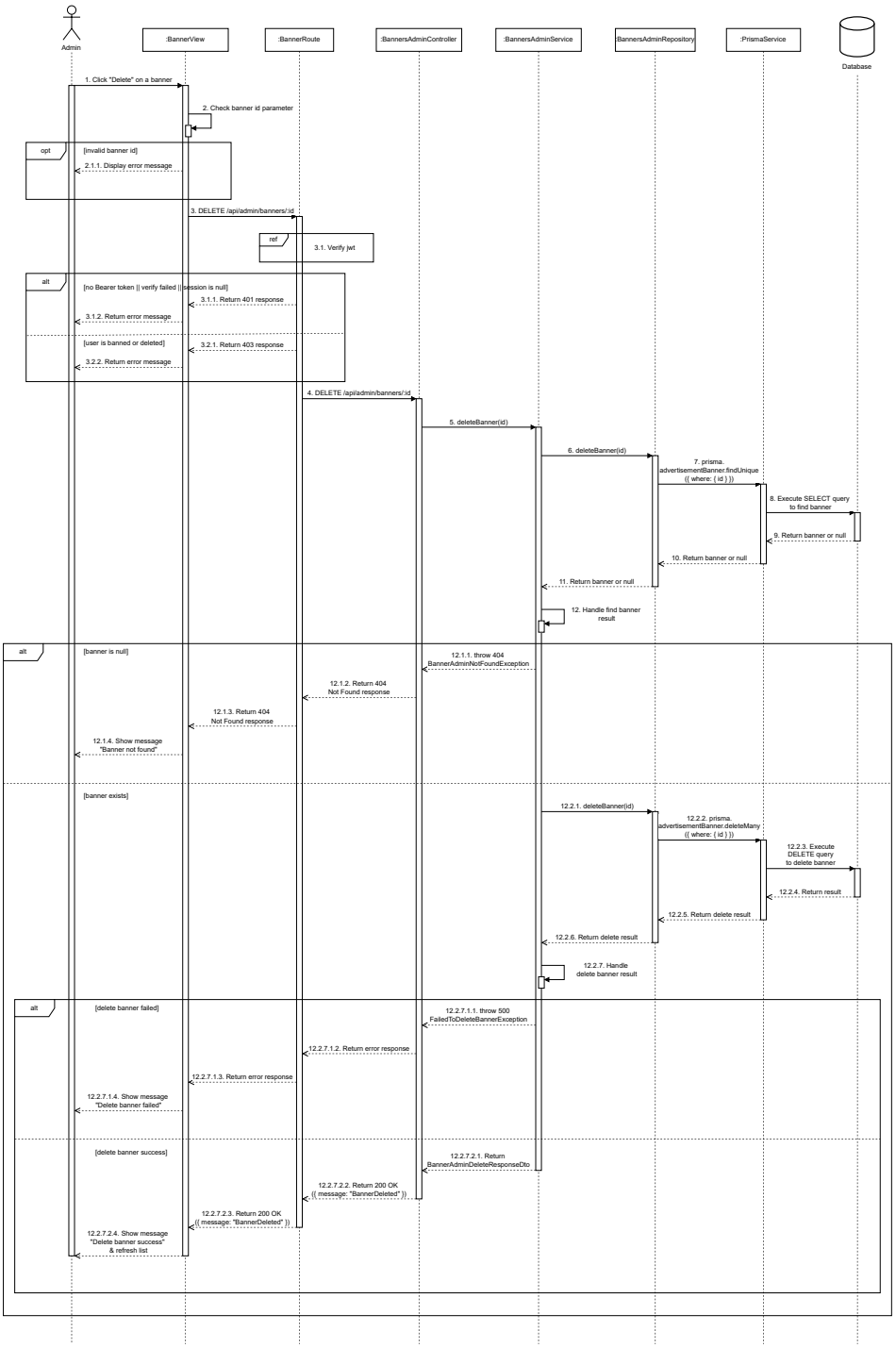

Figure 180: Sequence diagram of UC-37.05 Delete Banner

3.33 Skill Management

Skill Management owns the master skill catalogue used by freelancer profiles and job postings. Skills are soft-deleted so that existing references remain valid, and a deleted skill can be restored.

3.33.1 Class Diagram

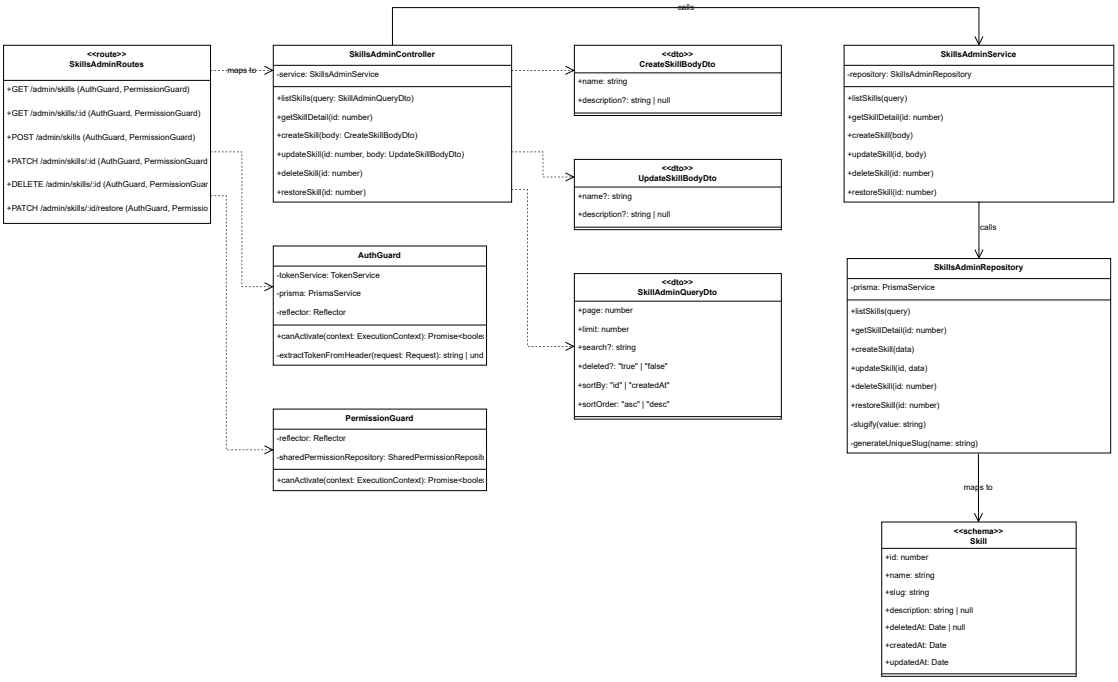

Figure 181: Class diagram of Skill Management

3.33.2 UC-38.01 – View Skill List

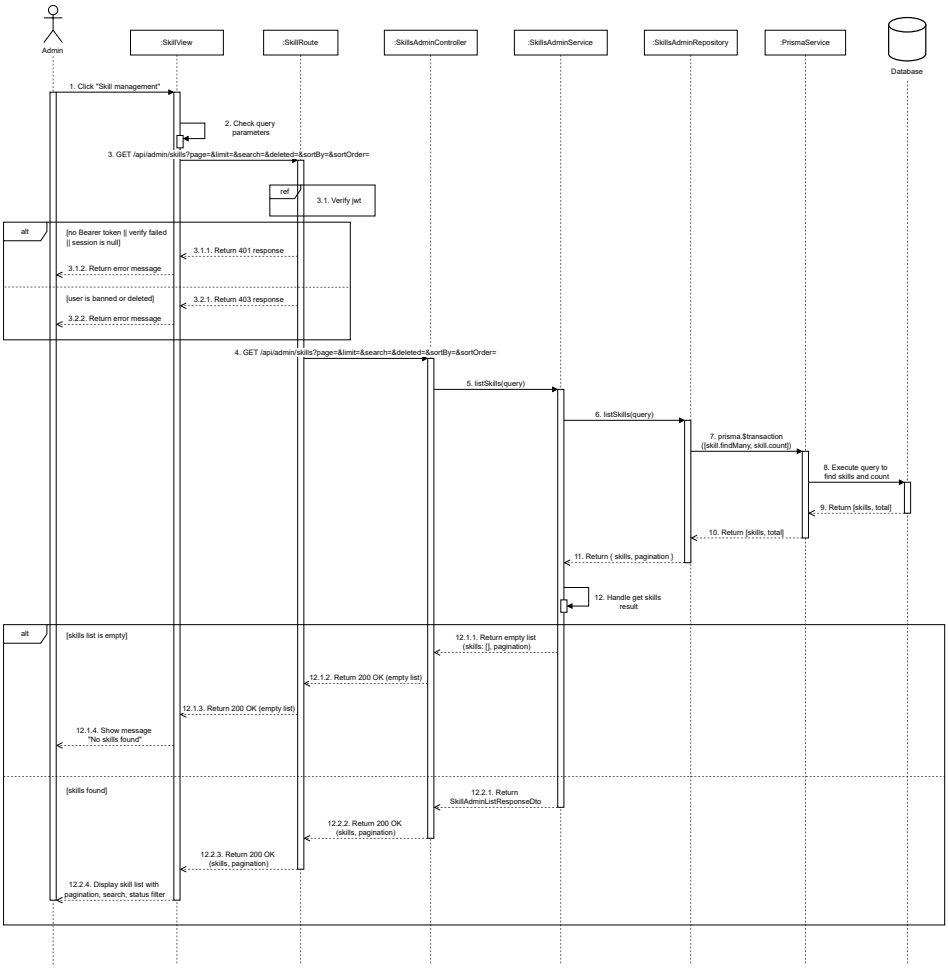

Figure 182: Sequence diagram of UC-38.01 View Skill List

3.33.3 UC-38.02 – View Skill Detail

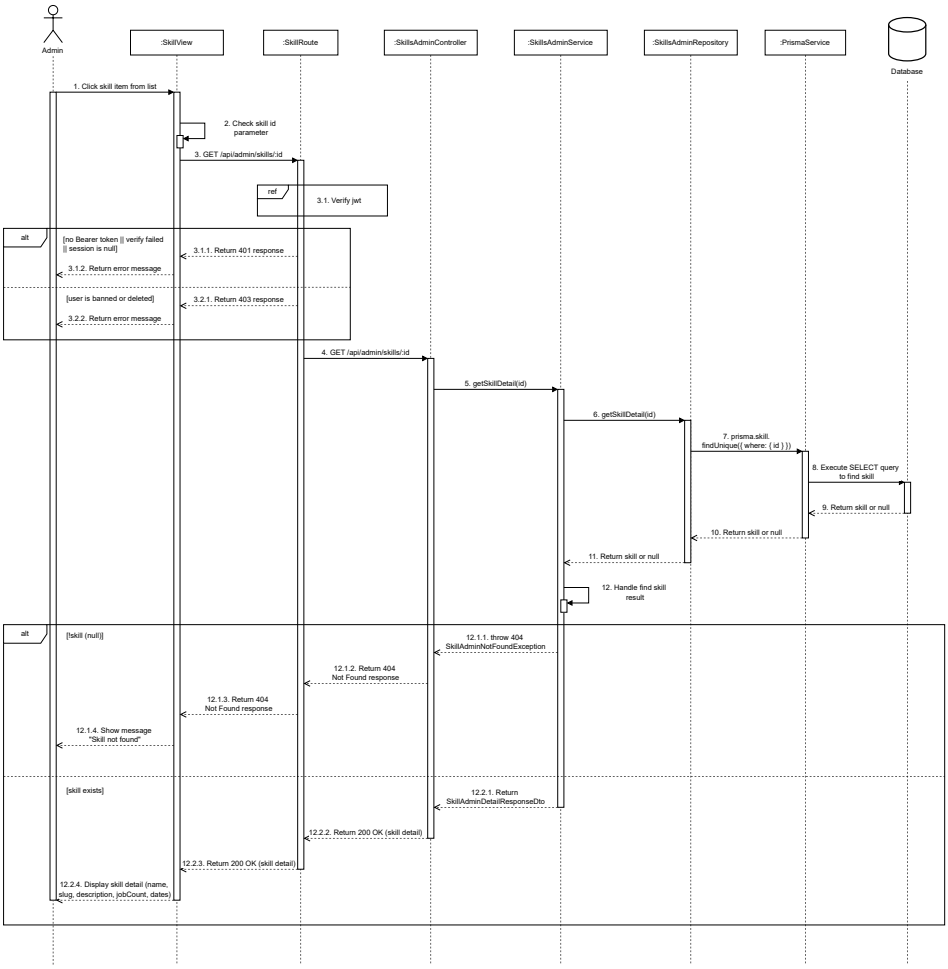

Figure 183: Sequence diagram of UC-38.02 View Skill Detail

3.33.4 UC-38.03 – Create Skill

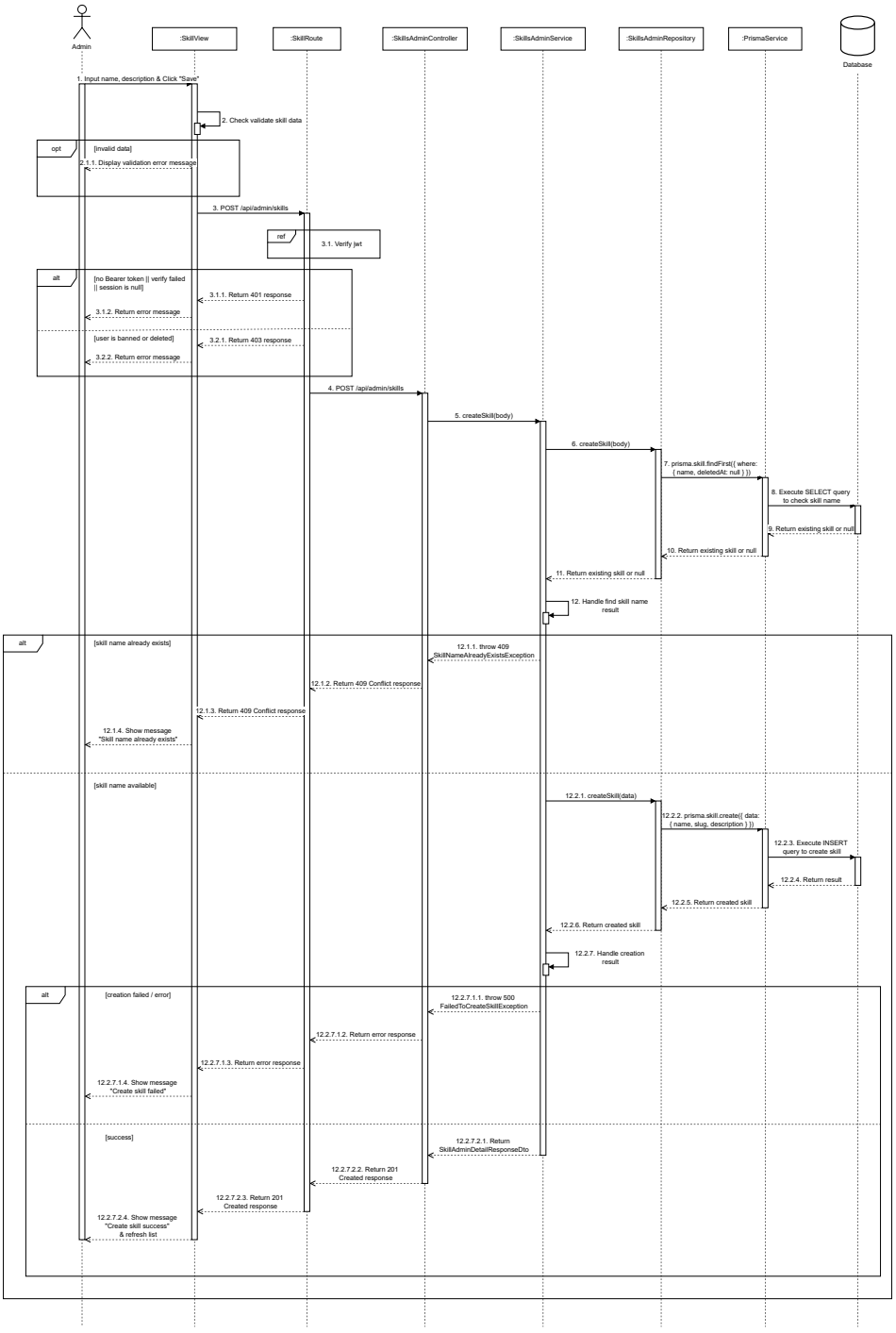

Figure 184: Sequence diagram of UC-38.03 Create Skill

3.33.5 UC-38.04 – Update Skill

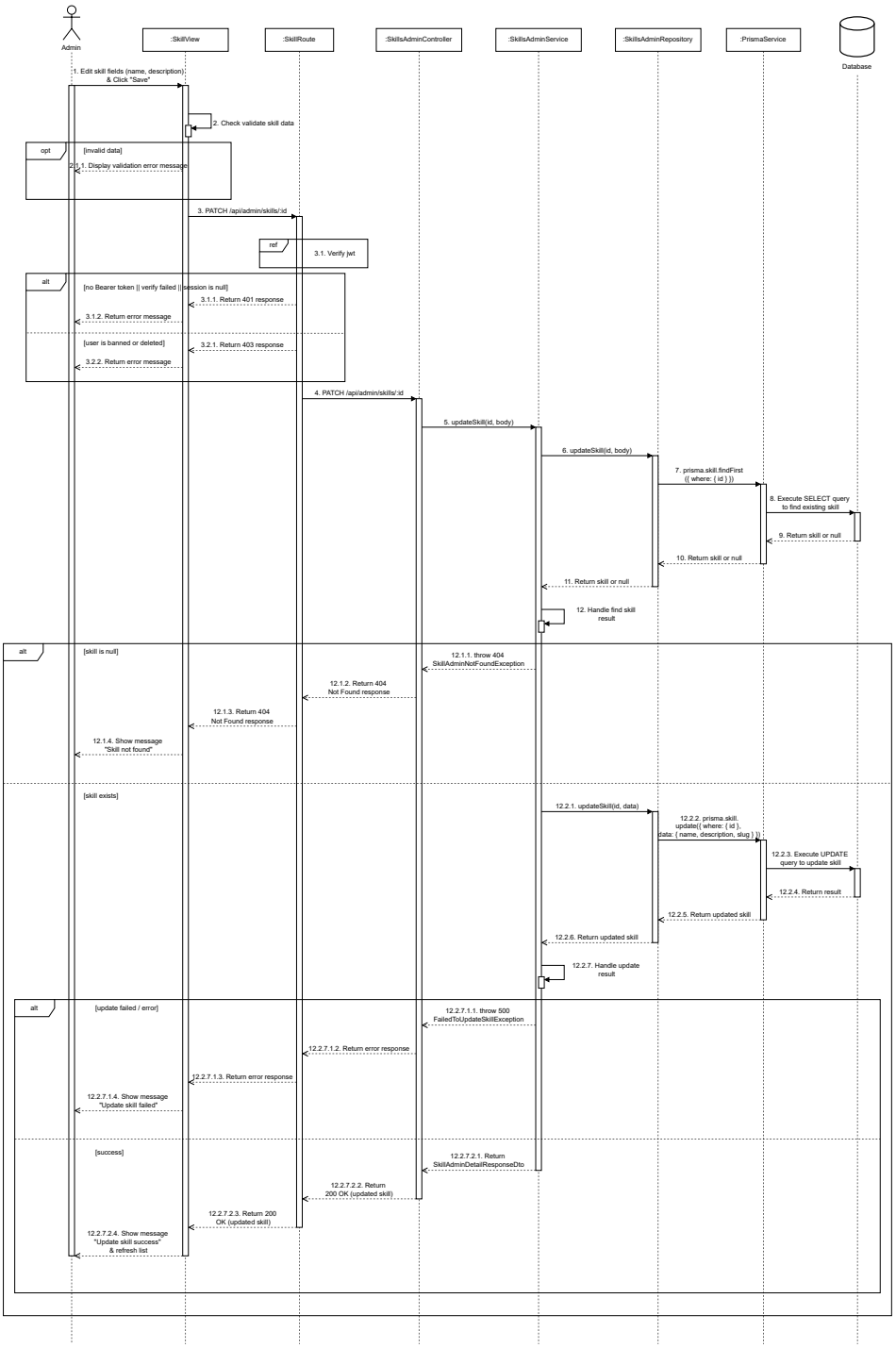

Figure 185: Sequence diagram of UC-38.04 Update Skill

3.33.6 UC-38.05 – Delete Skill

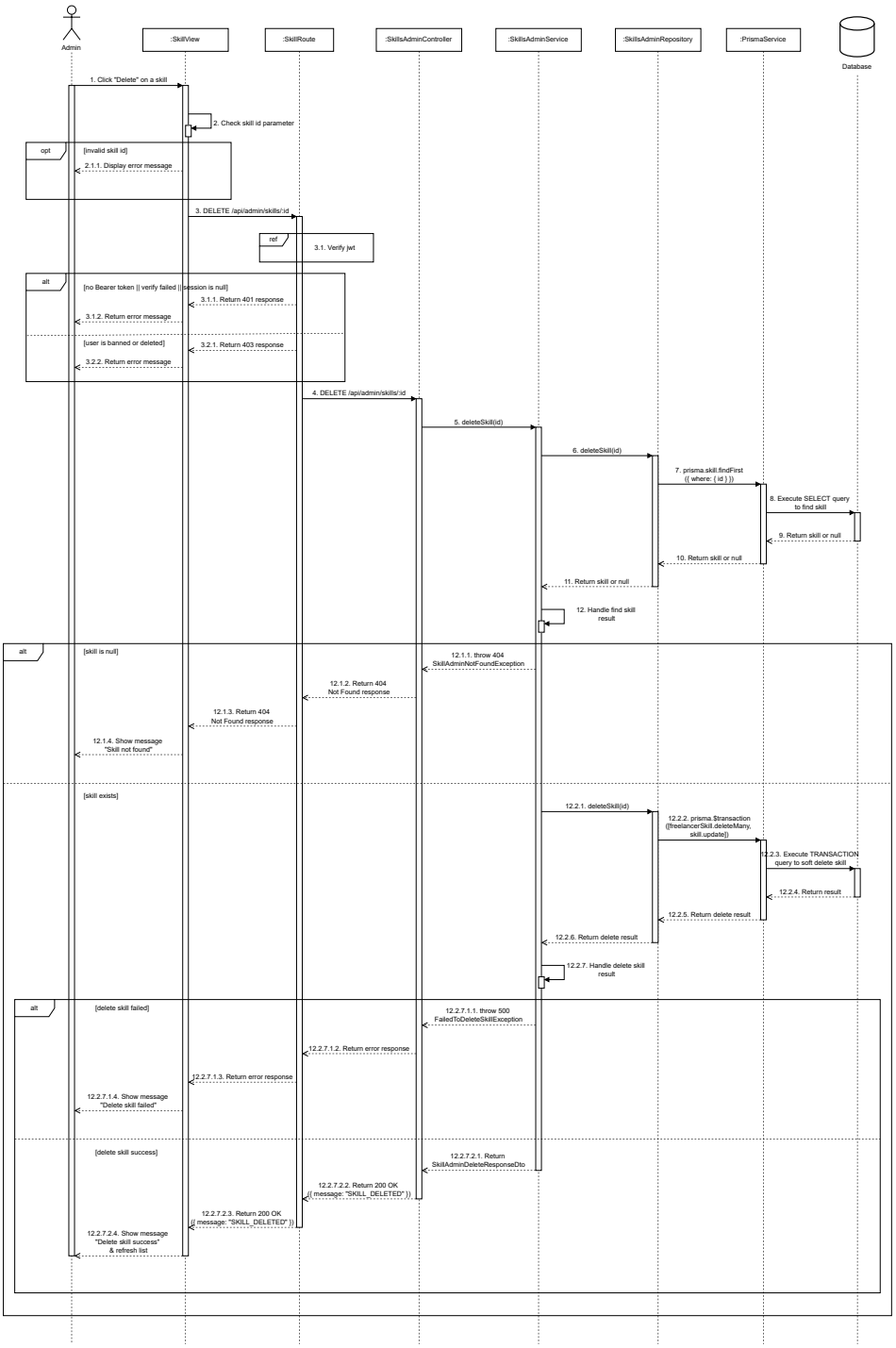

Figure 186: Sequence diagram of UC-38.05 Delete Skill

3.33.7 UC-38.06 – Restore Skill

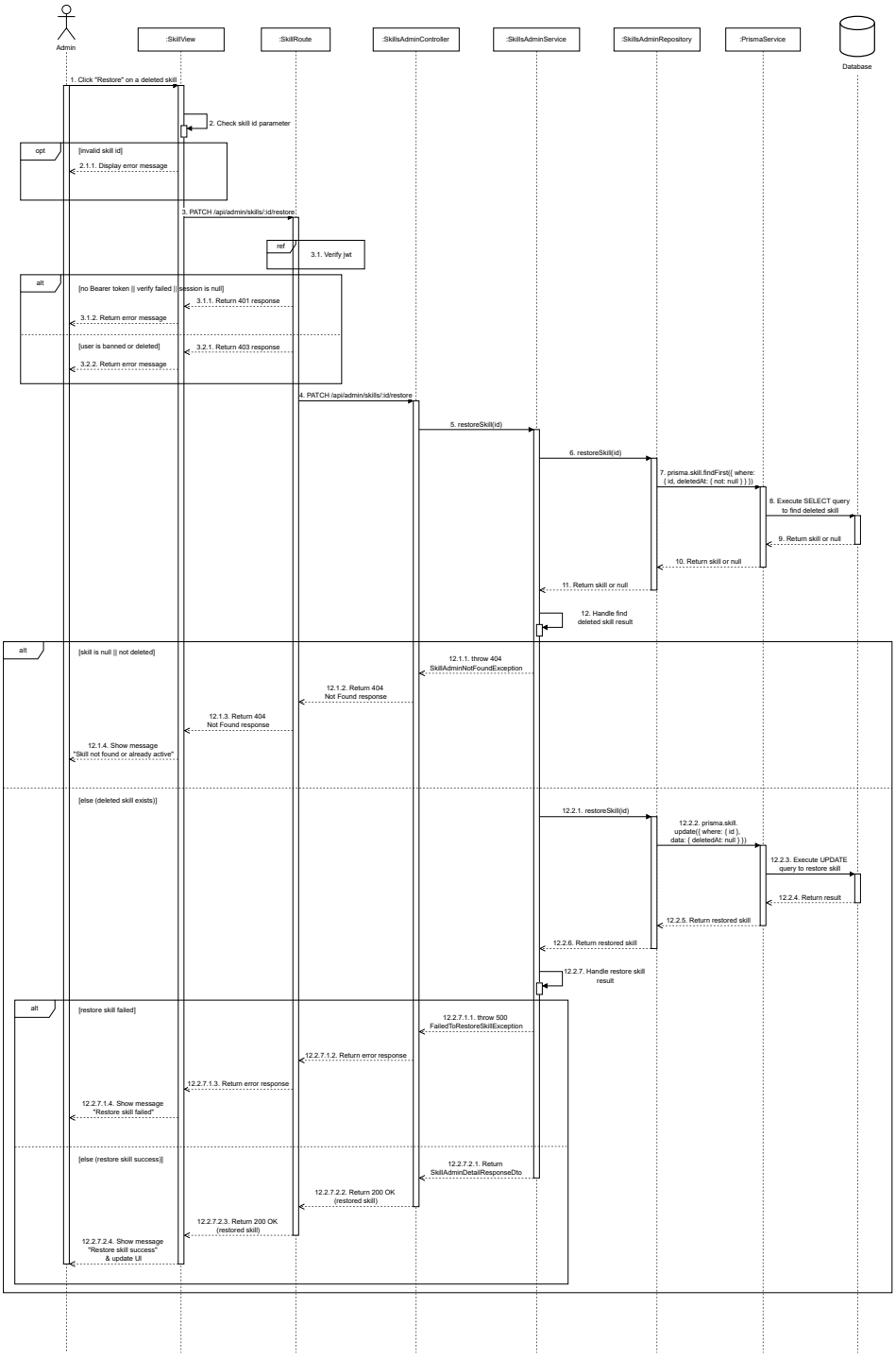

Figure 187: Sequence diagram of UC-38.06 Restore Skill

3.34 Profile Update Review

Administrators inspect pending profile revisions, compare current and proposed data, approve or reject a request, and view processed requests.

3.34.1 Class Diagram

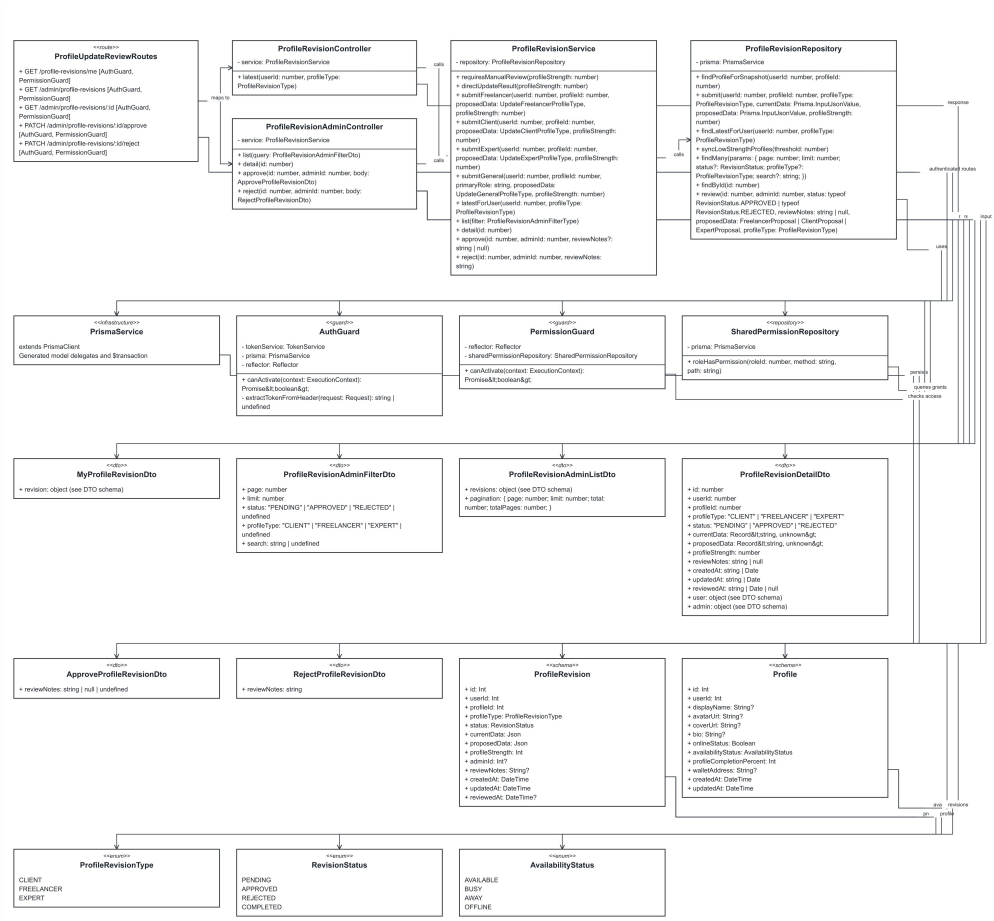

Figure 188: Class diagram of Profile Update Review

3.34.2 UC-47.01 – View Pending Profile Update Request List

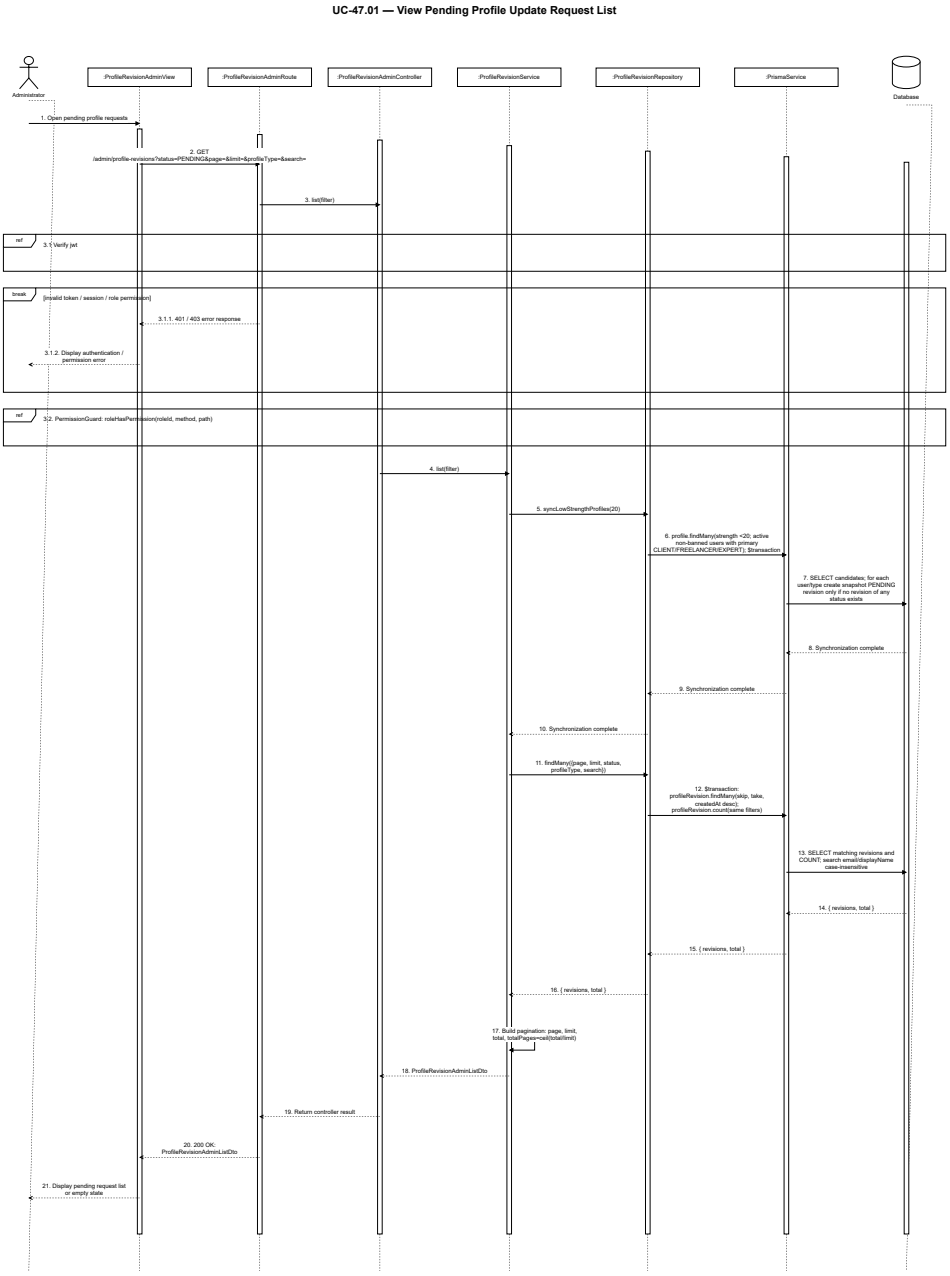

Figure 189: Sequence diagram of UC-47.01 View Pending Profile Update Request List

3.34.3 UC-47.02 – View Profile Update Request Detail

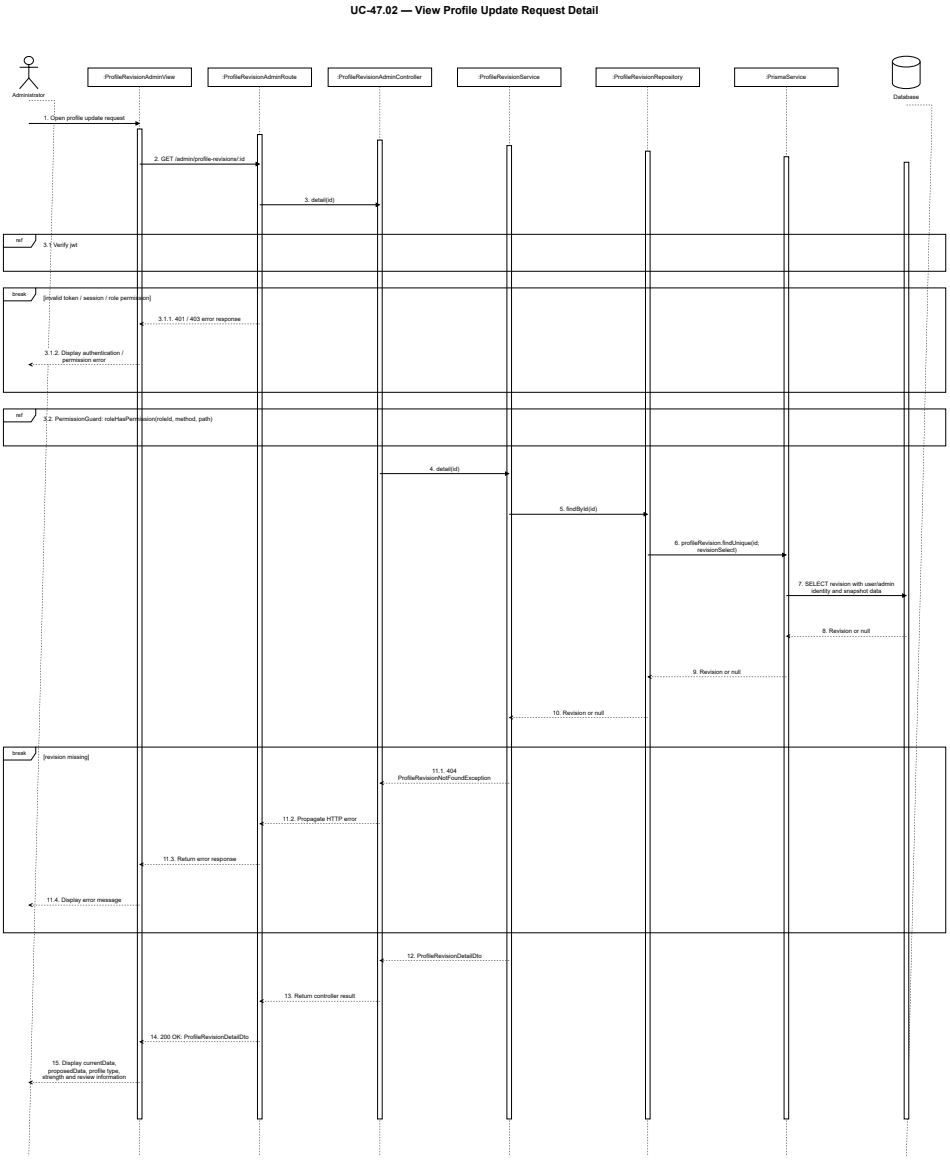

Figure 190: Sequence diagram of UC-47.02 View Profile Update Request Detail

3.34.4 UC-47.03 – Approve Profile Update Request

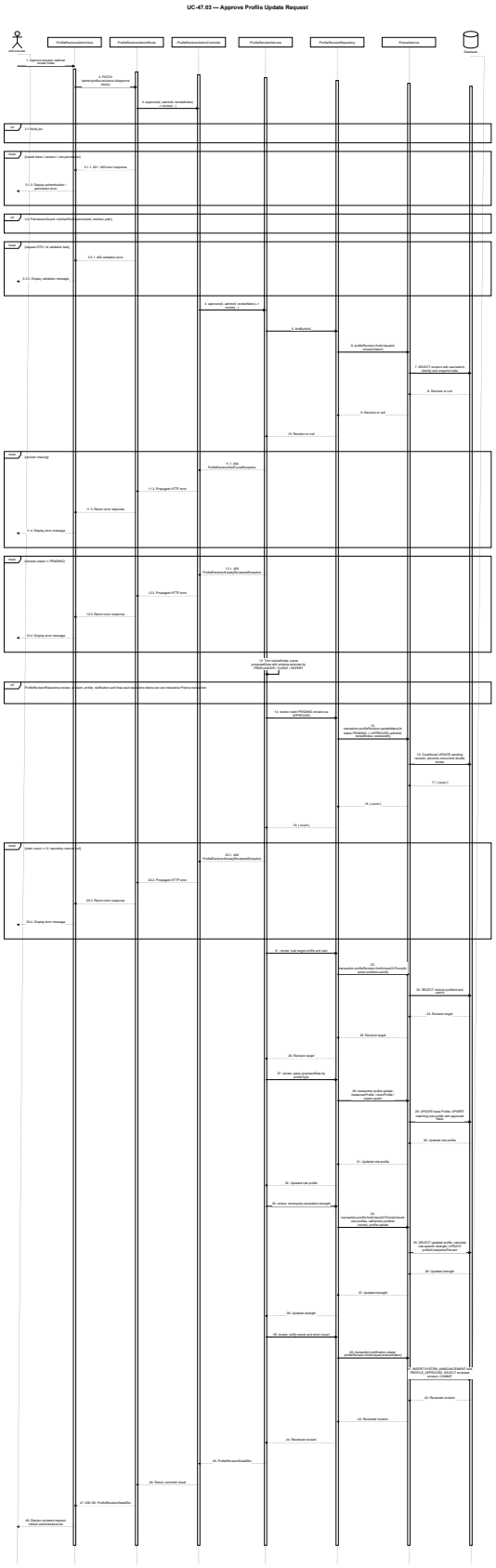

Figure 191: Sequence diagram of UC-47.03 Approve Profile Update Request

3.34.5 UC-47.04 – Reject Profile Update Request

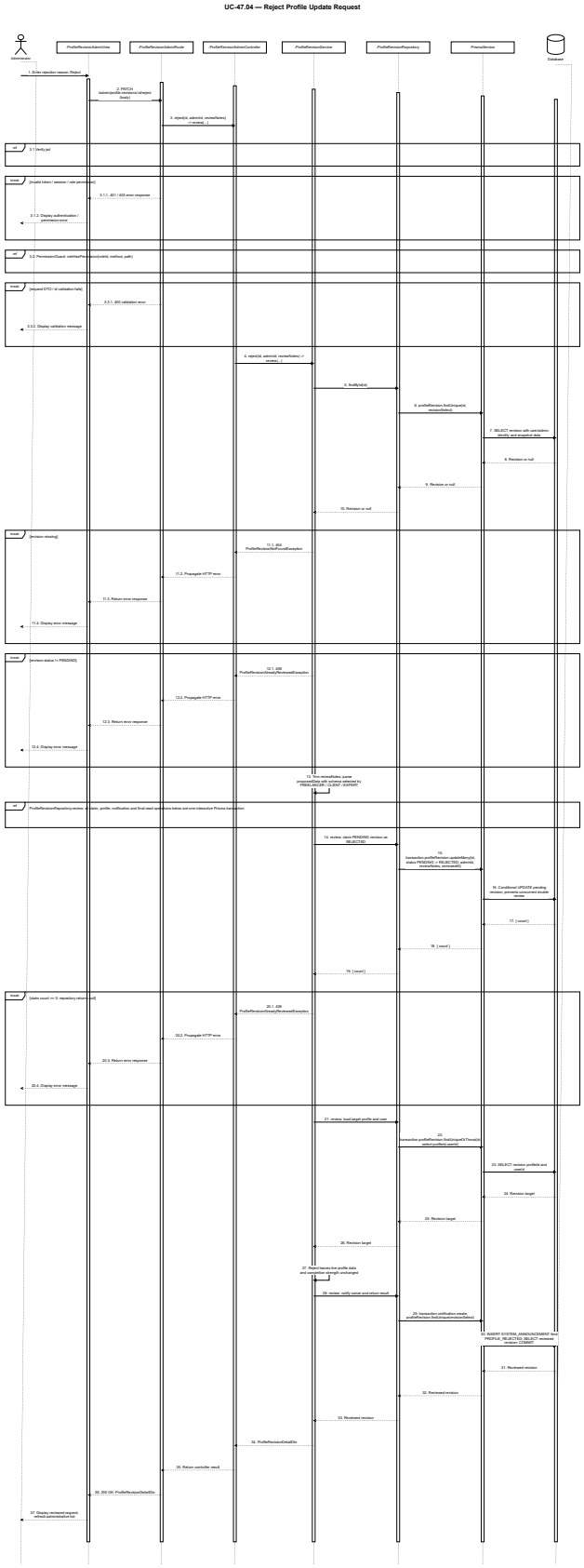

Figure 192: Sequence diagram of UC-47.04 Reject Profile Update Request

3.34.6 UC-47.05 – View Profile Update Request History

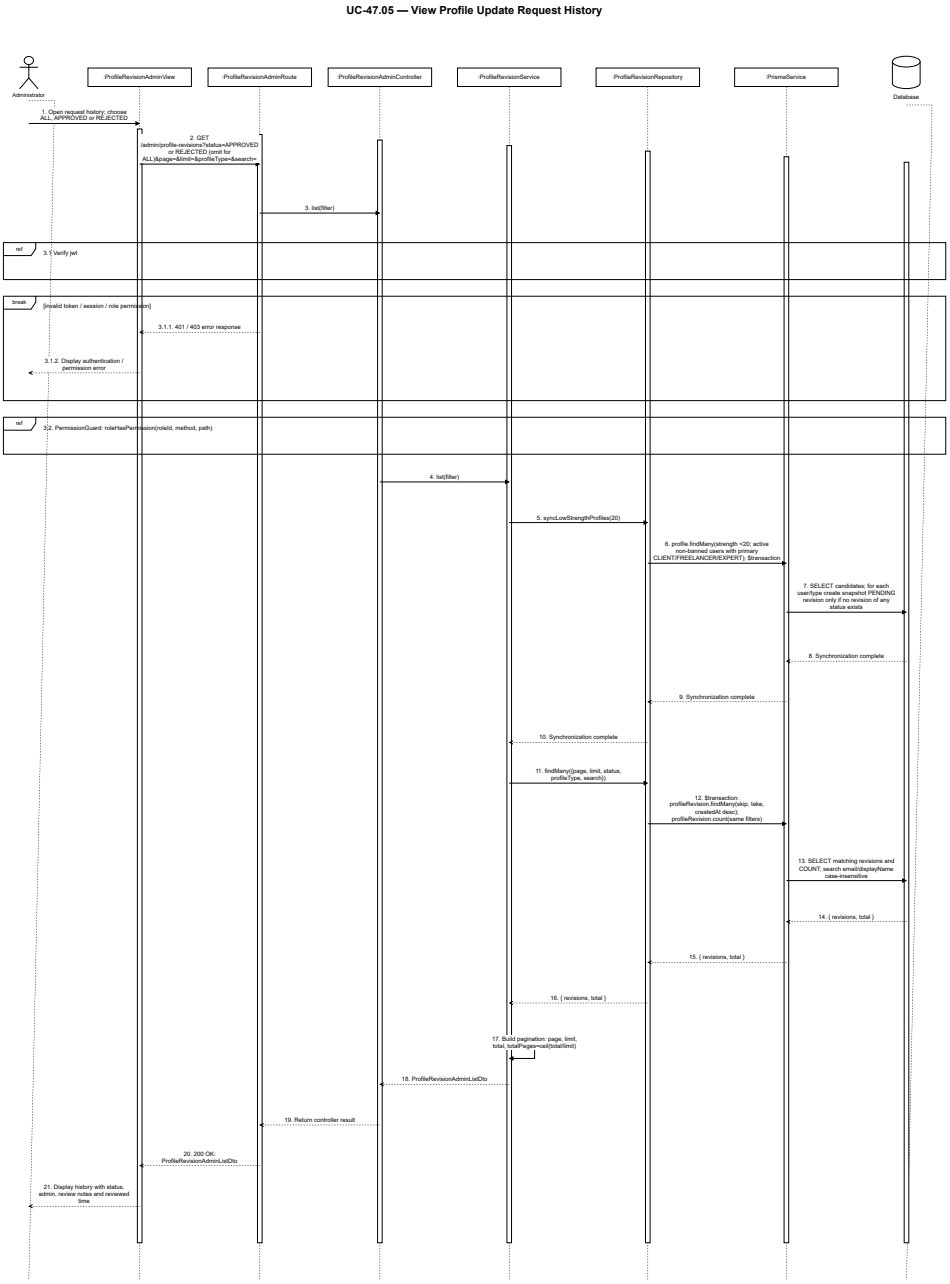

Figure 193: Sequence diagram of UC-47.05 View Profile Update Request History

3.35 Expert Profile Update

An expert retrieves and edits their own expert profile. Profile changes are routed through the revision policy when manual review is required.

3.35.1 Class Diagram

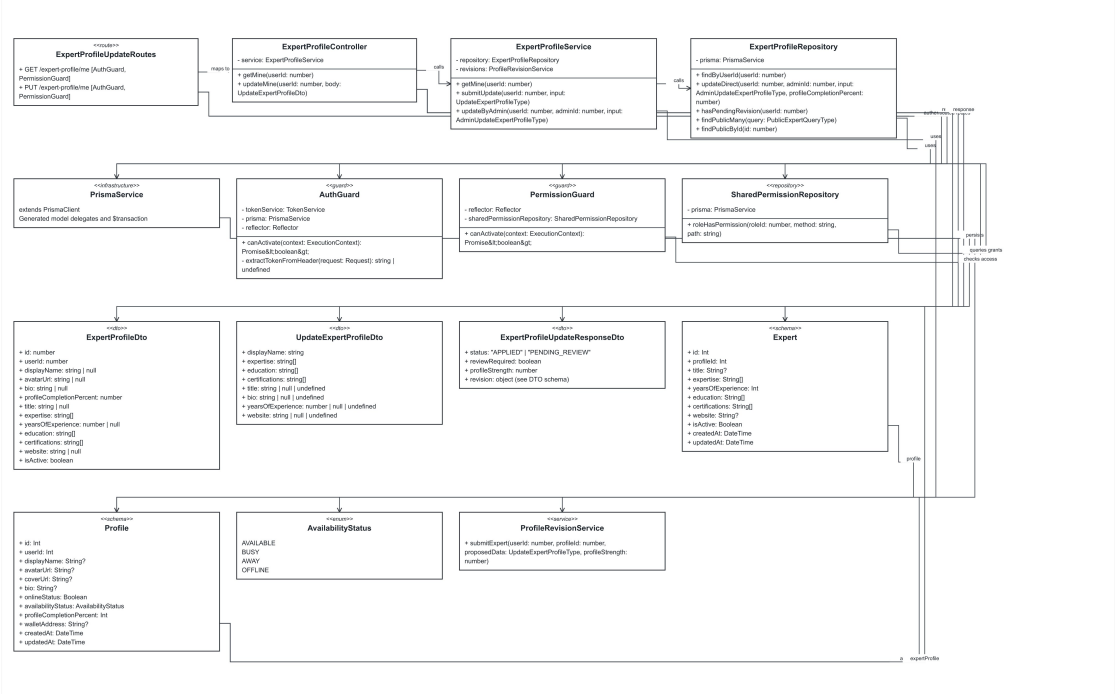

Figure 194: Class diagram of Expert Profile Update

3.35.2 UC-47.06 – Update Profile Expert

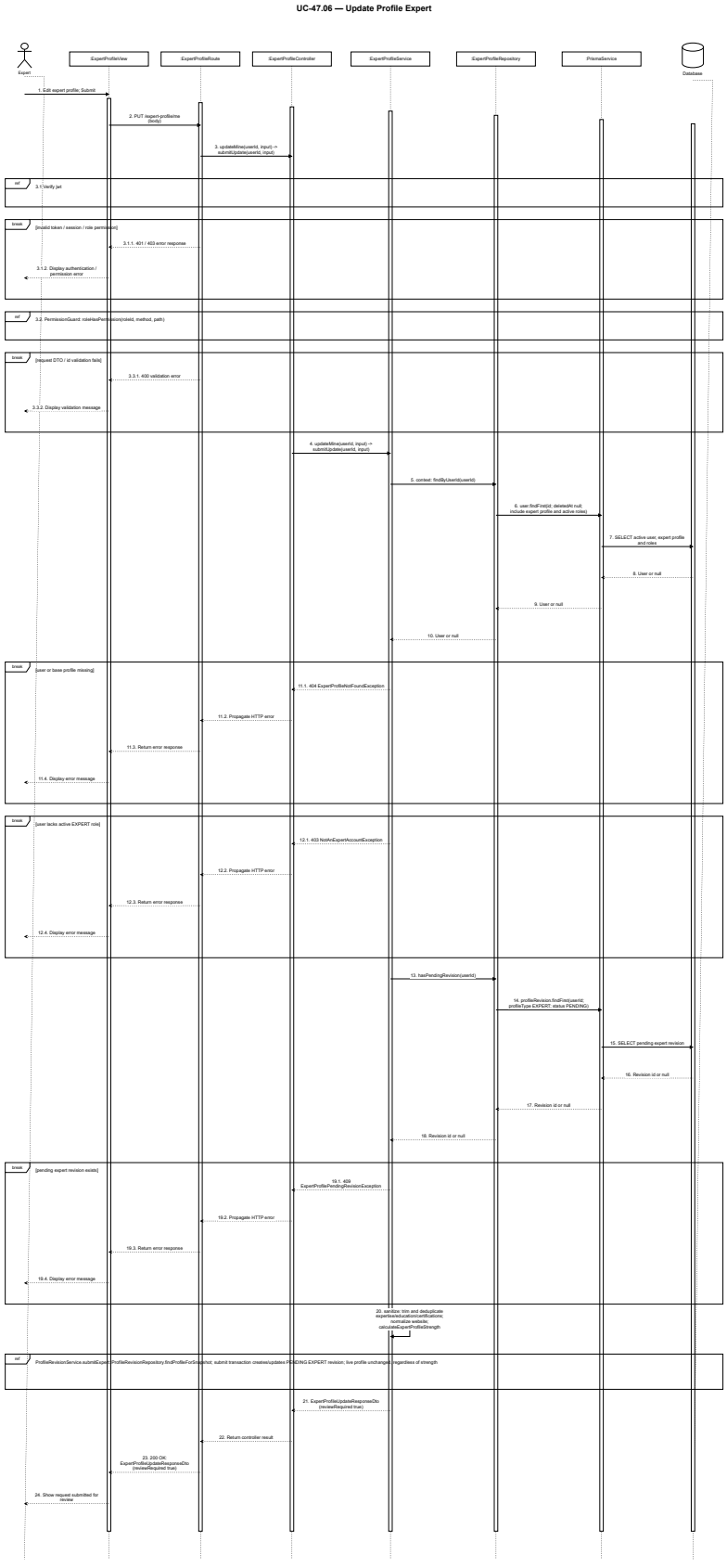

Figure 195: Sequence diagram of UC-47.06 Update Profile Expert

3.36 Freelancer CV Management

Freelancers upload, retrieve, replace, or delete the CV attached to their own profile. Optional analysis extracts skill suggestions from the stored CV.

3.36.1 Class Diagram

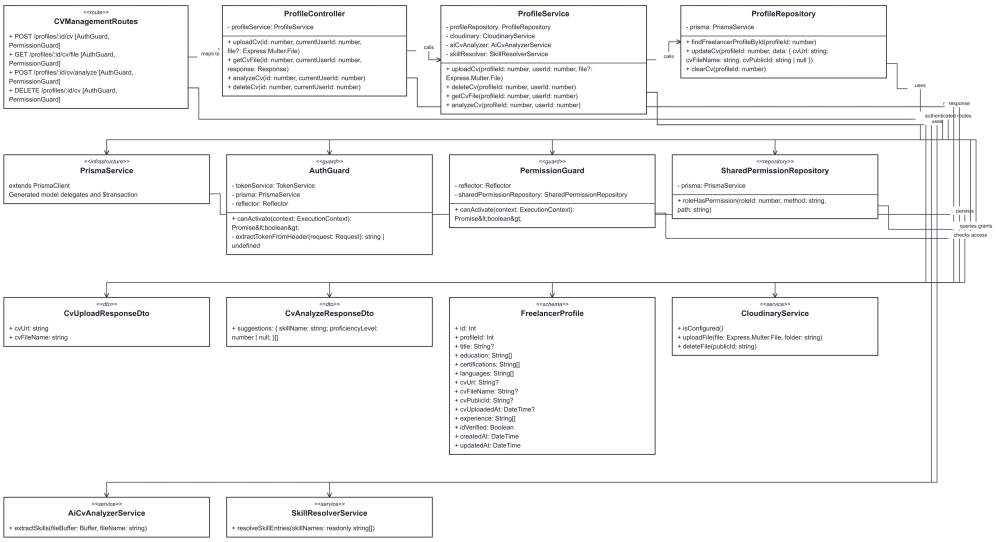

Figure 196: Class diagram of Freelancer CV Management

3.36.2 UC-07.03 – Upload CV

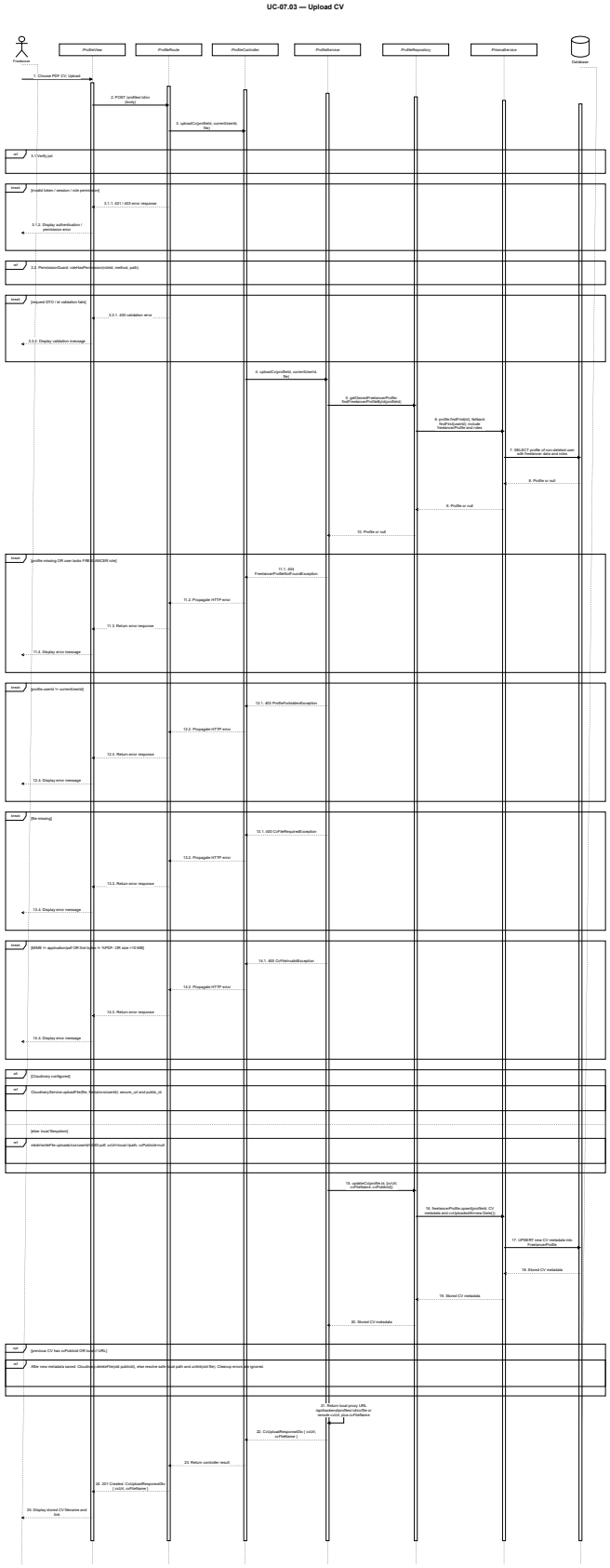

Figure 197: Sequence diagram of UC-07.03 Upload CV

3.36.3 UC-07.04 – View CV

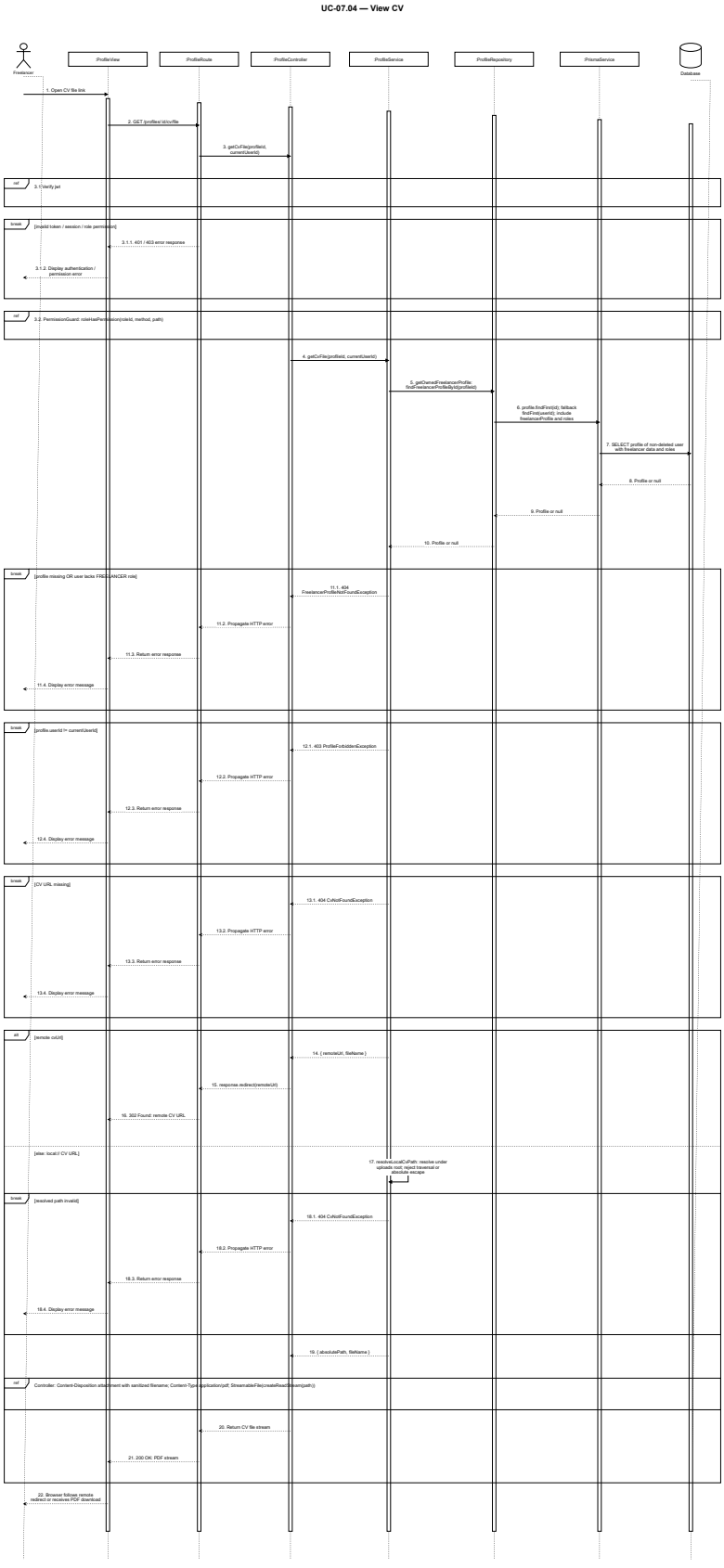

Figure 198: Sequence diagram of UC-07.04 View CV

3.36.4 UC-07.05 – Replace CV

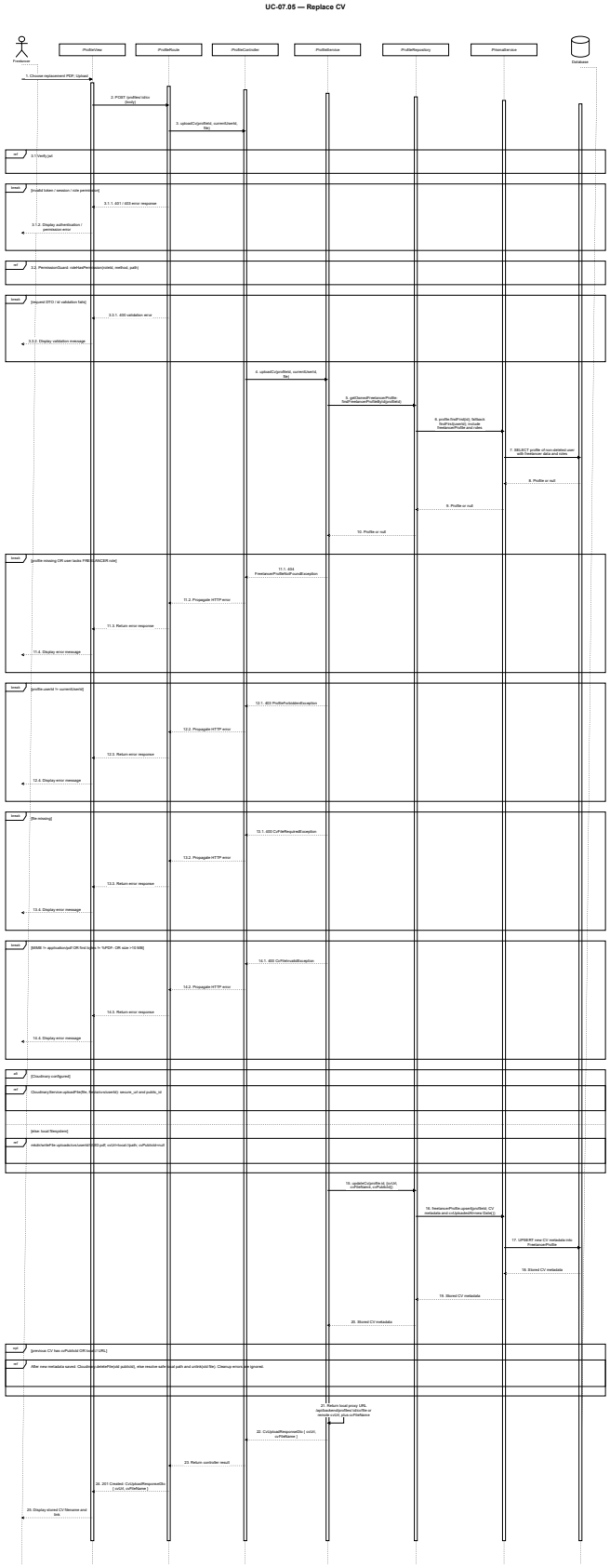

Figure 199: Sequence diagram of UC-07.05 Replace CV

3.36.5 UC-07.06 – Delete CV

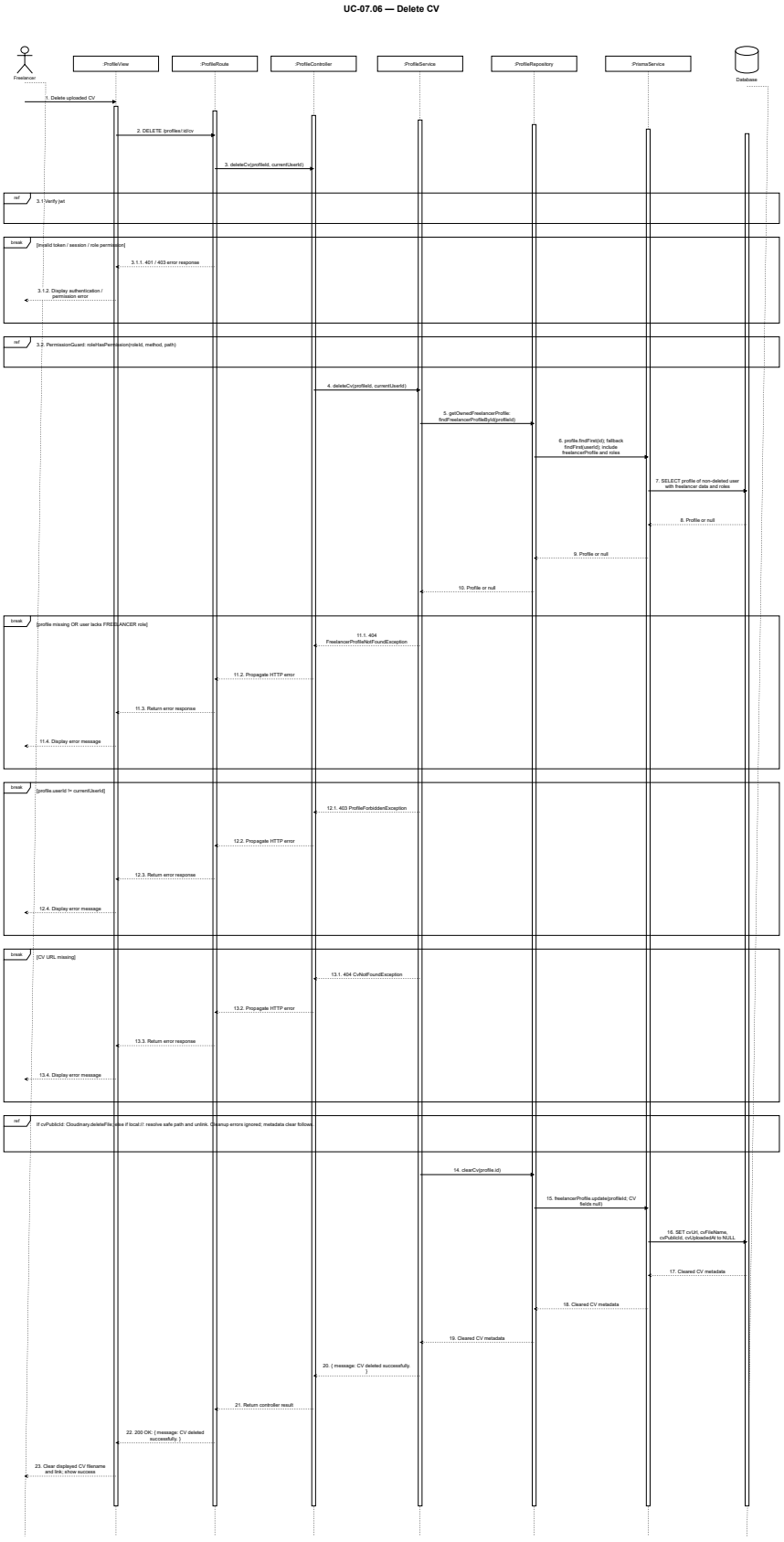

Figure 200: Sequence diagram of UC-07.06 Delete CV

3.37 Expert Profile Management

The administrator-side Expert Profile editor updates an expert account directly. Any pending self-service revision is superseded, and the account owner is notified of the administrative change.

3.37.1 Class Diagram

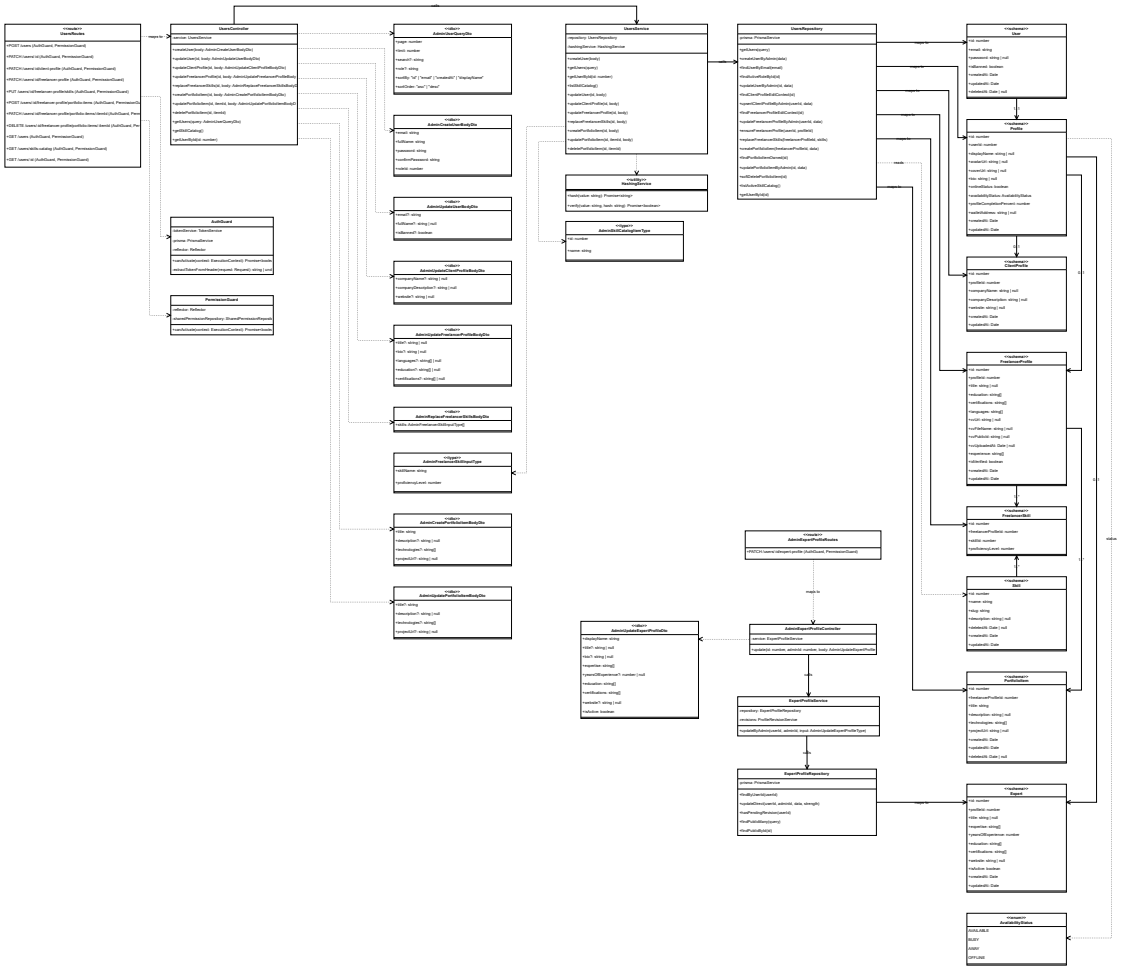

Figure 201: Class diagram of Expert Profile Management

3.37.2 UC-28.09 – Update Expert Profile

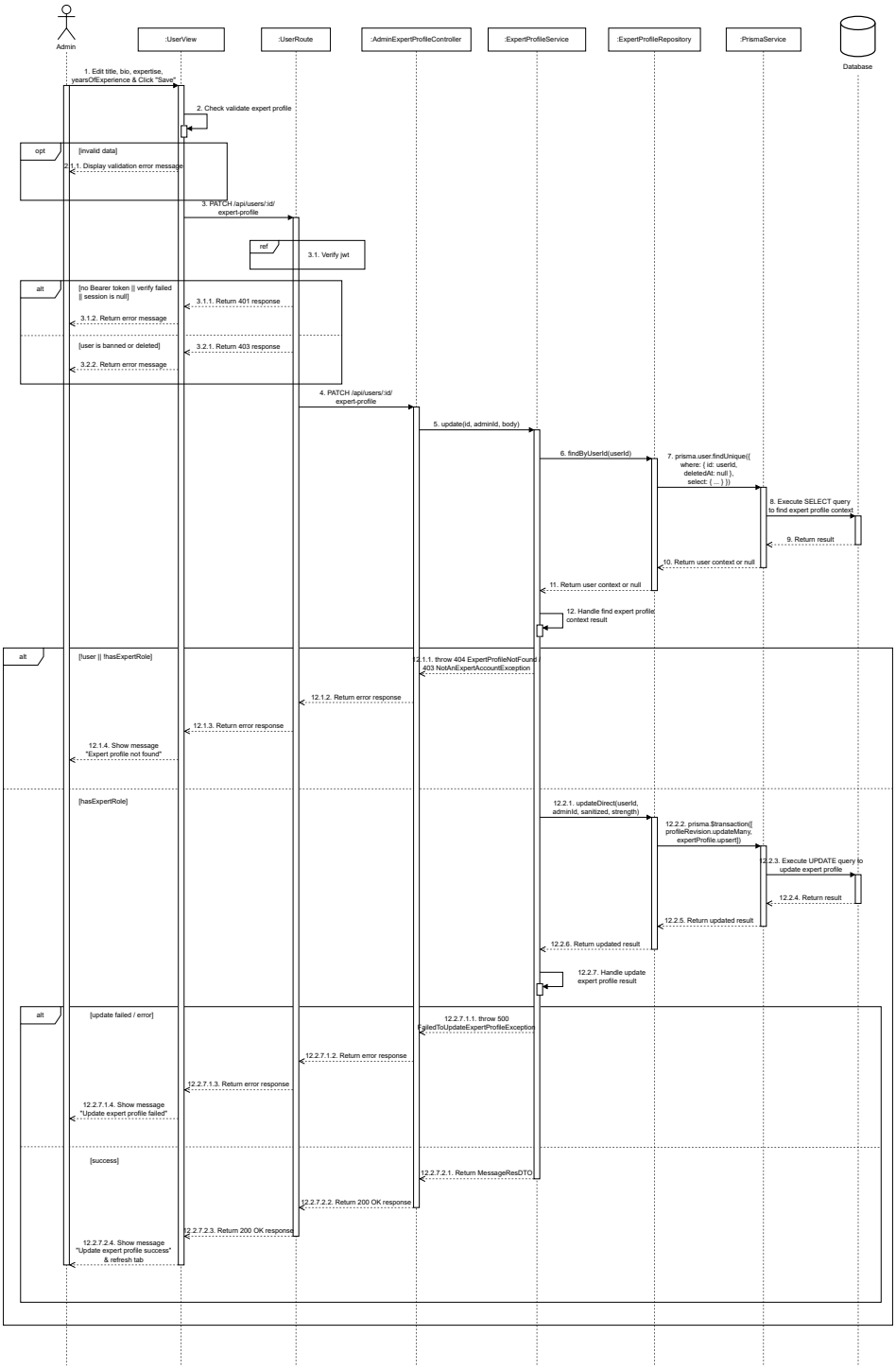

Figure 202: Sequence diagram of UC-28.09 Update Expert Profile

3.38 Job Category Management

Job Category Management maintains the taxonomy used by job postings. A category can be created, updated, soft-deleted, and restored; deletion is blocked while active jobs still reference the category.

3.38.1 Class Diagram

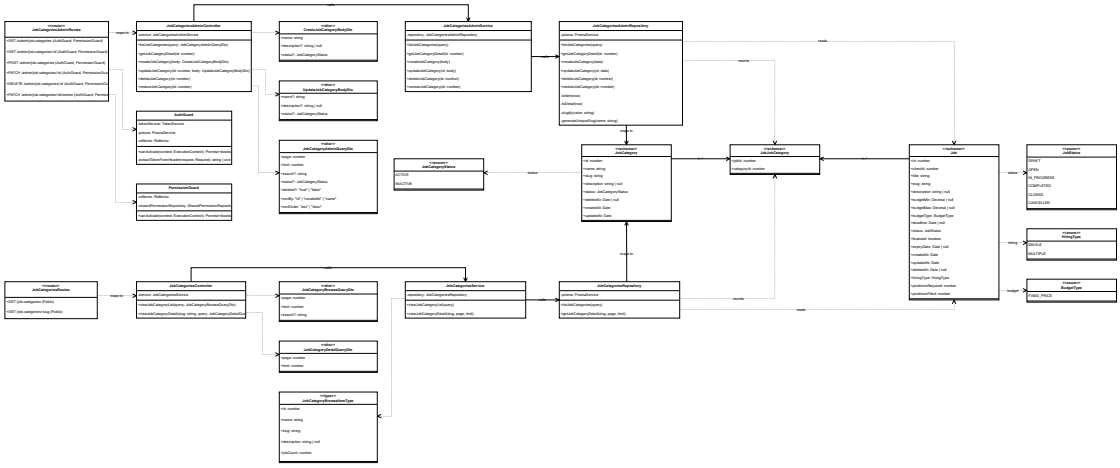

Figure 203: Class diagram of Job Category Management

3.38.2 UC-46.01 – View Job Category List

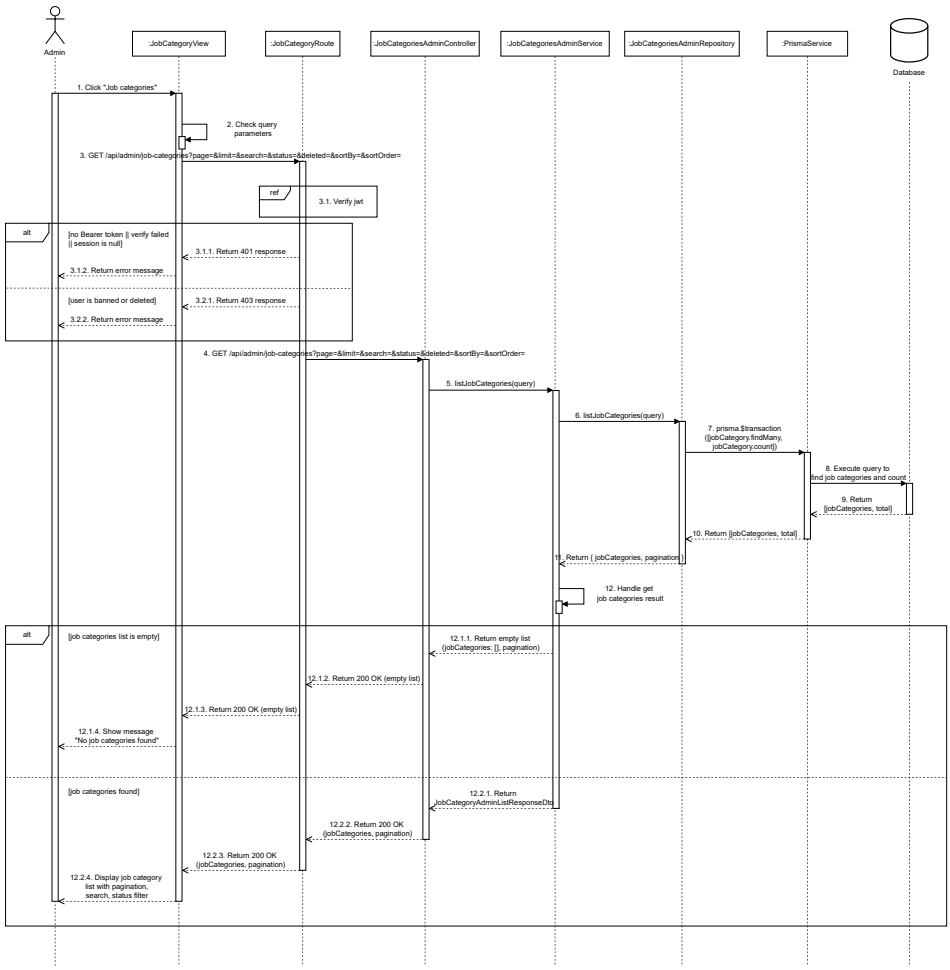

Figure 204: Sequence diagram of UC-46.01 View Job Category List

3.38.3 UC-46.02 – View Job Category Detail

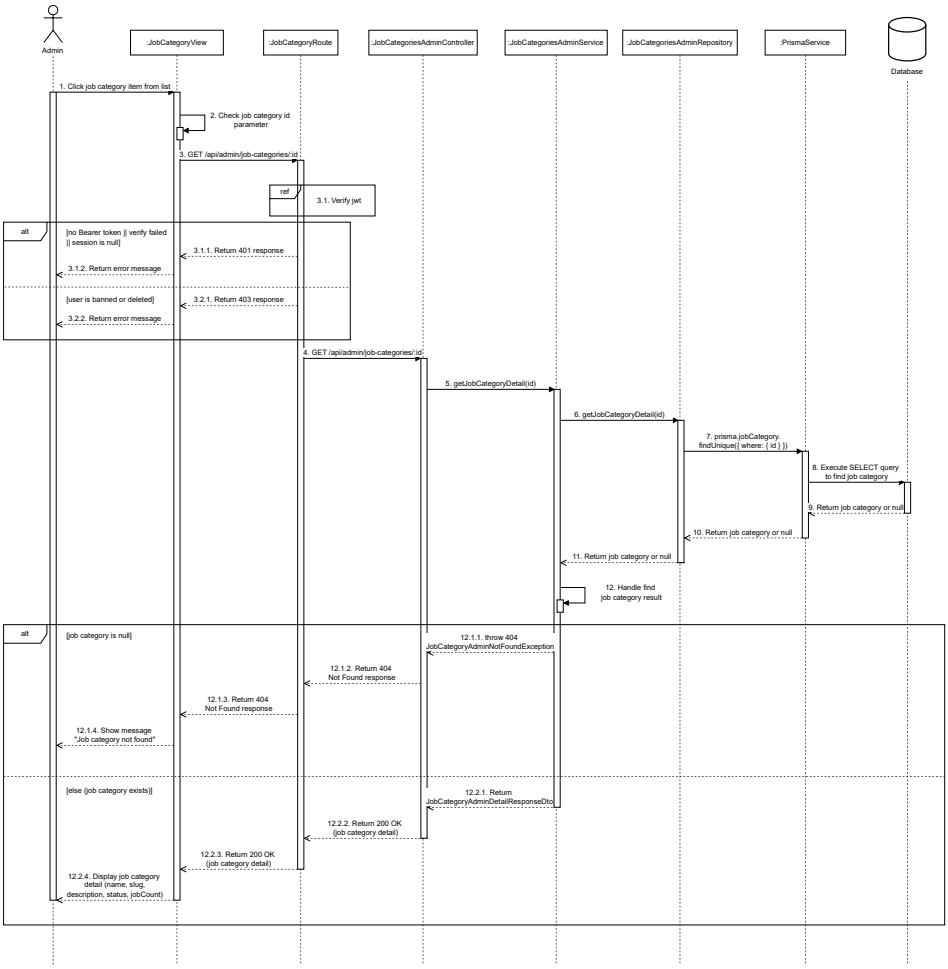

Figure 205: Sequence diagram of UC-46.02 View Job Category Detail

3.38.4 UC-46.03 – Create Job Category

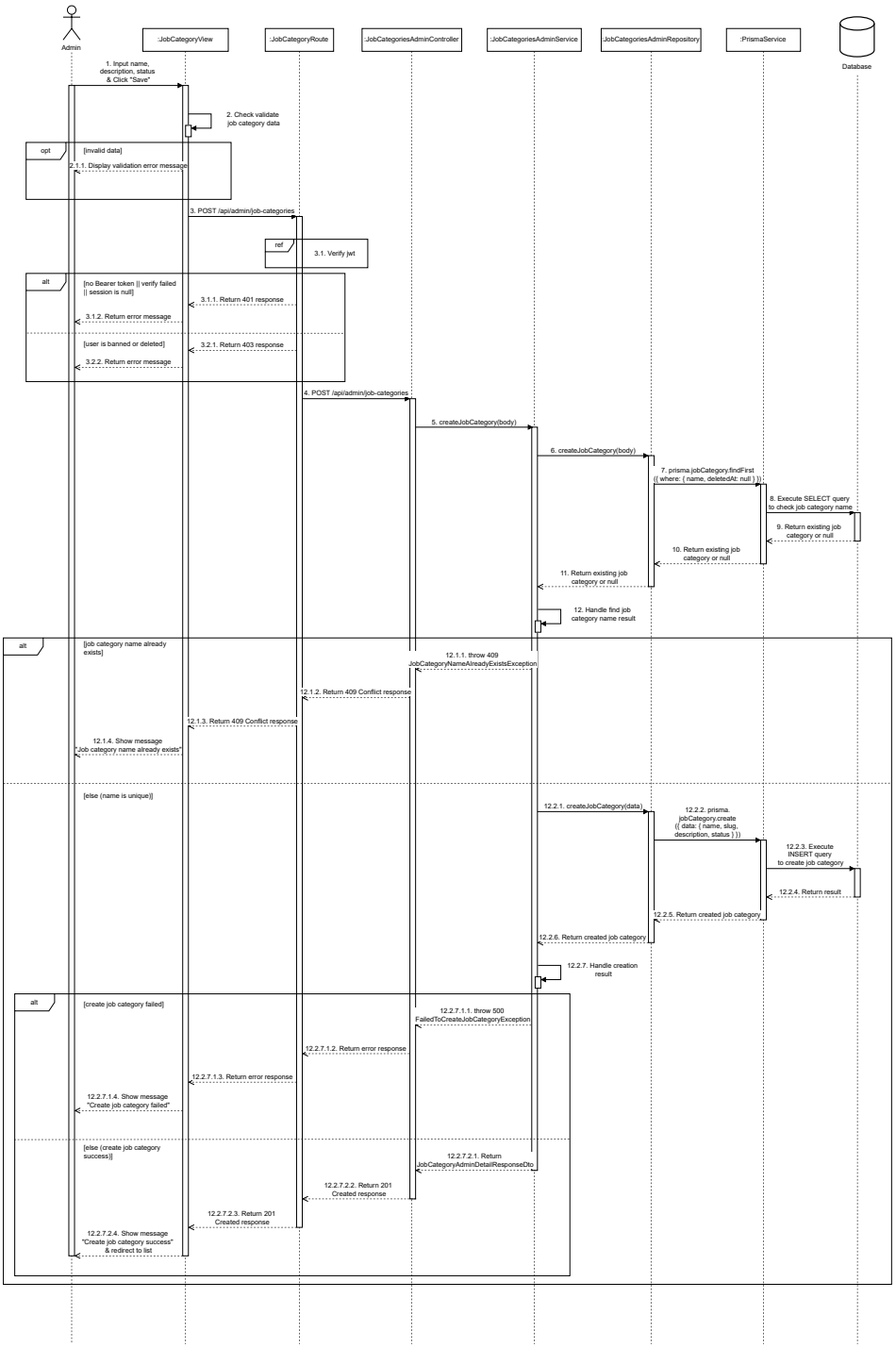

Figure 206: Sequence diagram of UC-46.03 Create Job Category

3.38.5 UC-46.04 – Update Job Category

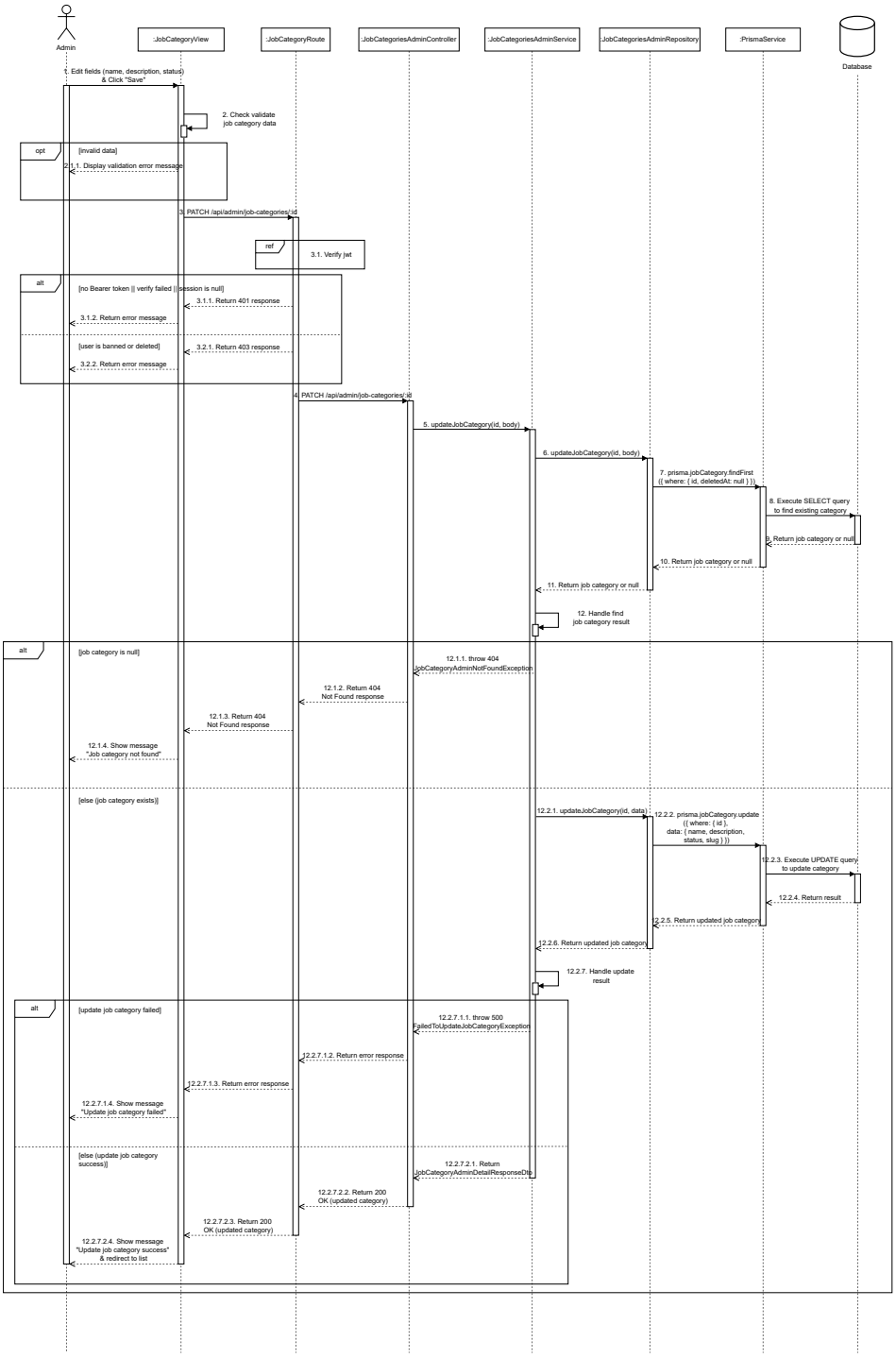

Figure 207: Sequence diagram of UC-46.04 Update Job Category

3.38.6 UC-46.05 – Delete Job Category

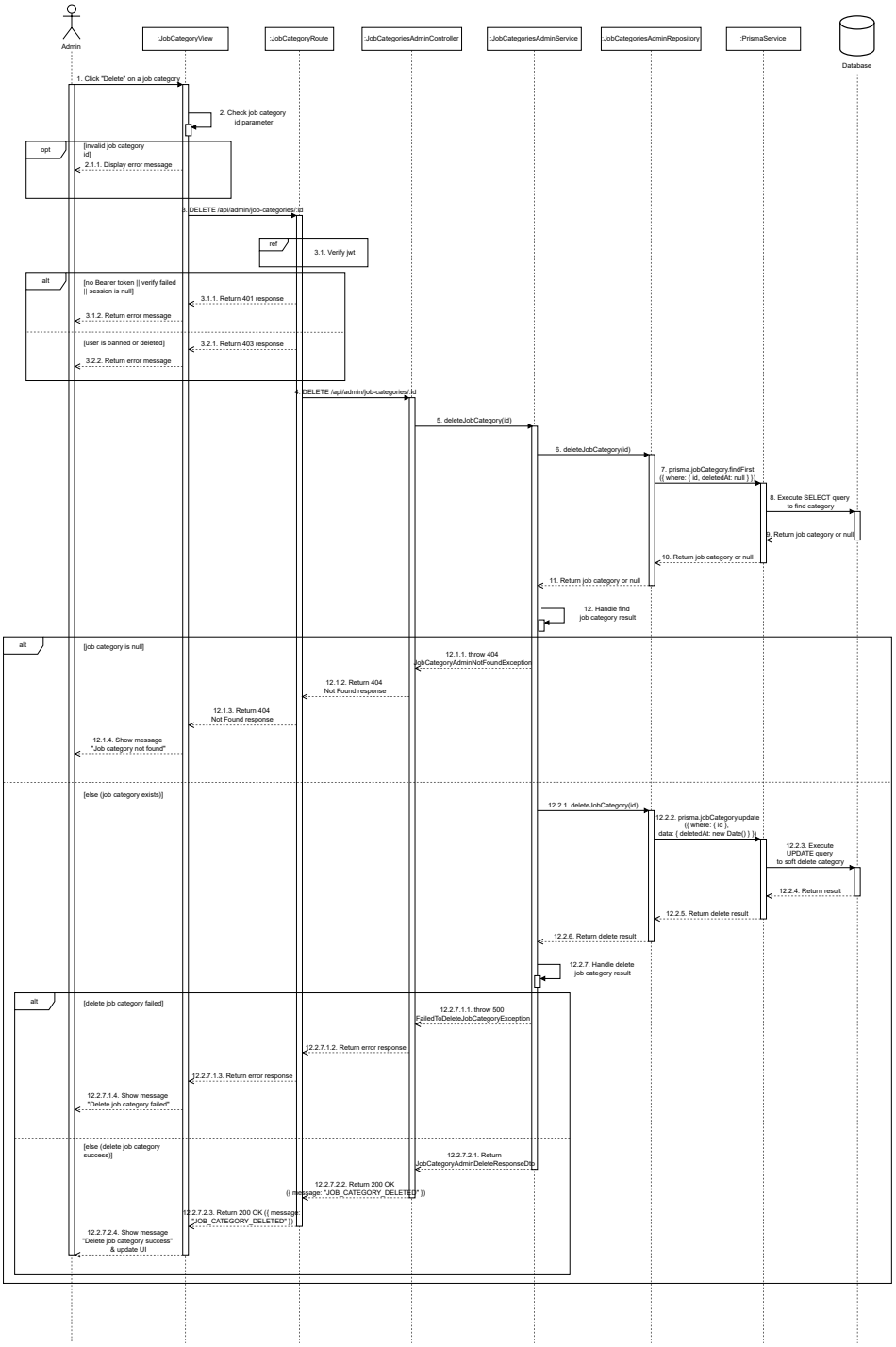

Figure 208: Sequence diagram of UC-46.05 Delete Job Category

4. Class Specification

4.1 AuthGuard Class

4.1.1 ClassMethod

No	Method	Description
01	`AuthGuard( reflector: Reflector, tokenService: TokenService, sessionRepo: SessionRepository)`	A method to initialize the authentication guard with dependency services. – `reflector`: `Reflector` – metadata reflector to check if a route is marked public. – `tokenService`: `TokenService` – token decoding and verification utility. – `sessionRepo`: `SessionRepository` – repository to query session active status. – *Returns:* `void` – constructor.
02	`canActivate(context: ExecutionContext)`	A method to inspect the execution context, extract the Bearer token, verify its validity and expiration, and verify that the session has not been revoked. – `context`: `ExecutionContext` – execution context containing HTTP request/response. – *Returns:* `Promise<boolean>` – `true` if authentication succeeds, throws `UnauthorizedException (401)` otherwise.
03	`extractTokenFromHeader( request: Request)`	A helper method to extract the Bearer token from the incoming HTTP Authorization header. – `request`: `Request` – current Express request object. – *Returns:* `string | null` – extracted raw token string, or `null` if not found.

4.1.2 ClassAttribute

No	Attribute	Type
01	`reflector`	`Reflector`
02	`tokenService`	`TokenService`
03	`sessionRepo`	`SessionRepository`

4.2 AuthController Class

4.2.1 ClassMethod

No	Method	Description
01	`AuthController(auth Service: AuthService)`	A method to initialize the controller with the authentication service. – `authService`: `AuthService` – service that owns the authentication use cases. – *Returns:* `void` – constructor, no return value.
02	`register(body: Register Dto)`	A method to create an email/password account after OTP verification. – `body`: `RegisterDto` – email, OTP code, password, full name, and initial role. – *Returns:* `Promise<RegisterResponseType>` – the created account without password fields. – Route: `POST /auth/register`; public endpoint.
03	`sendOTP(body: SendOTP BodyDTO)`	A method to generate and email an OTP for registration or password reset. – `body`: `SendOTPBodyDTO` – email and verification-code type. – *Returns:* `Promise<MessageResType>` – a success message. – Route: `POST /auth/otp`; public endpoint.
04	`login( body: LoginBodyDTO, userAgent: string, ip Address: string)`	A method to authenticate an email/password account and create a session. – `body`: `LoginBodyDTO` – email and password. – `userAgent`: `string` – device information used as session metadata. – `ipAddress`: `string` – client IP stored with the session. – *Returns:* `Promise<LoginResponseType>` – access and refresh tokens. – Route: `POST /auth/login`; public endpoint.
05	`refreshToken( body: RefreshTokenBodyDT O, userAgent: string, ip Address: string)`	A method to rotate a valid refresh token into a new token pair. – `body.refreshToken`: `string` – the current refresh token. – `userAgent`: `string` – device information for the replacement session. – `ipAddress`: `string` – client IP for the replacement session. – *Returns:* `Promise<LoginResponseType>` – the new access and refresh tokens. – Route: `POST /auth/refresh-token`; public endpoint.

No	Method	Description
06	`logout(body: LogoutBodyDTO)`	A method to terminate the session identified by a refresh token. – `body.refreshToken`: `string` – refresh token of the session to remove. – *Returns:* `Promise<MessageResType>` – a success message. – Route: `POST /auth/logout`; public endpoint.
07	`forgotPassword(body: ForgotPasswordBodyDTO)`	A method to reset a password after validating a password-reset OTP. – `body`: `ForgotPasswordBodyDTO` – email, OTP code, and new password. – *Returns:* `Promise<MessageResType>` – a success message. – Route: `POST /auth/forgot-password`; public endpoint.
08	`getGoogleLink( userAgent: string, ip: string)`	A method to build the Google OAuth consent URL. – `userAgent`: `string` – encoded in the OAuth state. – `ip`: `string` – encoded in the OAuth state. – *Returns:* `GetAuthorizationUrlResType` – the authorization URL. – Route: `GET /auth/google-link`; public endpoint.
09	`googleCallback( code: string, state: string, res: Response)`	A method to exchange the Google authorization code and redirect the browser. – `code`: `string` – authorization code returned by Google. – `state`: `string` – base64url payload with device and IP information. – `res`: `Response` – Express response used for the redirect. – *Returns:* `Promise<void>` – redirects to the web application with tokens or an error flag. – Route: `GET /auth/google/callback`; public endpoint.
10	`getMe(userId: number)`	A method to return the authenticated account and its roles. – `userId`: `number` – current user injected by `@UserActive('userId')`. – *Returns:* `Promise<GetMeResType>` – account, profile summary, and roles. – Route: `GET /auth/me`; authenticated endpoint.

No	Method	Description
11	`switchRole( userId: number, sessionId: number, body: SwitchRoleBodyDto)`	A method to make an already assigned role the primary role. – `userId`: `number` – current user. – `sessionId`: `number` – current session identifier. – `body.role`: role name to activate. – *Returns:* `Promise<SwitchRoleResponseType>` – a new access token for that role. – Route: `POST /auth/switch-role`; authenticated endpoint.
12	`joinRole( userId: number, sessionId: number, body: JoinRoleBodyDto)`	A method to assign an additional Client or Freelancer role. – `userId`: `number` – current user. – `sessionId`: `number` – current session identifier. – `body.role`: role name to join. – *Returns:* `Promise<JoinRoleResponseType>` – a new access token for the joined role. – Route: `POST /auth/join-role`; authenticated endpoint.

4.2.2 ClassAttribute

No	Attribute	Type
01	`authService`	`AuthService` – private readonly authentication service.
02	`logger`	`Logger` – private readonly NestJS logger for OAuth callback errors.

4.3 AuthService Class

4.3.1 ClassMethod

No	Method	Description
01	`AuthService( hashingService: HashingService, emailService: EmailService, sharedRoleRepository: SharedRoleRepository, authRepository: AuthRepository, tokenService: TokenService)`	A method to initialize the service and its Google OAuth client. – `hashingService`: `HashingService` – password hashing and verification. – `emailService`: `EmailService` – OTP delivery. – `sharedRoleRepository`: `SharedRoleRepository` – resolves system role identifiers. – `authRepository`: `AuthRepository` – account, OTP, session, and OAuth persistence. – `tokenService`: `TokenService` – JWT signing and verification. – *Returns:* `void` – constructor, no return value.
02	`register(body: RegisterBodyType)`	A method to validate registration data and create an account. – `body`: `RegisterBodyType` – email, OTP, password, full name, and role. – *Returns:* `Promise<UserType>` – the created account without sensitive fields. – Throws `EmailAlreadyExistsException`, `UserBannedException`, `InvalidVerificationCodeException`, or `OTPExpiredException`.
03	`sendOTP(body: SendOTPBodyType)`	A method to issue an OTP while enforcing the attempt window. – `body`: `SendOTPBodyType` – email and verification-code type. – *Returns:* `Promise<MessageResType>` – a success message. – Throws `EmailAlreadyExistsException`, `EmailNotFoundException`, or `TooManyAttemptsException`.
04	`login(body: LoginBodyType and session metadata)`	A method to verify credentials, ban state, and the primary role before issuing tokens. – `body`: `LoginBodyType` – email, password, user agent, and IP address. – *Returns:* `Promise<{ accessToken: string; refreshToken: string }>` – the token pair. – Throws `IncorrectEmailException`, `IncorrectPasswordException`, `UserBannedException`, or `RoleNotFoundException`.

No	Method	Description
05	`generateAccessAndRefreshTokens(payload)`	A private method to sign a refresh token, persist a session, and sign the access token. – `payload`: object – user id, role id, role name, user agent, and IP address. – *Returns:* `Promise<{ accessToken: string; refreshToken: string }>` – the new token pair. – The access token carries the persisted session id.
06	`refreshToken(payload)`	A method to validate and rotate a refresh token. – `payload`: object – refresh token, user agent, and IP address. – *Returns:* `Promise<{ accessToken: string; refreshToken: string }>` – the replacement token pair. – Throws `RefreshTokenRevokedException` or `UserBannedException`.
07	`logout(refreshToken: string)`	A method to remove the session that owns a refresh token. – `refreshToken`: `string` – token of the session to terminate. – *Returns:* `Promise<MessageResType>` – a success message. – Throws `RefreshTokenRevokedException` when the session is absent.
08	`forgotPassword(payload: ForgotPasswordBodyType)`	A method to validate a reset OTP, hash the new password, and remove the OTP. – `payload`: `ForgotPasswordBodyType` – email, OTP, and new password. – *Returns:* `Promise<MessageResType>` – a success message. – Throws `EmailNotFoundException`, `UserBannedException`, or OTP validation errors.
09	`validateVerificationCode(uniqueValue)`	A method to load an OTP and reject missing or expired codes. – `uniqueValue`: object – email, code, and verification type. – *Returns:* `Promise<EmailVerificationType>` – the matching OTP record. – Throws `InvalidVerificationCodeException` or `OTPExpiredException`.

No	Method	Description
10	`getMe(userId: number)`	A method to return the current account, profile summary, and roles. – `userId`: `number` – current account identifier. – *Returns:* `Promise<GetMeResType>` – the authenticated account view. – Throws `NotFoundException` when absent or `UserBannedException` when banned.
11	`switchRole(userId, sessionId, body)`	A method to change the primary role and issue a new access token. – `userId`: `number` – current account. – `sessionId`: `number` – current session. – `body`: `SwitchRoleBodyType` – target role. – *Returns:* `Promise<{ accessToken: string }>` – the role-scoped access token. – Throws `RoleNotFoundException`.
12	`joinRole(userId, sessionId, body)`	A method to assign an additional role and issue a new access token. – `userId`: `number` – current account. – `sessionId`: `number` – current session. – `body`: `JoinRoleBodyType` – role to join. – *Returns:* `Promise<{ accessToken: string }>` – the role-scoped access token. – Throws `RoleNotFoundException` or `RoleAlreadyAssignedException`.
13	`getAuthorizationUrl(payload)`	A method to build the Google OAuth consent URL with a base64url state. – `payload`: object – user agent and IP address. – *Returns:* `GetAuthorizationUrlResType` – the authorization URL.
14	`googleCallback(payload)`	A method to exchange a Google code, create or load the account, and issue tokens. – `payload`: object – Google authorization code and state. – *Returns:* `Promise<{ accessToken: string; refreshToken: string }>` – the token pair. – Creates a Client account and an OAuth account on first login; rejects banned accounts.

4.3.2 ClassAttribute

No	Attribute	Type
01	`hashingService`	`HashingService` – private readonly password utility.
02	`emailService`	`EmailService` – private readonly OTP delivery utility.
03	`sharedRoleRepository`	`SharedRoleRepository` – private readonly system-role lookup.
04	`authRepository`	`AuthRepository` – private readonly authentication persistence.
05	`tokenService`	`TokenService` – private readonly JWT utility.
06	`oauth2Client`	`OAuth2Client` – public Google OAuth client.
07	`logger`	`Logger` – private readonly NestJS logger.

4.4 AuthRepository Class

4.4.1 ClassMethod

No	Method	Description
01	`AuthRepository(prisma: PrismaService)`	A method to initialize the repository with the shared Prisma client. – `prisma`: `PrismaService` – database client. – *Returns:* `void` – constructor, no return value.
02	`findUserByEmail(email: string)`	A method to load a non-deleted account by email. – `email`: `string` – unique account email. – *Returns:* `Promise<UserType | null>` – the account without password timestamps, or `null`.
03	`findUserByEmailIncludeRoles(email: string)`	A method to load an account with its role assignments. – `email`: `string` – unique account email. – *Returns:* `Promise<UserType and roles | null>` – the account and ordered roles, or `null`.

No	Method	Description
04	`findVerificationCode( uniqueValue)`	A method to find an OTP by email, code, and type. – `uniqueValue`: object – email, code, and verification type. – *Returns:* `Promise<EmailVerificationType | null>` – the matching OTP, or `null`.
05	`findVerificationCodeBy EmailAndType(email, type)`	A method to load the current OTP record for an email and type. – `email`: `string` – account email. – `type`: `VerificationCodeType` – registration or password-reset code. – *Returns:* `Promise<EmailVerificationType | null>` – the OTP record, or `null`.
06	`createUserAndRegister( deleteOtp, user)`	A method to delete the consumed OTP and create the account in one transaction. – `deleteOtp`: object – OTP code, type, and email. – `user`: object – email, password, full name, and initial role. – *Returns:* `Promise<UserType>` – the created account.
07	`createUserAndRegister Google(user)`	A method to create a Google account with a Client role and profile. – `user`: object – email, full name, and role identifier. – *Returns:* `Promise<UserType and roles>` – the created account with roles.
08	`upsertVerificationCode( data)`	A method to create or replace an OTP record. – `data`: object – email, type, code, expiry, and attempt count. – *Returns:* `Promise<EmailVerificationType>` – the persisted OTP.
09	`findUserForLogin(email: string)`	A method to load an account and ordered roles for authentication. – `email`: `string` – account email. – *Returns:* `Promise<UserType and roles | null>` – the account, or `null`.
10	`createSession(data)`	A method to persist a refresh-token session. – `data`: object – user id, refresh token, device, IP address, and expiry. – *Returns:* `Promise<Session>` – the created session with its identifier.

No	Method	Description
11	`deleteSessionByRefresh Token(data)`	A method to delete a session by refresh token and user id. – `data`: object – refresh token and user id. – *Returns:* `Promise<Session>` – the deleted session.
12	`findUniqueRefreshToken IncludeUserRole(data)`	A method to load a session, its account, and the account roles by refresh token. – `data.refreshToken`: `string` – refresh token to resolve. – *Returns:* `Promise<Session and user roles | null>` – the session graph, or `null`.
13	`update(where, data)`	A method to update a non-deleted account. – `where`: object – account identifier. – `data`: `Partial<UserType>` – fields to change. – *Returns:* `Promise<UserType | null>` – the updated account.
14	`deleteVerifycationCode( uniqueValue)`	A method to delete an OTP after a successful password reset. – `uniqueValue`: object – email and verification type. – *Returns:* `Promise<EmailVerificationType>` – the deleted OTP.
15	`findUserById(userId: number)`	A method to load an account with profile and ordered roles. – `userId`: `number` – account identifier. – *Returns:* `Promise<UserType and profile and roles | null>` – the account view, or `null`.
16	`switchPrimaryRole(userId, roleName)`	A method to make an assigned role primary in one transaction. – `userId`: `number` – account identifier. – `roleName`: `string` – role to activate. – *Returns:* `Promise<RoleType | null>` – the activated role, or `null` when not assigned.
17	`joinRole(userId, role Name)`	A method to assign a Client or Freelancer role and create its profile row. – `userId`: `number` – account identifier. – `roleName`: `JoinRoleBodyType['role']` – role to join. – *Returns:* `Promise<{ status: 'joined' | ' already-assigned'; role: RoleType } | null>` – the assignment result, or `null` when the role is absent.

No	Method	Description
18	`createOauthAccount(data)`	A method to link a Google identity to an account. – `data`: object – user id, provider, and provider user id. – *Returns:* `Promise<OauthAccount>` – the created OAuth link.

4.4.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.5 RolesController Class

4.5.1 ClassMethod

No	Method	Description
01	`RolesController(rolesService: RolesService)`	A method to initialize the controller with the role service. – `rolesService`: `RolesService` – service that owns role use cases. – *Returns:* `void` – constructor, no return value.
02	`getRoles()`	A method to return all active roles. – *Returns:* `Promise<RoleListResponseType>` – the role list. – Route: `GET /roles`.
03	`getRoleById(id: number)`	A method to return one role with its permissions. – `id`: `number` – role identifier parsed by `ParseIntPipe`. – *Returns:* `Promise<RoleDetailResponseType>` – the role detail. – Route: `GET /roles/:id`.
04	`createRole(body: CreateRoleBodyDto)`	A method to create a custom role. – `body`: `CreateRoleBodyDto` – role name and optional description. – *Returns:* `Promise<CreateRoleResponseType>` – the created role. – Route: `POST /roles`.

No	Method	Description
05	`updateRole(id: number, body: UpdateRoleBodyDto)`	A method to update a role's name or description. – `id`: `number` – role identifier. – `body`: `UpdateRoleBodyDto` – optional name and description. – *Returns:* `Promise<UpdateRoleResponseType>` – the updated role. – Route: `PATCH /roles/:id`.
06	`setRolePermissions( id: number, body: SetRolePermissions BodyDto)`	A method to replace every permission assigned to a role. – `id`: `number` – role identifier. – `body.permissionIds`: `number[]` – the complete replacement set. – *Returns:* `Promise<SetRolePermissionsResponse Type>` – the updated role detail. – Route: `PUT /roles/:id/permissions`.
07	`deleteRole(id: number)`	A method to soft-delete an unused custom role. – `id`: `number` – role identifier. – *Returns:* `Promise<DeleteRoleResponseType>` – a success message. – Route: `DELETE /roles/:id`.

4.5.2 ClassAttribute

No	Attribute	Type
01	`rolesService`	`RolesService` – private readonly role service.

4.6 RolesService Class

4.6.1 ClassMethod

No	Method	Description
01	`RolesService(roles Repository: Roles Repository)`	A method to initialize the service with the role repository. – `rolesRepository`: `RolesRepository` – persistence dependency. – *Returns:* `void` – constructor, no return value.

No	Method	Description
02	`isSystemRoleName(name: string)`	A private method to detect the four built-in role names. – `name: string` – role name to test. – *Returns:* `boolean` – true for Admin, Client, Freelancer, or Expert.
03	`getRoles()`	A method to load the active role list. – *Returns:* `Promise<RoleListResponseType>` – the role list. – Failure is mapped to `FailedToLoadRolesException`.
04	`getRoleById(id: number)`	A method to load a role and its permissions. – `id: number` – role identifier. – *Returns:* `Promise<RoleDetailResponseType>` – the role detail. – Prisma errors are mapped to `FailedToLoadRoleDetailException`.
05	`createRole(body: CreateRoleBodyType)`	A method to validate the name and create a custom role. – `body: CreateRoleBodyType` – role name and optional description. – *Returns:* `Promise<CreateRoleResponseType>` – the created role. – Throws `CannotModifySystemRoleException` or `RoleAlreadyExistsException`.
06	`updateRole(id: number, body: UpdateRoleBodyType)`	A method to validate system-role and duplicate-name rules before updating. – `id: number` – role identifier. – `body: UpdateRoleBodyType` – optional name and description. – *Returns:* `Promise<UpdateRoleResponseType>` – the updated role. – Throws `CannotModifySystemRoleException` or `RoleAlreadyExistsException`.
07	`deleteRole(id: number)`	A method to soft-delete a custom role that is not assigned to any user. – `id: number` – role identifier. – *Returns:* `Promise<DeleteRoleResponseType>` – a success message. – Throws `CannotModifySystemRoleException` or `RoleInUseException`.

No	Method	Description
08	`setRolePermissions(id, body)`	A method to validate permission ids and replace the role permission set. – `id`: `number` – role identifier. – `body`: `SetRolePermissionsBodyType` – permission ids to assign. – *Returns:* `Promise<SetRolePermissionsResponseType>` – the updated role detail. – Throws `InvalidPermissionIdsException`.

4.6.2 ClassAttribute

No	Attribute	Type
01	`rolesRepository`	`RolesRepository` – private readonly persistence dependency.
02	`logger`	`Logger` – private readonly NestJS logger.

4.7 RolesRepository Class

4.7.1 ClassMethod

No	Method	Description
01	`RolesRepository(prisma: PrismaService)`	A method to initialize the repository with the shared Prisma client. – `prisma`: `PrismaService` – database client. – *Returns:* `void` – constructor, no return value.
02	`findAll()`	A method to load all non-deleted roles ordered by id. – *Returns:* `Promise<RoleListItemType[]>` – the role list.
03	`findById(id: number)`	A method to load a role and its active permissions. – `id`: `number` – role identifier. – *Returns:* `Promise<RoleDetailResponseType>` – the role detail. – Throws `RoleNotFoundException`.

No	Method	Description
04	`findActiveByName(name, excludeId?)`	A method to find an active role by case-insensitive name. – `name`: `string` – role name. – `excludeId`: `number | undefined` – optional role to exclude. – *Returns:* `Promise<RoleListItemType | null>` – the matching role, or `null`.
05	`createRole(body: CreateRoleBodyType)`	A method to insert a role. – `body`: `CreateRoleBodyType` – name and optional description. – *Returns:* `Promise<CreateRoleResponseType>` – the created role.
06	`updateRole(id: number, body: UpdateRoleBodyType)`	A method to update the provided role fields. – `id`: `number` – role identifier. – `body`: `UpdateRoleBodyType` – optional name and description. – *Returns:* `Promise<UpdateRoleResponseType>` – the updated role.
07	`softDeleteRole(id: number)`	A method to set `deletedAt` on a role. – `id`: `number` – role identifier. – *Returns:* `Promise<void>` – no value.
08	`countAssignedUsers(roleId: number)`	A method to count user-role assignments. – `roleId`: `number` – role identifier. – *Returns:* `Promise<number>` – the assignment count.
09	`findActivePermissionIds(ids: number[])`	A method to filter permission ids down to active permissions. – `ids`: `number[]` – requested permission ids. – *Returns:* `Promise<number[]>` – the active ids.
10	`replaceRolePermissions(roleId, permissionIds)`	A method to delete the old role permissions and insert the new set in a transaction. – `roleId`: `number` – role identifier. – `permissionIds`: `number[]` – replacement permission ids. – *Returns:* `Promise<void>` – no value.

4.7.2 ClassAttribute

No	Attribute	Type
01	prisma	PrismaService – private readonly database client.

4.8 ProfileController Class

This shared class supports Freelancer Profile Management, Freelancer Skill Management, Freelancer CV Management. Its relevant operations are specified together here.

4.8.1 ClassMethod

No	Method	Description
01	ProfileController(profileService: ProfileService)	Initializes the class with its required dependencies. – profileService: ProfileService – injected dependency. – *Returns:* void – constructor, no return value.
02	viewProfile(id: number)	Loads the public freelancer profile and verifies that the associated user has the freelancer role. – id: number – general profile identifier from the route. – *Returns:* An awaitable record with user, freelancerProfile, id, userId, displayName, avatarUrl, coverUrl, and the remaining selected fields. – Response schema: FreelancerProfileDetailDto. – Route: GET /profiles/:id.
03	updateProfile(id: number, currentUserId: number, body: UpdateFreelancerProfileDto)	Checks profile ownership and applies the profile revision policy before persisting changes. – id: number – general profile identifier from the route. – currentUserId: number – authenticated account identifier. – body: UpdateFreelancerProfileDto – validated input fields defined by the declared type. – *Returns:* An awaitable record with message, reviewRequired, profileStrength, revision. – Response schema: ProfileRevisionSubmissionDto. – Route: PUT /profiles/:id.

No	Method	Description
04	`uploadCv( id: number, currentUserId: number, file?: Express.Multer.File)`	Checks ownership, validates the CV, stores it, and updates its profile metadata. – `id`: `number` – general profile identifier from the route. – `currentUserId`: `number` – authenticated account identifier. – `file`: `Express.Multer.File` – uploaded file supplied to this operation; optional. – *Returns:* `Promise<{ cvUrl: string | null; cvFileName: string | null; }>` – Response schema: `CvUploadResponseDto`. – Route: `POST /profiles/:id/cv`.
05	`getCvFile( id: number, currentUserId: number, response: Response)`	Checks ownership and resolves the stored CV file or remote URL. – `id`: `number` – general profile identifier from the route. – `currentUserId`: `number` – authenticated account identifier. – `response`: `Response` – Express response used to set file headers or redirect. – *Returns:* `Promise<void | StreamableFile>` – Route: `GET /profiles/:id/cv/file`.
06	`analyzeCv(id: number, currentUserId: number)`	Analyzes the owned CV and returns skill suggestions. – `id`: `number` – general profile identifier from the route. – `currentUserId`: `number` – authenticated account identifier. – *Returns:* `Promise<{ suggestions: { skillName: string; proficiencyLevel: number | null; }[]; }>` – Response schema: `CvAnalyzeResponseDto`. – Route: `POST /profiles/:id/cv/analyze`.
07	`deleteCv(id: number, currentUserId: number)`	Removes the stored CV and clears its profile metadata. – `id`: `number` – general profile identifier from the route. – `currentUserId`: `number` – authenticated account identifier. – *Returns:* `Promise<{ message: string; }>` – Route: `DELETE /profiles/:id/cv`.

No	Method	Description
08	`getSkills(id: number)`	Loads the skill associations visible on a freelancer profile. – `id`: `number` – general profile identifier from the route. – *Returns:* An awaitable list of records with `skill`, `id`, `proficiencyLevel`, `freelancerProfileId`, `skillId`. – Route: `GET /profiles/:id/skills`.
09	`searchSkillSuggestions( search?: string)`	Searches the active skill catalog using the supplied text. – `search`: `string` – optional skill-name search text; optional. – *Returns:* `Promise<{ id: number; name: string; }[]>` – Route: `GET /profiles/skills/suggestions`.
10	`addSkill( id: number, currentUserId: number, body: AddFreelancerSkill Dto)`	Resolves a catalog skill, checks for an existing association, and adds a proficiency value to the owned profile. – `id`: `number` – general profile identifier from the route. – `currentUserId`: `number` – authenticated account identifier. – `body`: `AddFreelancerSkillDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `skill`, `id`, `proficiencyLevel`, `freelancerProfileId`, `skillId`. – Response schema: `AddFreelancerSkillResponse Dto`. – Route: `POST /profiles/:id/skills`.
11	`deleteSkill(id: number, currentUserId: number)`	Removes an owned freelancer-skill association while retaining the catalog skill. – `id`: `number` – freelancer-skill association identifier from the route. – `currentUserId`: `number` – authenticated account identifier. – *Returns:* `Promise<{ id: number; proficiency Level: number; freelancerProfileId: number; skillId: number; }>` – Route: `DELETE /profiles/skills/:id`.

4.8.2 ClassAttribute

No	Attribute	Type
01	`profileService`	`ProfileService` – private readonly injected dependency.

4.9 ProfileService Class

This shared class supports Freelancer Profile Management, Freelancer Skill Management, Freelancer CV Management. Its relevant operations are specified together here.

4.9.1 ClassMethod

No	Method	Description
01	`ProfileService( profileRepository: ProfileRepository, cloudinary: CloudinaryService, aiCvAnalyzer: AiCvAnalyzerService, skillResolver: SkillResolverService, profileRevisionService: ProfileRevisionService)`	Initializes the class with its required dependencies. – `profileRepository`: `ProfileRepository` – injected dependency. – `cloudinary`: `CloudinaryService` – injected dependency. – `aiCvAnalyzer`: `AiCvAnalyzerService` – injected dependency. – `skillResolver`: `SkillResolverService` – injected dependency. – `profileRevisionService`: `ProfileRevisionService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`uploadCv( profileId: number, userId: number, file?: Express.Multer.File)`	Checks ownership, validates the CV, stores it, and updates its profile metadata. – `profileId`: `number` – general profile identifier. – `userId`: `number` – account identifier for this operation. – `file`: `Express.Multer.File` – uploaded file supplied to this operation; optional. – *Returns:* `Promise<{ cvUrl: string | null; cvFileName: string | null; }>` – Throws `CvFileRequiredException`, `CvFileInvalidException`.
03	`deleteCv(profileId: number, userId: number)`	Removes the stored CV and clears its profile metadata. – `profileId`: `number` – general profile identifier. – `userId`: `number` – account identifier for this operation. – *Returns:* `Promise<{ message: string; }>` – Throws `CvNotFoundException`.

No	Method	Description
04	`getCvFile(profileId: number, userId: number)`	Checks ownership and resolves the stored CV file or remote URL. – `profileId`: `number` – general profile identifier. – `userId`: `number` – account identifier for this operation. – *Returns:* `Promise<{ remoteUrl: string; fileName: string; absolutePath?: undefined; } | { absolutePath: string; fileName: string; remoteUrl?: undefined; }>` – Throws `CvNotFoundException`.
05	`analyzeCv(profileId: number, userId: number)`	Analyzes the owned CV and returns skill suggestions. – `profileId`: `number` – general profile identifier. – `userId`: `number` – account identifier for this operation. – *Returns:* `Promise<{ suggestions: { skillName: string; proficiencyLevel: number | null; }[]; }>` – Throws `CvNotFoundException`, `CvAiUnavailableException`.
06	`viewProfile(profileId: number)`	Loads the public freelancer profile and verifies that the associated user has the freelancer role. – `profileId`: `number` – general profile identifier. – *Returns:* An awaitable record with `user`, `freelancerProfile`, `id`, `userId`, `displayName`, `avatarUrl`, `coverUrl`, and the remaining selected fields. – Throws `FreelancerProfileNotFoundException`.
07	`updateProfile( profileId: number, currentUserId: number, dto: UpdateFreelancerProfileType)`	Checks profile ownership and applies the profile revision policy before persisting changes. – `profileId`: `number` – general profile identifier. – `currentUserId`: `number` – authenticated account identifier. – `dto`: `UpdateFreelancerProfileType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `message`, `reviewRequired`, `profileStrength`, `revision`. – Throws `FreelancerProfileNotFoundException`, `ProfileForbiddenException`.

No	Method	Description
08	`getSkills(profileId: number)`	Loads the skill associations visible on a freelancer profile. – `profileId`: `number` – general profile identifier. – *Returns:* An awaitable list of records with `skill`, `id`, `proficiencyLevel`, `freelancerProfileId`, `skillId`. – Throws `FreelancerProfileNotFoundException`.
09	`searchSkillSuggestions( search?: string)`	Searches the active skill catalog using the supplied text. – `search`: `string` – optional skill-name search text; optional. – *Returns:* `Promise<{ id: number; name: string; }[]>`
10	`addSkill( profileId: number, currentUserId: number, dto: AddFreelancerSkill Type)`	Resolves a catalog skill, checks for an existing association, and adds a proficiency value to the owned profile. – `profileId`: `number` – general profile identifier. – `currentUserId`: `number` – authenticated account identifier. – `dto`: `AddFreelancerSkillType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `skill`, `id`, `proficiencyLevel`, `freelancerProfileId`, `skillId`. – Throws `FreelancerProfileNotFoundException`, `ProfileForbiddenException`, `FreelancerSkill DuplicateException`.
11	`deleteSkill(skillId: number, currentUserId: number)`	Removes an owned freelancer-skill association while retaining the catalog skill. – `skillId`: `number` – skill or association identifier used by this operation. – `currentUserId`: `number` – authenticated account identifier. – *Returns:* `Promise<{ id: number; proficiency Level: number; freelancerProfileId: number; skillId: number; }>` – Throws `FreelancerSkillNotFoundException`, `SkillForbiddenException`.

4.9.2 ClassAttribute

No	Attribute	Type
01	`profileRepository`	`ProfileRepository` – private readonly injected dependency.
02	`cloudinary`	`CloudinaryService` – private readonly injected dependency.
03	`aiCvAnalyzer`	`AiCvAnalyzerService` – private readonly injected dependency.
04	`skillResolver`	`SkillResolverService` – private readonly injected dependency.
05	`profileRevisionService`	`ProfileRevisionService` – private readonly injected dependency.

4.10 ProfileRepository Class

This shared class supports Freelancer Profile Management, Freelancer Skill Management, Freelancer CV Management. Its relevant operations are specified together here.

4.10.1 ClassMethod

No	Method	Description
01	`ProfileRepository(prisma: PrismaService)`	Initializes the class with its required dependencies. – `prisma`: `PrismaService` – injected database client. – *Returns:* `void` – constructor, no return value.
02	`findFreelancerProfileById(profileId: number)`	Retrieves profile data, freelancer information, and user roles used for existence and ownership checks. – `profileId`: `number` – general profile identifier. – *Returns:* An awaitable record with `user`, `freelancerProfile`, `id`, `userId`, `displayName`, `avatarUrl`, `coverUrl`, and the remaining selected fields; `null` when absent.

No	Method	Description
03	`updateFreelancerProfile( profileId: number, data: { displayName: string; title: string; bio?: string | null; availabilityStatus?: AvailabilityStatus; education?: string[] | null; certifications?: string[] | null; languages?: string[] | null; experience?: string[] | null; })`	Updates freelancer and general profile fields and recalculates profile completion. – `profileId`: `number` – general profile identifier. – `data`: `{ displayName: string; title: string; bio?: string | null; availability Status?: AvailabilityStatus; education?: string[] | null; certifications?: string[] | null; languages?: string[] | null; experience?: string[] | null; }` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `freelancer Profile`, `id`, `userId`, `displayName`, `avatarUrl`, `coverUrl`, `bio`, and the remaining selected fields; `null` when absent.
04	`updateCv( profileId: number, data: { cvUrl: string; cvFileName: string; cv PublicId: string | null })`	Upserts stored CV metadata on a freelancer profile. – `profileId`: `number` – general profile identifier. – `data`: `{ cvUrl: string; cvFileName: string; cvPublicId: string | null }` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `profileId`, `title`, `education`, `certifications`, and the remaining selected fields.
05	`clearCv(profileId: number)`	Clears the freelancer profile's CV metadata. – `profileId`: `number` – general profile identifier. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `profileId`, `title`, `education`, `certifications`, and the remaining selected fields.
06	`findSkillsByProfileId( profileId: number)`	Queries active catalog-skill associations for a freelancer profile. – `profileId`: `number` – general profile identifier. – *Returns:* An awaitable list of records with `skill`, `id`, `proficiencyLevel`, `freelancerProfileId`, `skillId`.
07	`searchActiveCatalog Skills(search?: string)`	Searches non-deleted catalog skills by name and returns matching identifiers and names. – `search`: `string` – optional skill-name search text; optional. – *Returns:* `Promise<{ id: number; name: string; }[]>`

No	Method	Description
08	`findOrCreateCatalogSkill( name: string)`	Returns a matching active catalog skill or creates one with a unique slug. – `name`: `string` – catalog skill name. – *Returns:* `Promise<{ id: number; createdAt: Date; updatedAt: Date; deletedAt: Date | null; name: string; description: string | null; slug: string; }>`
09	`findSkillByProfileIdAnd SkillId( profileId: number, skillId: number)`	Finds an existing freelancer-skill association for duplicate checking. – `profileId`: `number` – general profile identifier. – `skillId`: `number` – skill or association identifier used by this operation. – *Returns:* `Promise<{ id: number; proficiency Level: number; freelancerProfileId: number; skillId: number; } | null>`
10	`addSkillToProfile( profileId: number, skillId: number, proficiencyLevel: number)`	Ensures a freelancer profile exists and inserts a skill association. – `profileId`: `number` – general profile identifier. – `skillId`: `number` – skill or association identifier used by this operation. – `proficiencyLevel`: `number` – skill proficiency level. – *Returns:* An awaitable record with `skill`, `id`, `proficiencyLevel`, `freelancerProfileId`, `skillId`.
11	`findSkillById(skillId: number)`	Loads a skill association with the profile data needed to identify its owner. – `skillId`: `number` – skill or association identifier used by this operation. – *Returns:* An awaitable record with `freelancer Profile`, `id`, `proficiencyLevel`, `freelancer ProfileId`, `skillId`; `null` when absent.
12	`deleteSkill(skillId: number)`	Removes an owned freelancer-skill association while retaining the catalog skill. – `skillId`: `number` – skill or association identifier used by this operation. – *Returns:* `Promise<{ id: number; proficiency Level: number; freelancerProfileId: number; skillId: number; }>`

4.10.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.11 PortfolioController Class

4.11.1 ClassMethod

No	Method	Description
01	`PortfolioController( portfolioService: PortfolioService)`	Initializes the class with its required dependencies. – `portfolioService`: `PortfolioService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`getPortfolios(id: number)`	Loads the active portfolio items for a freelancer profile. – `id`: `number` – general profile identifier from the route. – *Returns:* An awaitable list of records with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields. – Response schema: `PortfolioItemListResponseDto`. – Route: `GET /profiles/:id/portfolios`.
03	`addPortfolio( currentUserId: number, id: number, body: AddPortfolioDto)`	Creates a portfolio item on the current user's freelancer profile. – `currentUserId`: `number` – authenticated account identifier. – `id`: `number` – general profile identifier from the route. – `body`: `AddPortfolioDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields. – Response schema: `AddPortfolioResponseDto`. – Route: `POST /profiles/:id/portfolios`.

No	Method	Description
04	`getPortfolioDetail(id: number)`	Loads one active portfolio item. – `id: number` – portfolio item identifier from the route. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields. – Response schema: `PortfolioItemResponseDto`. – Route: `GET /profiles/portfolios/:id`.
05	`updatePortfolio( currentUserId: number, id: number, body: UpdatePortfolio Dto)`	Updates an owned portfolio item using the supplied fields. – `currentUserId: number` – authenticated account identifier. – `id: number` – portfolio item identifier from the route. – `body: UpdatePortfolioDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields. – Response schema: `UpdatePortfolioResponseDto`. – Route: `PUT /profiles/portfolios/:id`.
06	`deletePortfolio(current UserId: number, id: number)`	Marks an owned portfolio item as deleted. – `currentUserId: number` – authenticated account identifier. – `id: number` – portfolio item identifier from the route. – *Returns:* `Promise<{ message: "Success. PortfolioDeleted"; }>` – Route: `DELETE /profiles/portfolios/:id`.

4.11.2 ClassAttribute

No	Attribute	Type
01	`portfolioService`	`PortfolioService` – private readonly injected dependency.

4.12 PortfolioService Class

4.12.1 ClassMethod

No	Method	Description
01	`PortfolioService( portfolioRepository: PortfolioRepository)`	Initializes the class with its required dependencies. – `portfolioRepository`: `PortfolioRepository` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`getPortfolios(profileId: number)`	Loads the active portfolio items for a freelancer profile. – `profileId`: `number` – general profile identifier. – *Returns:* An awaitable list of records with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields. – Throws `FreelancerProfileNotFoundException`, `UnableToLoadPortfoliosException`.
03	`addPortfolio( profileId: number, currentUserId: number, data: AddPortfolioType)`	Creates a portfolio item on the current user's freelancer profile. – `profileId`: `number` – general profile identifier. – `currentUserId`: `number` – authenticated account identifier. – `data`: `AddPortfolioType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields. – Throws `FreelancerProfileNotFoundException`, `PortfolioForbiddenException`, `UnableToLoad PortfoliosException`.
04	`getPortfolioDetail( portfolioId: number)`	Loads one active portfolio item. – `portfolioId`: `number` – portfolio item identifier. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields. – Throws `PortfolioNotFoundException`, `UnableTo LoadPortfoliosException`.

No	Method	Description
05	updatePortfolio(portfolioId: number, currentUserId: number, data: UpdatePortfolio Type)	Updates an owned portfolio item using the supplied fields. – portfolioId: number – portfolio item identifier. – currentUserId: number – authenticated account identifier. – data: UpdatePortfolioType – validated input fields defined by the declared type. – *Returns:* An awaitable record with id, createdAt, updatedAt, deletedAt, title, description, freelancerProfileId, and the remaining selected fields. – Throws PortfolioNotFoundException, Portfolio ForbiddenException, UnableToUpdatePortfolio Exception.
06	deletePortfolio(portfolio Id: number, currentUser Id: number)	Marks an owned portfolio item as deleted. – portfolioId: number – portfolio item identifier. – currentUserId: number – authenticated account identifier. – *Returns:* Promise<{ message: "Success. PortfolioDeleted"; }> – Throws PortfolioNotFoundException, Portfolio ForbiddenException, UnableToDeletePortfolio Exception.

4.12.2 ClassAttribute

No	Attribute	Type
01	portfolioRepository	PortfolioRepository – private readonly injected dependency.

4.13 PortfolioRepository Class

4.13.1 ClassMethod

No	Method	Description
01	PortfolioRepository(prisma: PrismaService)	Initializes the class with its required dependencies. – prisma: PrismaService – injected database client. – *Returns:* void – constructor, no return value.

No	Method	Description
02	`findFreelancerProfileById(profileId: number)`	Retrieves profile data, freelancer information, and user roles used for existence and ownership checks. – `profileId`: `number` – general profile identifier. – *Returns:* An awaitable record with `user`, `freelancerProfile`, `id`, `userId`, `displayName`, `avatarUrl`, `coverUrl`, and the remaining selected fields; `null` when absent.
03	`findPortfoliosByFreelancerProfileId(freelancerProfileId: number)`	Queries portfolio items that have not been soft-deleted. – `freelancerProfileId`: `number` – freelancer profile identifier. – *Returns:* An awaitable list of records with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields.
04	`addPortfolioToProfile(profileId: number, data: AddPortfolioType)`	Creates a portfolio item in a transaction, creating the freelancer profile if required. – `profileId`: `number` – general profile identifier. – `data`: `AddPortfolioType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields.
05	`findPortfolioById(portfolioId: number)`	Queries one portfolio item that has not been soft-deleted. – `portfolioId`: `number` – portfolio item identifier. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields; `null` when absent.
06	`findPortfolioWithProfile(portfolioId: number)`	Loads a portfolio item together with the profile used to verify ownership. – `portfolioId`: `number` – portfolio item identifier. – *Returns:* An awaitable record with `freelancerProfile`, `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, and the remaining selected fields; `null` when absent.

No	Method	Description
07	`updatePortfolio( portfolioId: number, data: UpdatePortfolio Type)`	Updates an owned portfolio item using the supplied fields. – `portfolioId`: `number` – portfolio item identifier. – `data`: `UpdatePortfolioType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields.
08	`deletePortfolio(portfolio Id: number)`	Marks an owned portfolio item as deleted. – `portfolioId`: `number` – portfolio item identifier. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `title`, `description`, `freelancerProfileId`, and the remaining selected fields.

4.13.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.14 PermissionsController Class

4.14.1 ClassMethod

No	Method	Description
01	`PermissionsController( permissionsService: PermissionsService)`	A method to initialize the controller with the permission service. – `permissionsService`: `PermissionsService` – service that owns permission queries. – *Returns:* `void` – constructor, no return value.
02	`getPermissions(query: PermissionFilterDto)`	A method to return a filtered page of route permissions. – `query`: `PermissionFilterDto` – page, limit, search, method, module, and sort options. – *Returns:* `Promise<PermissionListResponseType>` – permissions, pagination, and module names. – Route: `GET /permissions`.

No	Method	Description
03	`getPermissionById(id: number)`	A method to return one permission. – `id: number` – permission identifier parsed by `ParseIntPipe`. – *Returns:* `Promise<PermissionDetailResponseType>` – the permission detail. – Route: `GET /permissions/:id`.

4.14.2 ClassAttribute

No	Attribute	Type
01	`permissionsService`	`PermissionsService` – private readonly permission service.

4.15 PermissionsService Class

4.15.1 ClassMethod

No	Method	Description
01	`PermissionsService(permissionsRepository: PermissionsRepository)`	A method to initialize the service with the permission repository. – `permissionsRepository: PermissionsRepository` – persistence dependency. – *Returns:* `void` – constructor, no return value.
02	`getPermissions(filter: PermissionFilterType)`	A method to load a permission page and compute pagination metadata. – `filter: PermissionFilterType` – page, limit, search, method, module, and sort options. – *Returns:* `Promise<PermissionListResponseType>` – permissions, pagination, and module names. – Failure is mapped to `FailedToLoadPermissionsException`.
03	`getPermissionById(id: number)`	A method to load one permission. – `id: number` – permission identifier. – *Returns:* `Promise<PermissionDetailResponseType>` – the permission detail. – Prisma errors are mapped to `FailedToLoadPermissionDetailException`.

4.15.2 ClassAttribute

No	Attribute	Type
01	`permissionsRepository`	`PermissionsRepository` – private readonly persistence dependency.
02	`logger`	`Logger` – private readonly NestJS logger.

4.16 PermissionsRepository Class

4.16.1 ClassMethod

No	Method	Description
01	`PermissionsRepository( prisma: PrismaService)`	A method to initialize the repository with the shared Prisma client. – `prisma`: `PrismaService` – database client. – *Returns:* `void` – constructor, no return value.
02	`findAll(filter: PermissionFilterType)`	A method to query permissions, the matching total, and the distinct module list. – `filter`: `PermissionFilterType` – page, limit, search, method, module, and sort options. – *Returns:* `Promise<{ permissions: PermissionListItemType[]; total: number; modules: string[] }>` – the page and supporting data. – Search matches name, path, or module case-insensitively.
03	`findById(id: number)`	A method to load one active permission. – `id`: `number` – permission identifier. – *Returns:* `Promise<PermissionDetailResponseType>` – the permission detail. – Throws `PermissionNotFoundException`.

4.16.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.17 SessionsController Class

4.17.1 ClassMethod

No	Method	Description
01	`SessionsController( sessionsService: SessionsService)`	A method to initialize the controller with the session service. – `sessionsService`: `SessionsService` – service that owns session queries. – *Returns:* `void` – constructor, no return value.
02	`getMySessions( userId: number, sessionId: number, query: SessionFilterDto)`	A method to return the current user's sessions. – `userId`: `number` – current user. – `sessionId`: `number` – current session, used to mark the active row. – `query`: `SessionFilterDto` – page, limit, search, and sort options. – *Returns:* `Promise<SessionListResponseType>` – sessions and pagination. – Route: `GET /sessions`; authenticated endpoint.
03	`getMySessionById( userId: number, sessionId: number, id: number)`	A method to return one owned session. – `userId`: `number` – current user. – `sessionId`: `number` – current session, used to mark the active row. – `id`: `number` – session identifier. – *Returns:* `Promise<SessionDetailResponseType>` – the session detail. – Route: `GET /sessions/:id`; authenticated endpoint.
04	`revokeMySession( userId: number, sessionId: number, id: number)`	A method to revoke one owned session. – `userId`: `number` – current user. – `sessionId`: `number` – current session, used to compute `loggedOut`. – `id`: `number` – session identifier. – *Returns:* `Promise<RevokeSessionResponseType>` – a success message and logout flag. – Route: `DELETE /sessions/:id`; authenticated endpoint.

4.17.2 ClassAttribute

No	Attribute	Type
01	`sessionsService`	`SessionsService` – private readonly session service.

4.18 SessionsService Class

4.18.1 ClassMethod

No	Method	Description
01	`SessionsService(sessionsRepository: SessionsRepository)`	A method to initialize the service with the session repository. – `sessionsRepository`: `SessionsRepository` – persistence dependency. – *Returns:* `void` – constructor, no return value.
02	`getMySessions(userId, filter, currentSessionId?)`	A method to load a page of the user's sessions. – `userId`: `number` – current user. – `filter`: `SessionFilterType` – page, limit, search, and sort options. – `currentSessionId`: `number | undefined` – session to mark as current. – *Returns:* `Promise<SessionListResponseType>` – sessions and pagination. – Failure is mapped to `FailedToLoadSessionsException`.
03	`getMySessionById(id, userId, currentSessionId?)`	A method to load one session owned by the user. – `id`: `number` – session identifier. – `userId`: `number` – current user. – `currentSessionId`: `number | undefined` – session to mark as current. – *Returns:* `Promise<SessionDetailResponseType>` – the session detail. – Prisma errors are mapped to `FailedToLoadSessionDetailException`.

No	Method	Description
04	`revokeMySession(id, userId, currentSessionId?)`	A method to delete an active session owned by the user. – `id`: `number` – session identifier. – `userId`: `number` – current user. – `currentSessionId`: `number | undefined` – current session used to compute `loggedOut`. – *Returns:* `Promise<RevokeSessionResponseType>` – a success message and logout flag. – Prisma errors are mapped to `FailedToRevokeSessionException`.

4.18.2 ClassAttribute

No	Attribute	Type
01	`sessionsRepository`	`SessionsRepository` – private readonly persistence dependency.
02	`logger`	`Logger` – private readonly NestJS logger.

4.19 SessionsRepository Class

4.19.1 ClassMethod

No	Method	Description
01	`SessionsRepository(prisma: PrismaService)`	A method to initialize the repository with the shared Prisma client. – `prisma`: `PrismaService` – database client. – *Returns:* `void` – constructor, no return value.
02	`findAllByUserId(userId, filter, currentSessionId?)`	A method to query the user's sessions and mark the current one. – `userId`: `number` – owner identifier. – `filter`: `SessionFilterType` – page, limit, search, and sort options. – `currentSessionId`: `number | undefined` – session to mark as current. – *Returns:* `Promise<{ sessions: SessionListItemType[]; total: number }>` – the page and total. – Search matches device information or IP address.

No	Method	Description
03	`findByIdForUser(id, userId, currentSessionId?)`	A method to load one session owned by the user. – `id`: `number` – session identifier. – `userId`: `number` – owner identifier. – `currentSessionId`: `number | undefined` – session to mark as current. – *Returns:* `Promise<SessionDetailResponseType>` – the session detail. – Throws `SessionNotFoundException`.
04	`deleteActiveByIdForUser(id, userId)`	A method to delete an unexpired session owned by the user. – `id`: `number` – session identifier. – `userId`: `number` – owner identifier. – *Returns:* `Promise<void>` – no value. – Throws `SessionNotFoundException` or `SessionAlreadyExpiredException`.

4.19.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.20 Forum Category Administration (Shared Classes)

UC-14.01–14.02 and UC-17.01–17.06 use the shared `ForumAdminController`, `ForumAdminServi` and `ForumAdminRepository`. Their complete specifications, including the category list, detail, create, update, delete, and restore endpoints, are given once in the shared forum administration class specifications for Sprint 3 (see the `ForumAdminController` subsection).

4.21 IdentityVerificationController Class

4.21.1 ClassMethod

No	Method	Description
01	`IdentityVerification Controller( service: AccountProfile Service)`	Initializes the class with its required dependencies. – `service`: `AccountProfileService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`upload( userId: number, body: UploadIdentity DocumentDto, file?: Express.Multer. File)`	Validates and stores identity evidence and creates its verification record. – `userId`: `number` – account identifier supplied by authentication or the route. – `body`: `UploadIdentityDocumentDto` – validated input fields defined by the declared type. – `file`: `Express.Multer.File` – uploaded file supplied to this operation; optional. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `status`, `adminId`, `review Notes`, and the remaining selected fields. – Response schema: `IdentityVerificationDocument Dto`. – Route: `POST /identity-verifications/ documents`.
03	`aiCheck( userId: number, body: UploadIdentity DocumentDto, files?: Express.Multer. File[])`	Checks submitted identity images and stores the verification evidence and AI result. – `userId`: `number` – account identifier supplied by authentication or the route. – `body`: `UploadIdentityDocumentDto` – validated input fields defined by the declared type. – `files`: `Express.Multer.File[]` – uploaded file supplied to this operation; optional. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `status`, `adminId`, `review Notes`, and the remaining selected fields. – Response schema: `IdentityVerificationDocument Dto`. – Route: `POST /identity-verifications/ai-check`.

No	Method	Description
04	`livenessPose(userId: number, file?: Express.Multer.File)`	Requests pose checking for a submitted liveness image. – `userId`: `number` – account identifier supplied by authentication or the route. – `file`: `Express.Multer.File` – uploaded file supplied to this operation; optional. – *Returns:* An awaitable record with `face_found`, `yaw`, `pitch`, `roll`, `blink`. – Route: `POST /identity-verifications/liveness-pose`.
05	`getStatus(userId: number)`	Returns the latest verification state and document history. – `userId`: `number` – account identifier supplied by authentication or the route. – *Returns:* An awaitable record with `status`, `documents`. – Response schema: `IdentityVerificationStatusDto`. – Route: `GET /identity-verifications/status`.
06	`getFile( userId: number, id: number, side: string | undefined, response: Response)`	Resolves an identity document file belonging to the authenticated user. – `userId`: `number` – account identifier supplied by authentication or the route. – `id`: `number` – target record identifier. – `side`: `string | undefined` – side supplied to the operation. – `response`: `Response` – Express response used to set file headers or redirect. – *Returns:* `Promise<StreamableFile>` – Route: `GET /identity-verifications/documents/:id/file`.

4.21.2 ClassAttribute

No	Attribute	Type
01	`service`	`AccountProfileService` – private readonly injected dependency.

4.22 AccountProfileService Class

This shared class supports Identity Verification, Client Profile Management, Social Link Management, Favorite Freelancer Management, Following Management, Account Profile Management. Its relevant operations are specified together here.

4.22.1 ClassMethod

No	Method	Description
01	`AccountProfileService(repository: AccountProfileRepository, profileRevisionService: ProfileRevisionService, cloudinary: CloudinaryService, hashing: HashingService, identityAi: IdentityAiService)`	Initializes the class with its required dependencies. – `repository`: `AccountProfileRepository` – injected dependency. – `profileRevisionService`: `ProfileRevisionService` – injected dependency. – `cloudinary`: `CloudinaryService` – injected dependency. – `hashing`: `HashingService` – injected dependency. – `identityAi`: `IdentityAiService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`getGeneralProfile(userId: number)`	Loads general account and profile information for the authenticated user. – `userId`: `number` – account identifier for this operation. – *Returns:* An awaitable record with `email`, `avatarUrl`, `roles`, `id`, `userId`, `displayName`, `coverUrl`, and the remaining selected fields. – Throws `ProfileNotFoundException`.
03	`updateGeneralProfile(userId: number, input: UpdateGeneralProfileType)`	Submits or applies edits to general profile information according to the revision policy. – `userId`: `number` – account identifier for this operation. – `input`: `UpdateGeneralProfileType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `message`, `reviewRequired`, `profileStrength`, `revision`. – Throws `ProfileNotFoundException`.

No	Method	Description
04	`changePassword(userId: number, input: ChangePasswordType)`	Verifies the existing password and stores the newly hashed password. – `userId`: `number` – account identifier for this operation. – `input`: `ChangePasswordType` – validated input fields defined by the declared type. – *Returns:* `Promise<{ message: string; }>` – Throws `ProfileNotFoundException`, `PasswordUnavailableException`, `CurrentPasswordIncorrectException`.
05	`uploadAvatar(userId: number, file?: Express.Multer.File)`	Validates and stores an avatar file, then updates the profile reference. – `userId`: `number` – account identifier for this operation. – `file`: `Express.Multer.File` – uploaded file supplied to this operation; optional. – *Returns:* `Promise<{ avatarUrl: string; }>` – Throws `ProfileNotFoundException`, `AvatarFileRequiredException`, `AvatarFileInvalidException`.
06	`getAvatarFile(userId: number)`	Resolves the stored avatar for delivery. – `userId`: `number` – account identifier for this operation. – *Returns:* `Promise<{ absolutePath: string; fileName: string; contentType: string; }>` – Throws `AvatarNotFoundException`.
07	`uploadIdentityDocument( userId: number, documentType: DocumentTypeType, file?: Express.Multer.File)`	Validates and stores identity evidence and creates its verification record. – `userId`: `number` – account identifier for this operation. – `documentType`: `DocumentTypeType` – supported identity-document category. – `file`: `Express.Multer.File` – uploaded file supplied to this operation; optional. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `status`, `adminId`, `reviewNotes`, and the remaining selected fields. – Throws `IdentityFileRequiredException`, `IdentityFileInvalidException`.

No	Method	Description
08	`submitIdentityAiCheck( userId: number, documentType: DocumentTypeType, files: { front?: Express.Multer.File; back?: Express.Multer.File; selfie?: Express.Multer.File; liveness: Express.Multer.File[]; })`	Checks submitted identity images and stores the verification evidence and AI result. – `userId`: `number` – account identifier for this operation. – `documentType`: `DocumentTypeType` – supported identity-document category. – `files`: `{ front?: Express.Multer.File; back?: Express.Multer.File; selfie?: Express.Multer.File; liveness: Express.Multer.File[]; }` – uploaded file supplied to this operation. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `status`, `adminId`, `reviewNotes`, and the remaining selected fields. – Throws `IdentityAiFrontRequiredException`, `IdentityAiSelfieRequiredException`, `IdentityAiLivenessRequiredException`, `IdentityAiFramesInvalidException`, `IdentityAlreadyVerifiedException`.
09	`livenessPose(userId: number, file?: Express.Multer.File)`	Requests pose checking for a submitted liveness image. – `userId`: `number` – account identifier for this operation. – `file`: `Express.Multer.File` – uploaded file supplied to this operation; optional. – *Returns:* An awaitable record with `face_found`, `yaw`, `pitch`, `roll`, `blink`. – Throws `IdentityAiFramesInvalidException`.
10	`getIdentityStatus(userId: number)`	Returns the latest verification state and document history. – `userId`: `number` – account identifier for this operation. – *Returns:* An awaitable record with `status`, `documents`.
11	`getIdentityDocumentFile( userId: number, id: number, side?: 'front' | 'back')`	Resolves an identity document file belonging to the authenticated user. – `userId`: `number` – account identifier for this operation. – `id`: `number` – target record identifier. – `side`: `'front' | 'back'` – side supplied to the operation; optional. – *Returns:* `Promise<{ absolutePath: string; fileName: string; }>` – Throws `IdentityDocumentNotFoundException`.

No	Method	Description
12	`getClientProfile(userId: number)`	Loads public client profile details and verifies the client role. – `userId`: `number` – account identifier for this operation. – *Returns:* An awaitable record with `user`, `client Profile`, `socialLinks`, `id`, `userId`, `displayName`, `avatarUrl`, and the remaining selected fields. – Throws `ClientProfileNotFoundException`.
13	`updateClientProfile( userId: number, input: UpdateClient ProfileType)`	Submits or applies changes to company information according to the revision policy. – `userId`: `number` – account identifier for this operation. – `input`: `UpdateClientProfileType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `message`, `review Required`, `profileStrength`, `revision`.
14	`getSocialLinks(userId: number)`	Retrieves social links associated with the current user's profile. – `userId`: `number` – account identifier for this operation. – *Returns:* An awaitable list of records with `id`, `profile Id`, `url`, `platform`. – Throws `ProfileNotFoundException`.
15	`addSocialLink(userId: number, input: AddSocial LinkType)`	Creates a profile social link after checking for a duplicate platform. – `userId`: `number` – account identifier for this operation. – `input`: `AddSocialLinkType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `profileId`, `url`, `platform`. – Throws `ProfileNotFoundException`, `SocialLink DuplicateException`.
16	`deleteSocialLink(userId: number, id: number)`	Deletes a social link after verifying its owner. – `userId`: `number` – account identifier for this operation. – `id`: `number` – target record identifier. – *Returns:* `Promise<{ message: string; }>` – Throws `SocialLinkNotFoundException`, `Account ProfileForbiddenException`.
17	`getFavorites(clientId: number)`	Returns the client's favourite freelancers. – `clientId`: `number` – client account identifier. – *Returns:* An awaitable list of records with `freelancer Id`, `createdAt`, `profile`.

No	Method	Description
18	addFavorite(clientId: number, freelancerId: number)	Creates a favourite relationship after client and target validation. – clientId: number – client account identifier. – freelancerId: number – freelancer account identifier. – *Returns:* Promise<{ message: string; }> – Throws FreelancerNotFoundException, FavoriteDuplicateException.
19	removeFavorite(clientId: number, freelancerId: number)	Removes a favourite relationship belonging to the client. – clientId: number – client account identifier. – freelancerId: number – freelancer account identifier. – *Returns:* Promise<{ message: string; }> – Throws FavoriteNotFoundException.
20	getFollowing(clientId: number)	Loads the current client's followed freelancers. – clientId: number – client account identifier. – *Returns:* An awaitable list of records with freelancerId, createdAt, profile.
21	discoverFreelancers(clientId: number)	Loads freelancer profiles available for discovery. – clientId: number – client account identifier. – *Returns:* An awaitable list of records with freelancerId, isFollowing, profile.
22	followFreelancer(clientId: number, freelancerId: number)	Creates a following relationship after role, target, and duplicate checks. – clientId: number – client account identifier. – freelancerId: number – freelancer account identifier. – *Returns:* Promise<{ message: string; }> – Throws FreelancerNotFoundException, FollowDuplicateException.
23	unfollowFreelancer(clientId: number, freelancerId: number)	Removes an existing following relationship owned by the client. – clientId: number – client account identifier. – freelancerId: number – freelancer account identifier. – *Returns:* Promise<{ message: string; }> – Throws FollowNotFoundException.

4.22.2 ClassAttribute

No	Attribute	Type
01	`repository`	`AccountProfileRepository` – private readonly injected dependency.
02	`profileRevisionService`	`ProfileRevisionService` – private readonly injected dependency.
03	`cloudinary`	`CloudinaryService` – private readonly injected dependency.
04	`hashing`	`HashingService` – private readonly injected dependency.
05	`identityAi`	`IdentityAiService` – private readonly injected dependency.

4.23 AccountProfileRepository Class

This shared class supports Identity Verification, Client Profile Management, Social Link Management, Favorite Freelancer Management, Following Management, Account Profile Management. Its relevant operations are specified together here.

4.23.1 ClassMethod

No	Method	Description
01	`AccountProfileRepository(prisma: PrismaService)`	Initializes the class with its required dependencies. – `prisma`: `PrismaService` – injected database client. – *Returns:* `void` – constructor, no return value.
02	`findUserWithRoles(userId: number)`	Loads a non-deleted account with its roles and profile for authorization checks. – `userId`: `number` – account identifier for this operation. – *Returns:* An awaitable record with `profile`, `userRoles`, `id`, `createdAt`, `updatedAt`, `email`, `password`, and the remaining selected fields; `null` when absent.
03	`findGeneralProfile(userId: number)`	Loads account email, role assignments, and general profile information. – `userId`: `number` – account identifier for this operation. – *Returns:* An awaitable record with `profile`, `id`, `email`, `userRoles`; `null` when absent.

No	Method	Description
04	`findPasswordCredential( userId: number)`	Loads the current password hash and profile identifier for a non-deleted account. – `userId`: `number` – account identifier for this operation. – *Returns:* An awaitable record with `profile`, `password`; `null` when absent.
05	`updateGeneralProfile( userId: number, input: UpdateGeneral ProfileType, primaryRole?: string)`	Submits or applies edits to general profile information according to the revision policy. – `userId`: `number` – account identifier for this operation. – `input`: `UpdateGeneralProfileType` – validated input fields defined by the declared type. – `primaryRole`: `string` – primary role used to select the profile policy; optional. – *Returns:* An awaitable record with `id`, `userId`, `displayName`, `avatarUrl`, `coverUrl`, `bio`, `online Status`, and the remaining selected fields.
06	`updatePassword(userId: number, password: string)`	Persists an already hashed account password. – `userId`: `number` – account identifier for this operation. – `password`: `string` – already hashed password persisted by the repository. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `email`, `password`, `isBanned`, `deletedAt`, and the remaining selected fields.
07	`updateAvatar(userId: number, avatarUrl: string)`	Updates the profile avatar URL and returns the stored URL and update time. – `userId`: `number` – account identifier for this operation. – `avatarUrl`: `string` – stored avatar URL. – *Returns:* An awaitable record with `avatarUrl`, `updatedAt`.
08	`findClientProfile(userId: number)`	Loads the client profile, social links, and account roles. – `userId`: `number` – account identifier for this operation. – *Returns:* An awaitable record with `user`, `client Profile`, `socialLinks`, `id`, `userId`, `displayName`, `avatarUrl`, and the remaining selected fields; `null` when absent.

No	Method	Description
09	`updateClientProfile( userId: number, input: UpdateClient ProfileType)`	Submits or applies changes to company information according to the revision policy. – `userId`: `number` – account identifier for this operation. – `input`: `UpdateClientProfileType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `clientProfile`, `socialLinks`, `id`, `userId`, `displayName`, `avatarUrl`, `coverUrl`, and the remaining selected fields.
10	`createIdentityDocument( userId: number, documentType: Document TypeType, fileUrl: string)`	Creates an identity document associated with the account and uploaded file. – `userId`: `number` – account identifier for this operation. – `documentType`: `DocumentTypeType` – supported identity-document category. – `fileUrl`: `string` – stored identity-document URL. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `status`, `adminId`, `review Notes`, and the remaining selected fields.
11	`createIdentityAiDocument(data: { userId: number; documentType: Document TypeType; fileUrl: string; backFileUrl: string | null; status: VerificationStatus; reviewNotes: string | null; duplicateOfId: number | null; aiStatus: string; aiScore: number | null; aiError: string | null; aiSummary: Prisma.InputJsonValue; ai SelfieSummary: Prisma. InputJsonValue; aiScanned At: Date; })`	Stores the identity document together with AI verification results and review metadata. – `data`: `{ userId: number; documentType: DocumentTypeType; fileUrl: string; backFile Url: string | null; status: Verification Status; reviewNotes: string | null; duplicateOfId: number | null; aiStatus: string; aiScore: number | null; aiError: string | null; aiSummary: Prisma.InputJson Value; aiSelfieSummary: Prisma.InputJson Value; aiScannedAt: Date; }` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `status`, `adminId`, `review Notes`, and the remaining selected fields.

No	Method	Description
12	`findApprovedIdentity DocumentByIdentityNumber( userId: number, identityNumber: string)`	Searches other accounts for an approved document with the same normalized identity number. – `userId: number` – account identifier for this operation. – `identityNumber: string` – identity number used for duplicate checking. – *Returns:* `Promise<{ id: number; }>`
13	`findIdentityDocuments( userId: number)`	Lists non-deleted identity documents for the account, newest first. – `userId: number` – account identifier for this operation. – *Returns:* An awaitable list of records with `id`, `userId`, `createdAt`, `deletedAt`, `status`, `adminId`, `review Notes`, and the remaining selected fields.
14	`findIdentityDocumentById( id: number)`	Loads a non-deleted identity document by its identifier. – `id: number` – target record identifier. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `status`, `adminId`, `review Notes`, and the remaining selected fields; `null` when absent.
15	`findSocialLinks(userId: number)`	Lists profile social links by platform; returns null if the profile does not exist. – `userId: number` – account identifier for this operation. – *Returns:* An awaitable list of records with `id`, `profile Id`, `url`, `platform`; `null` when absent.
16	`findSocialLinkByPlatform( userId: number, platform: AddSocialLink Type['platform'])`	Finds a profile social link for the specified platform to detect duplicates. – `userId: number` – account identifier for this operation. – `platform: AddSocialLinkType['platform']` – social platform identifier. – *Returns:* An awaitable record with `id`, `profileId`, `url`, `platform`; `null` when absent.
17	`createSocialLink(userId: number, input: AddSocial LinkType)`	Creates a social link on the existing account profile. – `userId: number` – account identifier for this operation. – `input: AddSocialLinkType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `profileId`, `url`, `platform`.

No	Method	Description
18	`findSocialLinkById(id: number)`	Loads a social link with its profile for ownership validation. – `id`: `number` – target record identifier. – *Returns:* An awaitable record with `profile`, `id`, `profileId`, `url`, `platform`; `null` when absent.
19	`deleteSocialLink(id: number)`	Deletes a social link after verifying its owner. – `id`: `number` – target record identifier. – *Returns:* An awaitable record with `id`, `profileId`, `url`, `platform`.
20	`findFavorite(clientId: number, freelancerId: number)`	Finds the favorite association for a client and freelancer pair. – `clientId`: `number` – client account identifier. – `freelancerId`: `number` – freelancer account identifier. – *Returns:* An awaitable record with `createdAt`, `clientId`, `freelancerId`; `null` when absent.
21	`createFavorite(clientId: number, freelancerId: number)`	Creates the client-to-freelancer favorite association. – `clientId`: `number` – client account identifier. – `freelancerId`: `number` – freelancer account identifier. – *Returns:* An awaitable record with `createdAt`, `clientId`, `freelancerId`.
22	`deleteFavorite(clientId: number, freelancerId: number)`	Deletes the specified favorite association. – `clientId`: `number` – client account identifier. – `freelancerId`: `number` – freelancer account identifier. – *Returns:* An awaitable record with `createdAt`, `clientId`, `freelancerId`.
23	`findFavorites(clientId: number)`	Lists favorites with freelancer profile and skill information, newest first. – `clientId`: `number` – client account identifier. – *Returns:* An awaitable list of records with `freelancer`, `createdAt`, `clientId`, `freelancerId`.
24	`findFollow(clientId: number, freelancerId: number)`	Finds the follow association for a client and freelancer pair. – `clientId`: `number` – client account identifier. – `freelancerId`: `number` – freelancer account identifier. – *Returns:* An awaitable record with `createdAt`, `clientId`, `freelancerId`; `null` when absent.

No	Method	Description
25	`createFollowWithNotification(clientId: number, freelancerId: number, followerName: string)`	Creates the follow association and a new-follower notification in one transaction. – `clientId`: `number` – client account identifier. – `freelancerId`: `number` – freelancer account identifier. – `followerName`: `string` – display name used in the follower notification. – *Returns:* `Promise<{ createdAt: Date; clientId: number; freelancerId: number; }>`
26	`deleteFollow(clientId: number, freelancerId: number)`	Deletes the specified follow association. – `clientId`: `number` – client account identifier. – `freelancerId`: `number` – freelancer account identifier. – *Returns:* An awaitable record with `createdAt`, `clientId`, `freelancerId`.
27	`findFollowing(clientId: number)`	Lists followed freelancers whose accounts are active, with profile and skill information. – `clientId`: `number` – client account identifier. – *Returns:* An awaitable list of records with `freelancer`, `createdAt`, `clientId`, `freelancerId`.
28	`findDiscoverableFreelancers(clientId: number)`	Lists eligible freelancer accounts with their profile, skills, and follow relationship to the current client. – `clientId`: `number` – client account identifier. – *Returns:* An awaitable list of records with `profile`, `id`, `followsAsFreelancer`.

4.23.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.24 ClientProfileController Class

4.24.1 ClassMethod

No	Method	Description
01	`ClientProfileController(service: AccountProfileService)`	Initializes the class with its required dependencies. – `service`: `AccountProfileService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`getDetail(userId: number)`	Loads public client profile details and verifies the client role. – `userId`: `number` – account identifier supplied by authentication or the route. – *Returns:* An awaitable record with `user`, `clientProfile`, `socialLinks`, `id`, `userId`, `displayName`, `avatarUrl`, and the remaining selected fields. – Response schema: `ClientProfileDetailDto`. – Route: `GET /clients/:userId`.
03	`update(userId: number, body: UpdateClientProfileDto)`	Submits or applies changes to company information according to the revision policy. – `userId`: `number` – account identifier supplied by authentication or the route. – `body`: `UpdateClientProfileDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `message`, `reviewRequired`, `profileStrength`, `revision`. – Response schema: `ProfileRevisionSubmissionDto`. – Route: `PUT /clients/me/profile`.

4.24.2 ClassAttribute

No	Attribute	Type
01	`service`	`AccountProfileService` – private readonly injected dependency.

4.25 SocialLinkController Class

4.25.1 ClassMethod

No	Method	Description
01	`SocialLinkController( service: AccountProfile Service)`	Initializes the class with its required dependencies. – `service`: `AccountProfileService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`getAll(userId: number)`	Retrieves social links associated with the current user's profile. – `userId`: `number` – account identifier supplied by authentication or the route. – *Returns:* An awaitable list of records with `id`, `profile Id`, `url`, `platform`. – Response schema: `SocialLinkListDto`. – Route: `GET /social-links`.
03	`add(userId: number, body: AddSocialLinkDto)`	Creates a profile social link after checking for a duplicate platform. – `userId`: `number` – account identifier supplied by authentication or the route. – `body`: `AddSocialLinkDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `profileId`, `url`, `platform`. – Response schema: `SocialLinkDto`. – Route: `POST /social-links`.
04	`remove(userId: number, id: number)`	Deletes a social link after verifying its owner. – `userId`: `number` – account identifier supplied by authentication or the route. – `id`: `number` – target record identifier. – *Returns:* `Promise<{ message: string; }>` – Route: `DELETE /social-links/:id`.

4.25.2 ClassAttribute

No	Attribute	Type
01	`service`	`AccountProfileService` – private readonly injected dependency.

4.26 FavoriteFreelancerController Class

4.26.1 ClassMethod

No	Method	Description
01	`FavoriteFreelancer Controller( service: AccountProfile Service)`	Initializes the class with its required dependencies. – `service`: `AccountProfileService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`getAll(userId: number)`	Returns the client's favourite freelancers. – `userId`: `number` – account identifier supplied by authentication or the route. – *Returns:* An awaitable list of records with `freelancer Id`, `createdAt`, `profile`. – Response schema: `FavoriteFreelancerListDto`. – Route: `GET /favorites/freelancers`.
03	`add(userId: number, freelancerId: number)`	Creates a favourite relationship after client and target validation. – `userId`: `number` – account identifier supplied by authentication or the route. – `freelancerId`: `number` – freelancer account identifier. – *Returns:* `Promise<{ message: string; }>` – Route: `POST /favorites/freelancers/: freelancerId`.
04	`remove(userId: number, freelancerId: number)`	Removes a favourite relationship belonging to the client. – `userId`: `number` – account identifier supplied by authentication or the route. – `freelancerId`: `number` – freelancer account identifier. – *Returns:* `Promise<{ message: string; }>` – Route: `DELETE /favorites/freelancers/: freelancerId`.

4.26.2 ClassAttribute

No	Attribute	Type
01	`service`	`AccountProfileService` – private readonly injected dependency.

4.27 UsersController Class

4.27.1 ClassMethod

No	Method	Description
01	`UsersController(service: UsersService)`	A method to initialize the controller with the user administration service. – `service`: `UsersService` – service that owns admin user use cases. – *Returns:* `void` – constructor, no return value.
02	`createUser(body: AdminCreateUserBodyDto)`	A method to create an account with an initial role. – `body`: `AdminCreateUserBodyDto` – email, password, full name, and role. – *Returns:* `Promise<AdminCreateUserResponseType>` – the created account. – Route: `POST /users`.
03	`updateUser(id: number, actorId: number, body: AdminUpdateUserBodyDto)`	A method to update email, display name, or ban state. – `id`: `number` – target user identifier. – `actorId`: `number` – current administrator, used for the self-ban check. – `body`: `AdminUpdateUserBodyDto` – optional email, full name, and ban flag. – *Returns:* `Promise<AdminUpdateUserResponseType>` – the updated account. – Route: `PATCH /users/:id`.
04	`updateClientProfile(id: number, body: AdminUpdateClientProfileBodyDto)`	A method to update the Client profile attached to an account. – `id`: `number` – target user identifier. – `body`: `AdminUpdateClientProfileBodyDto` – company name, description, and website. – *Returns:* `Promise<AdminClientProfileResponseType>` – the updated Client profile. – Route: `PATCH /users/:id/client-profile`.

No	Method	Description
05	`updateFreelancerProfile( id: number, body: AdminUpdate FreelancerProfileBodyDto)`	A method to update the Freelancer headline and biography. – `id: number` – target user identifier. – `body: AdminUpdateFreelancerProfileBodyDto` – title, bio, languages, education, and certifications. – *Returns:* `Promise<MessageResType>` – a success message. – Route: `PATCH /users/:id/freelancer-profile`.
06	`replaceFreelancerSkills( id: number, body: AdminReplace FreelancerSkillsBodyDto)`	A method to replace every skill on a Freelancer profile. – `id: number` – target user identifier. – `body.skills`: array of skill name and proficiency pairs. – *Returns:* `Promise<MessageResType>` – a success message. – Route: `PUT /users/:id/freelancer-profile/ skills`.
07	`createPortfolioItem( id: number, body: AdminCreate PortfolioItemBodyDto)`	A method to add a portfolio item to a Freelancer profile. – `id: number` – target user identifier. – `body: AdminCreatePortfolioItemBodyDto` – title, description, technologies, and project URL. – *Returns:* `Promise<MessageResType>` – a success message. – Route: `POST /users/:id/freelancer-profile/ portfolio-items`.
08	`updatePortfolioItem( id: number, itemId: number, body: AdminUpdate PortfolioItemBodyDto)`	A method to update one portfolio item owned by the Freelancer profile. – `id: number` – target user identifier. – `itemId: number` – portfolio item identifier. – `body: AdminUpdatePortfolioItemBodyDto` – optional item fields. – *Returns:* `Promise<MessageResType>` – a success message. – Route: `PATCH /users/:id/freelancer-profile/ portfolio-items/:itemId`.

No	Method	Description
09	`deletePortfolioItem( id: number, itemId: number)`	A method to soft-delete one portfolio item. – `id`: `number` – target user identifier. – `itemId`: `number` – portfolio item identifier. – *Returns:* `Promise<MessageResType>` – a success message. – Route: `DELETE /users/:id/freelancer-profile/portfolio-items/:itemId`.
10	`getUsers(query: AdminUserQueryDto)`	A method to return a filtered, paginated user list. – `query`: `AdminUserQueryDto` – page, limit, search, role, sort, and status filters. – *Returns:* `Promise<AdminUserListResponseType>` – users and pagination. – Route: `GET /users`.
11	`getSkillCatalog()`	A method to return the active skill catalogue for the profile editor. – *Returns:* `Promise<AdminSkillCatalogListType>` – active skills ordered by name. – Route: `GET /users/skills-catalog`.
12	`getUserById(id: number)`	A method to return a consolidated user detail. – `id`: `number` – target user identifier. – *Returns:* `Promise<AdminUserDetailResponseType>` – account, roles, and profile summaries. – Route: `GET /users/:id`.

4.27.2 ClassAttribute

No	Attribute	Type
01	`service`	`UsersService` – private readonly user administration service.

4.28 UsersService Class

4.28.1 ClassMethod

No	Method	Description
01	`UsersService( repository: UsersRepository, hashingService: HashingService)`	A method to initialize the service with persistence and password hashing. – `repository`: `UsersRepository` – user persistence dependency. – `hashingService`: `HashingService` – password hashing utility. – *Returns:* `void` – constructor, no return value.
02	`createUser(body: AdminCreateUserBodyType)`	A method to validate the email and role, hash the password, and create the account. – `body`: `AdminCreateUserBodyType` – email, password, full name, and role id. – *Returns:* `Promise<AdminCreateUserResponseType>` – the created account. – Throws `EmailAlreadyExistsException`, `CreateUserRoleNotFoundException`, `CannotAssignAdminRoleException`, or `FailedToCreateUserException`.
03	`getUsers(query: AdminUserQueryType)`	A method to load a user page. – `query`: `AdminUserQueryType` – page, limit, search, role, sort, and status filters. – *Returns:* `Promise<AdminUserListResponseType>` – users and pagination. – Failure is mapped to `FailedToGetUsersException`.
04	`getUserById(id: number)`	A method to load a consolidated account detail. – `id`: `number` – target user identifier. – *Returns:* `Promise<AdminUserDetailResponseType>` – account, roles, and profiles. – Throws `UserNotFoundException` when absent or `FailedToGetUserException` on an unexpected read failure.
05	`listSkillCatalog()`	A method to return the active skill catalogue. – *Returns:* `Promise<AdminSkillCatalogListType>` – active skills ordered by name.

No	Method	Description
06	`updateUser(id, actorId, body)`	A method to update account fields while enforcing the self-ban and email rules. – `id`: `number` – target user identifier. – `actorId`: `number` – current administrator. – `body`: `AdminUpdateUserBodyType` – optional email, full name, and ban flag. – *Returns:* `Promise<AdminUpdateUserResponse Type>` – the updated account. – Throws `CannotBanSelfException`, `EmailAlready ExistsException`, `UserNotFoundException`, or `FailedToUpdateUserException`.
07	`updateClientProfile(user Id, body)`	A method to validate the Client profile context and upsert the profile. – `userId`: `number` – target user identifier. – `body`: `AdminUpdateClientProfileBodyType` – company fields. – *Returns:* `Promise<AdminClientProfileResponse Type>` – the updated Client profile. – Throws `UserNotFoundException`, `NoClientRole ForClientProfileException`, or `FailedToUpdate ClientProfileException`.
08	`updateFreelancerProfile( userId, body)`	A method to validate the Freelancer context and upsert profile fields. – `userId`: `number` – target user identifier. – `body`: `AdminUpdateFreelancerProfileBodyType` – title, bio, languages, education, and certifications. – *Returns:* `Promise<MessageResType>` – a success message. – Throws `NoFreelancerRoleForProfileException` or `FailedToUpdateFreelancerProfileException`.
09	`replaceFreelancerSkills( userId, body)`	A method to replace all Freelancer skills and create missing catalogue skills. – `userId`: `number` – target user identifier. – `body`: `AdminReplaceFreelancerSkillsBodyType` – skill name and proficiency pairs. – *Returns:* `Promise<MessageResType>` – a success message. – Throws `NoFreelancerRoleForProfileException` or `FailedToSaveSkillsException`.

No	Method	Description
10	`createPortfolioItem(userId, body)`	A method to create a portfolio item on the Freelancer profile. – `userId`: `number` – target user identifier. – `body`: `AdminCreatePortfolioItemBodyType` – item fields. – *Returns:* `Promise<MessageResType>` – a success message. – Throws `FailedToCreatePortfolioItemException`.
11	`updatePortfolioItem(userId, itemId, body)`	A method to update an owned portfolio item. – `userId`: `number` – target user identifier. – `itemId`: `number` – portfolio item identifier. – `body`: `AdminUpdatePortfolioItemBodyType` – optional item fields. – *Returns:* `Promise<MessageResType>` – a success message. – Throws `PortfolioItemNotFoundException` or `FailedToUpdatePortfolioItemException`.
12	`deletePortfolioItem(userId, itemId)`	A method to soft-delete an owned portfolio item. – `userId`: `number` – target user identifier. – `itemId`: `number` – portfolio item identifier. – *Returns:* `Promise<MessageResType>` – a success message. – Throws `PortfolioItemNotFoundException` or `FailedToDeletePortfolioItemException`.
13	`getFreelancerProfileContext(userId)`	A private method to load the profile and role context for a Freelancer edit. – `userId`: `number` – target user identifier. – *Returns:* `Promise<context>` – profile id, Freelancer profile id, and role flag. – Throws `UserNotFoundException` or `NoFreelancerRoleForProfileException`.
14	`ensureFreelancerProfileRow(userId, context)`	A private method to create the Profile and FreelancerProfile rows when missing. – `userId`: `number` – target user identifier. – `context`: object – profile id and role context. – *Returns:* `Promise<number | null>` – the Freelancer profile id. – Used by skill and portfolio editors before writing child rows.

4.28.2 ClassAttribute

No	Attribute	Type
01	`repository`	`UsersRepository` – private readonly persistence dependency.
02	`hashingService`	`HashingService` – private readonly password utility.

4.29 UsersRepository Class

4.29.1 ClassMethod

No	Method	Description
01	`UsersRepository(prisma: PrismaService)`	A method to initialize the repository with the shared Prisma client. – `prisma`: `PrismaService` – database client. – *Returns:* `void` – constructor, no return value.
02	`getUsers(query: AdminUserQueryType)`	A method to query users with search, role, status, sorting, and pagination. – `query`: `AdminUserQueryType` – list filters. – *Returns:* `Promise<AdminUserListResponseType>` – users and pagination.
03	`createUserByAdmin(data)`	A method to create the account, profile, and primary role in one write. – `data`: object – email, hashed password, full name, role id, and role name. – *Returns:* `Promise<AdminCreateUserResponseType>` – the created account. – Creates an empty Expert profile when the role is Expert.
04	`findUserByEmail(email)`	A method to find a non-deleted account by email. – `email`: `string` – account email. – *Returns:* `Promise<{ id: number } | null>` – the account id, or `null`.
05	`findActiveRoleById(id)`	A method to load an active role by id. – `id`: `number` – role identifier. – *Returns:* `Promise<{ id: number; name: string } | null>` – the role, or `null`.

No	Method	Description
06	`updateUserByAdmin(id, data)`	A method to update account fields, profile name, and sessions on ban. – `id`: `number` – target account. – `data`: object – optional email, full name, and ban flag. – *Returns:* `Promise<AdminUpdateUserResponseType | null>` – the updated account, or `null` when absent. – Setting `isBanned` to true deletes every session in the same transaction.
07	`findClientProfileEdit Context(id)`	A method to load the profile context required by the Client editor. – `id`: `number` – target account. – *Returns:* `Promise<context | null>` – profile id, Client profile id, and role flag.
08	`upsertClientProfileBy Admin(userId, data)`	A method to create or update the Client profile. – `userId`: `number` – target account. – `data`: object – company name, description, and website. – *Returns:* `Promise<ClientProfile | null>` – the persisted Client profile.
09	`findFreelancerProfileEdit Context(id)`	A method to load the profile context required by the Freelancer editor. – `id`: `number` – target account. – *Returns:* `Promise<context | null>` – profile id, Freelancer profile id, and role flag.
10	`updateFreelancerProfileBy Admin(userId, data)`	A method to update Profile biography and Freelancer profile fields. – `userId`: `number` – target account. – `data`: object – title, bio, languages, education, and certifications. – *Returns:* `Promise<void>` – no value.
11	`ensureFreelancerProfile( userId, profileId)`	A method to create the missing Profile or FreelancerProfile row. – `userId`: `number` – target account. – `profileId`: `number | null` – existing profile id, or `null`. – *Returns:* `Promise<number | null>` – the Freelancer profile id.

No	Method	Description
12	`replaceFreelancerSkills( freelancerProfileId, skills)`	A method to replace all skills and create missing catalogue skills in a transaction. – `freelancerProfileId`: `number` – Freelancer profile identifier. – `skills`: array of skill name and proficiency pairs. – *Returns:* `Promise<void>` – no value. – Free-text names are matched case-insensitively; missing skills receive a generated slug.
13	`createPortfolioItem( freelancerProfileId, data)`	A method to insert a portfolio item. – `freelancerProfileId`: `number` – Freelancer profile identifier. – `data`: object – title, description, technologies, and project URL. – *Returns:* `Promise<PortfolioItem>` – the created item.
14	`findPortfolioItemOwned( id)`	A method to load a portfolio item with its owner and delete state. – `id`: `number` – portfolio item identifier. – *Returns:* `Promise<{ id: number; freelancer ProfileId: number; deletedAt: Date | null } | null>` – the ownership record, or `null`.
15	`updatePortfolioItemBy Admin(id, data)`	A method to update only the provided portfolio item fields. – `id`: `number` – portfolio item identifier. – `data`: object – optional item fields. – *Returns:* `Promise<PortfolioItem>` – the updated item.
16	`softDeletePortfolioItem( id)`	A method to set the portfolio item delete timestamp. – `id`: `number` – portfolio item identifier. – *Returns:* `Promise<PortfolioItem>` – the soft-deleted item.
17	`listActiveSkillCatalog()`	A method to return active skills ordered by name. – *Returns:* `Promise<AdminSkillCatalogListType>` – the catalogue.

No	Method	Description
18	`getUserById(id)`	A method to load the full account graph for the detail page. – `id`: `number` – target account. – *Returns:* `Promise<AdminUserDetailResponseType | null>` – the detail view, or `null`.

4.29.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.30 FollowingFreelancerController Class

4.30.1 ClassMethod

No	Method	Description
01	`FollowingFreelancerController(service: AccountProfileService)`	Initializes the class with its required dependencies. – `service`: `AccountProfileService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`discover(userId: number)`	Loads freelancer profiles available for discovery. – `userId`: `number` – account identifier supplied by authentication or the route. – *Returns:* An awaitable list of records with `freelancerId`, `isFollowing`, `profile`. – Response schema: `DiscoverFreelancerListDto`. – Route: `GET /following/freelancers/discover`.
03	`getFollowing(userId: number)`	Loads the current client's followed freelancers. – `userId`: `number` – account identifier supplied by authentication or the route. – *Returns:* An awaitable list of records with `freelancerId`, `createdAt`, `profile`. – Response schema: `FollowingFreelancerListDto`. – Route: `GET /following/freelancers`.

No	Method	Description
04	`followFreelancer(userId: number, freelancerId: number)`	Creates a following relationship after role, target, and duplicate checks. – `userId`: `number` – account identifier supplied by authentication or the route. – `freelancerId`: `number` – freelancer account identifier. – *Returns:* `Promise<{ message: string; }>` – Route: `POST /following/freelancers/:freelancerId`.
05	`unfollowFreelancer(userId: number, freelancerId: number)`	Removes an existing following relationship owned by the client. – `userId`: `number` – account identifier supplied by authentication or the route. – `freelancerId`: `number` – freelancer account identifier. – *Returns:* `Promise<{ message: string; }>` – Route: `DELETE /following/freelancers/:freelancerId`.

4.30.2 ClassAttribute

No	Attribute	Type
01	`service`	`AccountProfileService` – private readonly injected dependency.

4.31 NotificationController Class

4.31.1 ClassMethod

No	Method	Description
01	`NotificationController(service: NotificationService)`	Initializes the class with its required dependencies. – `service`: `NotificationService` – injected dependency. – *Returns:* `void` – constructor, no return value.

No	Method	Description
02	`getAll(userId: number)`	Returns the list associated with the authenticated user. – `userId`: `number` – account identifier supplied by authentication or the route. – *Returns:* An awaitable list of records with `id`, `userId`, `createdAt`, `deletedAt`, `title`, `data`, `type`, and the remaining selected fields. – Response schema: `NotificationListDto`. – Route: `GET /notifications`.
03	`markAllRead(userId: number)`	Marks all active unread notifications of the user as read. – `userId`: `number` – account identifier supplied by authentication or the route. – *Returns:* `Promise<{ message: "Success. NotificationsMarkedAsRead"; count: number; }>` – Route: `PATCH /notifications/read-all`.
04	`markRead(userId: number, id: number)`	Marks an owned notification as read. – `userId`: `number` – account identifier supplied by authentication or the route. – `id`: `number` – target record identifier. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `title`, `data`, `type`, and the remaining selected fields. – Response schema: `NotificationDto`. – Route: `PATCH /notifications/:id/read`.
05	`delete(userId: number, id: number)`	Checks notification ownership and marks the notification as deleted. – `userId`: `number` – account identifier supplied by authentication or the route. – `id`: `number` – target record identifier. – *Returns:* `Promise<{ message: "Success. NotificationDeleted"; }>` – Route: `DELETE /notifications/:id`.

4.31.2 ClassAttribute

No	Attribute	Type
01	service	NotificationService – private readonly injected dependency.

4.32 NotificationService Class

4.32.1 ClassMethod

No	Method	Description
01	NotificationService(repository: NotificationRepository)	Initializes the class with its required dependencies. – repository: NotificationRepository – injected dependency. – *Returns:* void – constructor, no return value.
02	getAll(userId: number)	Returns the list associated with the authenticated user. – userId: number – account identifier for this operation. – *Returns:* An awaitable list of records with id, userId, createdAt, deletedAt, title, data, type, and the remaining selected fields.
03	markRead(userId: number, notificationId: number)	Marks an owned notification as read. – userId: number – account identifier for this operation. – notificationId: number – notification identifier. – *Returns:* An awaitable record with id, userId, createdAt, deletedAt, title, data, type, and the remaining selected fields. – Throws NotificationNotFoundException.
04	markAllRead(userId: number)	Marks all active unread notifications of the user as read. – userId: number – account identifier for this operation. – *Returns:* Promise<{ message: "Success. NotificationsMarkedAsRead"; count: number; }>
05	delete(userId: number, notificationId: number)	Checks notification ownership and marks the notification as deleted. – userId: number – account identifier for this operation. – notificationId: number – notification identifier. – *Returns:* Promise<{ message: "Success. NotificationDeleted"; }> – Throws NotificationNotFoundException.

4.32.2 ClassAttribute

No	Attribute	Type
01	`repository`	`NotificationRepository` – private readonly injected dependency.

4.33 NotificationRepository Class

4.33.1 ClassMethod

No	Method	Description
01	`NotificationRepository( prisma: PrismaService)`	Initializes the class with its required dependencies. – `prisma`: `PrismaService` – injected database client. – *Returns:* `void` – constructor, no return value.
02	`findAll(userId: number)`	Lists non-deleted notifications for the account, newest first. – `userId`: `number` – account identifier for this operation. – *Returns:* An awaitable list of records with `id`, `userId`, `createdAt`, `deletedAt`, `title`, `data`, `type`, and the remaining selected fields.
03	`findById(userId: number, notificationId: number)`	Finds a non-deleted notification belonging to the specified account. – `userId`: `number` – account identifier for this operation. – `notificationId`: `number` – notification identifier. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `title`, `data`, `type`, and the remaining selected fields; `null` when absent.
04	`markRead(notificationId: number)`	Marks an owned notification as read. – `notificationId`: `number` – notification identifier. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `title`, `data`, `type`, and the remaining selected fields.
05	`markAllRead(userId: number)`	Marks all active unread notifications of the user as read. – `userId`: `number` – account identifier for this operation. – *Returns:* An awaitable record with `count`.

No	Method	Description
06	`softDelete(notification Id: number)`	Persists the deletion timestamp for the specified record. – `notificationId`: `number` – notification identifier. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `deletedAt`, `title`, `data`, `type`, and the remaining selected fields.

4.33.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.34 Following Management (Shared Classes)

`AccountProfileService` is specified in Section 4.22.

- `getFollowing()`.
- `discoverFreelancers()`.
- `followFreelancer()`.
- `unfollowFreelancer()`.

`AccountProfileRepository` is specified in Section 4.23.

- `findUserWithRoles()`.
- `findFollow()`.
- `createFollowWithNotification()`.
- `deleteFollow()`.
- `findFollowing()`.
- `findDiscoverableFreelancers()`.

4.35 Role Switching (Shared Classes)

`AuthController` is specified in Authentication in Sprint 1.

- `switchRole()`.
- `joinRole()`.

`AuthService` is specified in Authentication in Sprint 1.

- `switchRole()`.
- `joinRole()`.

`AuthRepository` is specified in Authentication in Sprint 1.

- `switchPrimaryRole()`.
- `joinRole()`.

4.36 BannersAdminController Class

4.36.1 ClassMethod

No	Method	Description
01	`BannersAdminController(service: BannersAdminService)`	A method to initialize the controller with the banner administration service. – `service`: `BannersAdminService` – service that owns banner use cases. – *Returns:* `void` – constructor, no return value.
02	`listBanners(query: BannerAdminQueryDto)`	A method to return a filtered, paginated banner list. – `query`: `BannerAdminQueryDto` – page, limit, search, position, active, and sort options. – *Returns:* `Promise<BannerAdminListResponseType>` – banners and pagination. – Route: `GET /admin/banners`.
03	`createBanner(body: CreateBannerBodyDto)`	A method to create an advertisement banner. – `body`: `CreateBannerBodyDto` – title, image URL, link, position, active state, and dates. – *Returns:* `Promise<BannerAdminDetailResponseType>` – the created banner. – Route: `POST /admin/banners`.
04	`updateBanner(id: number, body: UpdateBannerBodyDto)`	A method to update a banner. – `id`: `number` – banner identifier. – `body`: `UpdateBannerBodyDto` – optional banner fields. – *Returns:* `Promise<BannerAdminDetailResponseType>` – the updated banner. – Route: `PATCH /admin/banners/:id`.
05	`deleteBanner(id: number)`	A method to permanently delete a banner. – `id`: `number` – banner identifier. – *Returns:* `Promise<BannerAdminDeleteResponseType>` – a success message. – Route: `DELETE /admin/banners/:id`.

No	Method	Description
06	`uploadImage(file?: Express.Multer.File)`	A method to upload a banner image and return its URL. – `file`: `Express.Multer.File | undefined` – multipart image field named `file`. – *Returns:* `Promise<{ imageUrl: string; publicId: string }>` – the stored image reference. – Route: `POST /admin/banners/upload-image`; multipart field `file`; maximum 10 MB.
07	`getBannerDetail(id: number)`	A method to return one banner. – `id`: `number` – banner identifier. – *Returns:* `Promise<BannerAdminDetailResponseType>` – the banner detail. – Route: `GET /admin/banners/:id`.

4.36.2 ClassAttribute

No	Attribute	Type
01	`service`	`BannersAdminService` – private readonly banner administration service.

4.37 BannersAdminService Class

4.37.1 ClassMethod

No	Method	Description
01	`BannersAdminService(repository: BannersAdminRepository, cloudinaryService: CloudinaryService)`	A method to initialize the service with persistence and image storage. – `repository`: `BannersAdminRepository` – banner persistence dependency. – `cloudinaryService`: `CloudinaryService` – remote image storage. – *Returns:* `void` – constructor, no return value.

No	Method	Description
02	`uploadImage(file?: Express.Multer.File)`	A method to validate an image and store it on Cloudinary or the local upload directory. – `file: Express.Multer.File | undefined` – multipart image. – *Returns:* `Promise<{ imageUrl: string; publicId: string }>` – the stored image reference. – Throws `BannerImageRequiredException`, `BannerImageInvalidTypeException`, `BannerImageTooLargeException`, or `FailedToUploadBannerImageException`.
03	`listBanners(query: BannerAdminQueryType)`	A method to load a banner page. – `query: BannerAdminQueryType` – list filters. – *Returns:* `Promise<BannerAdminListResponseType>` – banners and pagination. – Failure is mapped to `FailedToLoadBannerListException`.
04	`getBannerDetail(id: number)`	A method to load one banner. – `id: number` – banner identifier. – *Returns:* `Promise<BannerAdminDetailResponseType>` – the banner detail. – Throws `BannerAdminNotFoundException` or `FailedToLoadBannerDetailException`.
05	`createBanner(body: BannerCreateBodyType)`	A method to create a banner after validating its position. – `body: BannerCreateBodyType` – banner fields. – *Returns:* `Promise<BannerAdminDetailResponseType>` – the created banner. – Throws `BannerPositionTakenException` or `FailedToCreateBannerException`.
06	`updateBanner(id: number, body: BannerUpdateBodyType)`	A method to update a banner after validating its position. – `id: number` – banner identifier. – `body: BannerUpdateBodyType` – optional banner fields. – *Returns:* `Promise<BannerAdminDetailResponseType>` – the updated banner. – Throws `BannerAdminNotFoundException`, `BannerPositionTakenException`, or `FailedToUpdateBannerException`.

No	Method	Description
07	`deleteBanner(id: number)`	A method to permanently delete a banner. – `id`: `number` – banner identifier. – *Returns:* `Promise<BannerAdminDeleteResponse Type>` – a success message. – Throws `BannerAdminNotFoundException` or `FailedToDeleteBannerException`.

4.37.2 ClassAttribute

No	Attribute	Type
01	`repository`	`BannersAdminRepository` – private readonly persistence dependency.
02	`cloudinaryService`	`CloudinaryService` – private readonly image storage dependency.

4.38 BannersAdminRepository Class

4.38.1 ClassMethod

No	Method	Description
01	`BannersAdminRepository(prisma: PrismaService)`	A method to initialize the repository with the shared Prisma client. – `prisma`: `PrismaService` – database client. – *Returns:* `void` – constructor, no return value.
02	`listBanners(query: BannerAdminQueryType)`	A method to query banners with search, position, active-state, and sorting filters. – `query`: `BannerAdminQueryType` – list filters. – *Returns:* `Promise<BannerAdminListResponse Type>` – banners and pagination.
03	`getBannerDetail(id: number)`	A method to load one banner. – `id`: `number` – banner identifier. – *Returns:* `Promise<BannerAdminDetailResponse Type | null>` – the banner, or `null`.

No	Method	Description
04	`createBanner(data: BannerCreateBodyType)`	A method to validate the position constraint and insert a banner. – `data`: `BannerCreateBodyType` – banner fields. – *Returns:* `Promise<BannerAdminDetailResponse Type>` – the created banner. – Throws `BannerPositionTakenException`.
05	`updateBanner(id, data)`	A method to validate the position constraint and update a banner. – `id`: `number` – banner identifier. – `data`: `BannerUpdateBodyType` – optional banner fields. – *Returns:* `Promise<BannerAdminDetailResponse Type>` – the updated banner. – Throws `BannerAdminNotFoundException` or `Banner PositionTakenException`.
06	`deleteBanner(id: number)`	A method to delete a banner and report whether a row was removed. – `id`: `number` – banner identifier. – *Returns:* `Promise<BannerAdminDeleteResponse Type>` – a success message. – Throws `BannerAdminNotFoundException`.

4.38.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.39 SkillsAdminController Class

4.39.1 ClassMethod

No	Method	Description
01	`SkillsAdminController( service: SkillsAdmin Service)`	A method to initialize the controller with the skill administration service. – `service`: `SkillsAdminService` – service that owns skill use cases. – *Returns:* `void` – constructor, no return value.
02	`listSkills(query: Skill AdminQueryDto)`	A method to return a filtered, paginated skill list. – `query`: `SkillAdminQueryDto` – page, limit, search, active, and sort options. – *Returns:* `Promise<SkillAdminListResponseType>` – skills and pagination. – Route: `GET /admin/skills`.
03	`createSkill(body: Create SkillBodyDto)`	A method to create a skill. – `body`: `CreateSkillBodyDto` – name and optional description. – *Returns:* `Promise<SkillAdminDetailResponse Type>` – the created skill. – Route: `POST /admin/skills`.
04	`updateSkill(id: number, body: UpdateSkillBody Dto)`	A method to update a skill. – `id`: `number` – skill identifier. – `body`: `UpdateSkillBodyDto` – optional name, description, and active state. – *Returns:* `Promise<SkillAdminDetailResponse Type>` – the updated skill. – Route: `PATCH /admin/skills/:id`.
05	`restoreSkill(id: number)`	A method to restore a soft-deleted skill. – `id`: `number` – skill identifier. – *Returns:* `Promise<SkillAdminDetailResponse Type>` – the restored skill. – Route: `PATCH /admin/skills/:id/restore`.
06	`deleteSkill(id: number)`	A method to soft-delete a skill. – `id`: `number` – skill identifier. – *Returns:* `Promise<SkillAdminDeleteResponse Type>` – a success message. – Route: `DELETE /admin/skills/:id`.

No	Method	Description
07	`getSkillDetail(id: number)`	A method to return one skill. – `id: number` – skill identifier. – *Returns:* `Promise<SkillAdminDetailResponseType>` – the skill detail. – Route: `GET /admin/skills/:id`.

4.39.2 ClassAttribute

No	Attribute	Type
01	`service`	`SkillsAdminService` – private readonly skill administration service.

4.40 SkillsAdminService Class

4.40.1 ClassMethod

No	Method	Description
01	`SkillsAdminService(repository: SkillsAdminRepository)`	A method to initialize the service with the skill repository. – `repository: SkillsAdminRepository` – persistence dependency. – *Returns:* `void` – constructor, no return value.
02	`listSkills(query: SkillAdminQueryType)`	A method to load a skill page. – `query: SkillAdminQueryType` – list filters. – *Returns:* `Promise<SkillAdminListResponseType>` – skills and pagination. – Failure is mapped to `FailedToLoadSkillListException`.
03	`getSkillDetail(id: number)`	A method to load one skill. – `id: number` – skill identifier. – *Returns:* `Promise<SkillAdminDetailResponseType>` – the skill detail. – Throws `SkillAdminNotFoundException` or `FailedToLoadSkillDetailException`.

No	Method	Description
04	`createSkill(body: AdminCreateSkillBodyType)`	A method to create a skill and handle a concurrent duplicate. – `body: AdminCreateSkillBodyType` – name and optional description. – *Returns:* `Promise<SkillAdminDetailResponseType>` – the created skill. – Throws `SkillNameAlreadyExistsException` or `FailedToCreateSkillException`.
05	`updateSkill(id, body)`	A method to update a skill and handle a concurrent duplicate. – `id: number` – skill identifier. – `body: AdminUpdateSkillBodyType` – optional name, description, and active state. – *Returns:* `Promise<SkillAdminDetailResponseType>` – the updated skill. – Throws `SkillAdminNotFoundException`, `SkillNameAlreadyExistsException`, or `FailedToUpdateSkillException`.
06	`deleteSkill(id: number)`	A method to soft-delete a skill. – `id: number` – skill identifier. – *Returns:* `Promise<SkillAdminDeleteResponseType>` – a success message. – Throws `SkillAdminNotFoundException` or `FailedToDeleteSkillException`.
07	`restoreSkill(id: number)`	A method to restore a soft-deleted skill. – `id: number` – skill identifier. – *Returns:* `Promise<SkillAdminDetailResponseType>` – the restored skill. – Throws `SkillAdminNotFoundException`, `SkillNameAlreadyExistsException`, or `FailedToRestoreSkillException`.

4.40.2 ClassAttribute

No	Attribute	Type
01	`repository`	`SkillsAdminRepository` – private readonly persistence dependency.

4.41 SkillsAdminRepository Class

4.41.1 ClassMethod

No	Method	Description
01	`SkillsAdminRepository(prisma: PrismaService)`	A method to initialize the repository with the shared Prisma client. – `prisma`: `PrismaService` – database client. – *Returns:* `void` – constructor, no return value.
02	`listSkills(query: SkillAdminQueryType)`	A method to query skills with search, active-state, sorting, and pagination. – `query`: `SkillAdminQueryType` – list filters. – *Returns:* `Promise<SkillAdminListResponseType>` – skills and pagination. – Search matches name, slug, or description.
03	`getSkillDetail(id: number)`	A method to load one skill. – `id`: `number` – skill identifier. – *Returns:* `Promise<SkillAdminDetailResponseType | null>` – the skill, or `null`.
04	`createSkill(data: AdminCreateSkillBodyType)`	A method to generate a unique slug and insert a skill. – `data`: `AdminCreateSkillBodyType` – name and optional description. – *Returns:* `Promise<SkillAdminDetailResponseType>` – the created skill. – Throws `SkillNameAlreadyExistsException`.
05	`updateSkill(id, data)`	A method to update a skill and regenerate its slug when the name changes. – `id`: `number` – skill identifier. – `data`: `AdminUpdateSkillBodyType` – optional name, description, and active state. – *Returns:* `Promise<SkillAdminDetailResponseType>` – the updated skill. – Throws `SkillAdminNotFoundException` or `SkillNameAlreadyExistsException`.
06	`deleteSkill(id: number)`	A method to set the skill delete timestamp. – `id`: `number` – skill identifier. – *Returns:* `Promise<SkillAdminDeleteResponseType>` – a success message. – Throws `SkillAdminNotFoundException`.

No	Method	Description
07	`restoreSkill(id: number)`	A method to clear the skill delete timestamp after checking name conflicts. – `id`: `number` – skill identifier. – *Returns:* `Promise<SkillAdminDetailResponse Type>` – the restored skill. – Throws `SkillAdminNotFoundException` or `Skill NameAlreadyExistsException`.
08	`generateUniqueSlug(name: string)`	A private method to derive a slug and append a numeric suffix on collision. – `name`: `string` – skill name. – *Returns:* `Promise<string>` – the unique slug.

4.41.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.42 ReviewController Class

This shared class supports Review Management, Review Response Management. Its relevant operations are specified together here.

4.42.1 ClassMethod

No	Method	Description
01	`ReviewController(service: ReviewService)`	Initializes the class with its required dependencies. – `service`: `ReviewService` – injected dependency. – *Returns:* `void` – constructor, no return value.

No	Method	Description
02	`list(userId: number, contractId: number)`	Loads the records associated with the supplied contract or administrative filters. – `userId`: `number` – account identifier supplied by authentication or the route. – `contractId`: `number` – contract identifier. – *Returns:* An awaitable list of records with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields. – Response schema: `ReviewListDto`. – Route: `GET /reviews/contracts/:contractId`.
03	`create( userId: number, contractId: number, body: CreateReviewDto)`	Persists a new record using the supplied, validated fields. – `userId`: `number` – account identifier supplied by authentication or the route. – `contractId`: `number` – contract identifier. – `body`: `CreateReviewDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields. – Response schema: `ReviewDto`. – Route: `POST /reviews/contracts/:contractId`.
04	`detail(userId: number, reviewId: number)`	Loads one record and enforces the corresponding visibility rules. – `userId`: `number` – account identifier supplied by authentication or the route. – `reviewId`: `number` – review identifier. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields. – Response schema: `ReviewDto`. – Route: `GET /reviews/:reviewId`.

No	Method	Description
05	`update( userId: number, reviewId: number, body: UpdateReviewDto)`	Applies the supplied changes to the selected record. – `userId`: `number` – account identifier supplied by authentication or the route. – `reviewId`: `number` – review identifier. – `body`: `UpdateReviewDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields. – Response schema: `ReviewDto`. – Route: `PATCH /reviews/:reviewId`.
06	`remove(userId: number, reviewId: number)`	Removes the selected record after the relevant service checks. – `userId`: `number` – account identifier supplied by authentication or the route. – `reviewId`: `number` – review identifier. – *Returns:* `Promise<{ message: "Success.Review Deleted"; }>` – Route: `DELETE /reviews/:reviewId`.
07	`respond( userId: number, reviewId: number, body: CreateReview ResponseDto)`	Creates a response to a review addressed to the current user. – `userId`: `number` – account identifier supplied by authentication or the route. – `reviewId`: `number` – review identifier. – `body`: `CreateReviewResponseDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `user`, `id`, `userId`, `createdAt`, `updatedAt`, `deletedAt`, `reviewId`, and the remaining selected fields. – Response schema: `ReviewResponseDto`. – Route: `POST /reviews/:reviewId/response`.

No	Method	Description
08	`updateResponse( userId: number, responseId: number, body: UpdateReview ResponseDto)`	Updates a review response owned by the current user. – `userId`: `number` – account identifier supplied by authentication or the route. – `responseId`: `number` – review response identifier. – `body`: `UpdateReviewResponseDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `user`, `id`, `userId`, `createdAt`, `updatedAt`, `deletedAt`, `reviewId`, and the remaining selected fields. – Response schema: `ReviewResponseDto`. – Route: `PATCH /reviews/responses/:responseId`.
09	`removeResponse(userId: number, responseId: number)`	Removes a review response owned by the current user. – `userId`: `number` – account identifier supplied by authentication or the route. – `responseId`: `number` – review response identifier. – *Returns:* `Promise<{ message: "Success.Review ResponseDeleted"; }>` – Route: `DELETE /reviews/responses/:responseId`.

4.42.2 ClassAttribute

No	Attribute	Type
01	`service`	`ReviewService` – private readonly injected dependency.

4.43 ReviewService Class

This shared class supports Review Management, Review Response Management. Its relevant operations are specified together here.

4.43.1 ClassMethod

No	Method	Description
01	`ReviewService(repository: ReviewRepository)`	Initializes the class with its required dependencies. – `repository`: `ReviewRepository` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`list(userId: number, contractId: number)`	Loads the records associated with the supplied contract or administrative filters. – `userId`: `number` – account identifier for this operation. – `contractId`: `number` – contract identifier. – *Returns:* An awaitable list of records with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields.
03	`detail(userId: number, reviewId: number)`	Loads one record and enforces the corresponding visibility rules. – `userId`: `number` – account identifier for this operation. – `reviewId`: `number` – review identifier. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields. – Throws `ReviewNotFoundException`.
04	`create( userId: number, contractId: number, input: CreateReviewType)`	Persists a new record using the supplied, validated fields. – `userId`: `number` – account identifier for this operation. – `contractId`: `number` – contract identifier. – `input`: `CreateReviewType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields. – Throws `ReviewContractNotCompletedException`, `ReviewAlreadyExistsException`.
05	`update( userId: number, reviewId: number, input: UpdateReviewType)`	Applies the supplied changes to the selected record. – `userId`: `number` – account identifier for this operation. – `reviewId`: `number` – review identifier. – `input`: `UpdateReviewType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields. – Throws `ReviewNotFoundException`, `ReviewForbiddenException`.

No	Method	Description
06	`remove(userId: number, reviewId: number)`	Removes the selected record after the relevant service checks. – `userId`: `number` – account identifier for this operation. – `reviewId`: `number` – review identifier. – *Returns:* `Promise<{ message: "Success.Review Deleted"; }>` – Throws `ReviewNotFoundException`, `Review ForbiddenException`.
07	`respond( userId: number, reviewId: number, responseText: string)`	Creates a response to a review addressed to the current user. – `userId`: `number` – account identifier for this operation. – `reviewId`: `number` – review identifier. – `responseText`: `string` – response text supplied to the operation. – *Returns:* An awaitable record with `user`, `id`, `userId`, `createdAt`, `updatedAt`, `deletedAt`, `reviewId`, and the remaining selected fields. – Throws `ReviewNotFoundException`, `Review ForbiddenException`, `ReviewResponseAlready ExistsException`.
08	`updateResponse( userId: number, responseId: number, responseText: string)`	Updates a review response owned by the current user. – `userId`: `number` – account identifier for this operation. – `responseId`: `number` – review response identifier. – `responseText`: `string` – response text supplied to the operation. – *Returns:* An awaitable record with `user`, `id`, `userId`, `createdAt`, `updatedAt`, `deletedAt`, `reviewId`, and the remaining selected fields. – Throws `ReviewResponseNotFoundException`, `ReviewForbiddenException`.
09	`removeResponse(userId: number, responseId: number)`	Removes a review response owned by the current user. – `userId`: `number` – account identifier for this operation. – `responseId`: `number` – review response identifier. – *Returns:* `Promise<{ message: "Success.Review ResponseDeleted"; }>` – Throws `ReviewResponseNotFoundException`, `ReviewForbiddenException`.

4.43.2 ClassAttribute

No	Attribute	Type
01	`repository`	`ReviewRepository` – private readonly injected dependency.

4.44 ReviewRepository Class

This shared class supports Review Management, Review Response Management. Its relevant operations are specified together here.

4.44.1 ClassMethod

No	Method	Description
01	`ReviewRepository(prisma: PrismaService)`	Initializes the class with its required dependencies. – `prisma`: `PrismaService` – injected database client. – *Returns:* `void` – constructor, no return value.
02	`findContract(contractId: number)`	Loads contract status and participants for review eligibility and visibility checks. – `contractId`: `number` – contract identifier. – *Returns:* An awaitable record with `id`, `status`, `clientId`, `freelancerId`, `client`, `freelancer`; `null` when absent.
03	`findById(reviewId: number, includeDeleted?: boolean)`	Loads one review for visibility and ownership checks. – `reviewId`: `number` – review identifier. – `includeDeleted`: `boolean` – whether to include soft-deleted reviews; defaults to false; optional. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields; `null` when absent.
04	`findByContractAndReviewer( contractId: number, reviewerId: number)`	Finds the existing review identifier for a contract and reviewer pair. – `contractId`: `number` – contract identifier. – `reviewerId`: `number` – reviewer id supplied to the operation. – *Returns:* An awaitable record with `id`; `null` when absent.

No	Method	Description
05	`findManyByContract( contractId: number)`	Lists reviews associated with the contract. – `contractId`: `number` – contract identifier. – *Returns:* An awaitable list of records with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields.
06	`create( contractId: number, reviewerId: number, revieweeId: number, reviewerName: string, input: CreateReviewType)`	Persists a new record using the supplied, validated fields. – `contractId`: `number` – contract identifier. – `reviewerId`: `number` – reviewer id supplied to the operation. – `revieweeId`: `number` – reviewee id supplied to the operation. – `reviewerName`: `string` – reviewer name supplied to the operation. – `input`: `CreateReviewType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields.
07	`update(reviewId: number, input: UpdateReviewType)`	Applies the supplied changes to the selected record. – `reviewId`: `number` – review identifier. – `input`: `UpdateReviewType` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields.
08	`softDelete(reviewId: number)`	Persists the deletion timestamp for the specified record. – `reviewId`: `number` – review identifier. – *Returns:* An awaitable record with `id`, `createdAt`, `updatedAt`, `deletedAt`, `contractId`, `reviewerId`, `revieweeId`, and the remaining selected fields.
09	`findResponseByReview( reviewId: number)`	Finds the response identifier and deletion state for the specified review. – `reviewId`: `number` – review identifier. – *Returns:* An awaitable record with `id`, `deletedAt`; `null` when absent.

No	Method	Description
10	`findResponseById(responseId: number)`	Loads a response identifier, author, and parent review for authorization checks. – `responseId`: `number` – review response identifier. – *Returns:* An awaitable record with `id`, `userId`, `reviewId`; `null` when absent.
11	`createResponse(reviewId: number, userId: number, reviewerId: number, contractId: number, responderName: string, responseText: string, deletedResponseId?: number)`	Persists a review response and the notification for the reviewer. – `reviewId`: `number` – review identifier. – `userId`: `number` – account identifier for this operation. – `reviewerId`: `number` – reviewer id supplied to the operation. – `contractId`: `number` – contract identifier. – `responderName`: `string` – responder name supplied to the operation. – `responseText`: `string` – response text supplied to the operation. – `deletedResponseId`: `number` – deleted response id supplied to the operation; optional. – *Returns:* An awaitable record with `user`, `id`, `userId`, `createdAt`, `updatedAt`, `deletedAt`, `reviewId`, and the remaining selected fields.
12	`updateResponse(responseId: number, responseText: string)`	Updates a review response owned by the current user. – `responseId`: `number` – review response identifier. – `responseText`: `string` – response text supplied to the operation. – *Returns:* An awaitable record with `user`, `id`, `userId`, `createdAt`, `updatedAt`, `deletedAt`, `reviewId`, and the remaining selected fields.
13	`softDeleteResponse(responseId: number)`	Persists a deletion timestamp on a review response. – `responseId`: `number` – review response identifier. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `deletedAt`, `reviewId`, `responseText`.

4.44.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.45 GeneralAccountProfileController Class

4.45.1 ClassMethod

No	Method	Description
01	`GeneralAccountProfileController(service: AccountProfileService)`	Initializes the class with its required dependencies. – `service`: `AccountProfileService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`get(userId: number)`	Loads general account and profile information for the authenticated user. – `userId`: `number` – account identifier supplied by authentication or the route. – *Returns:* An awaitable record with `email`, `avatarUrl`, `roles`, `id`, `userId`, `displayName`, `coverUrl`, and the remaining selected fields. – Response schema: `GeneralProfileDto`. – Route: `GET /account-profile`.
03	`update(userId: number, body: UpdateGeneralProfileDto)`	Submits or applies edits to general profile information according to the revision policy. – `userId`: `number` – account identifier supplied by authentication or the route. – `body`: `UpdateGeneralProfileDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `message`, `reviewRequired`, `profileStrength`, `revision`. – Response schema: `ProfileRevisionSubmissionDto`. – Route: `PUT /account-profile`.

No	Method	Description
04	changePassword(userId: number, body: ChangePasswordDto)	Verifies the existing password and stores the newly hashed password. – userId: number – account identifier supplied by authentication or the route. – body: ChangePasswordDto – validated input fields defined by the declared type. – *Returns:* Promise<{ message: string; }> – Route: PUT /account-profile/password.
05	uploadAvatar(userId: number, file?: Express.Multer.File)	Validates and stores an avatar file, then updates the profile reference. – userId: number – account identifier supplied by authentication or the route. – file: Express.Multer.File – uploaded file supplied to this operation; optional. – *Returns:* Promise<{ avatarUrl: string; }> – Response schema: AvatarUploadResponseDto. – Route: POST /account-profile/avatar.
06	getAvatarFile(userId: number, response: Response)	Resolves the stored avatar for delivery. – userId: number – account identifier supplied by authentication or the route. – response: Response – Express response used to set file headers or redirect. – *Returns:* Promise<StreamableFile> – Route: GET /account-profile/avatar/:userId/file.

4.45.2 ClassAttribute

No	Attribute	Type
01	service	AccountProfileService – private readonly injected dependency.

4.46 Account Profile Management (Shared Classes)

AccountProfileService is specified in Section 4.22.

- getGeneralProfile().
- updateGeneralProfile().
- changePassword().

- `uploadAvatar()`.
- `getAvatarFile()`.

  `AccountProfileRepository` is specified in Section 4.23.

- `findGeneralProfile()`.
- `findPasswordCredential()`.
- `updateGeneralProfile()`.
- `updatePassword()`.
- `updateAvatar()`.

4.47 AdminExpertProfileController Class

4.47.1 ClassMethod

No	Method	Description
01	`AdminExpertProfile Controller(service: ExpertProfileService)`	A method to initialize the controller with the expert profile service. – `service`: `ExpertProfileService` – service that owns expert profile use cases. – *Returns:* `void` – constructor, no return value.
02	`update( userId: number, adminId: number, body: AdminUpdateExpert ProfileDto)`	A method for an administrator to update an expert profile directly. – `userId`: `number` – expert account identifier. – `adminId`: `number` – current administrator, recorded on superseded revisions. – `body`: `AdminUpdateExpertProfileDto` – display name, bio, title, expertise, experience, education, certifications, website, and active state. – *Returns:* `Promise<MessageResType>` – a success message. – Route: `PATCH /users/:id/expert-profile`.

4.47.2 ClassAttribute

No	Attribute	Type
01	`service`	`ExpertProfileService` – private readonly expert profile service.

4.48 ExpertProfileService Class

4.48.1 ClassMethod

No	Method	Description
01	`ExpertProfileService( repository: ExpertProfileRepository, revisions: ProfileRevisionService)`	A method to initialize the service with persistence and revision handling. – `repository`: `ExpertProfileRepository` – expert profile persistence. – `revisions`: `ProfileRevisionService` – self-service revision workflow. – *Returns:* `void` – constructor, no return value.
02	`cleanList(values: string[])`	A private method to trim, de-duplicate, and preserve the first casing of a string list. – `values`: `string[]` – raw list values. – *Returns:* `string[]` – the cleaned list.
03	`sanitize(input)`	A private method to clean list fields and normalize the website value. – `input`: `UpdateExpertProfileType` – profile input. – *Returns:* `UpdateExpertProfileType` – the sanitized input.
04	`context(userId: number)`	A private method to load an expert account and verify its Expert role. – `userId`: `number` – target account. – *Returns:* `Promise<user context>` – account, profile, and roles. – Throws `ExpertProfileNotFoundException` or `NotAnExpertAccountException`.
05	`getMine(userId: number)`	A method to return the authenticated expert's profile. – `userId`: `number` – current expert account. – *Returns:* `Promise<ExpertProfileType>` – profile and expert fields. – Used by the self-service profile page, not by the admin endpoint.

No	Method	Description
06	`submitUpdate(userId, input)`	A method to submit a self-service expert profile revision. – `userId`: `number` – current expert account. – `input`: `UpdateExpertProfileType` – requested profile changes. – *Returns:* `Promise<revision result>` – the pending revision. – Throws `ExpertProfilePendingRevision Exception` when a revision is already pending.
07	`updateByAdmin(userId, adminId, input)`	A method to update an expert profile directly and supersede pending revisions. – `userId`: `number` – target expert account. – `adminId`: `number` – current administrator. – `input`: `AdminUpdateExpertProfileType` – administrative profile changes. – *Returns:* `Promise<MessageResType>` – a success message. – Throws `ExpertProfileNotFoundException` or `Not AnExpertAccountException`.

4.48.2 ClassAttribute

No	Attribute	Type
01	`repository`	`ExpertProfileRepository` – private readonly persistence dependency.
02	`revisions`	`ProfileRevisionService` – private readonly revision workflow dependency.

4.49 ExpertProfileRepository Class

4.49.1 ClassMethod

No	Method	Description
01	`ExpertProfileRepository( prisma: PrismaService)`	A method to initialize the repository with the shared Prisma client. – `prisma`: `PrismaService` – database client. – *Returns:* `void` – constructor, no return value.
02	`findByUserId(userId: number)`	A method to load an expert account with its profile and roles. – `userId`: `number` – target account. – *Returns:* `Promise<user context | null>` – the account graph, or `null`.
03	`updateDirect(userId, adminId, input, profile CompletionPercent)`	A method to supersede pending revisions, update the profile, and upsert the expert row in one transaction. – `userId`: `number` – target expert account. – `adminId`: `number` – reviewing administrator. – `input`: `AdminUpdateExpertProfileType` – sanitized profile changes. – `profileCompletionPercent`: `number` – recalculated completeness. – *Returns:* `Promise<void>` – no value. – Creates a system notification when a pending revision was superseded.
04	`hasPendingRevision(user Id: number)`	A method to check for a pending expert revision. – `userId`: `number` – target account. – *Returns:* `Promise<{ id: number } | null>` – the pending revision, or `null`.
05	`findPublicMany(query: PublicExpertQueryType)`	A method to query active public expert profiles. – `query`: `PublicExpertQueryType` – expertise, search, and pagination filters. – *Returns:* `Promise<public expert page>` – public expert cards.
06	`findPublicById(id: number)`	A method to load one active public expert profile. – `id`: `number` – expert identifier. – *Returns:* `Promise<public expert | null>` – the public profile, or `null`.

4.49.2 ClassAttribute

No	Attribute	Type
01	prisma	PrismaService – private readonly database client.

4.50 JobCategoriesAdminController Class

4.50.1 ClassMethod

No	Method	Description
01	JobCategoriesAdmin Controller(service: Job CategoriesAdminService)	A method to initialize the controller with the job category service. – service: JobCategoriesAdminService – service that owns job category use cases. – *Returns:* void – constructor, no return value.
02	listJobCategories(query: JobCategoryAdminQueryDto)	A method to return a filtered, paginated job category list. – query: JobCategoryAdminQueryDto – page, limit, search, status, deleted, and sort options. – *Returns:* Promise<JobCategoryAdminList ResponseType> – categories and pagination. – Route: GET /admin/job-categories.
03	createJobCategory(body: CreateJobCategoryBodyDto)	A method to create a job category. – body: CreateJobCategoryBodyDto – name, description, icon, and status. – *Returns:* Promise<JobCategoryAdminDetail ResponseType> – the created category. – Route: POST /admin/job-categories.
04	updateJobCategory(id: number, body: UpdateJob CategoryBodyDto)	A method to update a job category. – id: number – category identifier. – body: UpdateJobCategoryBodyDto – optional category fields. – *Returns:* Promise<JobCategoryAdminDetail ResponseType> – the updated category. – Route: PATCH /admin/job-categories/:id.
05	restoreJobCategory(id: number)	A method to restore a soft-deleted job category. – id: number – category identifier. – *Returns:* Promise<JobCategoryAdminDetail ResponseType> – the restored category. – Route: PATCH /admin/job-categories/:id/ restore.

No	Method	Description
06	`deleteJobCategory(id: number)`	A method to soft-delete a job category that has no active jobs. – `id: number` – category identifier. – *Returns:* `Promise<JobCategoryAdminDeleteResponseType>` – a success message. – Route: `DELETE /admin/job-categories/:id`.
07	`getJobCategoryDetail(id: number)`	A method to return one job category. – `id: number` – category identifier. – *Returns:* `Promise<JobCategoryAdminDetailResponseType>` – the category detail. – Route: `GET /admin/job-categories/:id`.

4.50.2 ClassAttribute

No	Attribute	Type
01	`service`	`JobCategoriesAdminService` – private readonly job category service.

4.51 JobCategoriesAdminService Class

4.51.1 ClassMethod

No	Method	Description
01	`JobCategoriesAdminService( repository: JobCategoriesAdminRepository)`	A method to initialize the service with the job category repository. – `repository: JobCategoriesAdminRepository` – persistence dependency. – *Returns:* `void` – constructor, no return value.
02	`listJobCategories(query: JobCategoryAdminQueryType)`	A method to load a job category page. – `query: JobCategoryAdminQueryType` – list filters. – *Returns:* `Promise<JobCategoryAdminListResponseType>` – categories and pagination. – Failure is mapped to `FailedToLoadJobCategoryListException`.

No	Method	Description
03	`getJobCategoryDetail(id: number)`	A method to load one job category. – `id: number` – category identifier. – *Returns:* `Promise<JobCategoryAdminDetail ResponseType>` – the category detail. – Throws `JobCategoryAdminNotFoundException` or `FailedToLoadJobCategoryDetailException`.
04	`createJobCategory(body: AdminCreateJobCategory BodyType)`	A method to create a category and handle a concurrent duplicate. – `body: AdminCreateJobCategoryBodyType` – category fields. – *Returns:* `Promise<JobCategoryAdminDetail ResponseType>` – the created category. – Throws `JobCategoryNameAlreadyExists Exception` or `FailedToCreateJobCategory Exception`.
05	`updateJobCategory(id, body)`	A method to update a category and handle a concurrent duplicate. – `id: number` – category identifier. – `body: AdminUpdateJobCategoryBodyType` – optional category fields. – *Returns:* `Promise<JobCategoryAdminDetail ResponseType>` – the updated category. – Throws `JobCategoryAdminNotFoundException`, `JobCategoryNameAlreadyExistsException`, or `FailedToUpdateJobCategoryException`.
06	`deleteJobCategory(id: number)`	A method to soft-delete a category after checking active job references. – `id: number` – category identifier. – *Returns:* `Promise<JobCategoryAdminDelete ResponseType>` – a success message. – Throws `JobCategoryAdminNotFoundException`, `JobCategoryHasJobsException`, or `FailedToDelete JobCategoryException`.

No	Method	Description
07	`restoreJobCategory(id: number)`	A method to restore a soft-deleted category. – `id: number` – category identifier. – *Returns:* `Promise<JobCategoryAdminDetail ResponseType>` – the restored category. – Throws `JobCategoryAdminNotFoundException`, `JobCategoryNameAlreadyExistsException`, or `FailedToRestoreJobCategoryException`.

4.51.2 ClassAttribute

No	Attribute	Type
01	`repository`	`JobCategoriesAdminRepository` – private readonly persistence dependency.

4.52 JobCategoriesAdminRepository Class

4.52.1 ClassMethod

No	Method	Description
01	`JobCategoriesAdmin Repository(prisma: PrismaService)`	A method to initialize the repository with the shared Prisma client. – `prisma: PrismaService` – database client. – *Returns:* `void` – constructor, no return value.
02	`listJobCategories(query: JobCategoryAdminQuery Type)`	A method to query categories with search, status, deleted-state, sorting, and pagination. – `query: JobCategoryAdminQueryType` – list filters. – *Returns:* `Promise<JobCategoryAdminList ResponseType>` – categories and pagination. – Search matches name, slug, or description.
03	`getJobCategoryDetail(id: number)`	A method to load one category with its job count. – `id: number` – category identifier. – *Returns:* `Promise<JobCategoryAdminDetail ResponseType | null>` – the category, or `null`.

No	Method	Description
04	`createJobCategory(data)`	A method to generate a unique slug and insert a category. – `data`: `AdminCreateJobCategoryBodyType` – category fields. – *Returns:* `Promise<JobCategoryAdminDetailResponseType>` – the created category. – Throws `JobCategoryNameAlreadyExistsException`.
05	`updateJobCategory(id, data)`	A method to update a category and regenerate its slug when the name changes. – `id`: `number` – category identifier. – `data`: `AdminUpdateJobCategoryBodyType` – optional category fields. – *Returns:* `Promise<JobCategoryAdminDetailResponseType>` – the updated category. – Throws `JobCategoryAdminNotFoundException` or `JobCategoryNameAlreadyExistsException`.
06	`deleteJobCategory(id: number)`	A method to check active job references and set the delete timestamp. – `id`: `number` – category identifier. – *Returns:* `Promise<JobCategoryAdminDeleteResponseType>` – a success message. – Throws `JobCategoryAdminNotFoundException` or `JobCategoryHasJobsException`.
07	`restoreJobCategory(id: number)`	A method to clear the delete timestamp after checking name conflicts. – `id`: `number` – category identifier. – *Returns:* `Promise<JobCategoryAdminDetailResponseType>` – the restored category. – Throws `JobCategoryAdminNotFoundException` or `JobCategoryNameAlreadyExistsException`.
08	`generateUniqueSlug(name: string)`	A private method to derive a slug and append a numeric suffix on collision. – `name`: `string` – category name. – *Returns:* `Promise<string>` – the unique slug.

4.52.2 ClassAttribute

No	Attribute	Type
01	`prisma`	`PrismaService` – private readonly database client.

4.53 ProfileRevisionController Class

4.53.1 ClassMethod

No	Method	Description
01	`ProfileRevision Controller(service: ProfileRevisionService)`	Initializes the class with its required dependencies. – `service`: `ProfileRevisionService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`latest(userId: number, profileType: Profile RevisionType)`	Returns the most recent revision for a user and profile type. – `userId`: `number` – account identifier supplied by authentication or the route. – `profileType`: `ProfileRevisionType` – role-specific profile category. – *Returns:* An awaitable record with `revision`. – Response schema: `MyProfileRevisionDto`. – Route: `GET /profile-revisions/me`.

4.53.2 ClassAttribute

No	Attribute	Type
01	`service`	`ProfileRevisionService` – private readonly injected dependency.

4.54 ProfileRevisionAdminController Class

4.54.1 ClassMethod

No	Method	Description
01	`ProfileRevisionAdmin Controller( service: ProfileRevision Service)`	Initializes the class with its required dependencies. – `service`: `ProfileRevisionService` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`list(query: Profile RevisionAdminFilterDto)`	Synchronizes low-strength profiles, then loads filtered revisions and pagination metadata. – `query`: `ProfileRevisionAdminFilterDto` – query criteria, including the filters and pagination fields declared by the type. – *Returns:* An awaitable record with `revisions`, `pagination`. – Response schema: `ProfileRevisionAdminListDto`. – Route: `GET /admin/profile-revisions`.
03	`detail(id: number)`	Loads revision details and fails when the revision does not exist. – `id`: `number` – target record identifier. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `user`, `profileId`, `profile Type`, and the remaining selected fields. – Response schema: `ProfileRevisionDetailDto`. – Route: `GET /admin/profile-revisions/:id`.
04	`approve( id: number, adminId: number, body: ApproveProfile RevisionDto)`	Approves a pending revision and applies its proposed profile changes. – `id`: `number` – target record identifier. – `adminId`: `number` – administrator identifier recorded for the review. – `body`: `ApproveProfileRevisionDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `user`, `profileId`, `profile Type`, and the remaining selected fields. – Response schema: `ProfileRevisionDetailDto`. – Route: `PATCH /admin/profile-revisions/:id/ approve`.

No	Method	Description
05	`reject( id: number, adminId: number, body: RejectProfile RevisionDto)`	Rejects a pending revision with review notes and preserves the current profile. – `id`: `number` – target record identifier. – `adminId`: `number` – administrator identifier recorded for the review. – `body`: `RejectProfileRevisionDto` – validated input fields defined by the declared type. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `user`, `profileId`, `profile Type`, and the remaining selected fields. – Response schema: `ProfileRevisionDetailDto`. – Route: `PATCH /admin/profile-revisions/:id/ reject`.

4.54.2 ClassAttribute

No	Attribute	Type
01	`service`	`ProfileRevisionService` – private readonly injected dependency.

4.55 ProfileRevisionService Class

4.55.1 ClassMethod

No	Method	Description
01	`ProfileRevisionService( repository: Profile RevisionRepository)`	Initializes the class with its required dependencies. – `repository`: `ProfileRevisionRepository` – injected dependency. – *Returns:* `void` – constructor, no return value.
02	`requiresManualReview( profileStrength: number)`	Compares profile strength with the manual-review threshold. – `profileStrength`: `number` – calculated profile completion percentage. – *Returns:* `boolean`

No	Method	Description
03	`directUpdateResult( profileStrength: number)`	Builds the result of an update that does not require manual review. – `profileStrength`: `number` – calculated profile completion percentage. – *Returns:* `{ message: "Success.Profile Updated"; reviewRequired: boolean; profile Strength: number; revision: null; }`
04	`submitFreelancer( userId: number, profileId: number, proposedData: Update FreelancerProfileType, profileStrength: number)`	Submits proposed freelancer-profile fields for review. – `userId`: `number` – account identifier for this operation. – `profileId`: `number` – general profile identifier. – `proposedData`: `UpdateFreelancerProfileType` – proposed role-specific profile changes. – `profileStrength`: `number` – calculated profile completion percentage. – *Returns:* An awaitable record with `message`, `review Required`, `profileStrength`, `revision`.
05	`submitClient( userId: number, profileId: number, proposedData: Update ClientProfileType, profileStrength: number)`	Submits proposed client-profile fields for review. – `userId`: `number` – account identifier for this operation. – `profileId`: `number` – general profile identifier. – `proposedData`: `UpdateClientProfileType` – proposed role-specific profile changes. – `profileStrength`: `number` – calculated profile completion percentage. – *Returns:* An awaitable record with `message`, `review Required`, `profileStrength`, `revision`.
06	`submitExpert( userId: number, profileId: number, proposedData: Update ExpertProfileType, profileStrength: number)`	Submits proposed expert-profile fields for review. – `userId`: `number` – account identifier for this operation. – `profileId`: `number` – general profile identifier. – `proposedData`: `UpdateExpertProfileType` – proposed role-specific profile changes. – `profileStrength`: `number` – calculated profile completion percentage. – *Returns:* An awaitable record with `message`, `review Required`, `profileStrength`, `revision`.

No	Method	Description
07	`submitGeneral( userId: number, profileId: number, primaryRole: string, proposedData: UpdateGeneralProfileType, profileStrength: number)`	Submits proposed general-profile fields for review. – `userId`: `number` – account identifier for this operation. – `profileId`: `number` – general profile identifier. – `primaryRole`: `string` – primary role used to select the profile policy. – `proposedData`: `UpdateGeneralProfileType` – proposed role-specific profile changes. – `profileStrength`: `number` – calculated profile completion percentage. – *Returns:* An awaitable record with `message`, `reviewRequired`, `profileStrength`, `revision`.
08	`latestForUser( userId: number, profileType: ProfileRevisionType)`	Returns the most recent revision for a user and profile type. – `userId`: `number` – account identifier for this operation. – `profileType`: `ProfileRevisionType` – role-specific profile category. – *Returns:* An awaitable record with `revision`.
09	`list(filter: ProfileRevisionAdminFilterType)`	Synchronizes low-strength profiles, then loads filtered revisions and pagination metadata. – `filter`: `ProfileRevisionAdminFilterType` – query criteria, including the filters and pagination fields declared by the type. – *Returns:* An awaitable record with `revisions`, `pagination`.
10	`detail(id: number)`	Loads revision details and fails when the revision does not exist. – `id`: `number` – target record identifier. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `user`, `profileId`, `profileType`, and the remaining selected fields. – Throws `ProfileRevisionNotFoundException`.

No	Method	Description
11	`approve( id: number, adminId: number, reviewNotes?: string | null)`	Approves a pending revision and applies its proposed profile changes. – `id`: `number` – target record identifier. – `adminId`: `number` – administrator identifier recorded for the review. – `reviewNotes`: `string | null` – administrator review comments; optional. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `user`, `profileId`, `profile Type`, and the remaining selected fields.
12	`reject(id: number, admin Id: number, reviewNotes: string)`	Rejects a pending revision with review notes and preserves the current profile. – `id`: `number` – target record identifier. – `adminId`: `number` – administrator identifier recorded for the review. – `reviewNotes`: `string` – administrator review comments. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `user`, `profileId`, `profile Type`, and the remaining selected fields.

4.55.2 ClassAttribute

No	Attribute	Type
01	`repository`	`ProfileRevisionRepository` – private readonly injected dependency.
02	`MANUAL_REVIEW_STRENGTH_THRESHOLD`	`number` – static readonly threshold, currently 20.

4.56 ProfileRevisionRepository Class

4.56.1 ClassMethod

No	Method	Description
01	`ProfileRevision Repository(prisma: PrismaService)`	Initializes the class with its required dependencies. – `prisma`: `PrismaService` – injected database client. – *Returns:* `void` – constructor, no return value.
02	`findProfileForSnapshot( userId: number, profile Id: number)`	Loads a profile owned by the account, including role-specific fields used to snapshot current data. – `userId`: `number` – account identifier for this operation. – `profileId`: `number` – general profile identifier. – *Returns:* An awaitable record with `freelancer Profile`, `clientProfile`, `expertProfile`, `id`, `user Id`, `displayName`, `avatarUrl`, and the remaining selected fields; `null` when absent.
03	`submit( userId: number, profileId: number, profileType: Profile RevisionType, currentData: Prisma. InputJsonValue, proposedData: Prisma. InputJsonValue, profileStrength: number)`	Saves the current and proposed profile data as a pending revision and supersedes existing pending requests. – `userId`: `number` – account identifier for this operation. – `profileId`: `number` – general profile identifier. – `profileType`: `ProfileRevisionType` – role-specific profile category. – `currentData`: `Prisma.InputJsonValue` – snapshot of the current live profile. – `proposedData`: `Prisma.InputJsonValue` – proposed role-specific profile changes. – `profileStrength`: `number` – calculated profile completion percentage. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `user`, `profileId`, `profile Type`, and the remaining selected fields.
04	`findLatestForUser( userId: number, profileType: Profile RevisionType)`	Loads the latest revision for the account and profile type. – `userId`: `number` – account identifier for this operation. – `profileType`: `ProfileRevisionType` – role-specific profile category. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `user`, `profileId`, `profile Type`, and the remaining selected fields; `null` when absent.
05	`syncLowStrengthProfiles( threshold: number)`	Ensures low-strength profiles have pending review requests where required. – `threshold`: `number` – profile strength threshold for manual review. – *Returns:* `Promise<void>`

No	Method	Description
06	`findMany( params: { page: number; limit: number; status?: RevisionStatus; profile Type?: ProfileRevision Type; search?: string; } )`	Loads a filtered page of revisions and the total matching count. – `params: { page: number; limit: number; status?: RevisionStatus; profileType?: ProfileRevisionType; search?: string; }` – query criteria, including the filters and pagination fields declared by the type. – *Returns:* An awaitable record with `revisions`, `total`.
07	`findById(id: number)`	Loads a revision with its account and profile summary. – `id: number` – target record identifier. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `user`, `profileId`, `profile Type`, and the remaining selected fields; `null` when absent.
08	`review( id: number, adminId: number, status: typeof Revision Status.APPROVED | typeof RevisionStatus.REJECTED, reviewNotes: string | null, proposedData: Freelancer Proposal | ClientProposal | ExpertProposal, profileType: Profile RevisionType)`	Processes a pending revision transactionally; approval applies proposed profile fields, while rejection preserves the live profile. – `id: number` – target record identifier. – `adminId: number` – administrator identifier recorded for the review. – `status: typeof RevisionStatus.APPROVED | typeof RevisionStatus.REJECTED` – target revision review status. – `reviewNotes: string | null` – administrator review comments. – `proposedData: FreelancerProposal | Client Proposal | ExpertProposal` – proposed role-specific profile changes. – `profileType: ProfileRevisionType` – role-specific profile category. – *Returns:* An awaitable record with `id`, `userId`, `createdAt`, `updatedAt`, `user`, `profileId`, `profile Type`, and the remaining selected fields; `null` when absent.

4.56.2 ClassAttribute

No	Attribute	Type
01	prisma	PrismaService – private readonly database client.

4.57 ExpertProfileController Class

4.57.1 ClassMethod

No	Method	Description
01	ExpertProfileController(service: ExpertProfile Service)	Initializes the class with its required dependencies. – service: ExpertProfileService – injected dependency. – *Returns:* void – constructor, no return value.
02	getMine(userId: number)	Loads the current expert profile and calculates the fields exposed by the self-service profile response. – userId: number – account identifier supplied by authentication or the route. – *Returns:* An awaitable record with id, userId, displayName, avatarUrl, bio, profileCompletion Percent, title, and the remaining selected fields. – Response schema: ExpertProfileDto. – Route: GET /expert-profile/me.
03	updateMine(userId: number, body: Update ExpertProfileDto)	Validates the expert profile and submits proposed changes through the profile revision workflow. – userId: number – account identifier supplied by authentication or the route. – body: UpdateExpertProfileDto – validated input fields defined by the declared type. – *Returns:* An awaitable record with message, review Required, profileStrength, revision. – Response schema: ExpertProfileUpdateResponse Dto. – Route: PUT /expert-profile/me.

4.57.2 ClassAttribute

No	Attribute	Type
01	service	ExpertProfileService – private readonly injected dependency.

4.58 Expert Profile Update (Shared Classes)

ExpertProfileService is specified in Expert Profile Administration in Sprint 5.

- getMine().
- submitUpdate().
- updateByAdmin().

ExpertProfileRepository is specified in Expert Profile Administration in Sprint 5.

- findByUserId().
- updateDirect().
- hasPendingRevision().
- findPublicMany().
- findPublicById().

4.59 Freelancer CV Management (Shared Classes)

ProfileController is specified in Section 4.8.

- uploadCv().
- getCvFile().
- analyzeCv().
- deleteCv().

ProfileService is specified in Section 4.9.

- uploadCv().
- deleteCv().
- getCvFile().
- analyzeCv().

ProfileRepository is specified in Section 4.10.

- findFreelancerProfileById().
- updateCv().
- clearCv().